With pain in your heart…

It is rare that someone would not know Mr Saros, the car mechanic professional living in the small town of Borvar, in Southern Hungary. This fame is because, in the fast-developing city, there are just two oversized garages next to several tiny businesses of questionable service quality and legal background; he owns one of those garages. Both sites offer a wide range of services from car electronics to mechanical repairs and two-year official technical supervision needed for licencing. Mr Saros now has to retire due to a health issue. He is forced to pass on the business he created "without a single cent of debt with his own hands". The building stands on a corner lot far more immense than needed at a location of very intensive traffic.

Among the family members, only **Janos** (John), the 20-year-old grandson of Mr Saros with basic training in car mechanics, is interested in owning the business. However, as he is just one of the seven grandchildren of Mr Saros, the "Maestro" would give him the company only if Janos compensated the other six grandchildren adequately. Janos, who possesses no mentionable capital, agrees by saying: "Do not worry, Grandpa. We show how much the building, the machines, and the tools are worth now. Then, as the business prospers, I will pay the proportional amount to all the others during the next 5-6 years."

Tamas (Thomas), who was the second-in-command next to Mr Saros during the last decade, is also interested. As the "Maestro" had to visit doctors more and more often, he took over the operative management nearly completely. Still, it was always Mr Saros to say the last word regarding strategic decisions. Tamas believes that he can get a loan for the purchase if needed with the help of his uncle supervising the local bank affiliate.

Once the local newspaper published that the "highly respected and well-known" Mr Saros was about to retire, a **tire repair network** with a country-wide chain asked for a meeting with the family. They plan to set up a tire repair and storage facility at the site.

Two top managers of **WellBuilt Co.** recently showed up in the city. They negotiated with several landowners, among them Mr Saros. Rumours say that the firm aims to build a shopping mall like in other nearby cities.

The municipality is troubled about the future of the Saros Garage as many students of the local secondary school of mechanics regularly go for an internship at the garage. The major stated to a journalist that the municipality would even "be ready for sacrifices" to upkeep the current level of education, though their "possibilities are minimal".

1. Which possible buyers do you believe would offer the highest sum for the firm?
2. How would you choose among the offers if you were Mr Saros?
3. What other parties could be interested in buying the business?
4. How would your answers change if you learned that the real estate was only rented to the firm by Mr Saros and legally consisted of two separate real estates?
5. How would the situation be different if the real estate were rented from the municipality and not from the owner?
6. How would the choice differ if a bank were the shop's owner due to a liquidation process?

Price, value, valuation

Péter Juhász, PhD, CFA

For sale: Successful cheesecake cafe in Budapest

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Full Sale

Asking Price: HUF 100 million (Native Currency: HUF 100,000,000)

Reason: Owners are relocating.

☑ Includes physical assets worth HUF 100 million

☑ Interested to connect with advisors

Established	5-10 year(s)
Employees	5 - 10
Legal Entity	Private Limited Company
Reported Sales	HUF 100 million
Run Rate Sales	HUF 90 million
EBITDA Margin	20 %

https://www.smergers.com/business/cafe-for-sale-in-budapest-hungary/1dnv5/

Prescriptive (normative) science

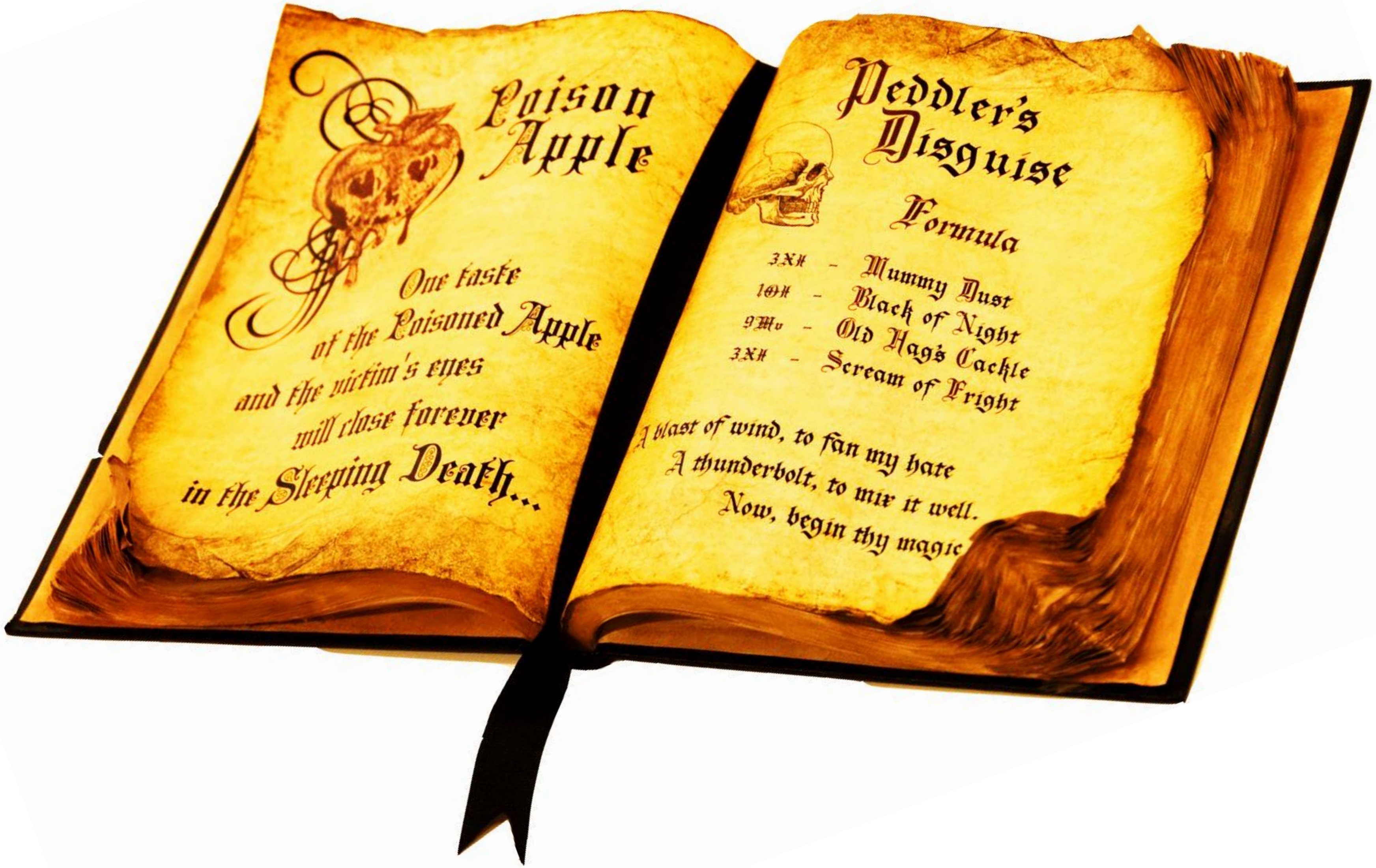
- „That has to be like this." (mathematics)

Descriptive science

- „That is like this." (botany, sociology)

This semester is a compromise.

On the stories



How are those different?

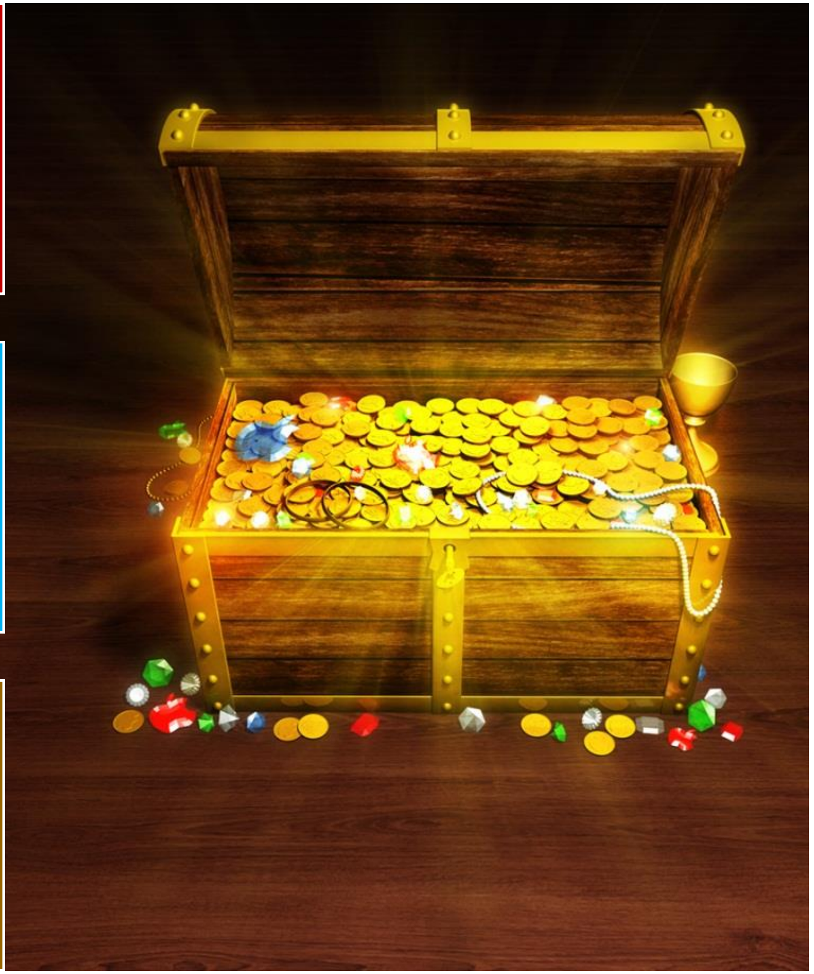
Project valuation	Business valuation	Equity valuation
Today	At a given date	In 1 year (target price)
Limited horizon	Endless operation	Endless operation
One single risk	Multitude of risks	Risk of market pricing
Cross-effects	Financial statements	Technical analysis
Accept or refuse?	**How much?**	**Sell or buy?**

~~Fair market value~~

Realistic transaction price range

What creates value (utility)?

Money	Problem-solving
Joy of work	Respecting traditions
Recognition	Power



Value for the customer

Pays the **price**, receives **utility**.

Alternatives: competitive solutions.

Value for the shareholders?

Provides **capital** and receives **return.**

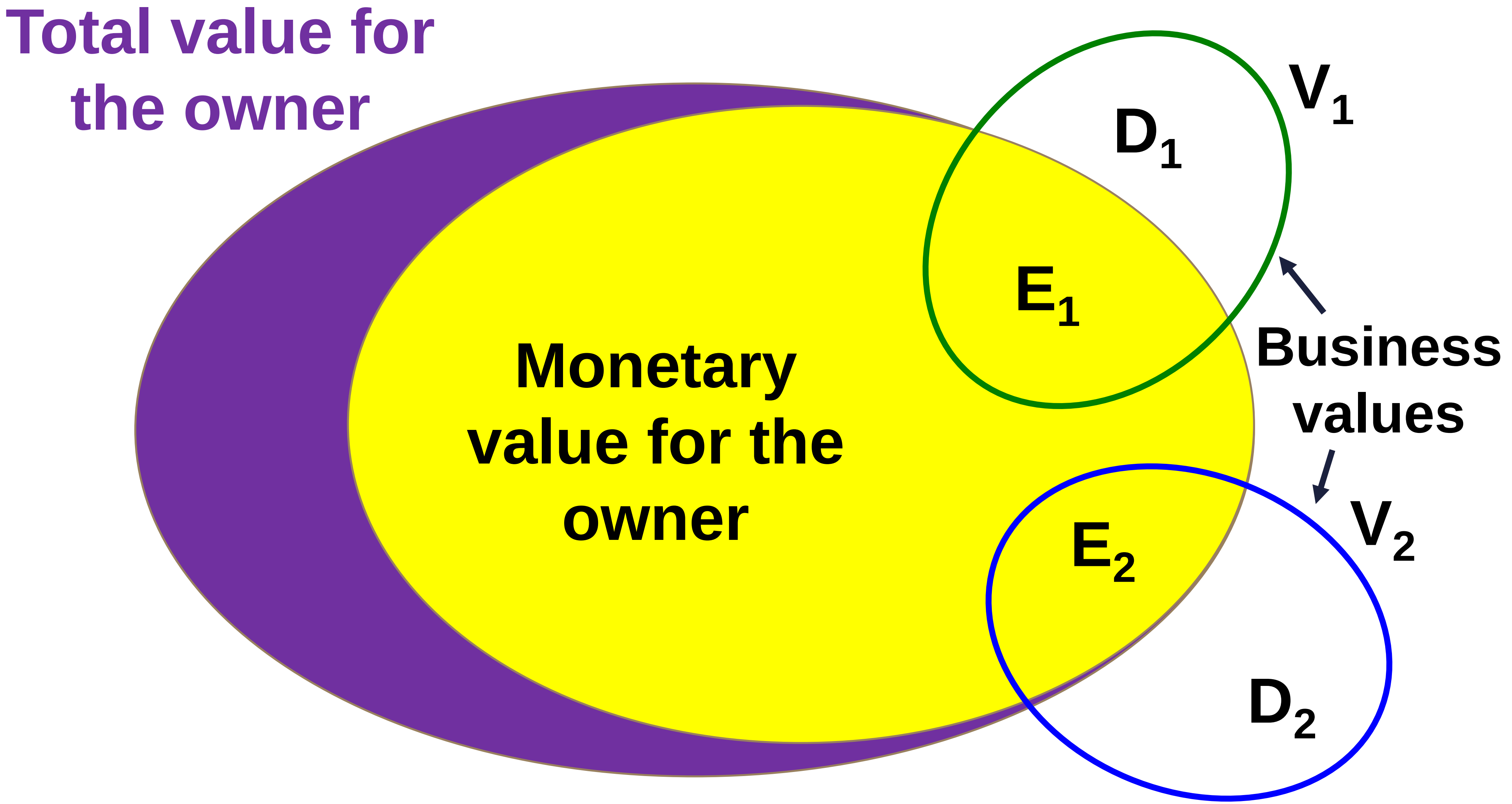
Alternatives: competitive investment opportunities.

Value for the stakeholders?

We only measure in money from the aspect of the owners.

„The value like beauty is in the eye of the beholder."

Value dimensions



Why do we need the value?

- Share trades
- IPO
- Privatisation
- Strategic decisions
- Management performance
- M&A
- Taxation
- Inheritance, divorce

Who needs to understand business valuation?

- Equity analysts
- Investment bankers
- Private equity professionals
- Fund managers
- Privatisation officers
- Business owners
- All top managers

What is worth to study?

The process of valuation

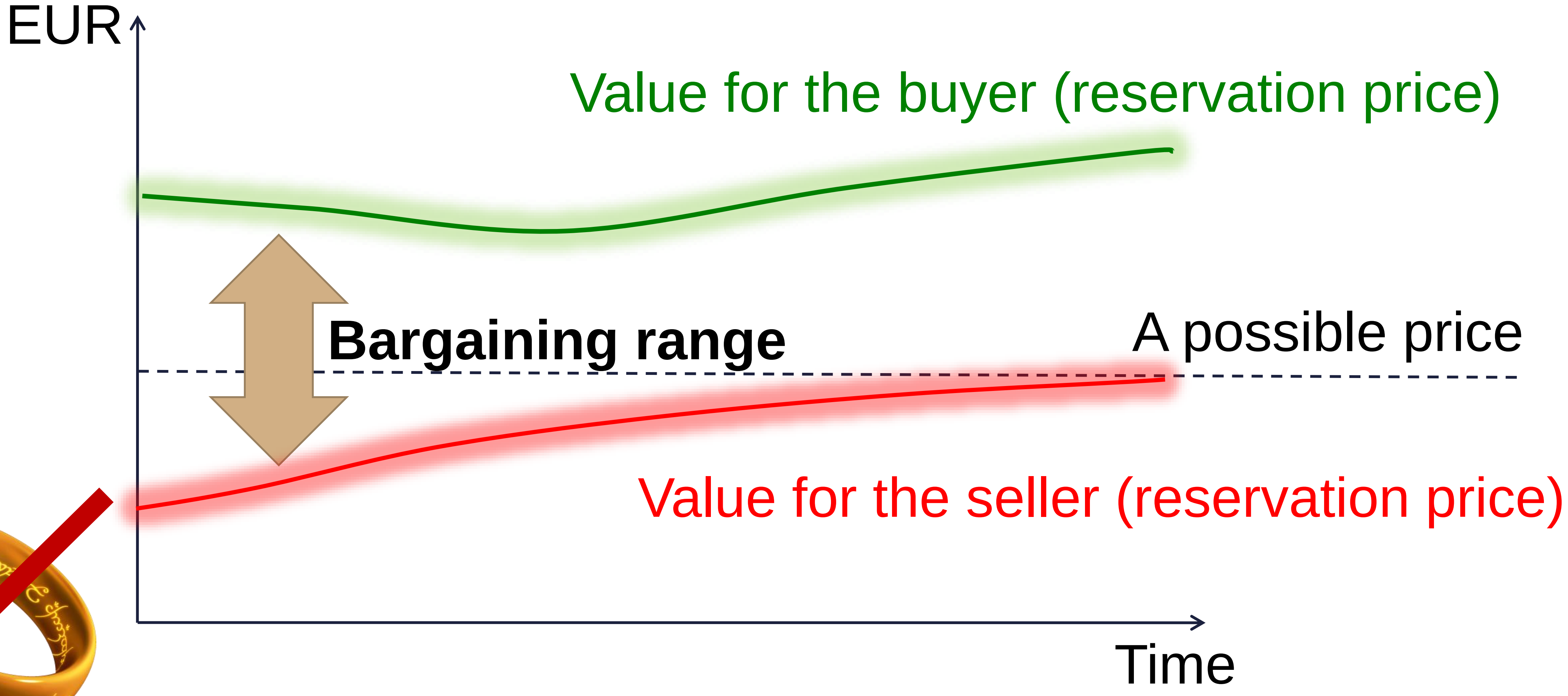
Interpreting financial statements

Basics of financial planning

Value based management

But, of course, valuation is also useful …

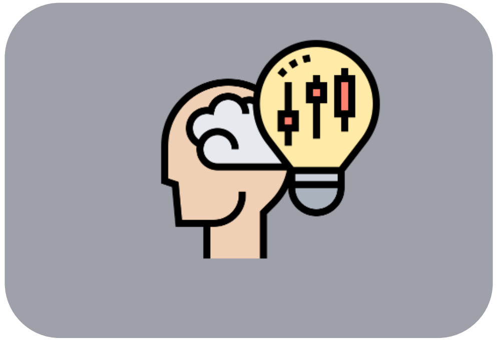
There is no „one value above them all”



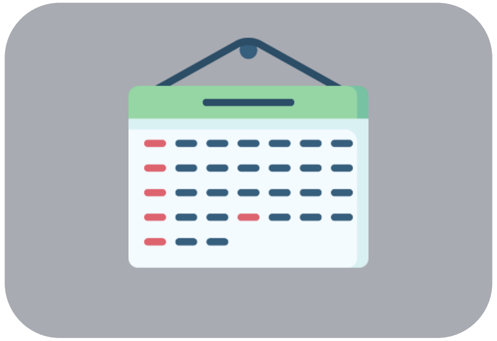
To specify in advance

What?

For whom?



Why?



For what date?

Why would the values differ?

Expectations (macro assumptions)

Makings (knowledge, connections, synergies)

Time horizon (private businesses)

Investment alternatives (not everything is public, family issues)

Risk averseness (required return, level of diversification)

Compromises

Reality…	Classroom…
Investors with unique traits (special ownership and interest links)	„Typical” investor (small shareholder, professional and financial investor)
Imperfect markets (no fresh capital)	Efficient markets (CAPM)
Information asymmetry	Perfect information
Personal expectations (rationality?)	„Market” expectations, expert estimations
That is what is being offered.	That is the result of the formula.

The aim drives the result of the valuation

Buyout

(firm unchanged)

Strategic decisions

(alternative future paths)

M&A

(synergies)

Liquidation

(operation ends)

Valuation approaches

- Income-based (dividend, DCF, capitalisation)
- Multiple-based (comparative transactions, market multiples)
- Asset-based (recreation, liquidation, forced liquidation)
- Real options? (not a valuation approach, rather a conversion tool)

Corporate finance

Managerial accounting

Financial accounting

Investments

Microeconomics

Performance measurement

Connected fields of science

Business economics

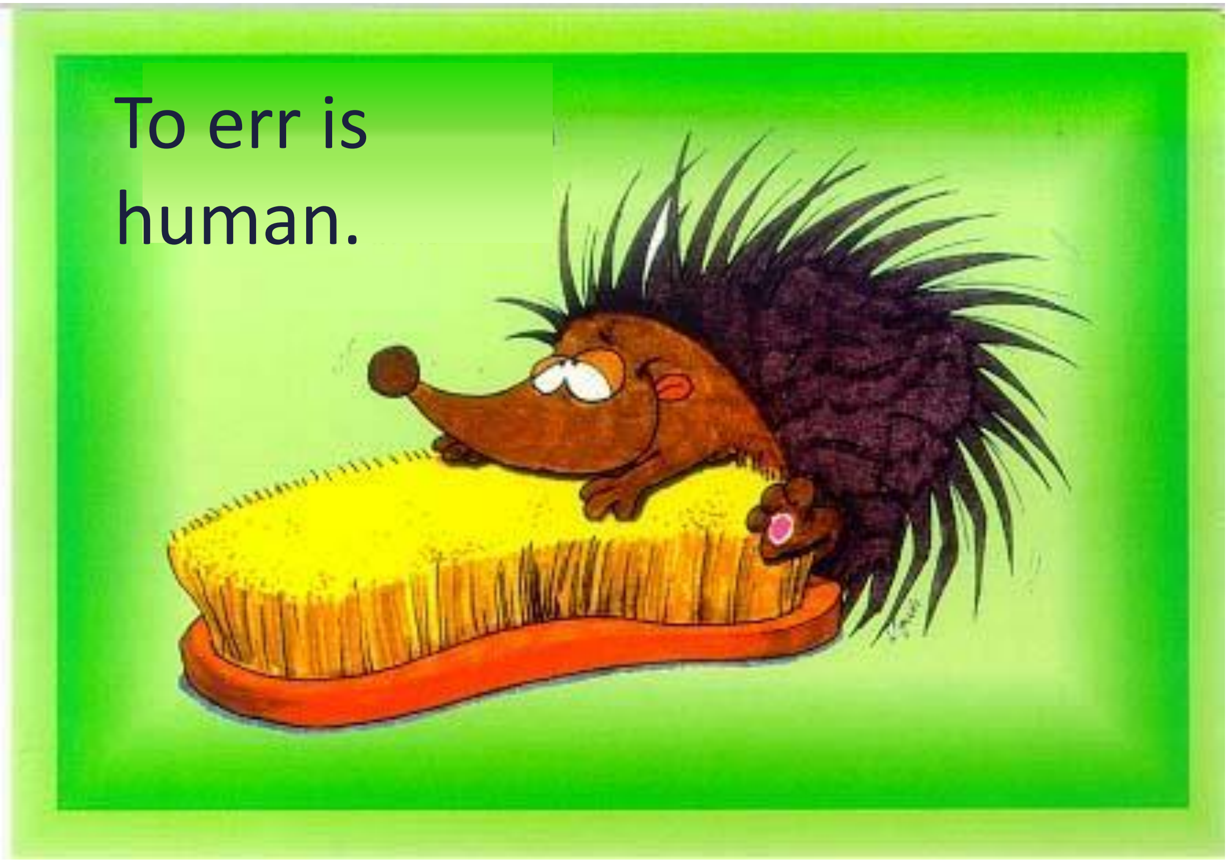
Value chain management

Strategic management

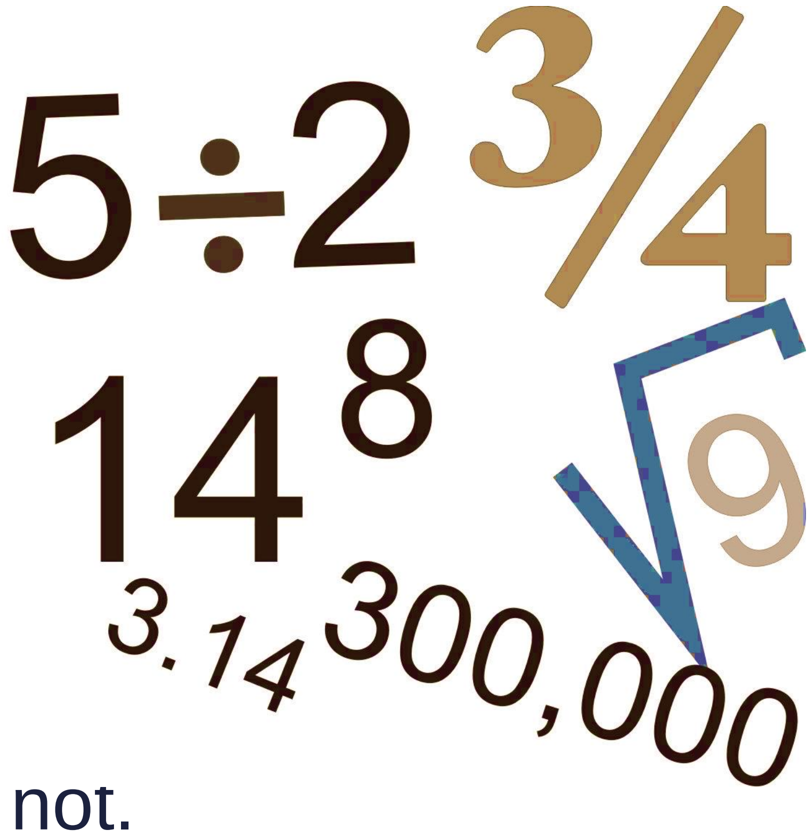
Organisation and leadership

Economic law

Misbeliefs about valuation



1. As valuation is quantitative, it has to be objective too.



The price is objective, the value not.

2. The result of a well-prepared valuation is timeless.

3. A good valuation leads to an exact result.

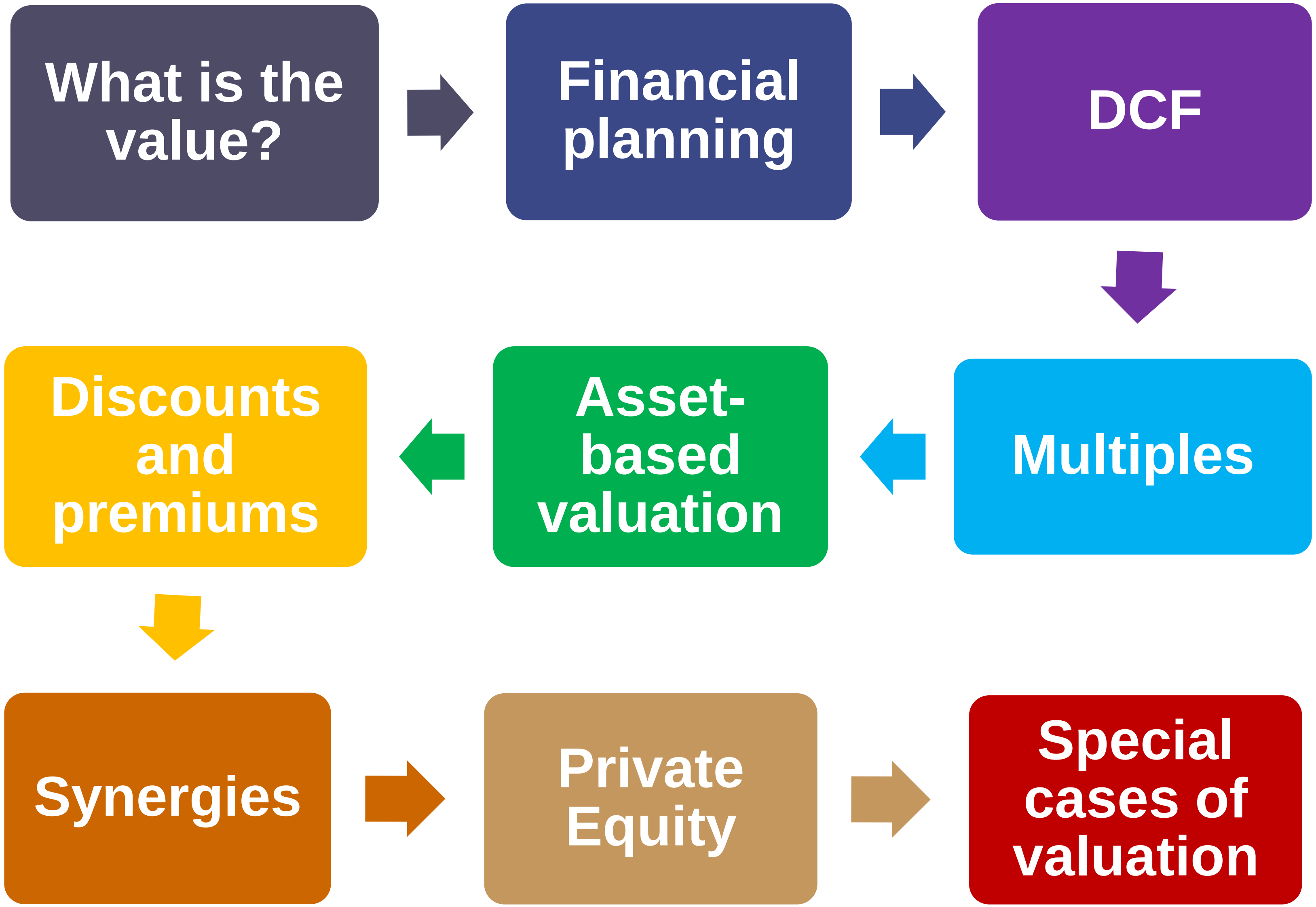
4. The more the quantified information, the more exact the valuation result is.

5. You may earn money by valuation only on an inefficient market .

The value has not to be paid.
The price reflects the value for none.

6. It is only the result of the valuation that counts. Methods are secondary.

Our schedule for the quarter



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Thank you for your attention. Any questions?

Discounted Cash Flow (DCF) based methods

Péter Juhász, PhD, CFA

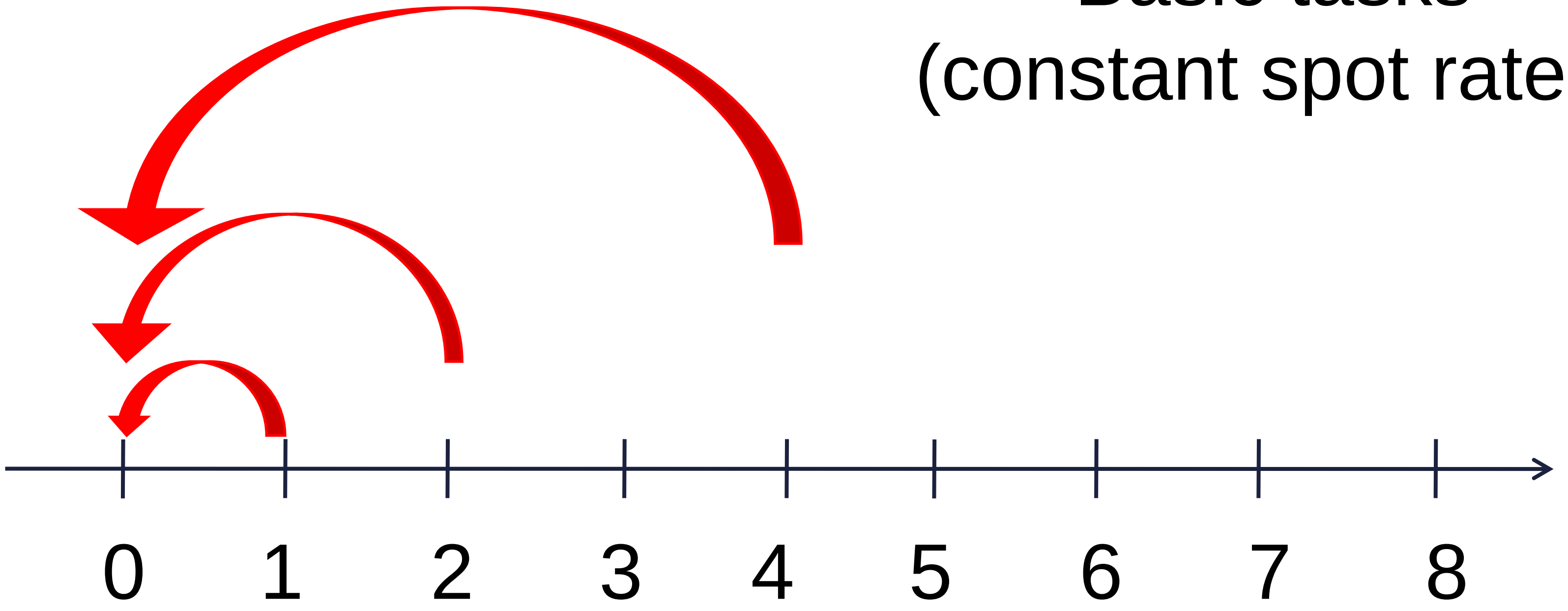
Corporate finance and reality

Corporate finance: flat yield curve, spot rates, required rates given

Reality: variable yield curve, forward rates, required rates are to estimated

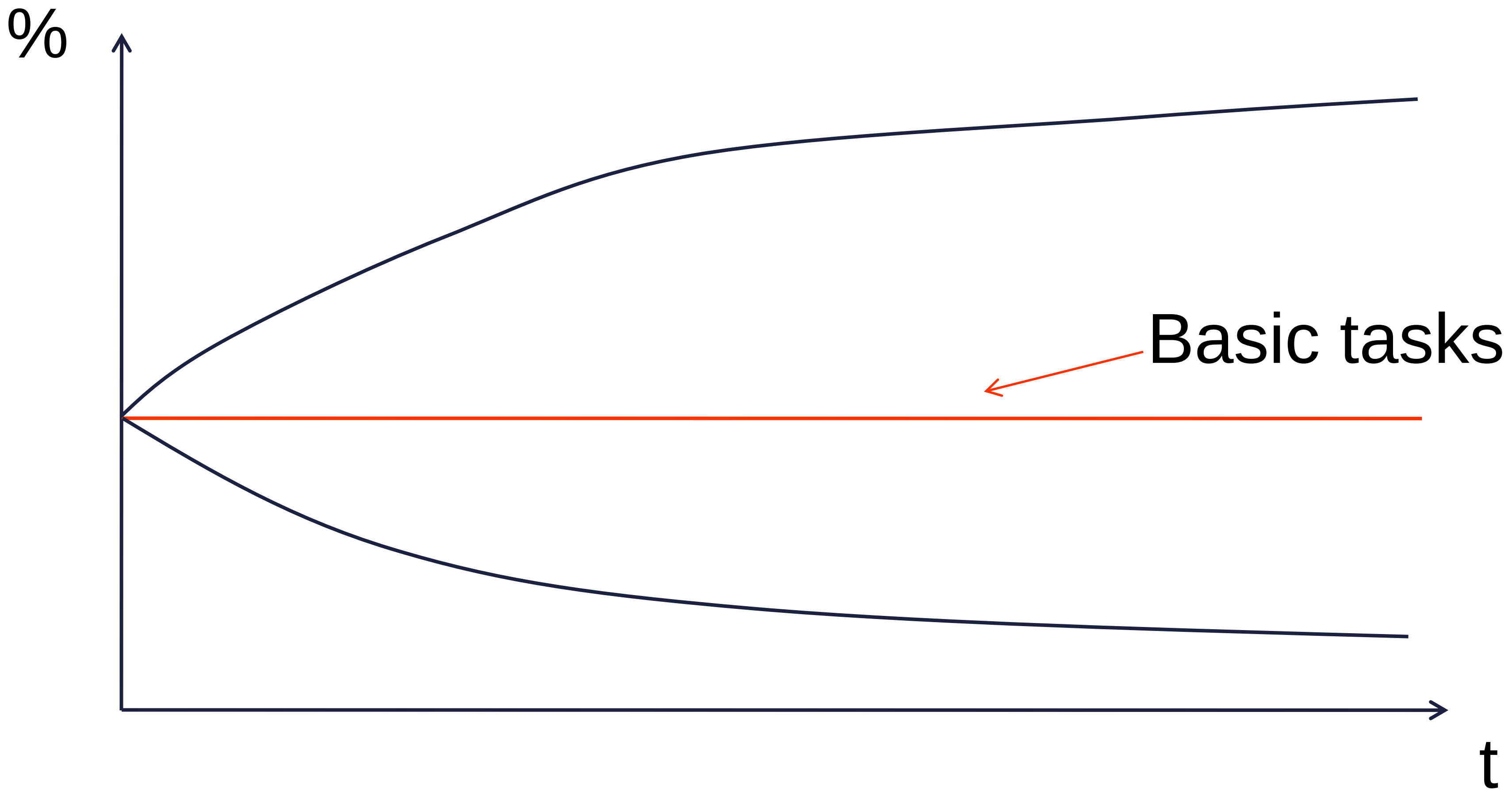
Calculating the present value 1

Basic tasks
(constant spot rates)

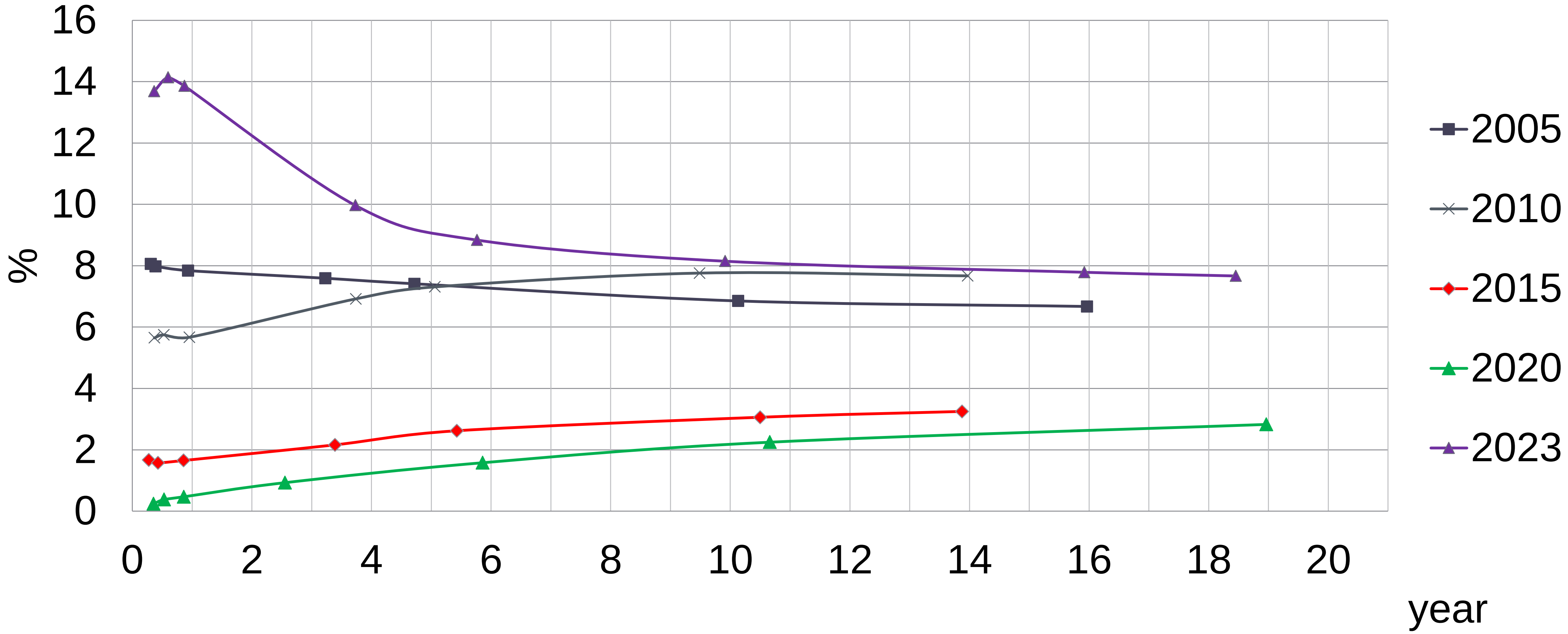


$$\frac{1}{(1+r)} \qquad \frac{1}{(1+r)^2} \qquad \frac{1}{(1+r)^4}$$

The form of the yield curve



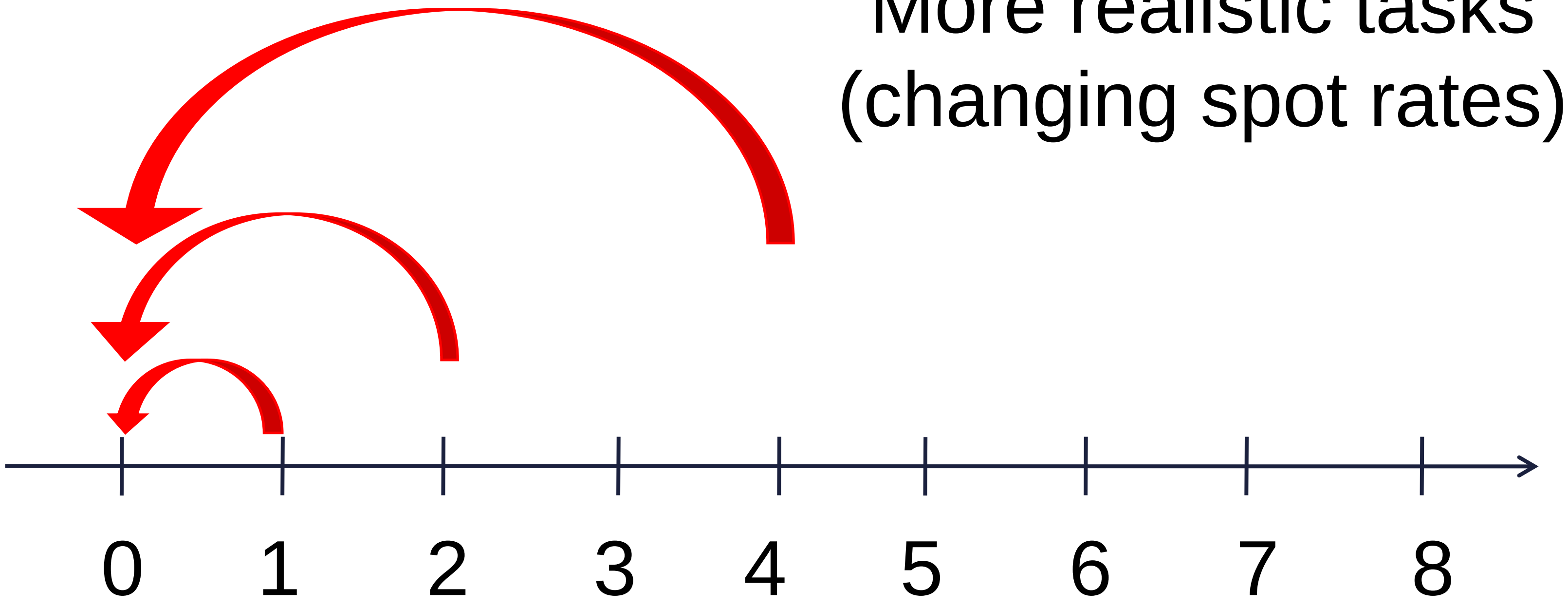
Required spot rates in Hungary (15 February)



Source: www.akk.hu

Calculating the present value 2

More realistic tasks
(changing spot rates)

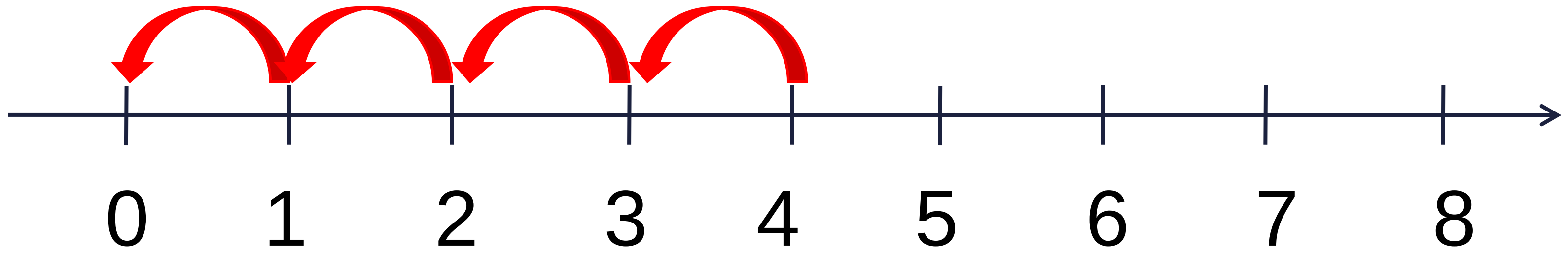


$$\frac{1}{(1+r_1)} \quad \frac{1}{(1+r_2)^2} \quad \frac{1}{(1+r_4)^4}$$

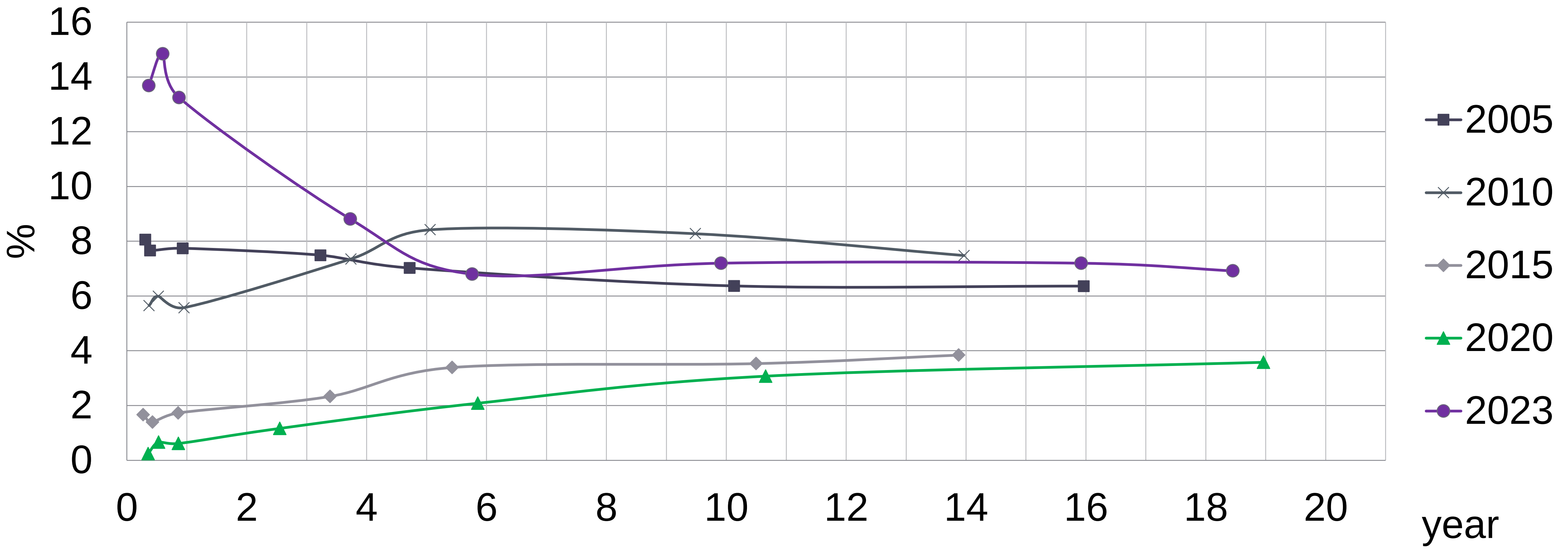
Calculating the present value 3

Reality
(forward rates)

$$\frac{1}{(1+{}_0f_1)} * \frac{1}{(1+{}_1f_2)} * \ldots * \frac{1}{(1+{}_3f_4)}$$



Required 1-year forward rates in Hungary (15 February)



A simple task

Year	0	1	2	3	4
CF		100.00	150.00	200.00	30.00
Spot r		8.00%	9.00%	10.00%	11.00%

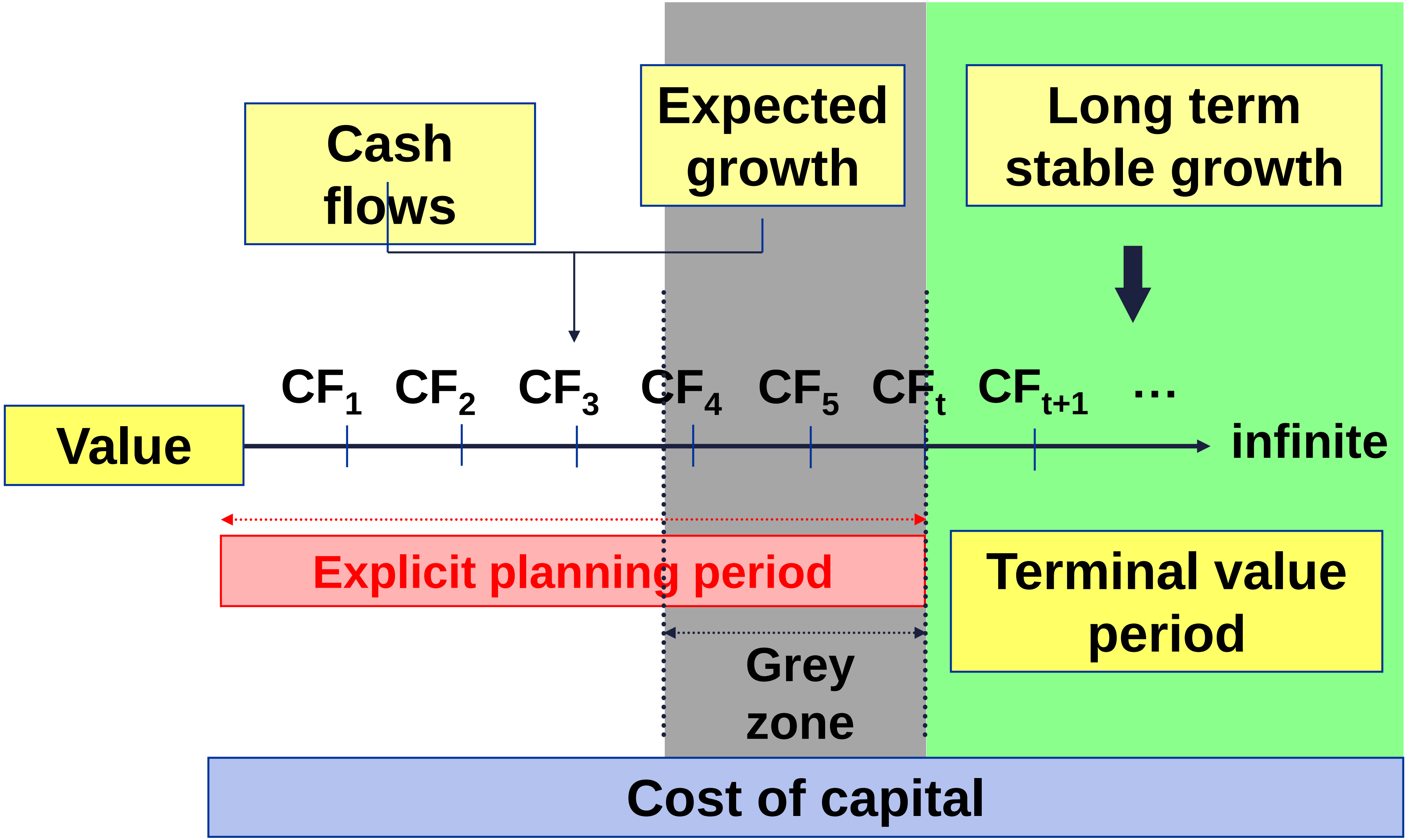
PV		92.59	126.25	150.26	19.76
Sum PV	388.87				

Forward r 1 year		8.00%	10.01%	12.03%	14.05%
Value boy		388.87	319.98	202.01	26.30

The general DCF model 1

$$NPV = \sum_{i=0}^{\infty} \frac{CF_i}{\prod_{j=1}^{i}(1+r_j)}$$

The general DCF model 2



Durations

Explicit period

The longer the better, min 5-7 years

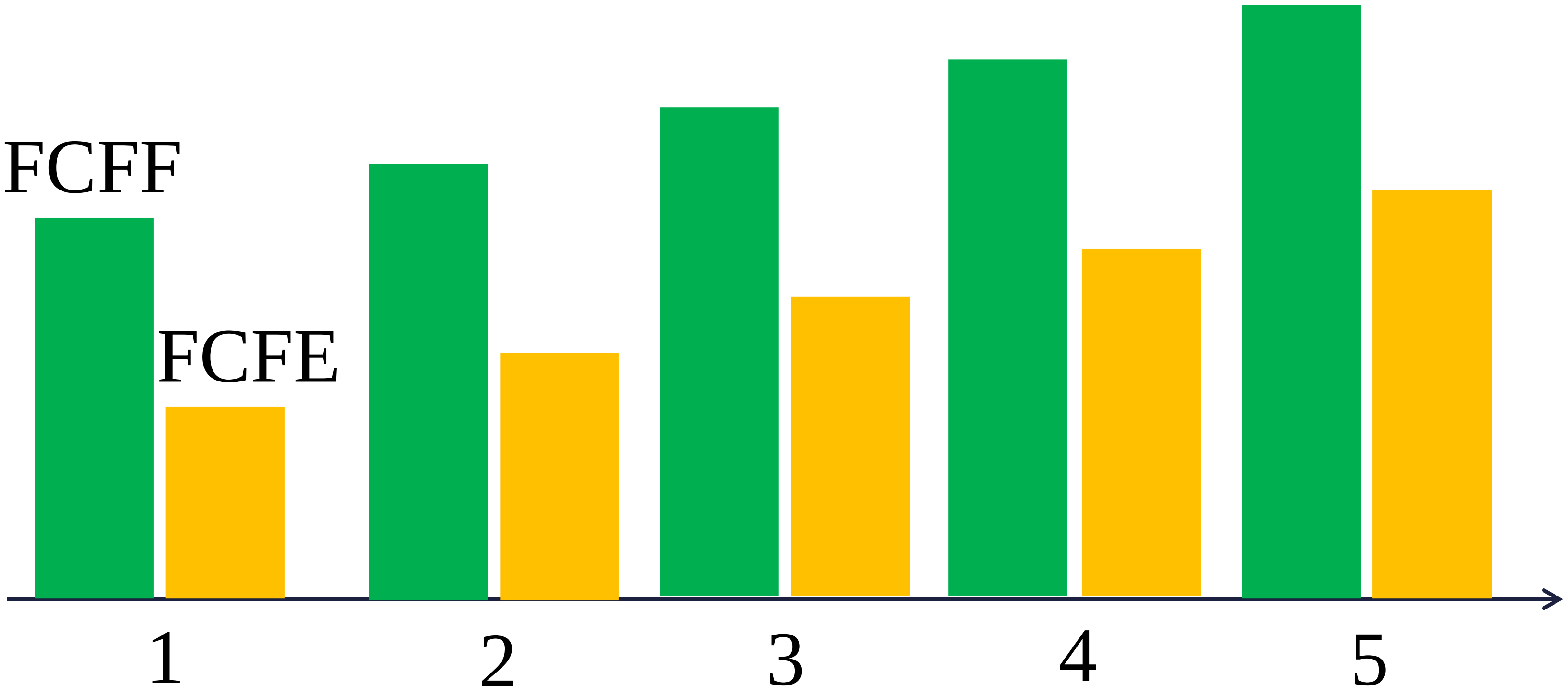
Terminal Value period

Only to start when everything will be stable

Grey (twilight) zone

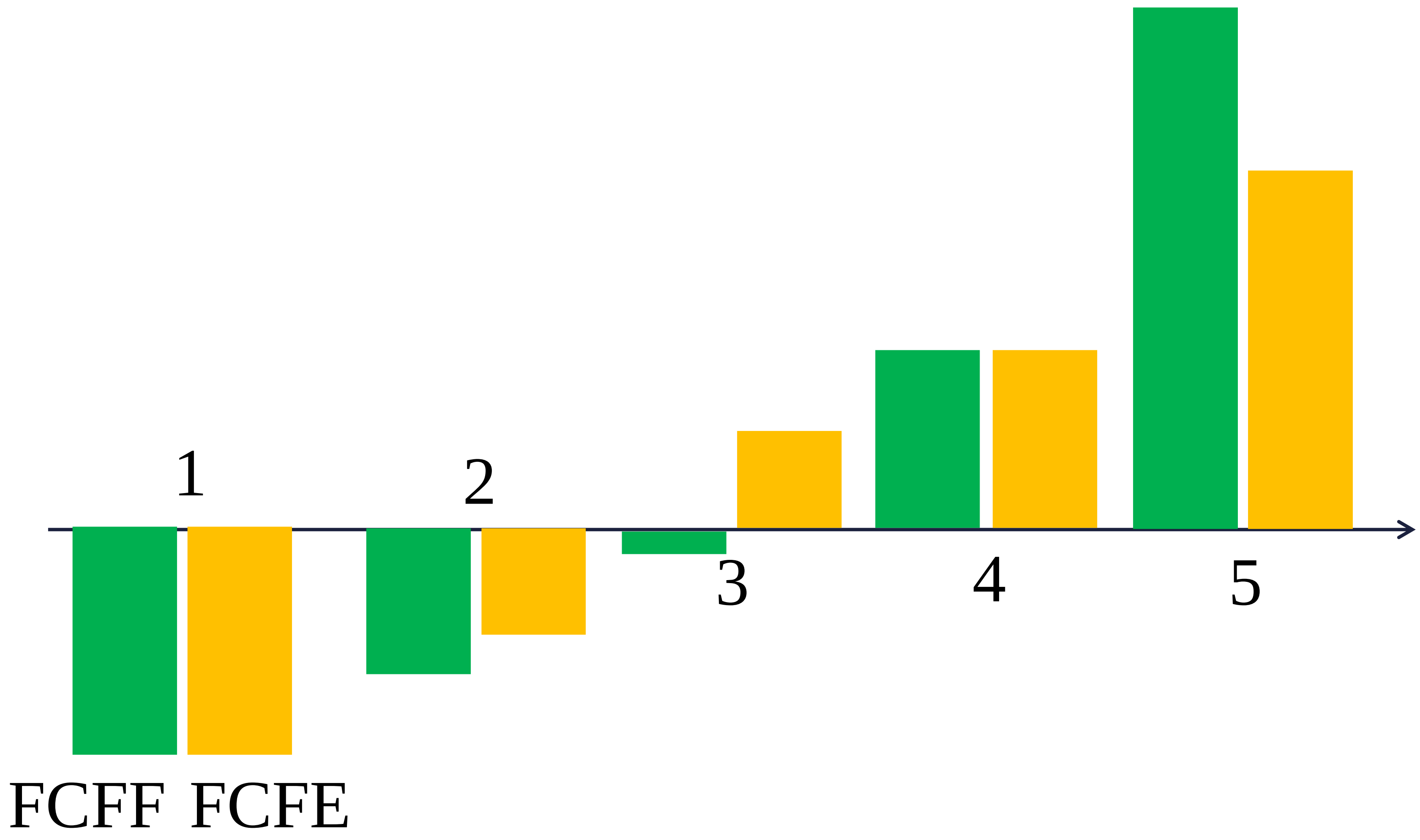
In-between the earlier two, may use a simplified (value driver only) model

What kind of DCF plan is this? 1

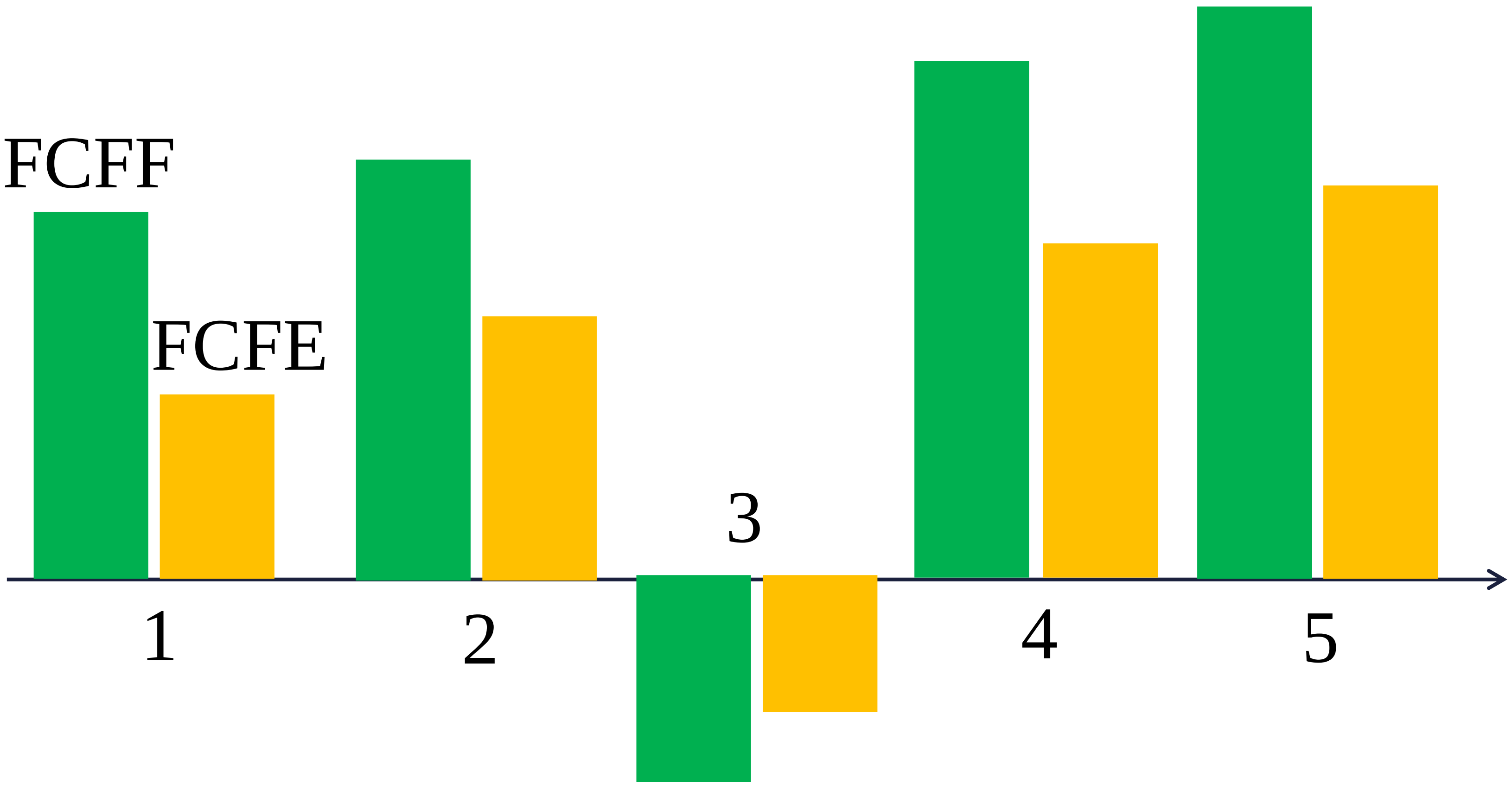


What kind of DCF plan is this? 2

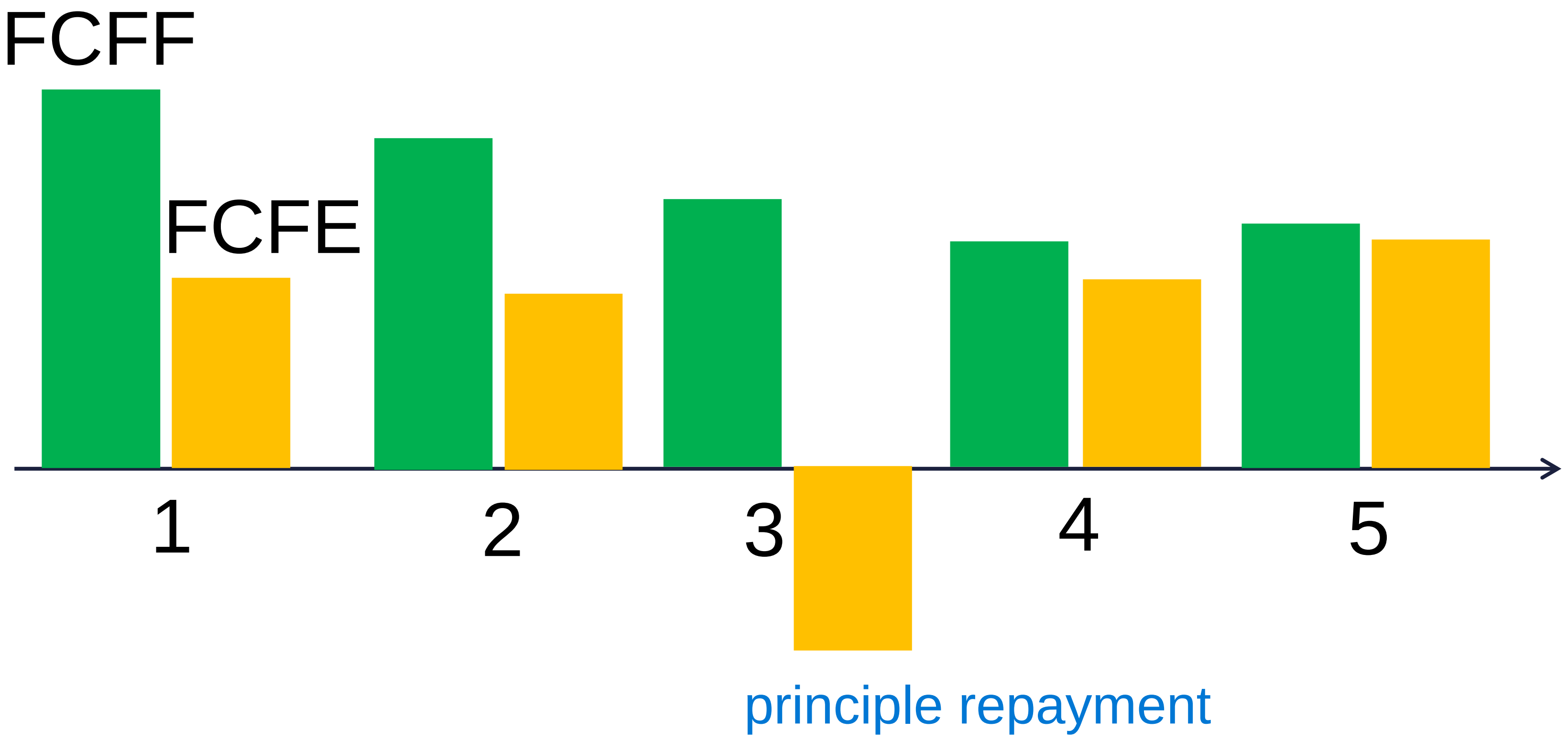
The difference includes d interest



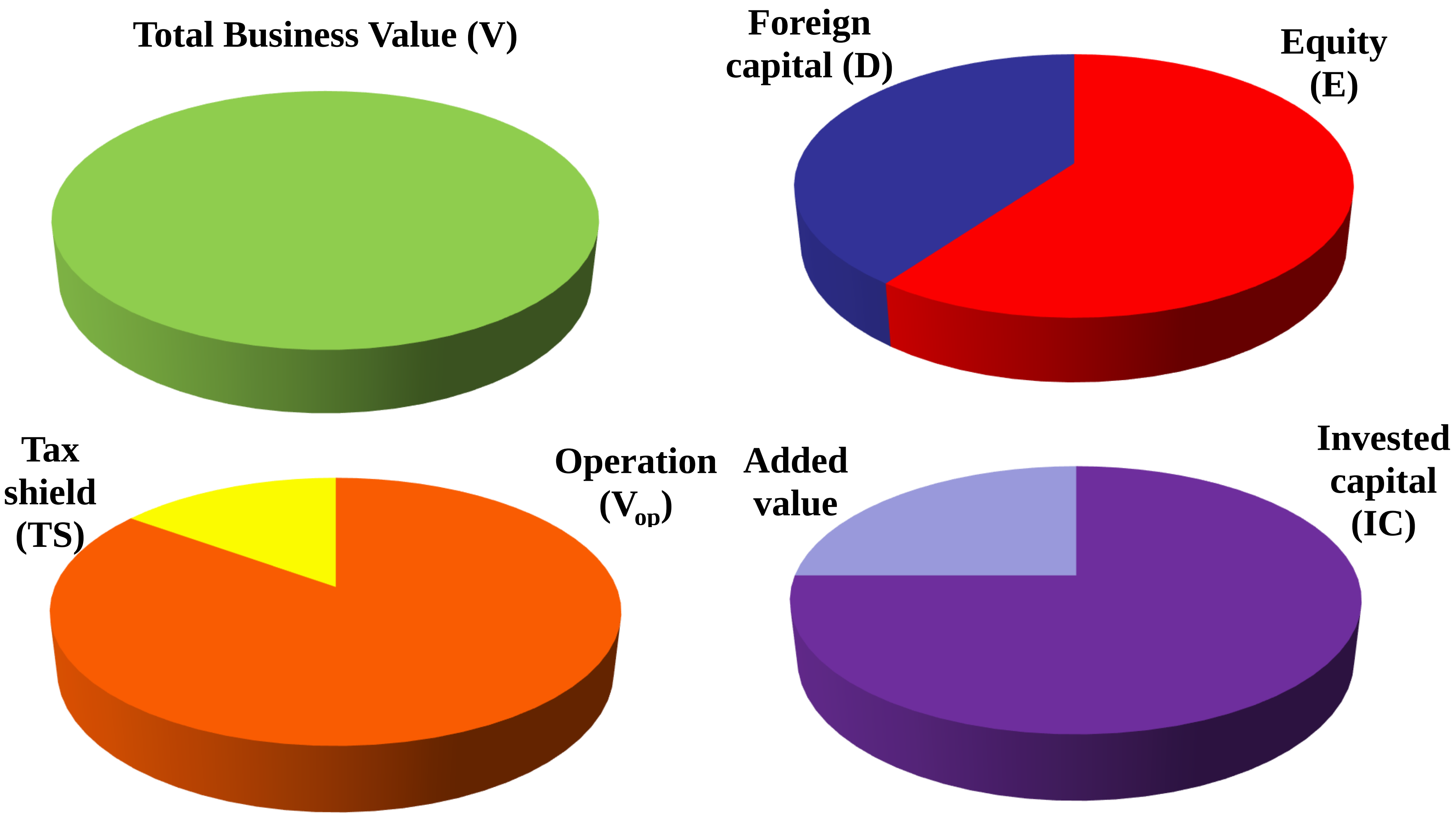
What kind of DCF plan is this? 3



What kind of DCF plan is this? 4



The division of the business value



Entity models

- FCFF
- APV
- EVA

Equity models

- Dividend models
- FCFE
- RI (Residual Income)

DCF valuation methods

Entity valuation (V)
where E = V_{op} + NOA - D

FCFF

$$\sum \frac{FCFF_t}{\prod_{i=1}^{t}(1+WACC_i)}$$

APV

$$\sum \frac{FCFF_t}{\prod_{i=1}^{t}(1+r_{At})} + PV(TS)$$

EVA

$$\sum \frac{EVA_t}{\prod_{i=1}^{t}(1+WACC_t)} + IC_0$$

Direct Equity models (E)

Dividend Discount Models

FCFE

$$\sum \frac{FCFE_t}{\prod_{i=1}^{t}(1+r_{Ei})}$$

RI

$$\sum \frac{RI_t}{\prod_{i=1}^{t}(1+r_{Ei})} + St_0$$

Entity or Equity models?

All DCF models lead to the same result.
(This is the essence of Miller-Modigliani theorems.)

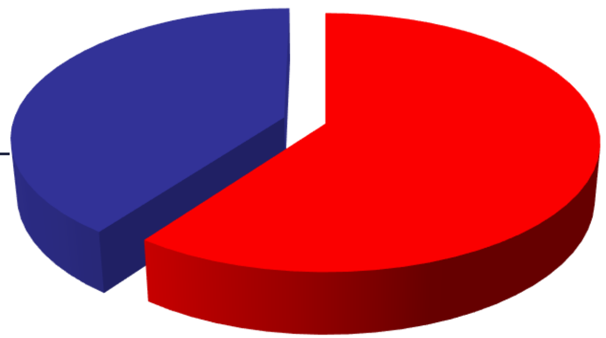
Value of operation (V_{op})

+ Value of non-operative assets

Total firm value (V)

- Value of Foreign capital (D)

= Value of equity (E)



Use rather entity models

we usually focus on the value creation process,

those can be used for Value Based Management (motivation),

multi-business firms can be analysed,

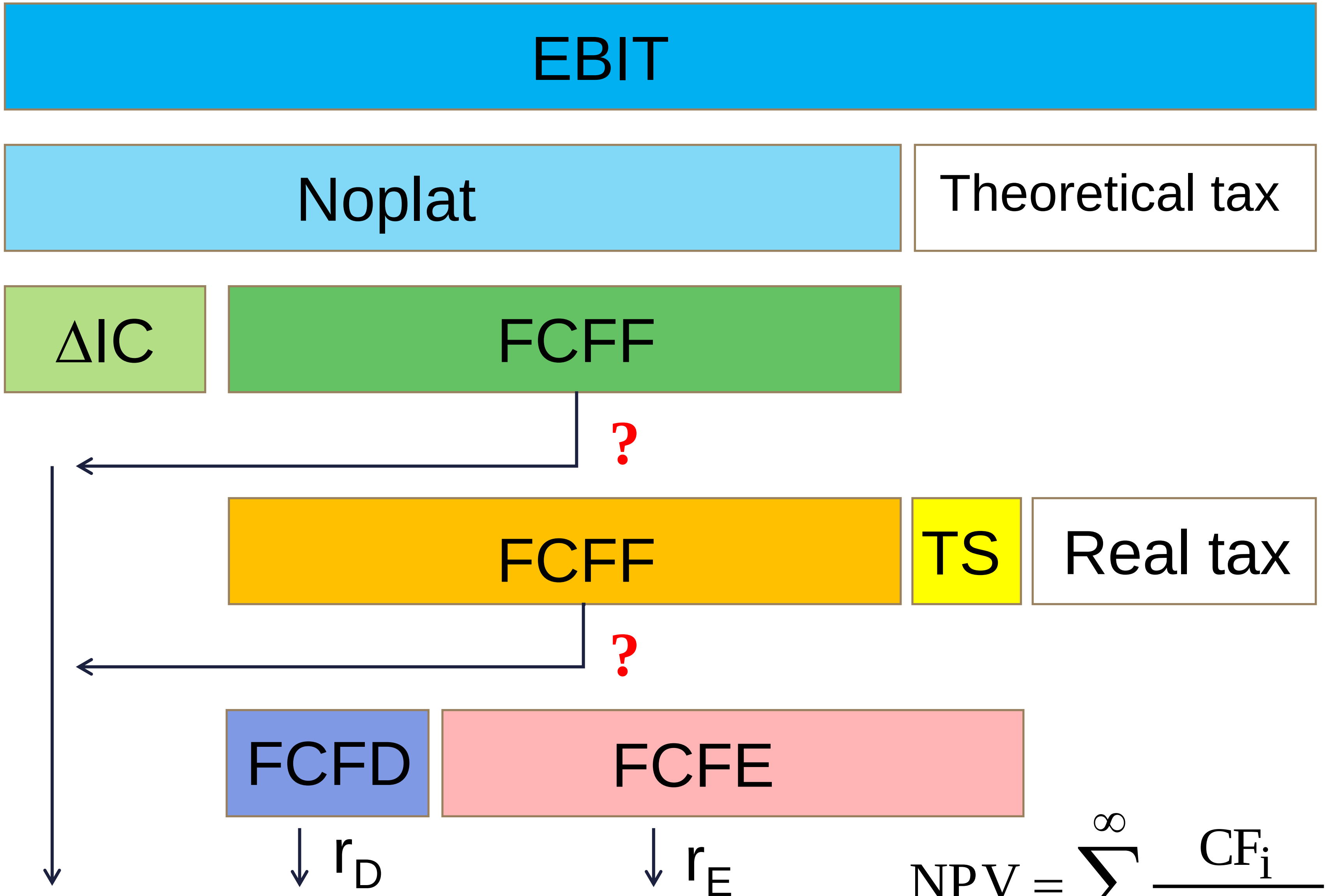
offer more freedom to model financing alternatives.

Steps of DCF valuations

1. **Estimating free cash flows**
2. **Estimating cost of capital**
3. Calculating terminal value
4. Calculating current value
5. Sensitivity tests, Scenario analysis
6. Considering other circumstances

Rules of Cash flow „matching”

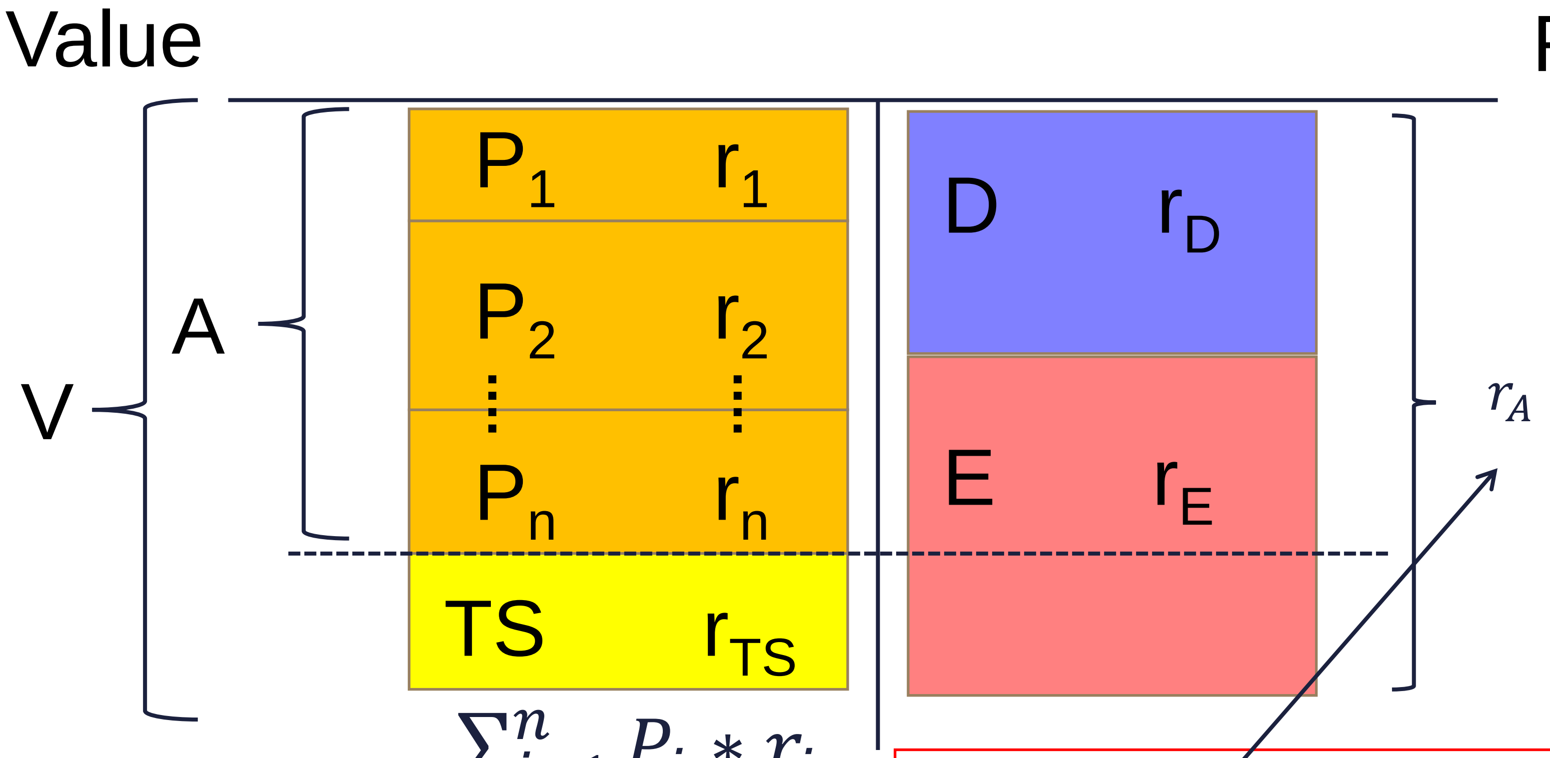
- Expiry
- Currency
- Nominal/real
- Effects of financing (TS)
- Risk



$V = D + E$

$$NPV = \sum_{i=0}^{\infty} \frac{CF_i}{(1+r_i)^i}$$

Operative cost of capital



McKinsy book
Lecture
Dont use D book

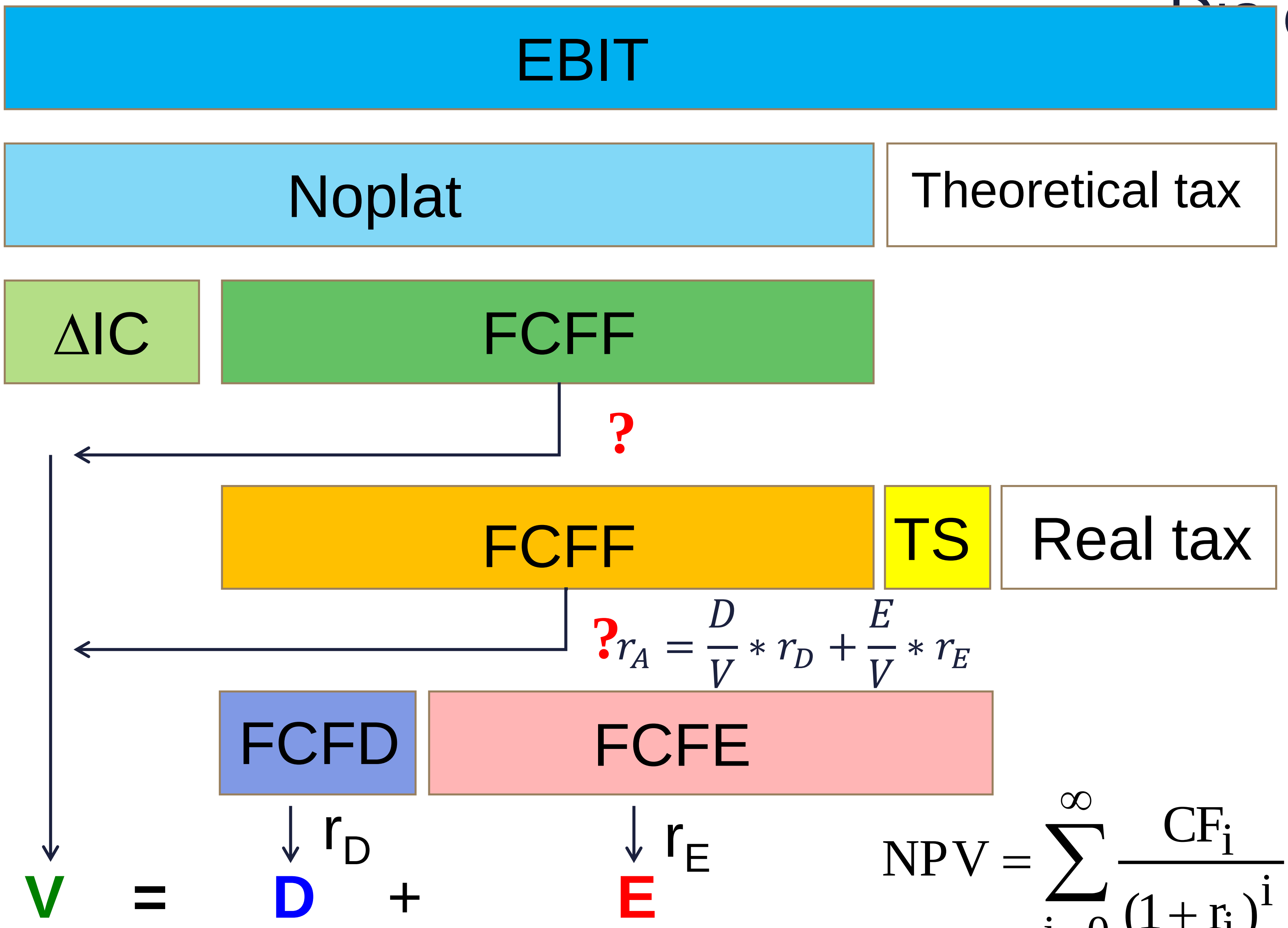
$$r_A = \frac{\sum_{i=1}^{n} P_i * r_i}{A}$$

$$r_V = \frac{A}{V} * r_A + \frac{TS}{V} * r_{TS}$$

If TS=0 or r_{TS}=r_A:

$$\boldsymbol{r_V = r_A = \frac{D}{V} * r_D + \frac{E}{V} * r_E}$$

Dia címsor



$$r_A = \frac{D}{V} * r_D + \frac{E}{V} * r_E$$

$$NPV = \sum_{i=0}^{\infty} \frac{CF_i}{(1+r_i)^i}$$

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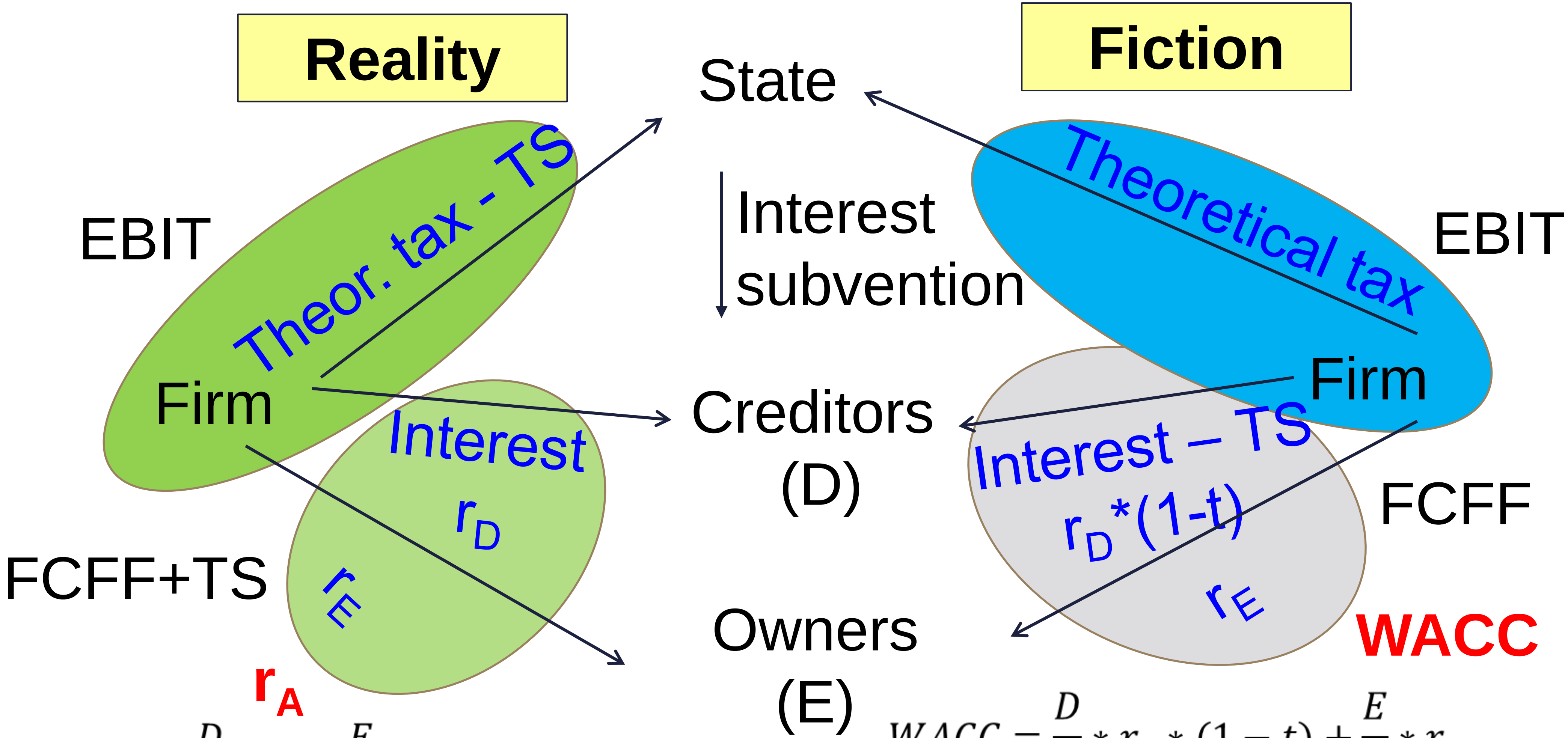
Weighted Average Cost of Capital (WACC)



$$r_A = \frac{D}{V} * r_D + \frac{E}{V} * r_E$$

$$WACC = \frac{D}{V} * r_D * (1 - t) + \frac{E}{V} * r_E$$

Dia címsor

EBIT

Noplat | Theoretical tax

ΔIC | FCFF

$$WACC = \frac{D}{V} * r_D * (1 - t) + \frac{E}{V} * r_E$$

FCFF | TS | Real tax

$$r_A = \frac{D}{V} * r_D + \frac{E}{V} * r_E$$

FCFD | FCFE

$$V = \underset{\uparrow r_D}{D} + \underset{\uparrow r_E}{E}$$

$$NPV = \sum_{i=0}^{\infty} \frac{CF_i}{(1+r_i)^i}$$

FCFF method

$$V = \sum \frac{FCFF_t}{\prod_{i=1}^{t}(1+WACC_i)}$$

FCFF: CF available to all financiers. (CF independent of financing)

V - D = E

FCFE method

$$E = \sum \frac{FCFE_t}{\prod_{i=1}^{t} (1 + r_{Ei})}$$

FCFE: CF available to shareholders. (CF depends on the financing structure, can be redrawn freely.)

E + D = V

Limits of FCFF valuation:

1) shows the value for a given financing structure,

2) financing effects can not be separated.

APV = V assuming equity only financing + Present value of tax shields **[PV (TS)]**

$$V_{Op} = \sum \frac{FCFF_t}{\prod_{i=1}^{t}(1+r_{A_i})}$$

What rate to use for discounting tax shield?
During this course, we assume tax shield is linked to operation $\Rightarrow$ $\mathbf{r_A}$

Limitations of FCFF valuation:

1) The seemingly high proportion of terminal value may create the impression that value is created after the explicit period.

2) does not show when the value is created.

Economic profit = added profit over the required level

$$\mathbf{EVA_i = IC_{i-1} * (ROIC_i - WACC_i)} = NOPLAT_i – IC_{i-1} * WACC_i$$

$$\mathbf{RI_i = BV(E)_{i-1} * (ROE_i - r_{Ei})} = \text{Net Profit} – BV(E)_{i-1} * r_{Ei}$$

EVA/RI method

$$V = \sum \frac{EVA_t}{\prod_{i=1}^{t}(1+WACC_i)} + IC_0$$

In the long run ROIC = WACC (ROE = r_E), thus EVA = RI = 0. The added value in TV in such a case is zero.

$$E = \sum \frac{RI_t}{\prod_{i=1}^{t}(1+r_{Ei})} + BV(E)_0$$

A simple example

What is the value of the firm if it is expected to generate 1500 Noplat next year, ROIC will remain stable at 15%, growth will amount to 5%, operative cost of capital will stay 10%, D/V rate will stick to 50%, r_D=6%, and tax rate is 20%?

Warm-up calculations

Noplat	1500.00
ROIC	15.00%
IC	10000.00

ROIC=RONIC	15.00%
g	5.00%
K	33.33%
FCFF	1000.00

r_A	10.00%
D/V	50.00%
r_D	6.00%
t	20.00%
r_E	14.00%
WACC	9.40%

FCFF method

Noplat	1 500.00
ΔIC	-500.00
FCFF	1 000.00

WACC	9.40%
g	5.00%

V	22 727.27

$$\sum \frac{\mathrm{FCFF_t}}{(1+\mathrm{WACC})^{t}}$$

$$V = \frac{FCFF}{WACC - g}$$

$$V = \frac{1000}{9.4\% - 5\%}$$

FCFE method

V	22727.27
D	11363.64
FCFF	1 000.00
$D*r_D$	-681.82
$D*r_D*t$	+136.36
DD	+568.18
FCFE	1 022.73
r_E	14.00%
E	11363,64
V=D+E	22727.27

$$\sum \frac{FCFE_t}{(1+r_E)^t}$$

$$E = \frac{FCFE}{r_E - g}$$

$$E = \frac{1022,73}{14\% - 5\%}$$

APV method

FCFF	1 000.00
$D*r_D*t$	136.36
r_A	10.00%
g	5.00%

$FCFF//r_A$	20 000.00
$TS//r_A$	2 727.27
V	22 727.27

$$\sum \frac{FCFF_t}{(1+r_A)^t} + PV(TS)$$

$$V = \frac{FCFF}{r_A - g} + \frac{D * r_D * t}{r_A - g}$$

$$V = \frac{1000,00}{10\% - 5\%} + \frac{136,36}{10\% - 5\%}$$

DCF valuation methods

Method	To discount	Discount rate	Notes
Total firm DCF	FCFF	WACC	Useful for multiple business units and set D/V ratio firms
Economic profit	EVA / RI	WACC / r_E	Shows when the value will be created.
APV	FCFF and tax shield (TS)	Operative cost of capital (r_A)	Separates the effect of financing, handy for M&As.
Direct equity valuation	FCFE	Required rate on equity(r_E)	Detailed, hectic financial plan available, bottom-up CF, financial service industry

Which should I choose?

1. What to do if business units have no stand-alone financing? FCFF
2. How risky is this business plan? EVA (RI)
3. What to do if it is very hard to estimate the cost of capital for this firm? APV
4. What is the value of the shares given this outlook?

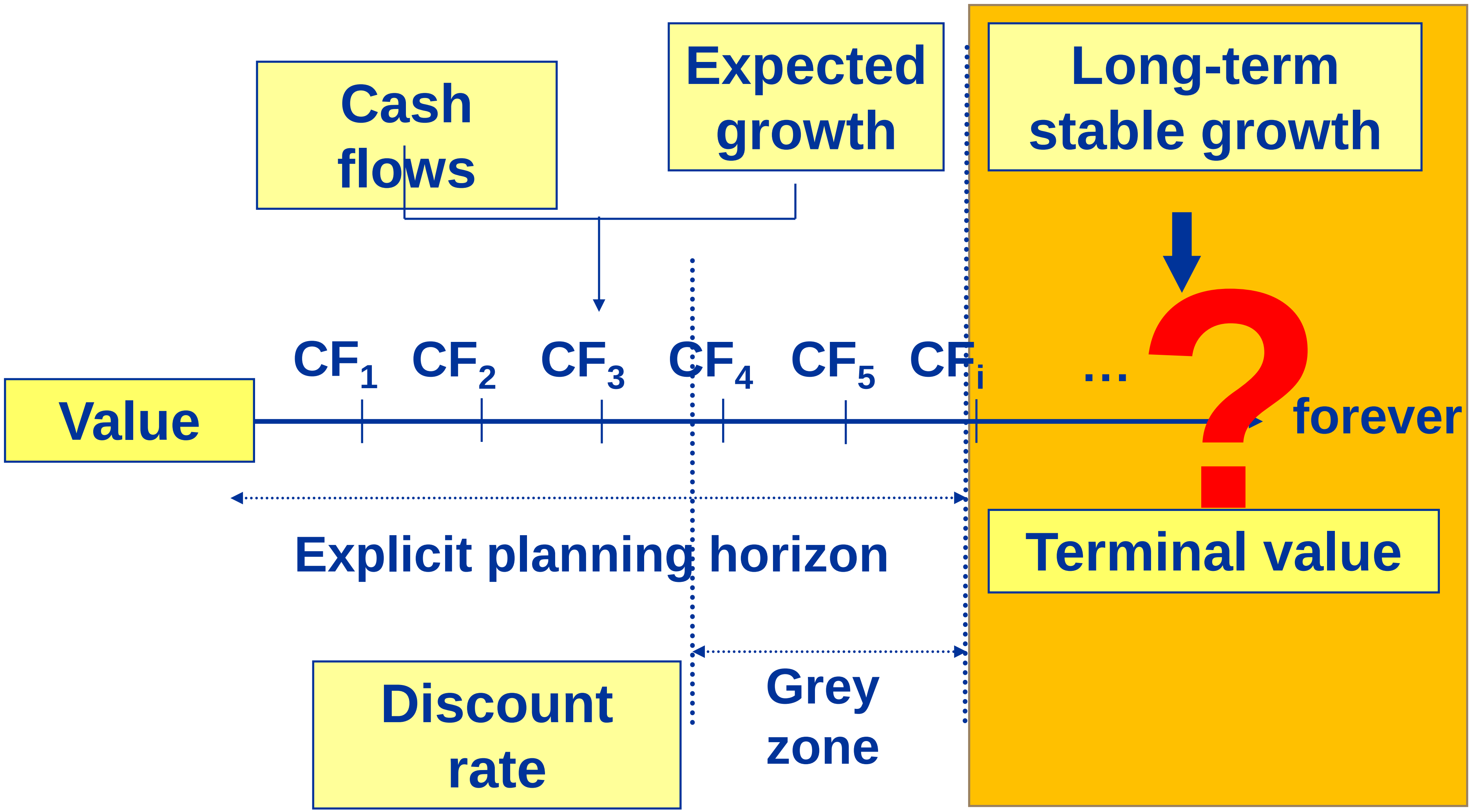
Thank you for your attention. Any questions?

peter.juhasz@uni-corvinus.hu

Estimating the terminal value

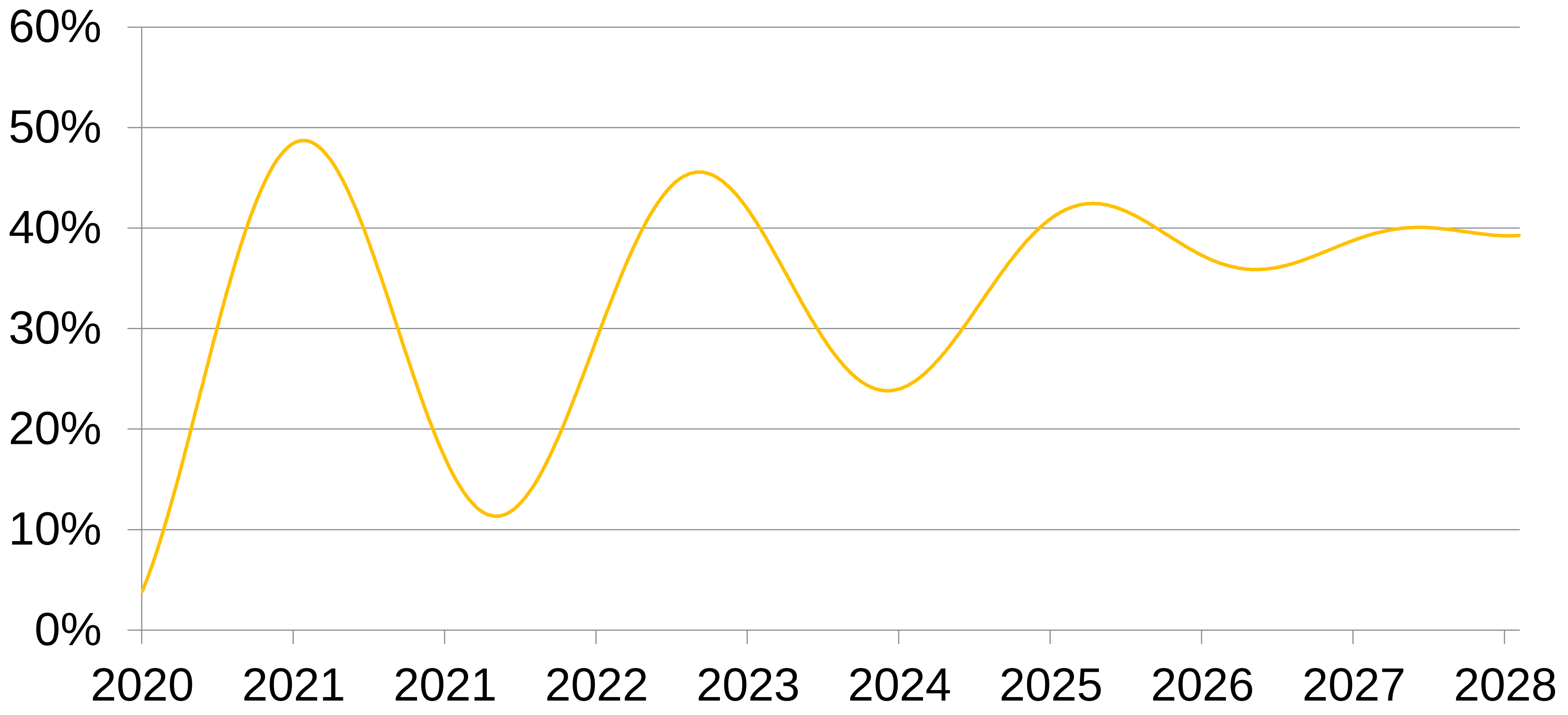
Péter Juhász, PhD, CFA

The general DCF model



How long should the explicit period last?

By the start of the terminal value period all inputs should stabilise.



The lifecycle of the ROIC and the g

Depend on the capabilities of the firm and the dynamics of the industry.

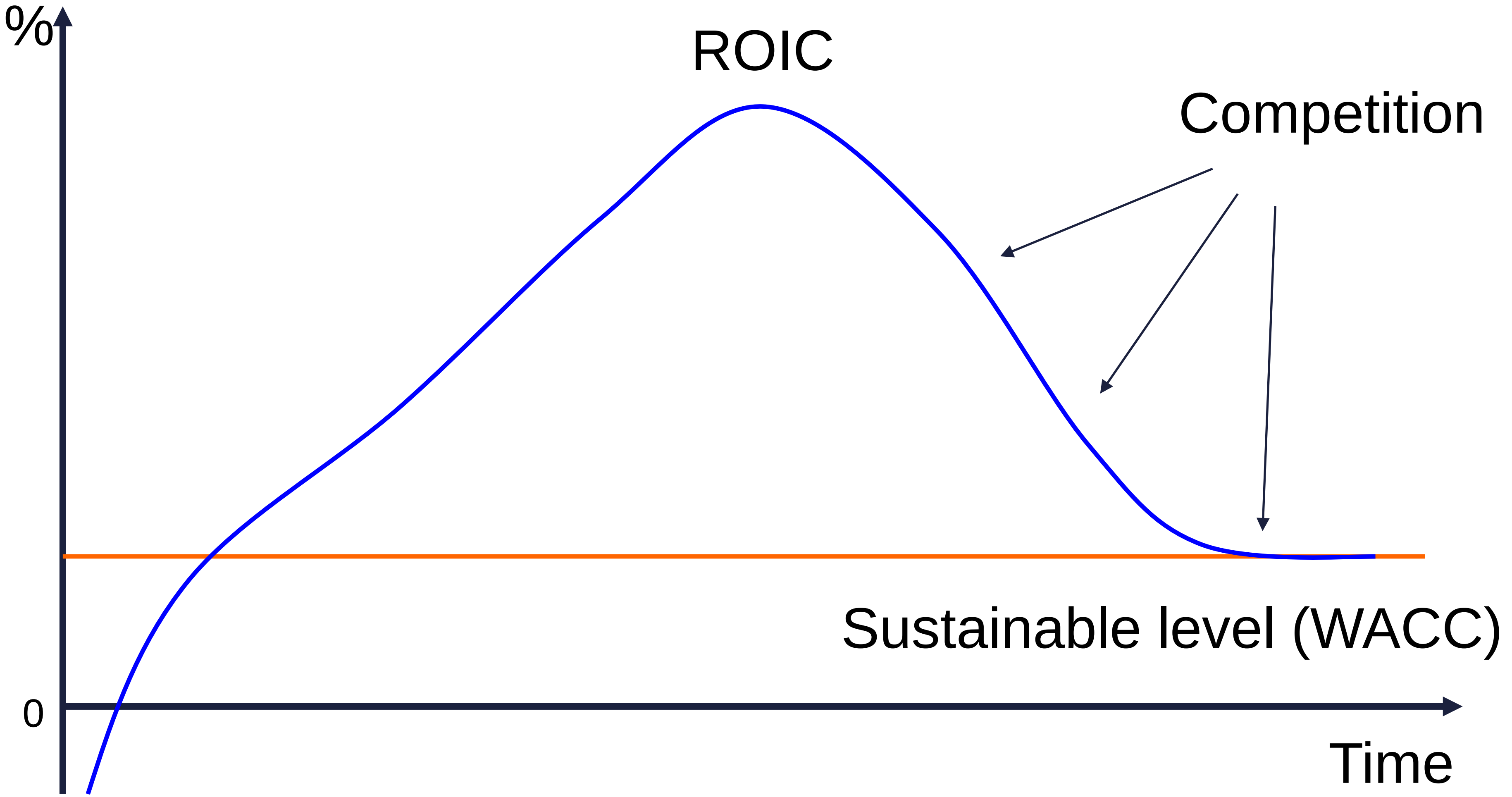
Those converge to the industry average.

Those mutually depend on each other (g generates value only if ROIC>WACC).

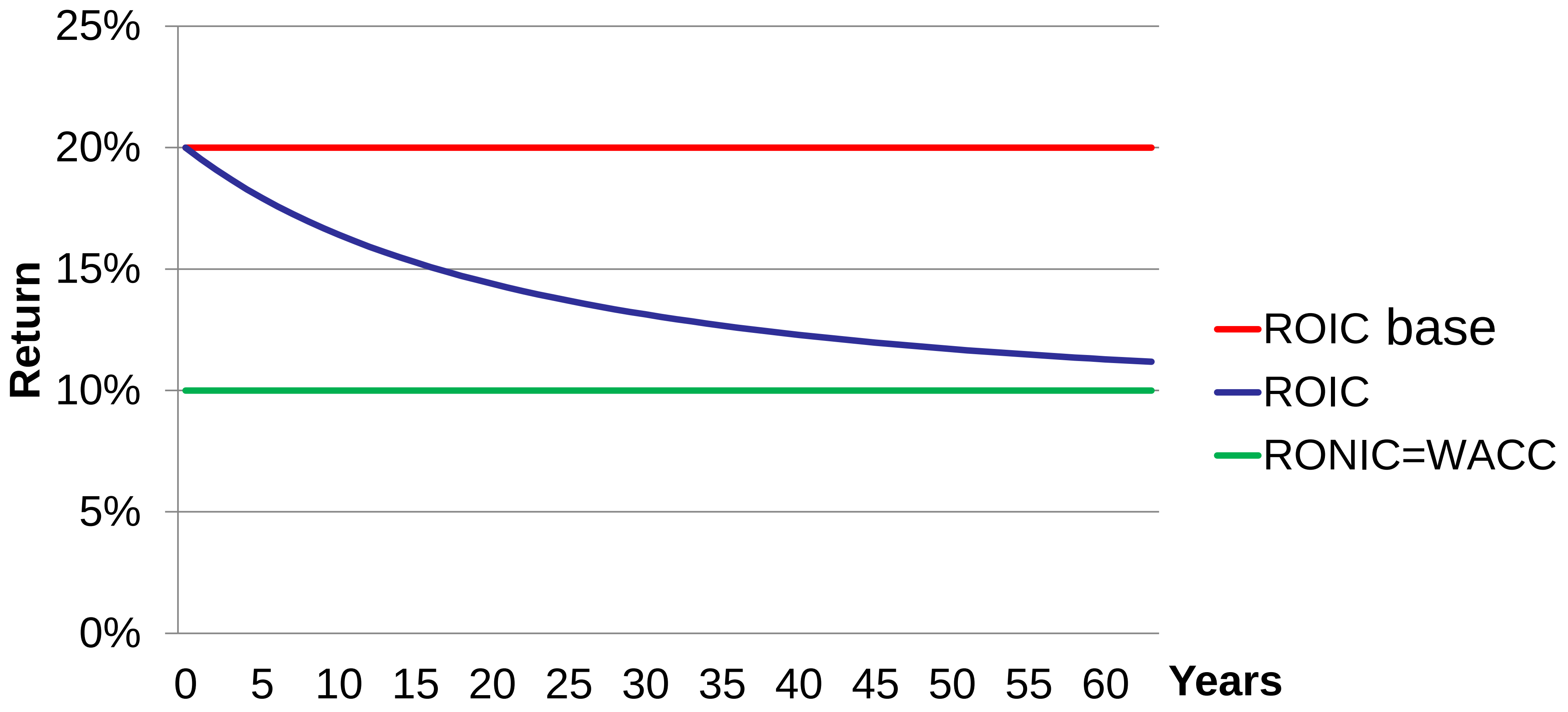
We plan for further details only if those improve the quality of the prediction.

(maintainability and precision of assumptions).

Changes in value creation over time



Connections among ROIC, RONIC, and WACC



ROIC remaining high 1

$$ROIC=(1-t)*(\text{Unit price}-\text{Unit cost})*\frac{\text{Quantity}}{IC}$$

Premium price?

ROIC remaining high 2

$$ROIC=(1-t)*(\text{Unit price}-\text{Unit cost})*\frac{\text{Quantity}}{IC}$$

Lower costs?

ROIC remaining high 3

$$ROIC=(1-t)*(Unit\ price-Unit\ cost)*\frac{Quantity}{IC}$$

More efficient capital management?

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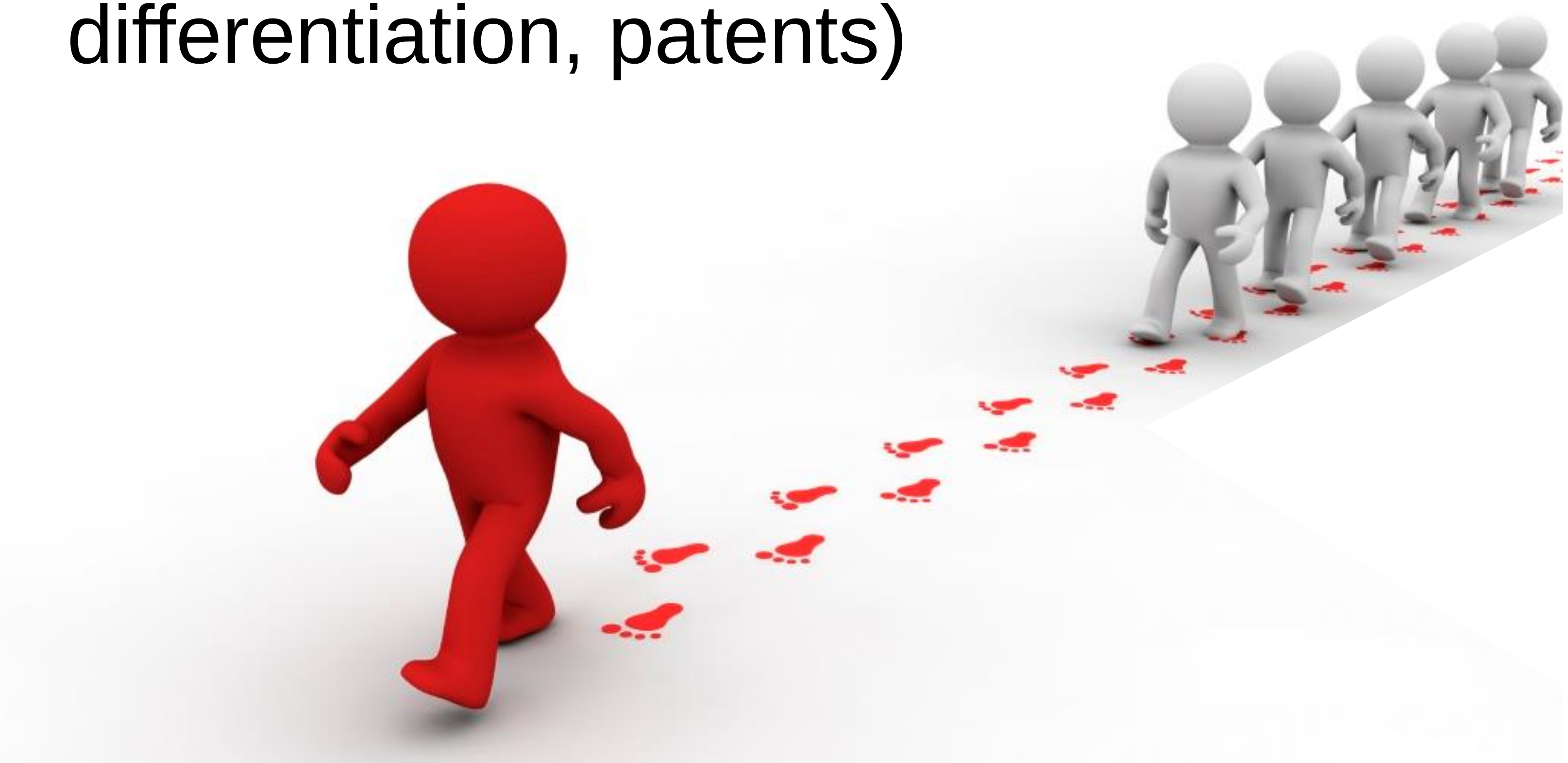
ROIC remaining high 4

To maintain your competitive advantage in the long run, you have to hinder copying. (state protection, exclusive contracts, licence, differentiation, patents)



1. FCFF

$$TV = \sum_{t=1}^{\infty} \frac{FCFF_t}{\prod_{i=1}^{t}(1+WACC_i)} = \frac{FCFF_{S\tan dard}}{WACC^* - g^*_{FCFF}}$$

Assumption: WACC* > g_{FCFF}* !

Is that always true?

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$WACC^* < ? > g_{FCFF}^*$

If $r_f > g_{GDP}$:

$$WACC > r_D > r_f > g_{GDP} >= g_{Firm}$$

If $WACC < g_{Firm}$, this is not a long-term stable status yet.

$$V_{TV} = \frac{NOPLAT_i - \boldsymbol{InvCF}_i}{WACC^* - g^*_{NOPLAT}} = \frac{NOPLAT(1-K)}{WACC^* - g^*_{NOPLAT}}$$

WACC* > g_{NOPLAT}* !

Inv CF = ?

= NOPLAT – FCFF = NOPLAT*K =

= <u>Net</u> capital investment (ΔIC)

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$$V_{TV} = \frac{NOPLAT\left(1 - \frac{g_{NOPLAT}}{RONIC}\right)}{WACC^* - g^*_{NOPLAT}}$$

WACC* > g_{NOPLAT}* < RONIC!

WACC* <?> RONIC

3. APV

$$TV = \frac{FCFF}{r_A - g} + \frac{D * r_D * t}{r_A - g}$$

4. FCFE

$$TV_E = \frac{FCFE}{r_E - g}$$

5. Economic profit

$$\mathrm{EVA} = \mathrm{NOPLAT} - \mathrm{IC} * \mathrm{WACC}$$

$$= \mathrm{IC} * (\mathrm{ROIC} - \mathrm{WACC})$$

$$EVA_{TV} = \left(\frac{\mathrm{EVA}_{\mathrm{T+1}}}{WACC} + \frac{NOPLAT_{\mathrm{T+1}} * \frac{g}{RONIC} * (RONIC - WACC)}{WACC * (WACC - g)} \right)$$

How did we get that?

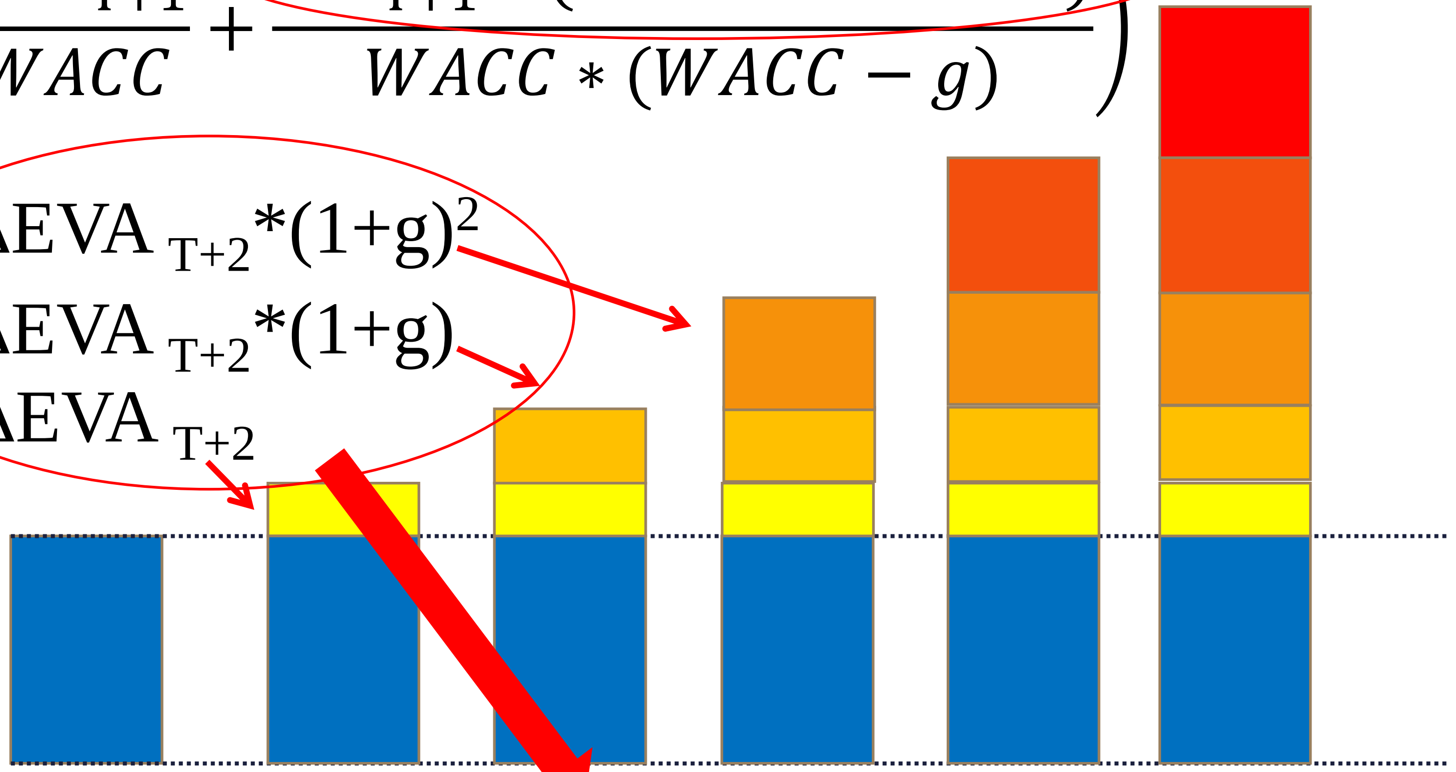
ΔEVA_{T+2}

$$EVA_{TV_T} = \left(\frac{\text{EVA}_{T+1}}{WACC} + \frac{\Delta IC_{T+1} * (RONIC - WACC)}{WACC * (WACC - g)} \right)$$

$\Delta \text{EVA}_{T+2} * (1+g)^2$

$\Delta \text{EVA}_{T+2} * (1+g)$

ΔEVA_{T+2}



EVA_{T+1}

$$\frac{EVA_{T+1}}{WACC} + \frac{\Delta EVA_{T+2}}{WACC} * \frac{1}{WACC - g}$$

Good news: convergence

Due to the competition in the long run: WACC = RONIC

$$\boldsymbol{V_{TV}} = \frac{NOPLAT(1 - \frac{g}{RONIC})}{WACC - g} = \frac{NOPLAT(1 - \frac{g}{WACC})}{WACC - g} =$$

$$\frac{NOPLAT(\frac{WACC}{WACC} - \frac{g}{WACC})}{WACC - g} = \frac{NOPLAT(\frac{WACC - g}{WACC})}{WACC - g} =$$

$$= \frac{NOPLAT}{WACC}$$

Growth (g) plays no role.

What is the meaning of this formula?

$$\cancel{TV = \frac{NOPLAT}{WACC - g}}$$

Growth without investment.

4. Non-CF-based TV methods

Liquidation method

Liquidation or forced liquidation

Replacement methods

What has to and can be replaced from the market?

Multiple-based methods

Data on future market? Future comparable firms?

Typical mistakes in quantifying TV

Wrong choice of starting year
(Cycles, trends, standard operation, first TV year reflect maintainable performance)

Misprediction of ROIC and WACC
(Consider market growth, intensity of competition, and inflation.)

Mistake in discounting
(TV formulas provide the value for the beginning of the TV period.)

Using a real growth rate in a nominal model
(It is harder to believe but has no value effect if RONIC=WACC.)

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Predicting growth

- Historic data
- Management expectations
- Market/industry analysis
- Macroanalysis

What do we predict?

Sales, operational profitability, EBIT, Noplat, operational CF, …, net profit

Averaging yearly growth rates

Arithmetical or geometric mean?

0	**100**	
1	**120**	**+20%**
2	**96**	**-20%**

Arithmetical: (+20 + -20) / 2 = 0

Geometrical:

$$\sqrt{(1.2*0.8)} = 0.9797 \longrightarrow \textbf{yearly} -2\%$$

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Sustainable growth rate

g_{Equity} = ΔEquity/Equity =

= (Net profit – Dividend) / Equity =

= [Net profit * (1 – div. payoff ratio)] / Equity =

= (Net profit / Equity) * reinvestment rate =

= ROE * reinvestment rate

Equity and Debt are book values here!

$$ROE = ROIC + Debt/Equity*[ROIC - r_D*(1-t)]$$

What changes together with Sales?

Quantity: COGS, variable costs

Receivables, if payment conditions unchanged

Suppliers, as procurement growth (price effect may differ!), if payment terms unchanged

Short term liabilities? Partly. Wages? Taxes?

Debt? Not automatic, a new contract is needed.

Equity: g = ROE * reinvestment rate

Effects of growth

Profit and CF

Capital need and structure

Operational risk (r_A)

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Limits of growth

Can be financed from inner sources?

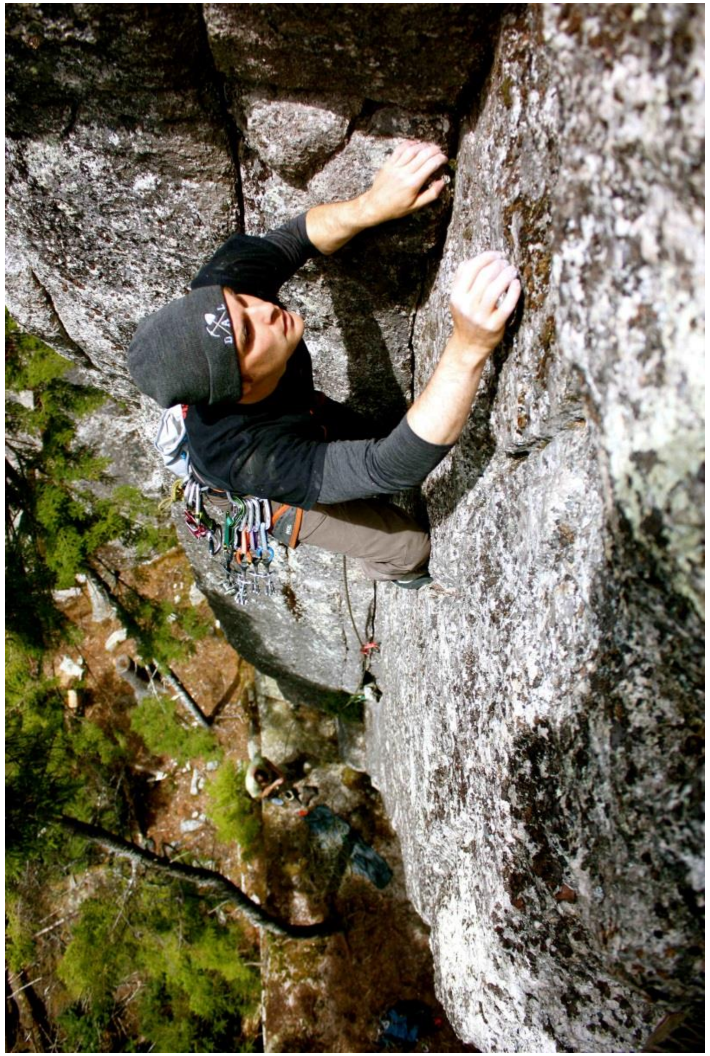
Economic (GDP/GNP) growth?

Growth of local and international markets?

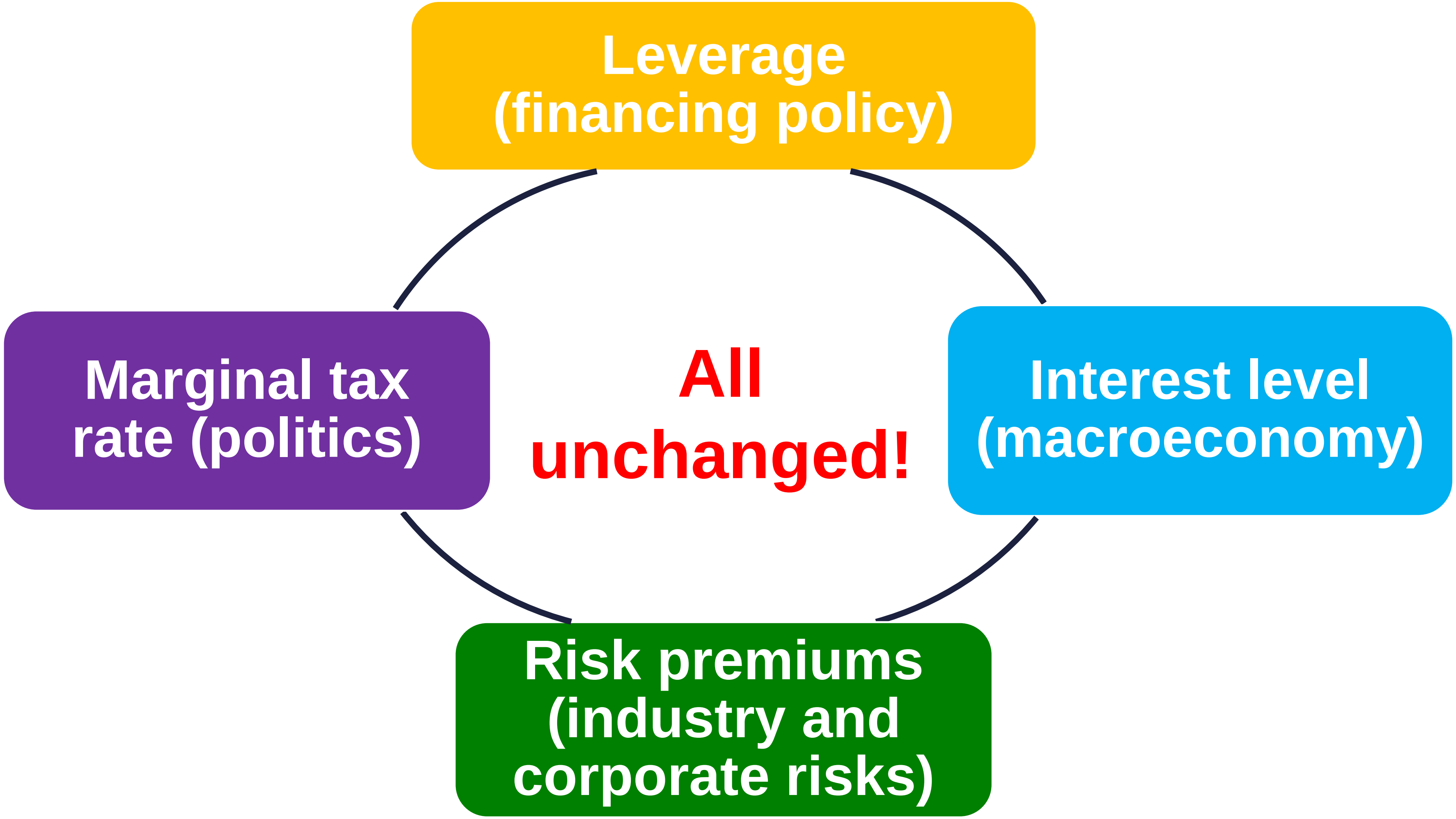
Issues linked to growth

- Nominal or real?
- Inflation stable?
- What currency?
- What supports it?

(market outlook, firm size, current growth, competitive advantages, competitors)



Assumptions of constant WACC



If WACC is constant …

Leverage is stable and already optimal,

No competitive advantage or drawback,

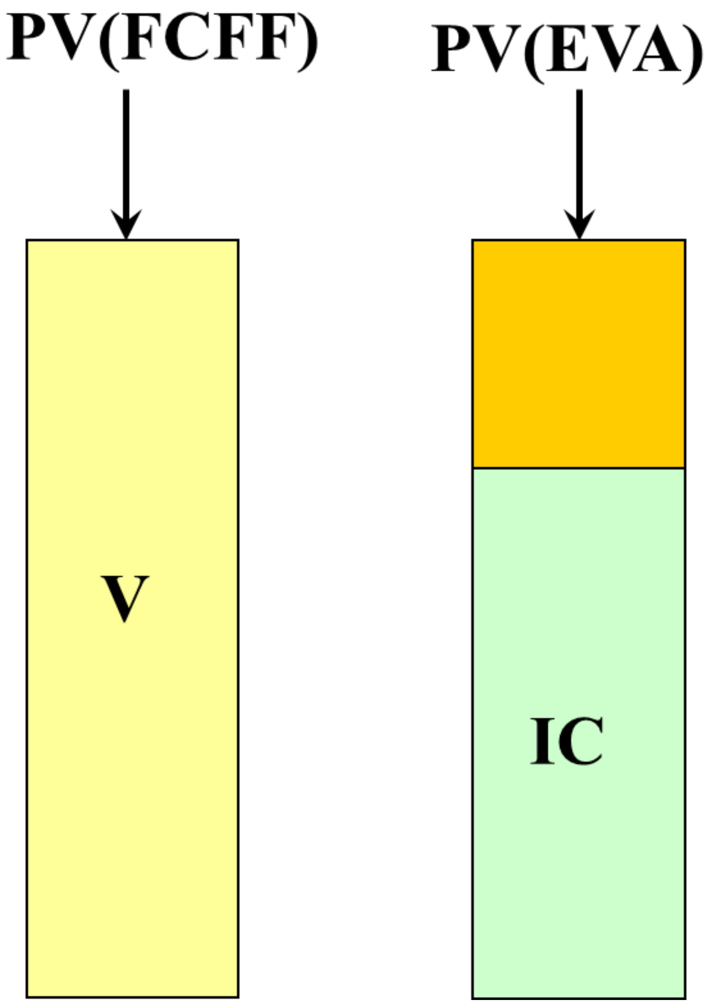
Industry and regulation unchanged.

The meaning of TV for different methods

FCFF/FCFE: The value of firm/equity at the start of the TV period.

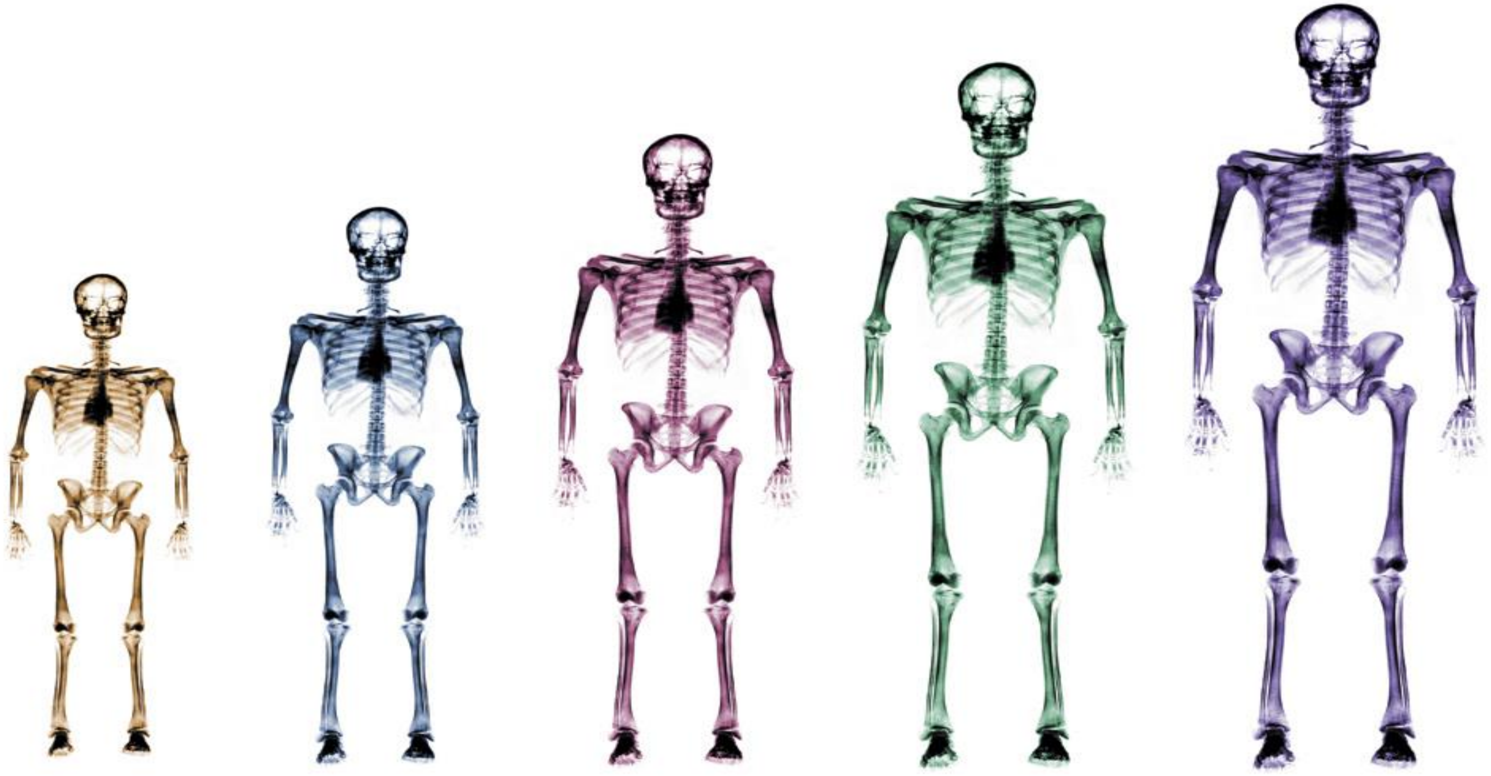
To get this value, you have to invest IC/book Equity, so that is there without any risk.

EVA/RI: The added value created by the firm during the TV period.



Lessons of TV 1

The length of the explicit period should have no value effect. We should extend the explicit period until all inputs reach their stable level.



Lessons of TV 2

Even in the case of RONIC=WACC it is not the same what the ROIC is at the beginning of the TV. If ROIC>>WACC, there will be an economic profit generated for very long time.

Only the TVs of EVA/RI models show what part of the beginning of TV value would be generated during the TV period.

IC

FCFF

EVA/RI

What is important? 1

ROIC	10.00%	10.00%
RONIC	10.00%	10.00%
g	3.00%	7.00%
Noplat	200.00	200.00
K	30.00%	70.00%
FCFF	140.00	60.00
EVA	0.00	0.00
EVA/WACC	0.00	0.00
TV_{FCFF}=V	2 000.00	2 000.00
TV_{EVA}	0.00	0.00
TV_{EVA}/V	0.00%	0.00%

WACC=10.00%

IC_{TV}=2000.00

What is important? 2

ROIC	15.00%	15.00%
RONIC	10.00%	10.00%
g	3.00%	7.00%
Noplat	300.00	300.00
K	30.00%	70.00%
FCFF	210.00	90.00
EVA	100.00	100.00
EVA/WACC	1 000.00	1 000.00
$TV_{FCFF}=V$	3 000.00	3 000.00
TV_{EVA}	1 000.00	1 000.00
TV_{EVA}/V	33.33%	33.33%

WACC=10.00%

IC_{TV}=2000.00

What is important? 3

ROIC	10.00%	10.00%
RONIC	15.00%	15.00%
g	3.00%	7.00%
Noplat	200.00	200.00
K	20.00%	46.67%
FCFF	160.00	106.67
EVA	0.00	0.00
EVA/WACC	0.00	0.00
TV_{FCFF}=V	2 285.71	3 555.56
TV_{EVA}	285.71	1 555.56
TV_{EVA}/V	12.50%	43.75%

WACC=10.00%

IC_{TV}=2000.00

What is important? 4

ROIC	15.00%	15.00%
RONIC	15.00%	15.00%
g	3.00%	7.00%
Noplat	300.00	300.00
K	20.00%	46.67%
FCFF	240.00	160.00
EVA	100.00	100.00
EVA/WACC	1 000.00	1 000.00
$TV_{FCFF}=V$	3 428.57	5 333.33
TV_{EVA}	1 428.57	3 333.33
TV_{EVA}/V	41.67%	62.50%

WACC=10.00%

IC_{TV}=2000.00

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Thank you for your attention. Any questions?

peter.juhasz@uni-corvinus.hu

Estimating the cost of capital

Péter Juhász, PhD, CFA

Matching the CF and the cost of capital

- **Currency**
- **Effects of financing (TS)**
- **Inflation**
- **Maturity**
- **Risk**

Estimating the required rate

Return of similar investments

- How is that similar?
- How do we know it is similar?

Measuring risk, calculating required rates

- CAPM
- Alternative methods (APT)

Whose risk is to be measured?

Investor	Risk	Metric	Requi-rement	Firm value
Private investor (one firm)	Total	Std deviation	40%	100/0.40= 250
Venture capital (one sector)	Sector portfolio +	Sector index beta	25%	100/0.25= 400
Hungarian investment found	Country portfolio +	Country index beta	15%	100/0.15= 666
Investment bank	Global portfolio +	Global index beta	10%	100/0.10=1000

Measuring the cost of capital

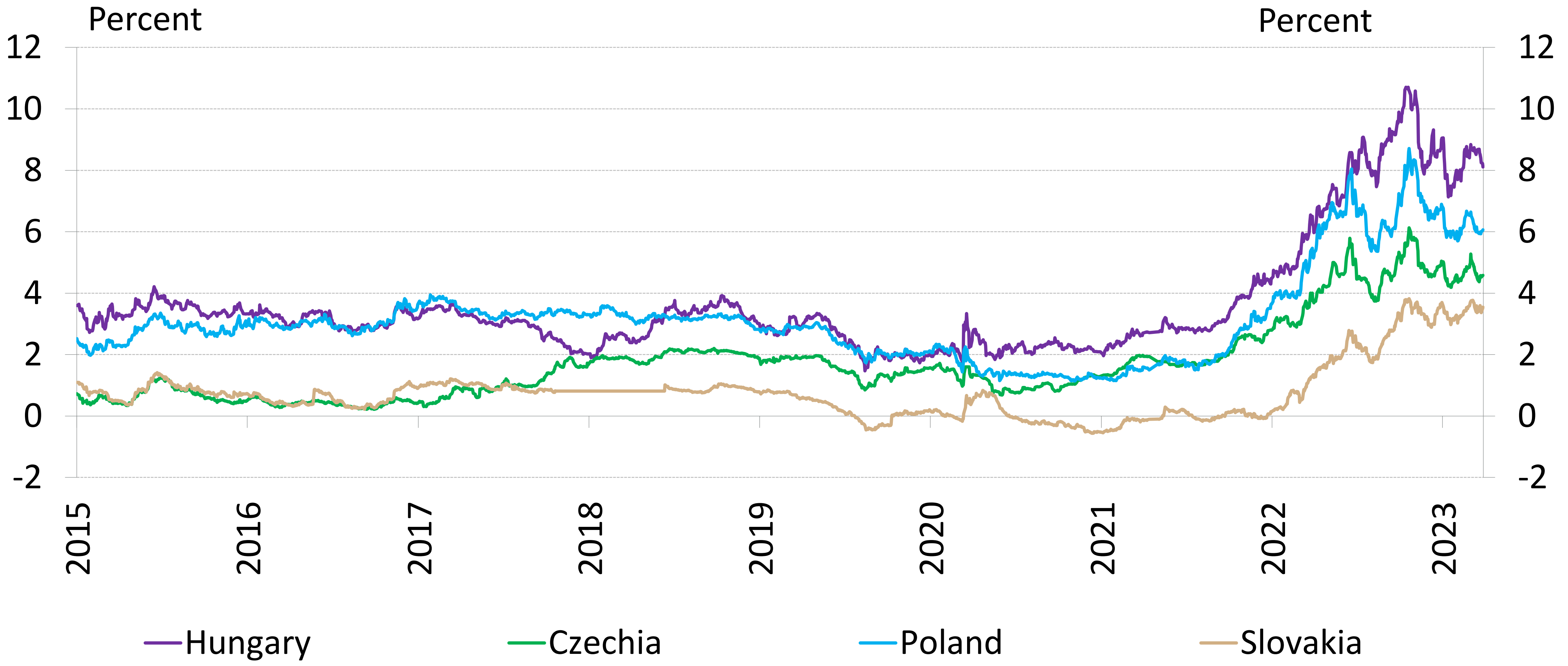
Cost of Debt	Historical risk premiums
	Credit rating premiums
	CAPM and debt beta
Cost of Equity	CAPM
	APT
	Implied return
	Build-up method

Cost of capital inputs 1 – Risk free rate

Free only from repayment risk; other (e.g. market) risks still exist.

- Real or nominal?
- Which country? (Switzerland vs Greece)
- Yield curves (spot/forward, never constant)
- Exchange rate predictions (parities)
- Consistent with inflation predictions?

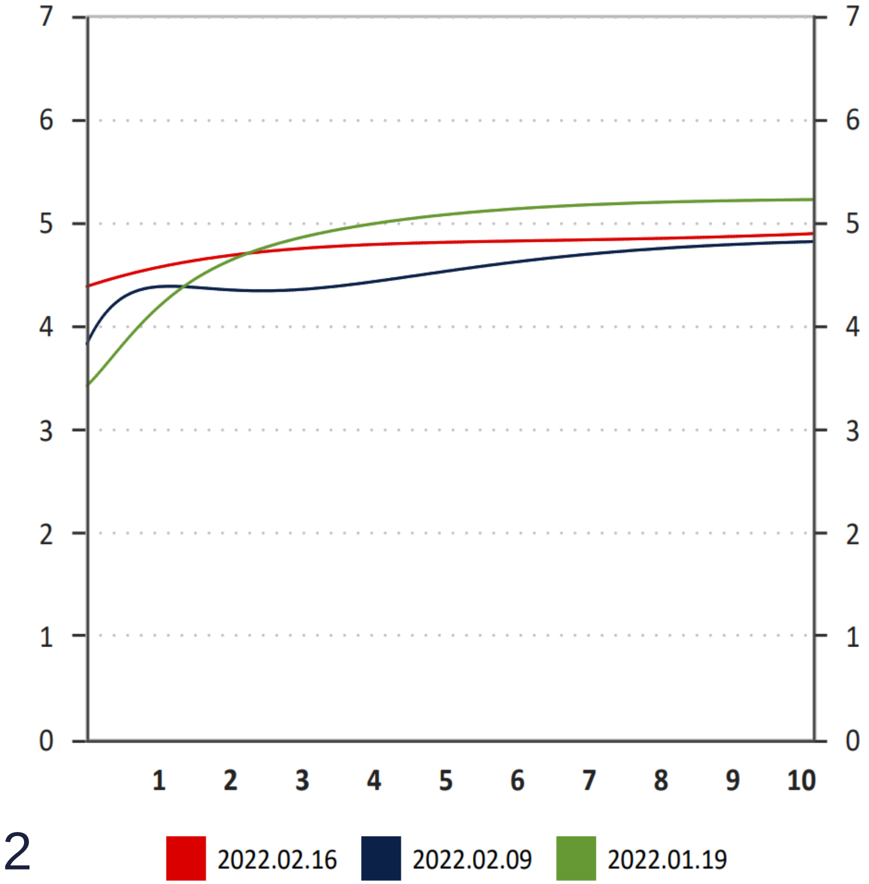
10-year government yields in CEE countries



Source: MNB, March 2023

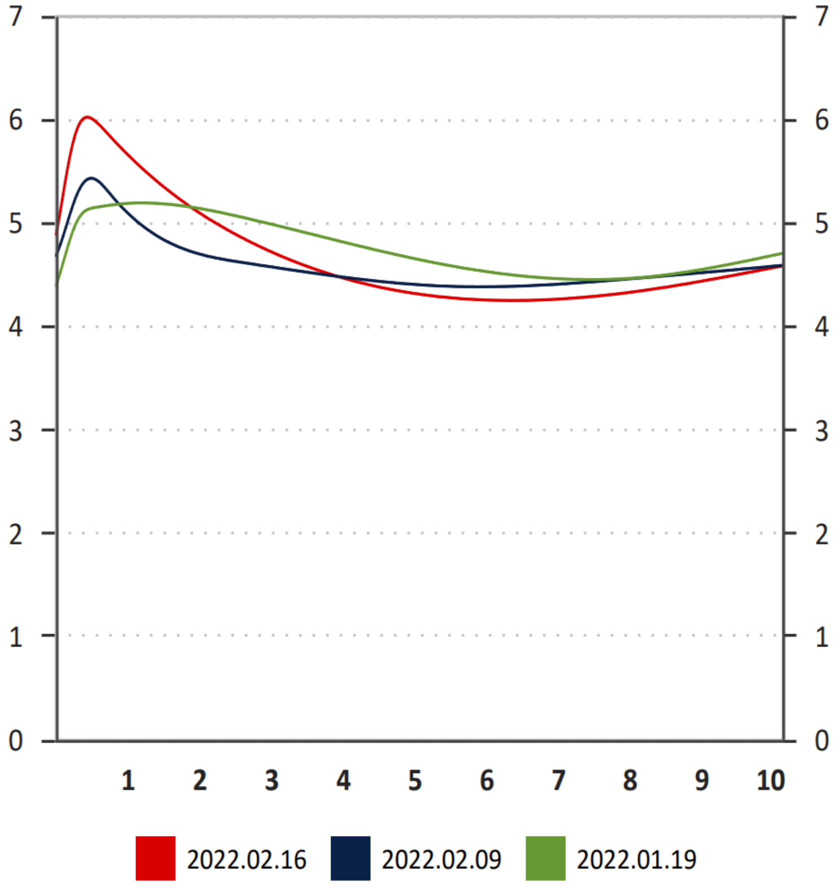
Estimating the forward rates

3-months gov. bonds



Forrás: MNB, MÁK
Megjegyzés: az MNB által Svensson módszerrel becsült kamatok. A vízszintes tengely lejáratokat mutat, években.

3-months interbank rates



Megjegyzés: az MNB által, bankközi pénzpiaci (FRA) és kamatswap jegyzések felhasználásával, spline módszerrel becsült kamatok. A vízszintes tengely lejáratokat mutat, években.

Source:
MNB, 22 February 2022

Cost of capital inputs 2 – Cost of debt

Former and existing debt contracts	Implied debt return of own or similar firms' bonds
Historical regressions (β_D)	Credit rating (risk free rate + premium)

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S&P credit rating premiums*

Rating	2010	2012	2014	2016	2018	2020	2022	2023
AAA	0.50%	0.65%	0.40%	0.75%	0.54%	0.63%	0.69%	0.69%
AA	0.75%	1.15%	0.70%	1.00%	0.72%	0.78%	0.85%	0.85%
A+	1.00%	1.30%	0.85%	1.10%	0.90%	0.98%	1.07%	1.23%
A	1.25%	1.40%	1.00%	1.25%	0.99%	1.08%	1.18%	1.42%
A-	1.50%	1.65%	1.30%	1.75%	1.13%	1.22%	1.33%	1.62%
BBB	2.00%	2.50%	2.00%	2.25%	1.27%	1.56%	1.71%	2.00%
BB+	3.50%	3.75%	3.00%	3.25%	1.98%	2.00%	2.31%	2.42%
BB	4.00%	4.75%	4.00%	4.25%	2.38%	2.40%	2.77%	3.13%
B+	4.25%	5.50%	5.50%	5.50%	2.98%	3.51%	4.05%	4.55%
B	5.25%	6.00%	6.50%	6.50%	3.57%	4.21%	4.86%	5.26%
B-	5.50%	6.75%	7.25%	7.50%	4.37%	5.15%	5.94%	7.37%
CCC	8.50%	8.75%	8.75%	9.00%	8.64%	8.20%	9.46%	11.57%
CC	10.00%	9.50%	9.50%	12.00%	10.63%	8.64%	9.97%	15.78%
C	12.00%	10.50%	10.50%	16.00%	13.95%	11.34%	13.09%	17.50%
D	15.00%	12.00%	12.00%	20.00%	18.60%	15.12%	17.44%	20.00%

*Huge manufacturing firms, USD

Source: Damodaran, January 2010-2023

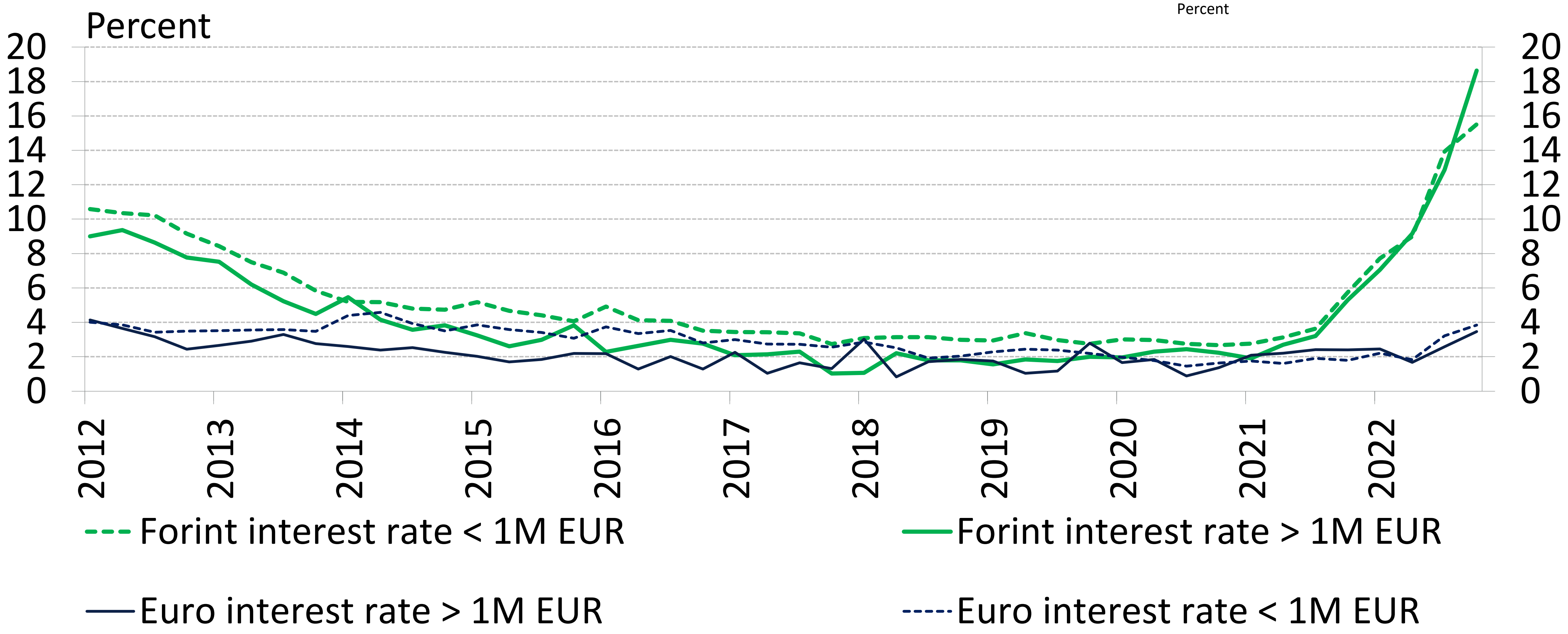
Reuters credit rating premiums*

Rating	1 y	2 y	3 y	5 y	7 y	10 y	30 y
Aaa/AAA	14	16	27	40	56	68	90
Aa1/AA+	22	30	31	48	64	77	99
Aa2/AA	24	37	39	54	67	80	103
Aa3/AA-	25	39	40	58	71	81	109
A1/A+	43	48	52	65	79	93	117
A2/A	46	51	54	67	81	95	121
A3/A-	50	54	57	72	84	98	124
Baa1/BBB+	62	72	80	92	121	141	170
Baa2/BBB	65	80	88	97	128	151	177
Baa3/BBB-	72	85	90	102	134	159	183
Ba1/BB+	185	195	205	215	235	255	275
Ba2/BB	195	205	215	225	245	265	285
Ba3/BB-	205	215	225	235	255	275	295
B1/B+	265	275	285	315	355	395	445
B2/B	275	285	295	325	365	405	455
B3/B-	285	295	305	335	375	415	465
Caa/CCC+	450	460	470	495	505	515	545
USA gov. bond rate	4.74	4.71	4.68	4.63	4.60	4.59	4.56

*Banks. USD.
January 2006

Source:
Bondsonline.com

Hungarian corporate debt rates*



*New lendings

Source: Inflation report, MNB, December 2022

Cost of capital inputs 3 – Tax rate

Nominal (if no information available)

Effective (historic)

Marginal (extensional investments)

Combined (growing firm)

We need predictions.

Corporate tax rate in the CEE region

%	2008	2010	2012	2014	2016	2018	2020	2021	2022
Albania	10	10	10	15	15	15	15	15	15
Bosnia and Herzegovina	10	10	10	10	10	10	10	10	10
Bulgaria	10	10	10	10	10	10	10	10	10
Czech Republic	21	19	19	19	19	19	19	19	19
Estonia	21	21	21	21	20	20	20	20	20
Belarus	24	24	18	18	18	18	18	18	18
Croatia	20	20	20	20	20	18	18	18	18
Poland	19	19	19	19	19	19	19	19	19
Hungary	**16**	**19**	**19**	**19**	**19**	**9**	**9**	**9**	9
Russia	24	20	20	20	20	20	20	20	20
Romania	16	16	16	16	16	16	16	16	16
Serbia	10	10	10	15	15	15	15	15	15
Slovakia	19	19	19	22	22	21	21	21	21
Slovenia	22	20	18	17	17	19	19	19	19
Ukraine	25	25	21	18	18	18	18	18	18

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Points of reference

	2008	2010	2012	2014	2016	2018	2020	2022
Austria	25.00%	25.00%	25.00%	25.00%	25.00%	25.00%	25.00%	25.00%
Cyprus	10.00%	10.00%	10.00%	12.50%	12.50%	12.50%	12.50%	12.50%
United States	40.00%	40.00%	40.00%	40.00%	40.00%	27.00%	27.00%	25.00%
United Kingdom	30.00%	28.00%	24.00%	21.00%	20.00%	19.00%	19.00%	19.00%
France	33.33%	33.33%	33.33%	33.33%	33.30%	33.00%	33.00%	25.00%
Greece	25.00%	24.00%	20.00%	26.00%	29.00%	29.00%	29.00%	22.00%
Ireland	12.50%	12.50%	12.50%	12.50%	12.50%	12.50%	12.50%	12.50%
Japan	40.69%	40.69%	38.01%	35.64%	30.86%	30.86%	30.86%	23.20%
China	25.00%	25.00%	25.00%	25.00%	25.00%	25.00%	25.00%	25.00%
Germany	29.51%	29.41%	29.48%	29.58%	29.72%	30.00%	30.00%	30.00%
North America aver.	36.75%	35.50%	33.00%	33.25%	33.25%	26.75%	26.75%	26.75%*
Asia average	27.99%	23.96%	22.89%	21.91%	21.92%	21.21%	21.21%	21.09%*
EU average	23.29%	23.04%	22.60%	21.34%	22.09%	21.29%	21.29%	21.12%*
Global average	**26.10%**	**24.69%**	**24.40%**	**23.57%**	**23.62%**	**24.05%**	**24.03%**	**23.79%***

Source: KPMG. quoted by: Damodaran, 2023 *Avg for 2021

Take care!

	2008	2010	2012	2014	2016	2018	2020	2022
Bahrain	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
United Arab Emirates	55.00%	55.00%	55.00%	55.00%	55.00%	55.00%	55.00%	0.00%
Kuwait	55.00%	15.00%	15.00%	15.00%	15.00%	15.00%	15.00%	15.00%

Source: KPMG, quoted by: Damodaran, 2023

Cost of capital inputs 4 – Cost of equity

- Interview (What do owners want?)
- Expert opinion (required rates for other firms)
- Regression methods: CAPM, APT
- Built-up method

Risks included in equity

Cost of equity contains two types of risk:

Risk of operation:

$$\boxed{r_A = r_f + \beta_A (r_M - r_f)}$$

Risk of financing:

$$\boxed{\beta_E = \beta_A + \frac{D}{E}(\beta_A - \beta_D)}$$

$$\boxed{r_E = r_A + \frac{D}{E}(r_A - r_D)}$$

$$\boxed{r_E = r_f + \beta_E (r_M - r_f)}$$

Historical regressions:

$$r_{Asset} = a + b * r_M$$

$$b \text{ (sloop)} = covar\ (r_{Asset}, r_M) / \sigma^2_M = „\beta_E”$$

Market portfolio?

Time frame?

Yield type?

Currency?

Exchange rate corrections?

Steps of beta estimation

Collect industry player data (r_E, β_E, r_D, β_D)

Estimate unleveraged cost of capital individually

Calculate the average of individual values (r_A, β_A)

Re-leverage r_A and β_A in line with the target's financing structure

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Market risk premium applied*

Country	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Austria	6.0	5.8	5.5	5.6	5.3	6.6	6.4	6.3	5.9	6.0	6.6
Belgium	6.0	6.0	5.5	5.4	5.5	6.6	6.4	6.3	6.0	6.0	7.0
UK	5.0	5.0	5.0	5.0	5.0	6.2	5.9	6.0	5.8	5.7	6.0
France	6.0	5.9	6.0	5.5	5.5	6.7	6.4	6.1	6.0	5.7	6.3
Greece	6.0	16.5	7.4	15.0	12.4	17.6	16.9	15.3	14.6	6.4	12.2
Germany	5.0	5.0	5.4	5.1	5.0	5.9	5.2	5.7	5.7	5.5	5.9
Italy	5.5	5.5	5.5	5.2	5.5	6.7	6.4	6.1	6.1	6.0	6.0
Portugal	5.9	8.5	6.5	5.5	8.0	8.0	7.7	7.7	6.9	6.5	6.8
Spain	5.5	6.0	5.9	5.5	6.0	6.8	6.2	6.4	6.4	6.4	6.3
USA	5.5	5.0	5.5	5.3	5.0	5.7	5.2	5.5	5.4	5.5	5.5
Switzerland	5.5	5.0	5.7	5.0	5.0	7.5	7.2	6.0	5.9	5.1	5.9
Hungary	7.0	8.7	8.9	8.9	8.0	8.6	8.4	8.5	7.1	7.1	9.0

* r_M-r_f, median

Source: Fernandez, 2012-2021

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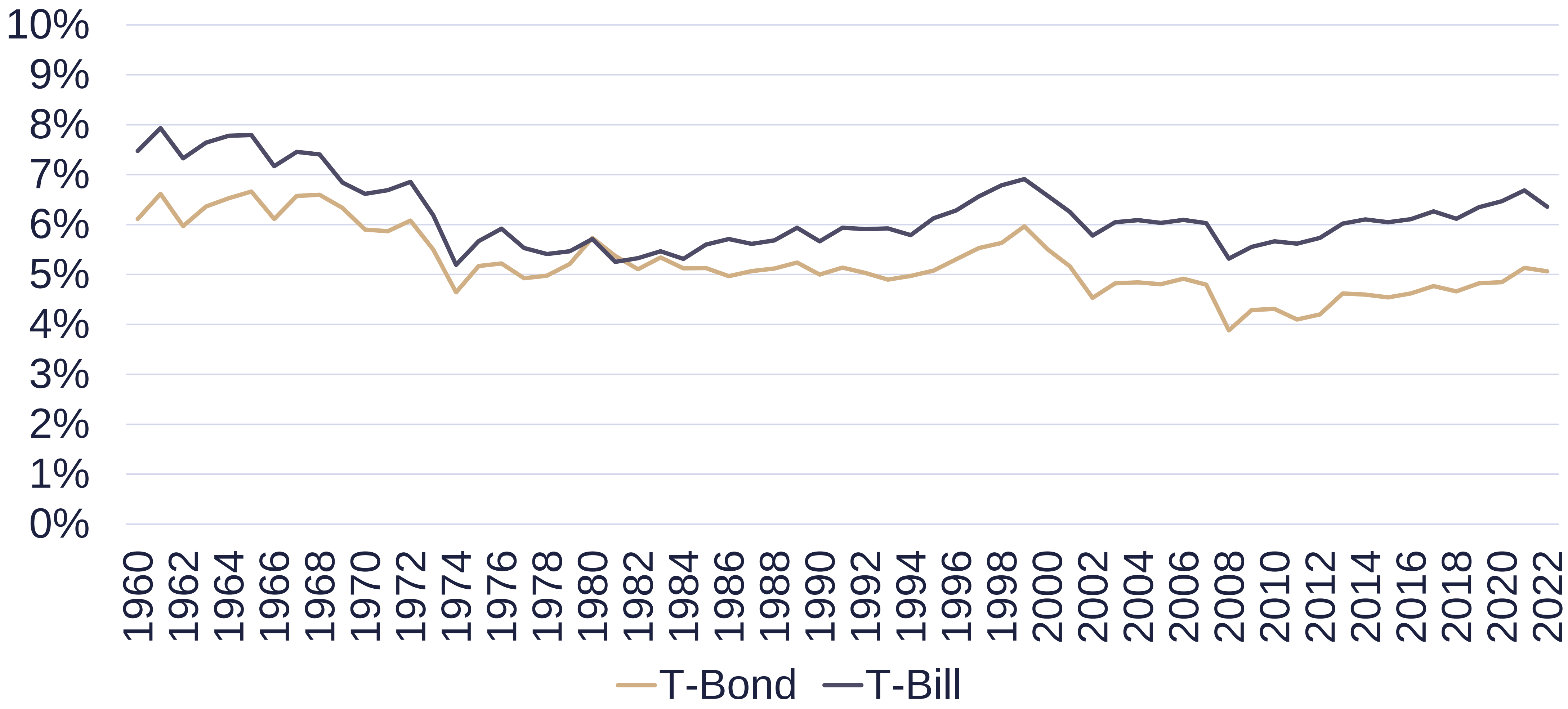
Historical market premium*



Source: Damodaran, March 2023

Historical market premium*

*r_M-r_f, %, USD, difference between **S&P500** and US government bonds, between the given year and **2013**, geometric average of yearly rates

Gov. bond type	**2002**	**2004**	**2006**	**2008**	**2009**	**2010**	**2011**	**2012**
Long term	1.70	2.50	2.30	0.90	11.40	7.60	6.00	13.60
Mid term	3.10	3.80	3.70	2.70	13.40	9.70	8.00	15.40
Short term	4.40	4.90	4.70	4.10	14.80	11.00	9.00	15.90

Source: Ibbotson SBBI, 2013

Fundamental beta

- Industry (cicles, basic vs luxury) – „$\beta_{industry}$”
- Operative risks (cost structure: variable/fix cost,
  - Δ EBIT / Δ Sales – DOL) – „$\beta_{strategy}$”
- – $\Phi(\beta_{industry}, \beta_{strategy}) \Rightarrow$ „β_A”
- Financing risks (taxation, D/E) $\Rightarrow$ „β_E”

Cost of capital inputs 6 – Leverage ratio

Historical data	Predicted trend
Target (announced) level	Iteration

Industry betas (Western EU)

Industry	N	β_E	β_A	D/V
Semiconductor	42	1,53	1,31	15,47%
Apparel	118	1,10	0,97	12,96%
Steel	57	1,13	0,83	30,55%
Air Transport	37	1,25	0,63	52,98%
Green & Renewable Energy	63	0,91	0,60	37,02%
Oil/Gas (Production and Exploration)	102	0,90	0,58	38,45%
Farming/Agriculture	50	0,83	0,52	41,02%
Utility (Water)	10	0,59	0,33	50,72%
Food Wholesalers	15	0,80	0,20	79,51%

Source: Damodaran online, March 2023

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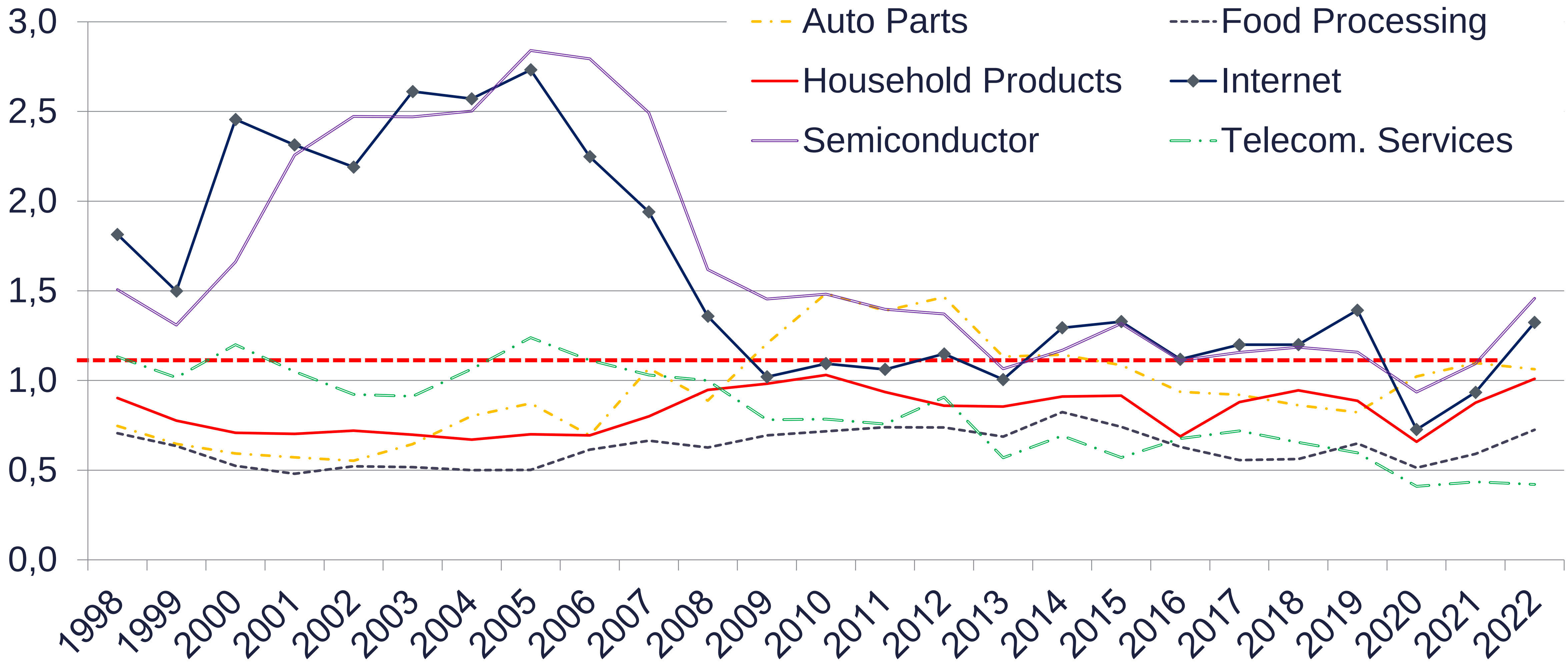
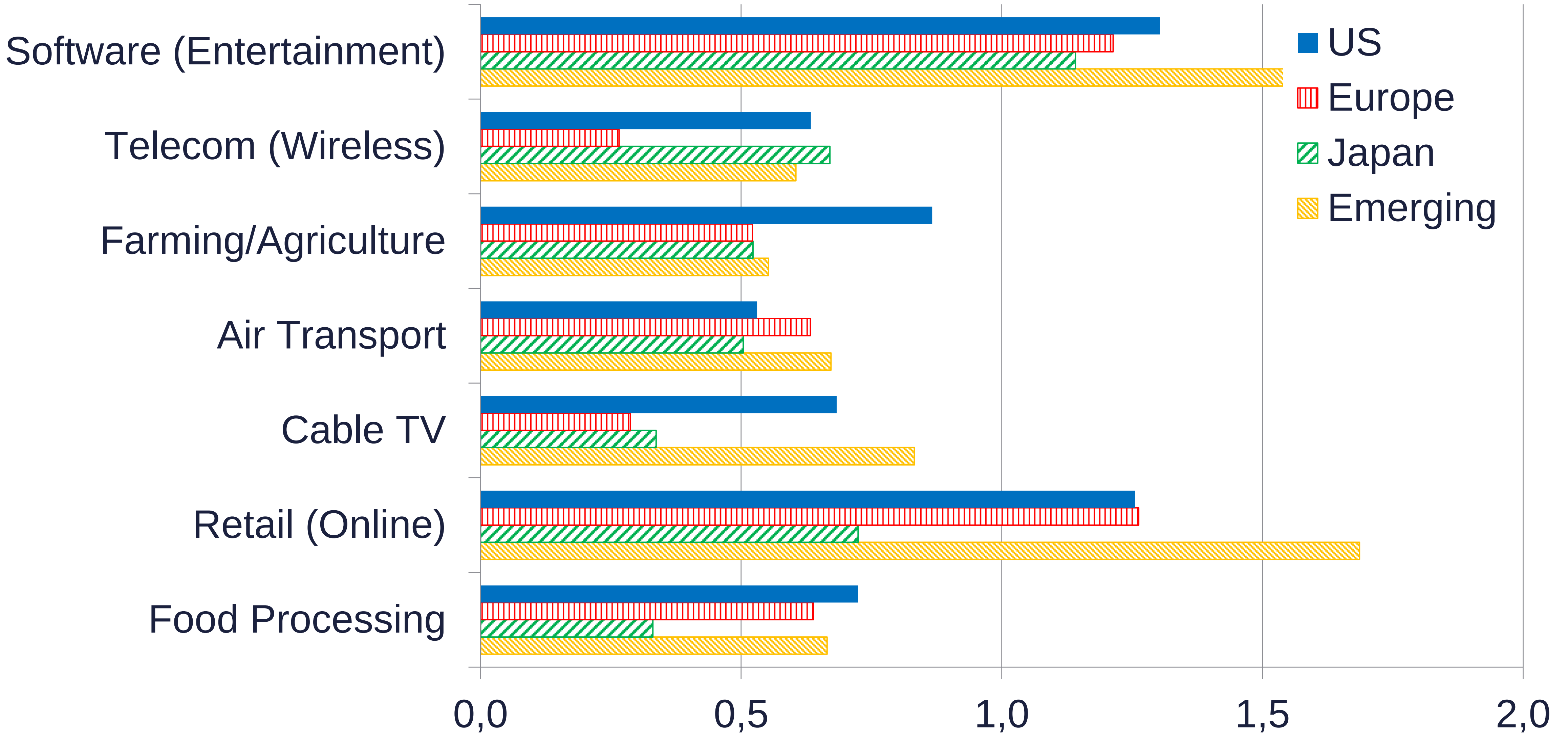
Stability of β_A* over time



*in the USA

Source: Damodaran online, March 2023

Geographic stability of β_A



Source: Damodaran online, March 2023

Estimating the cost of equity using the build-up method

Estimating required rate on equity:

$$\varepsilon(r_A) = r_f + MRP + RP_I + RP_S + RP_O$$

where

$E(r_E)$ = required rate of equity,

r_f = risk-free rate of return,

MRP = market risk premium,

RP_I = industry risk premium,

RP_S = size risk premium,

RP_O = other firm specific risk premium.

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A sample

Government bond rates of developed countries	1.0%
Country risk premium of Hungary	1.5%
Inflation difference	1.0%
Risk free rate (r_f)	**3.5%**
Market risk premium (MRP: r_M-r_f)	**5.5%**
Industry premium	1.5%
Size and other firm specific premiums	2.0%
Required return of assets (r_A)	**12.5%**

No need for betas.

Stability of size premium*

*%, USD, the yield premium of the given sized firms compared to huge companies, between the given year and **2013.**

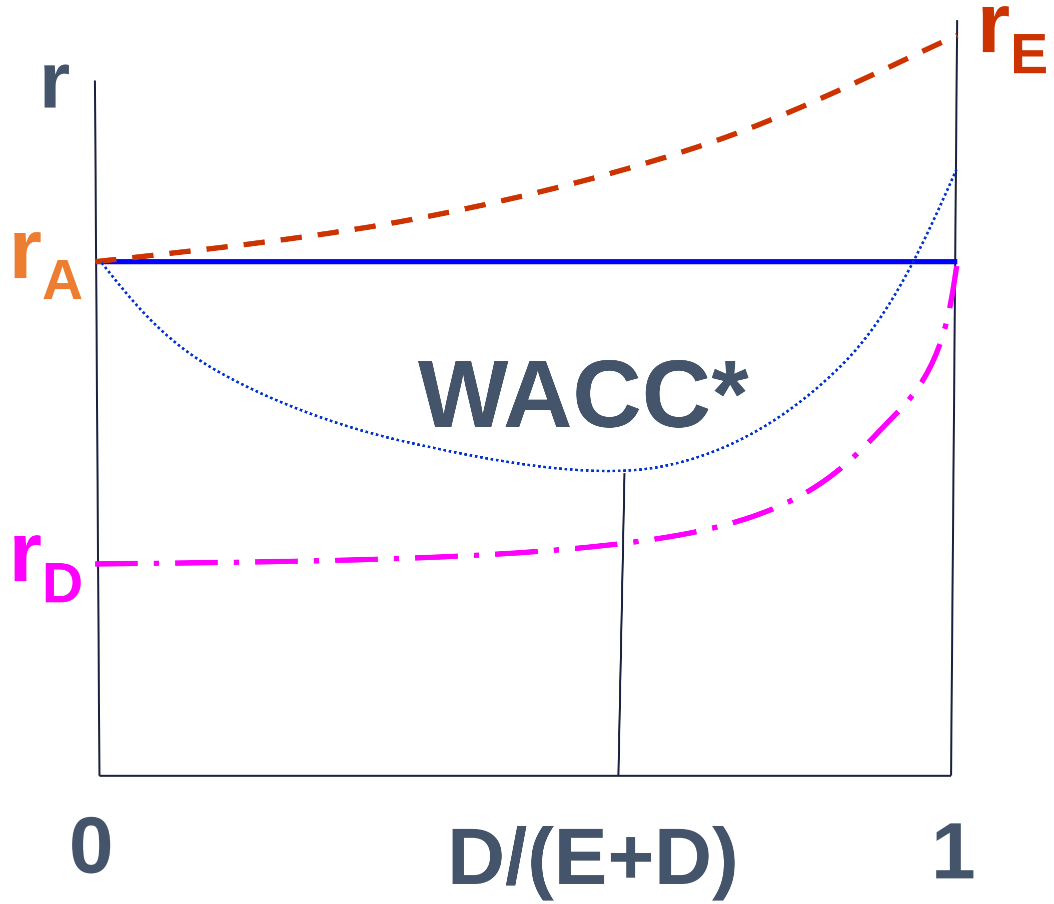
Capitalisation**	2001	2002	2003	2004	2005	2006	2007	2008
Medium	5.0	4.5	4.0	3.5	3.1	2.7	3.7	4.6
Small	7.0	4.9	4.4	3.2	2.5	2.7	3.6	5.5
Micro	10.8	7.1	5.9	2.1	1.9	2.4	3.0	6.4

** Categories include NYSE, NYSE Amex, and NASDAQ firms' 3-5, 6-8, 9-10 deciles.

Source: Ibbotson SBBI, 2013.

$$WACC = \frac{D}{V} * r_D * (1 - t) + \frac{E}{V} * r_E$$

But this formula misses the effects of financial distress.



* With financial distress effects

Can you reach the optimum?

- Curves are not continuous,
- Only your direct neighbourhood can be tracked.
- The amount of debt could be limited.
- Curves may deviate across financiers.
- Curves change over time.

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Thank you for your attention. Any questions?

peter.juhasz@uni-corvinus.hu

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Special challenges of DCF-based valuation

Péter Juhász, PhD, CFA

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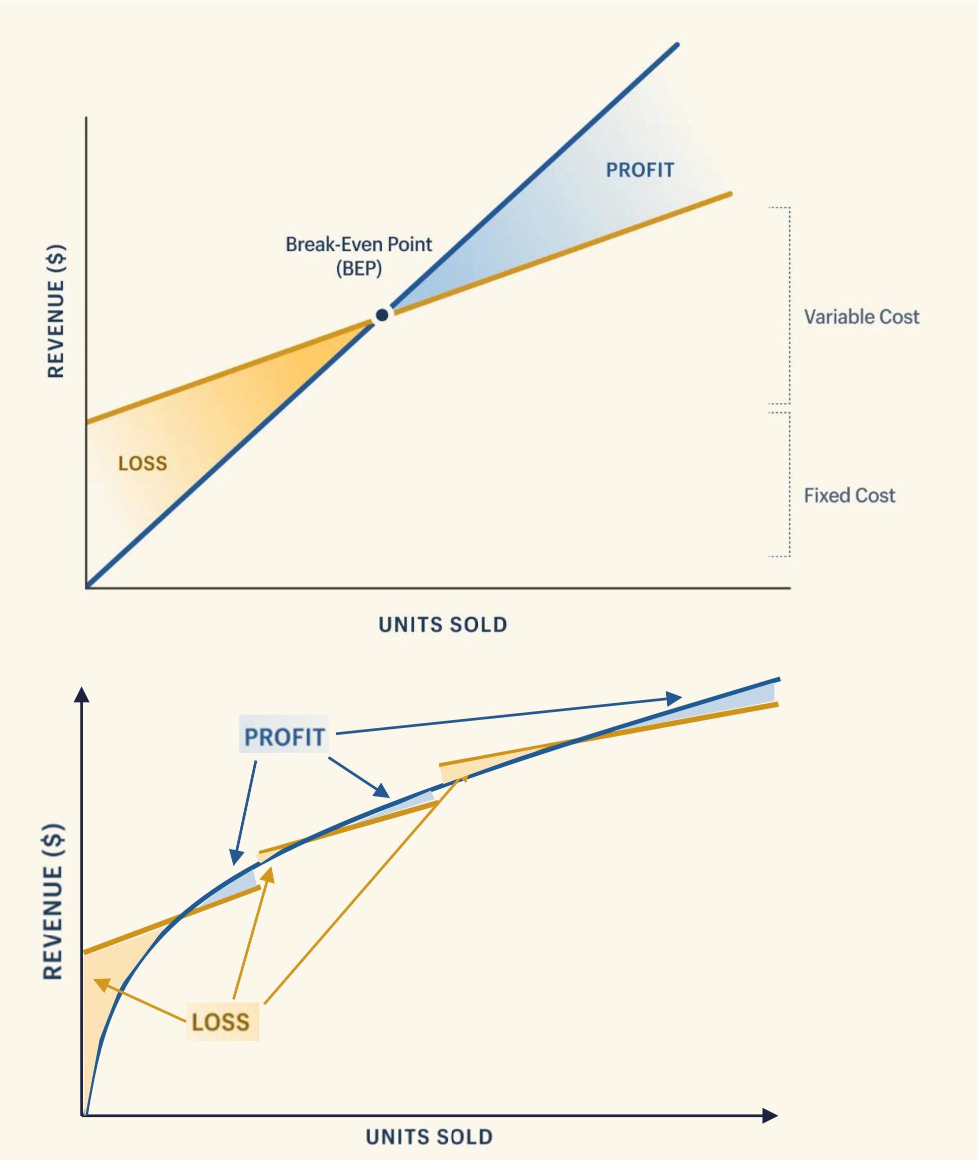
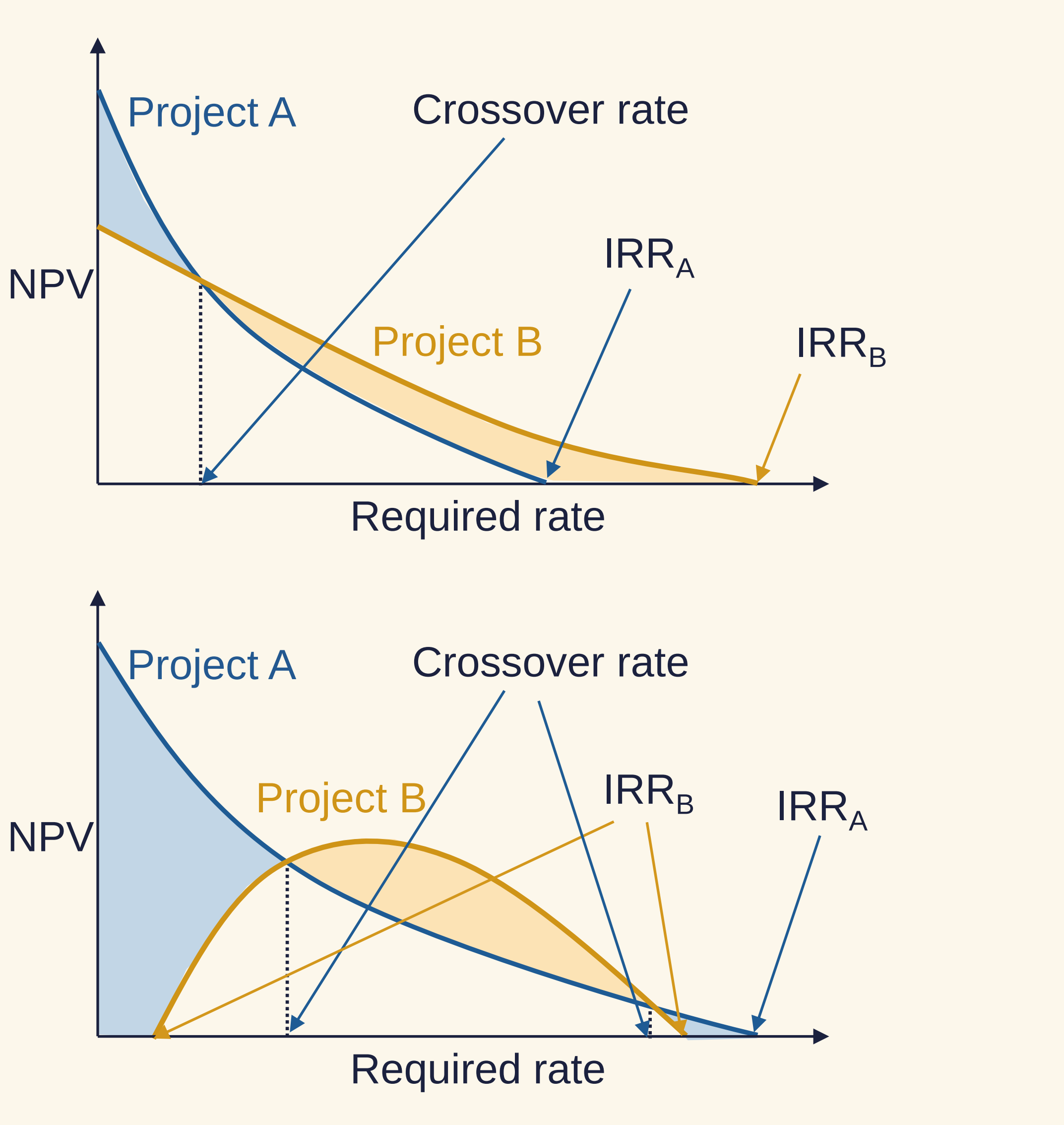
Modelling & Simulation

Breakeven analysis



1. For what volumes would the operation be profitable?
2. Contrasts additional revenues and expenses
3. Focuses on a given period only
4. In the long run all costs are variable
5. At what price would we take the order?

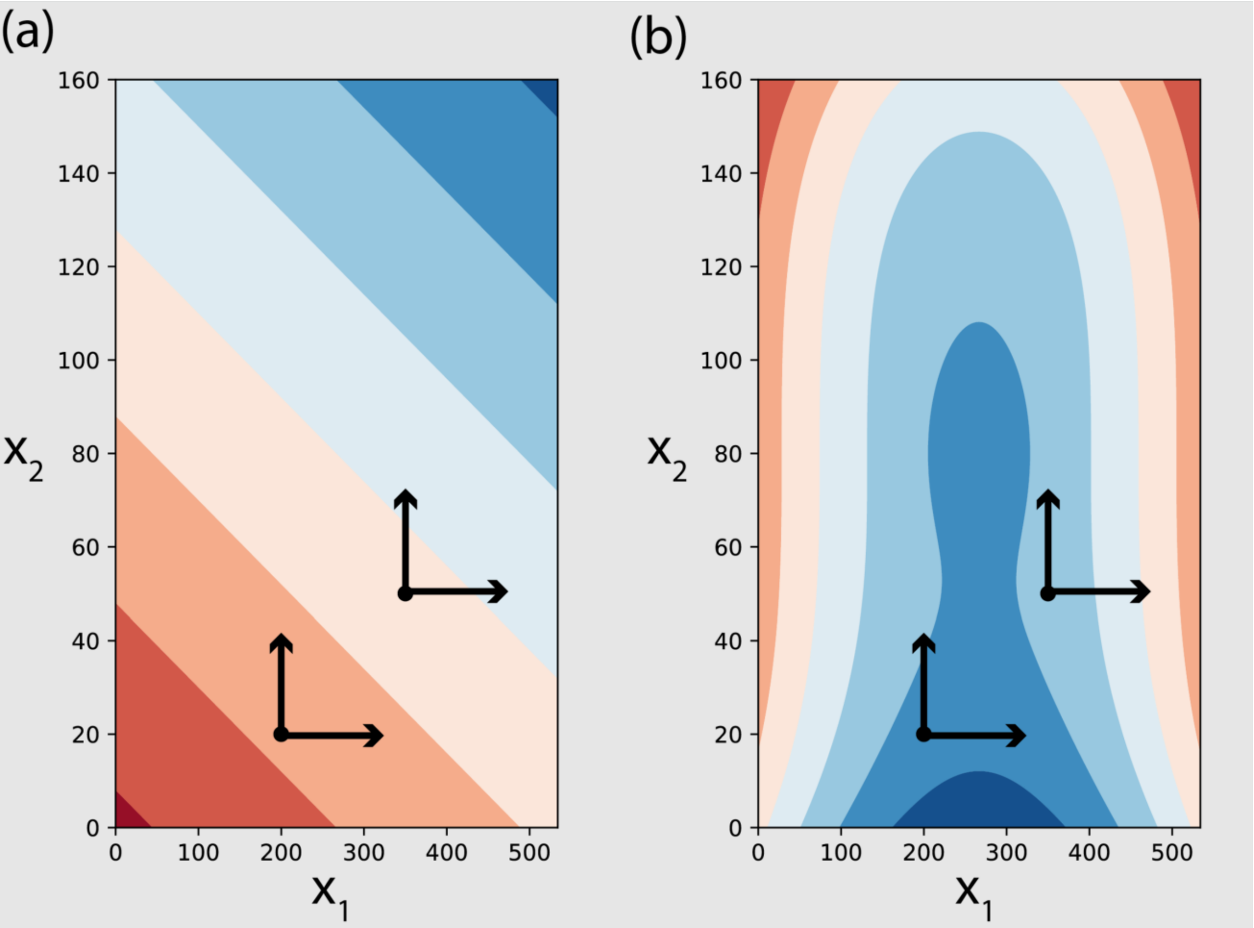
Crossover rate analysis

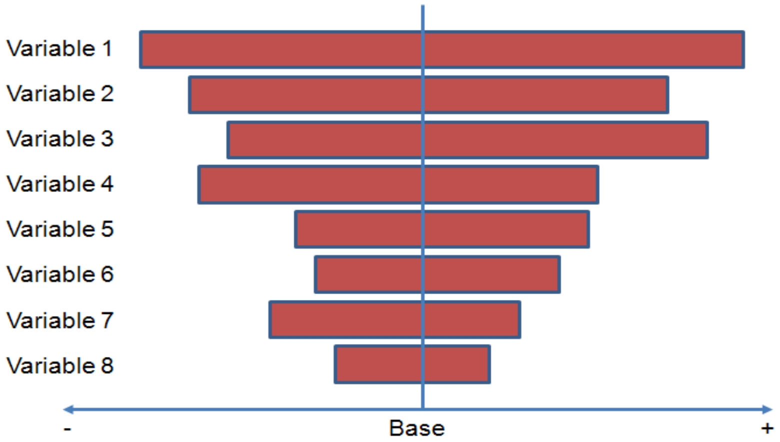


1. Contrasting project NPVs at multiple required rates
2. If the CF distribution across time differs the reaction of NPVs differ
3. Visualises the risk in cost of capital estimation
4. Does not consider project length

Sensitivity analysis

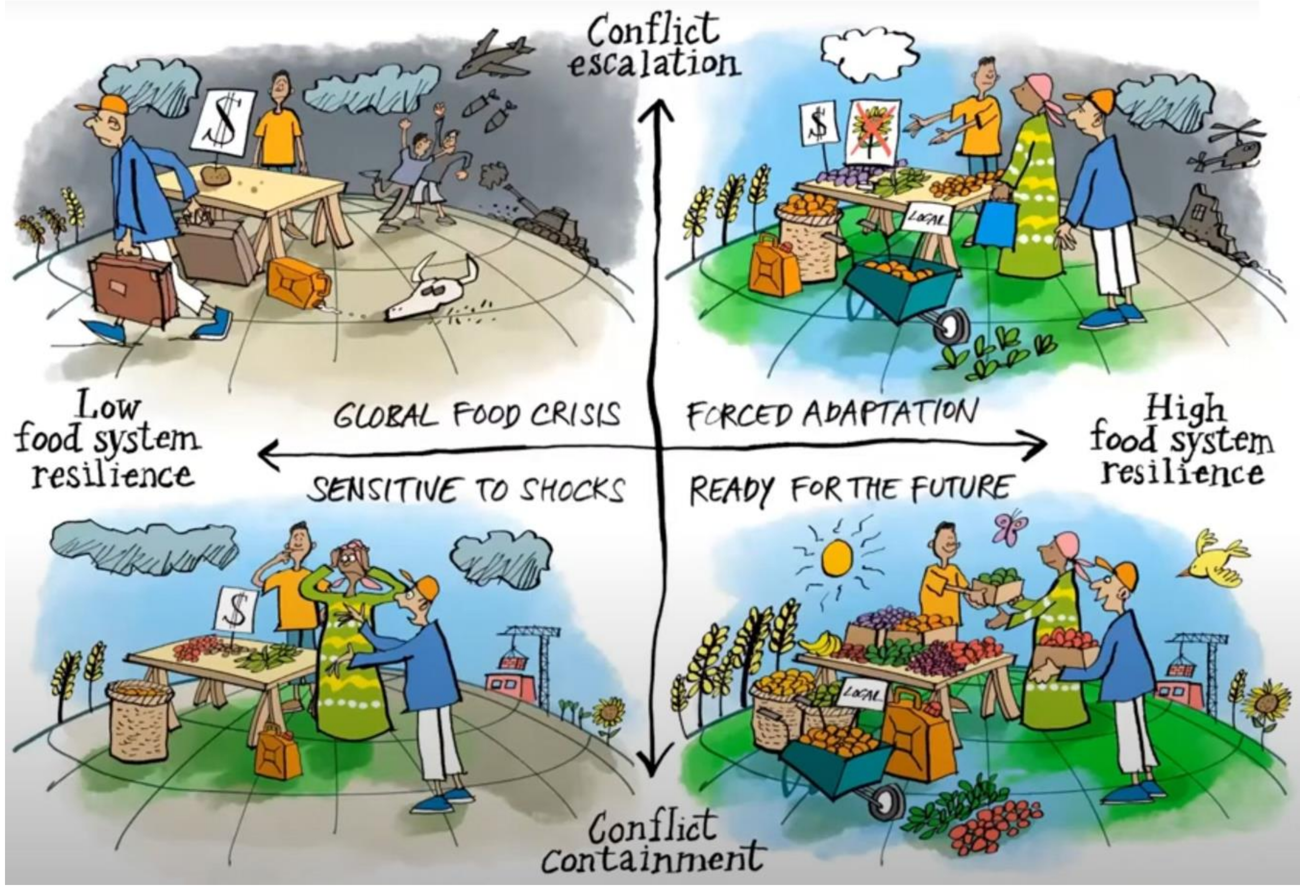
1. Substitutes expected values by the extremes of the likely value range
2. Identifies key variables to be estimated at highest percision





Scenario analysis

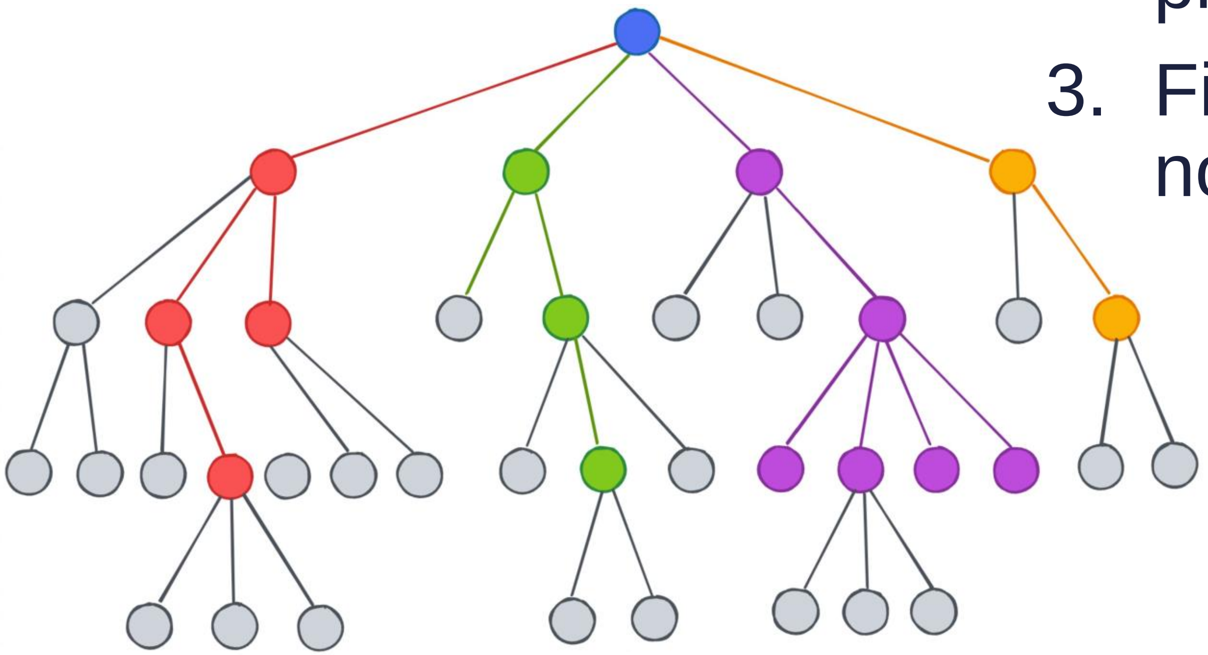
1. List possible believable future states of word.
2. Pick the 3-5 most likely of those and descrivbe them in detail.
3. Choose the best startegy for each of the scenarios.
4. Weight the outcomes for each scenario by the probability of the scenario to happen.



https://www.nlfoodpartnership.com/insights/Scenario_Analysis_and_Webinar_on_Food_Security_Impacts_of_the_Ukraine_War/

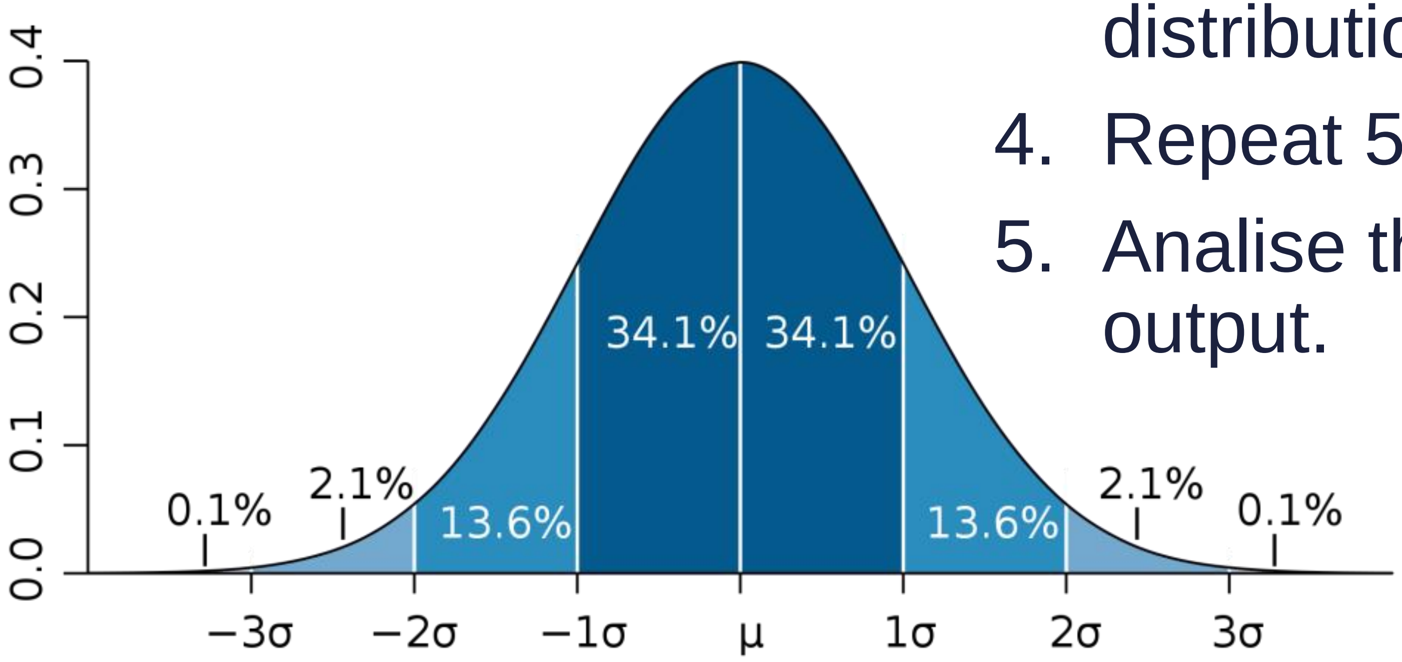
Decision trees

1. Find the decision points for the given scenario.
2. Describe choices of the decisionmaker (could be the nature!). List outcomes, probabilities, consequences.
3. Find the best choice for each of the nods backwards.



Monte Carlo simulation

1. Find the key input parameters.
2. Describe the distribution of those inputs.
3. For each of the variable pick a random realisation from the distribution.
4. Repeat 5,000 to 100,000 times.
5. Analise the distribution of the model output.



Real options

Opportunities that gain value for an increase in risk

Poor projects are often offered for sale as real options

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Multi-business firms

Multi-business companies

Business units have different risks so CFs can not be merged

Invested capital and required rate for Bus can often be estimated only.

We must address the transfer prices

What would be the market price?

HQ may have a negative value

Is there any synergy? Where does it come from?

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Valuing a multi-BU firm 1

Perpetuity-like FCFF=100, g=5%, D/V=50%, r_D=5% t=20%, r_A=10,5%, thus: WACC=10%.

Business value:

$$V = \frac{FCFF}{(WACC - g)} = \frac{100}{(10\% - 5\%)} = 2000$$

Valuing a multi-BU firm 2

BU A FCFF=30, BU B FCFF=90, HQ FCFF=-20.

A superficial estimation:

$$V_A = \frac{30}{(10\% - 5\%)} = 600$$

$$V_B = \frac{90}{(10\% - 5\%)} = 1800$$

$$V_K = \frac{-20}{(10\% - 5\%)} = -400$$

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Valuing a multi-BU firm 3

Detailed analysis:

	A	B	HQ	Average*
FCFF	30.00	90.00	-20.00	100.00
g	7.00%	4.33%	5.00%	5.00%
r_A	10.00%	10.66%	10.50%	10.50%
WACC	9.50%	10.16%	10.00%	10.00%

*FCFFs used as weights.

Valuing a multi-BU firm 4

Enhanced valuation:

$$V_A = \frac{30}{(9{,}5\% - 7\%)} = 1200{,}00$$

$$V_B = \frac{90}{(10{,}16\% - 4{,}33\%)} = 1543{,}74$$

$$V_K = \frac{-20}{(10\% - 5\%)} = -400{,}00$$

$$\boxed{V=2343.74}$$

Valuing a multi-BU firm 5

If BU A would be a stand-alone firm, its FCFF were 20, while that of the HWQ -15 (synergy lost):

$$V_A = \frac{20}{(9{,}5\% - 7\%)} = 800{,}00$$

$$V_B = \frac{90}{(10{,}16\% - 4{,}33\%)} = 1543{,}74$$

$$V_K = \frac{-15}{(10\% - 5\%)} = -300{,}00$$

Valuing a multi-BU firm 6

At what price would we sell BU A?

$$V_A = V_{A+B+K} - V_{B+K}$$

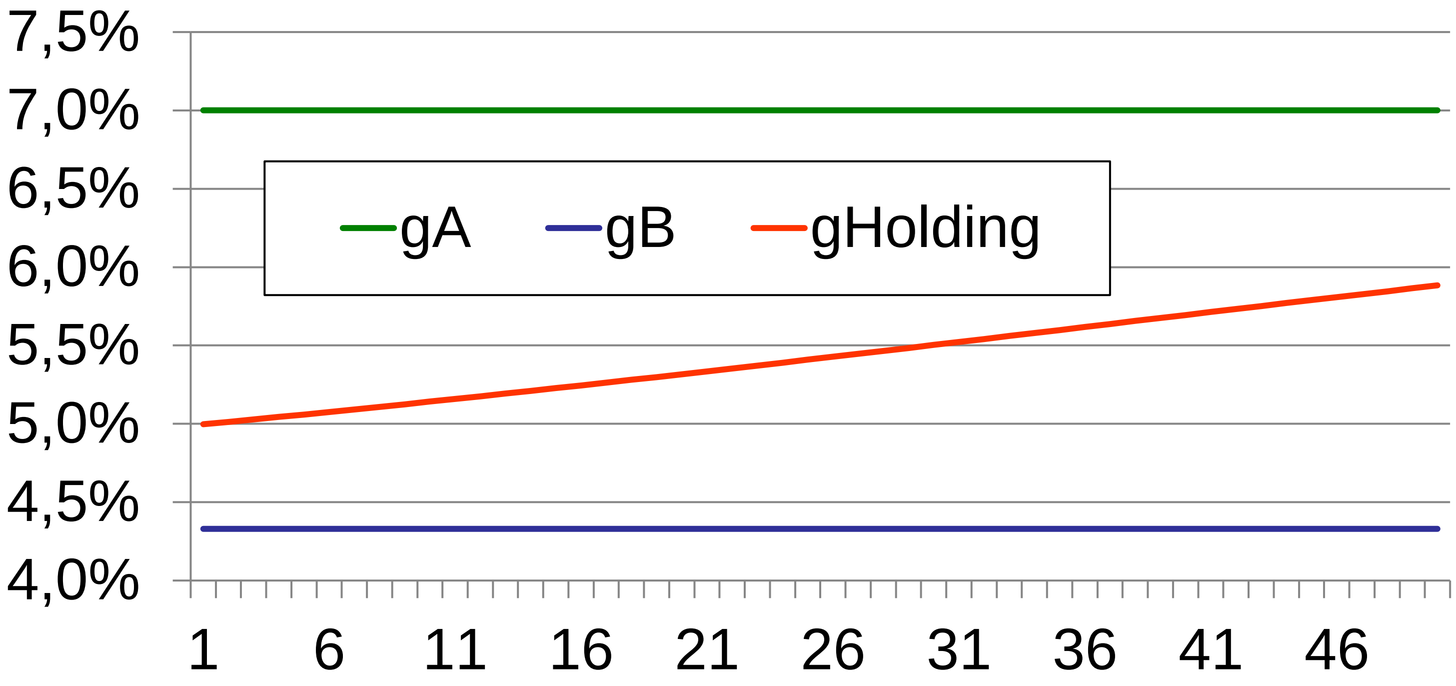
We do not need the value of BU A.

$$V_A = 2343{,}74 - (1543{,}74 - 300) = 1100$$

This is the value of the BU decreased by the costs emerging at the HQ.

Valuing a multi-BU firm 7

What is the actual growth rate of the firm if there would be no change in the operation?



Perpetuity-like BUs do not necessarily add up to a perpetuity.

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Inflation & FX rates

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Effects of the inflation

Inflation is not value neutral.

1. Not all items increase with inflation: D&A
2. Not all items are driven by the same inflation: selling price, procurement price, energy, wages, import, export
3. Effect of inflation may appear delayed: inventory

Valuation may only be done based on nominal plans.

Exchange rate risk

FX rate changes due to inflation rate differences are not to be managed; only purchase parity deviations should be addressed

Market hedging (forward, swap, option)

Corporate diversification across markets

Investor diversification

Seasonality

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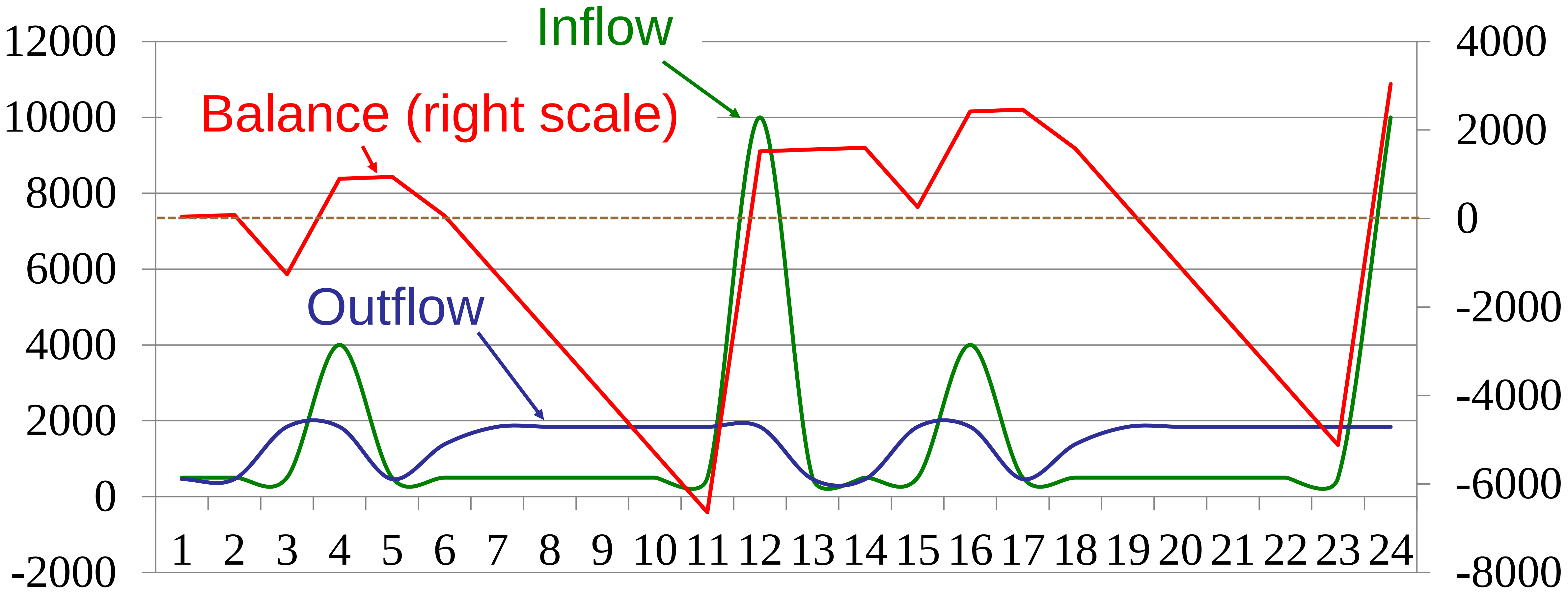
Seasonality

Are all CFs due at the same point in time?

Is that point the end of the year?

Seasonality effect on cash flows



At a rare of 1% the multiplier is 87.32%.

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Valuing liabilities

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Valuing liabilities

Bank loans

Bonds

Rental agreements

Lease

Convertible bonds

Management options

Minority interests

Preferred equity

Pension funds

Legal processes

Environmental obligations

Valuing debt

Generally, at book value (if a variable rate debt)

Fixed rate debt and subsidised debt calls for a separate valuation (May influence WACC calculation!)

In case of financial difficulties, the value of debt is directly linked to the value of the firm

Non-operational debt should not be considered in WACC

Other liabilities

Off-balance sheet items

E.g. legal processes: extent, probability, date

Dividend preferred shares

Like uncovered debt

Options

Management options, warrants, convertible debt

Minority interests

Needs valuation of affiliates

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Loss-making firms

The available information is not relevant

- Negative CF
- Growth rate is hard to estimate
- Tax calculation is challenging
- Continuous operations is not realistic (bankruptcy)

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Limits of applying the DCF approach 2

Young or new companies

the necessary information is not existing

Probabilities only ⇒ Venture capital

Differences: frequency of information service, required rate depends on the VC firm

Liquidity discount, extra risk due to fast growth

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Limits of applying the DCF approach 3

Private firms

the necessary information is not available

- More lax reporting rules than elsewhere (comparability?)
- Less information shared (Auditor? Analysts?)
- Exit is harder
- No market data (competitors data, betas)
- Management and ownership overlapping (salary vs dividend, non-operational assets)

A DCF „alternatives”

1. Use non-CF data
   (Dividend, net profit)
2. Use data from other firms
   (Multiple-based methods)
3. Use alternative approaches
   (going concern vs recreation or liquidation)

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Thank you for your attention. Any questions?

peter.juhasz@uni-corvinus.hu

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Multiple-based valuation methods

Péter Juhász, PhD, CFA

Multiple-based methods

Comparable transactions

Based on **earlier actual take-over transactions**

Distortions

The markets change over time
Synergic premiums

Market multiples

Based on **current stock exchange quotes**

Distortions

Market efficiency

Minority discount

Are all watches similar?

A simple example

What is the value of a book store with a sales of 200 M HUF if another book shop of 150 M HUF revenue has recently been sold for 160 M HUF?

P_1/S_1=160/150=1.07

P_2=1.07*200=<u>213.3 M HUF</u>

Pros and Cons

+

- Quick
- Simple
- Reflects the current market conditions

-

- Implicit assumptions
- Quality of data
- Comparability
- Bubbles
- Any value could be justified

The multiple dictionary

In multiples	In valuation
P	Price of one single share
E (EPS)	After tax profit/number of shares
P/E = Price / EPS	E / Profit after tax
EV (Enterprise value)	E + D - Cash
EV/BV (Book Value)	(V – Cash) / IC
P/BV	E / Book value of equity
P/S = Price / Sales per share	E / Sales
EV/S	(V – Cash) / Sales
CF	Operative CF (usually)

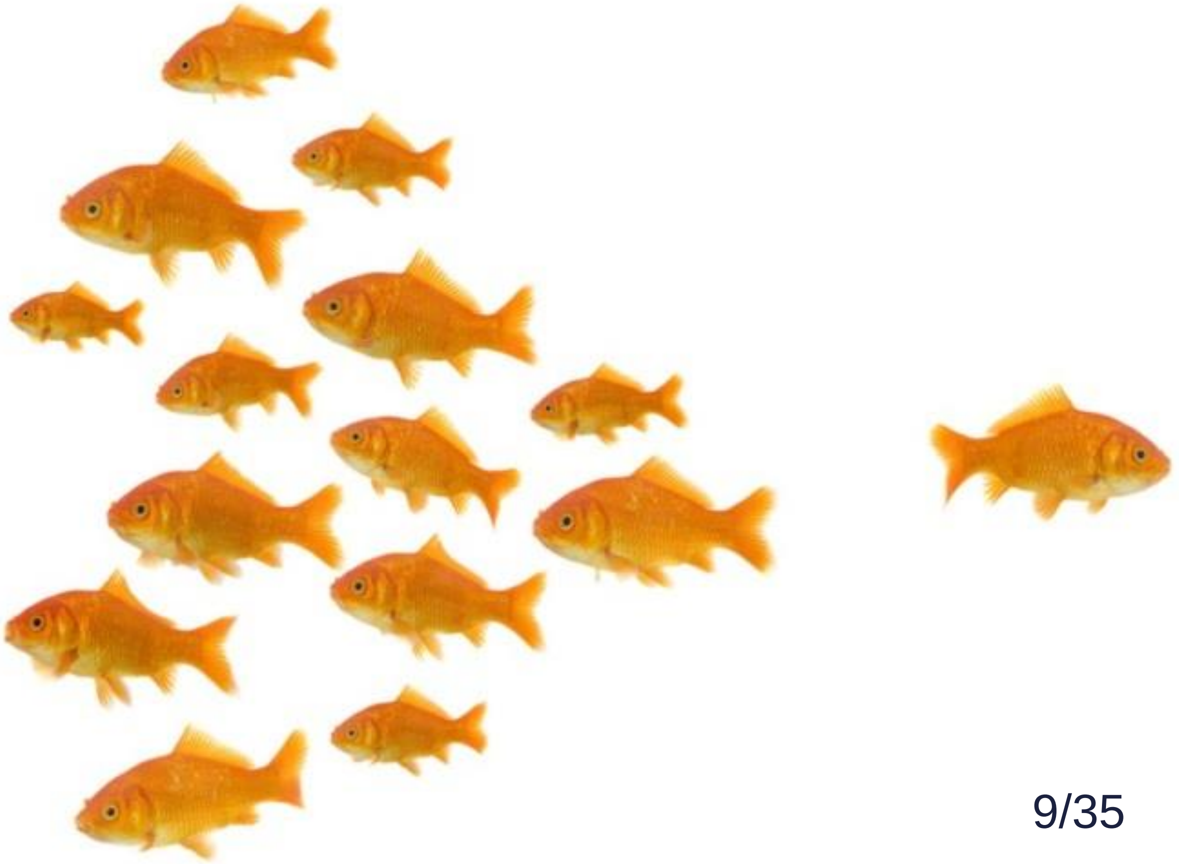
Grouping the multiples

Profit-based

Revenue-based

Asset-based

Industry specific



Defining a multiple

Simple	Clear	Accessible
Easy-to-understand	Properly defined (last, average, expected, extraordinary)	We have the data needed

Choosing the right ratio

Relevance

Either the numerator or the denominator should be some kind of value.

Consistency

The numerator and the denominator should be of the same dimension (V or E).

Uniformity

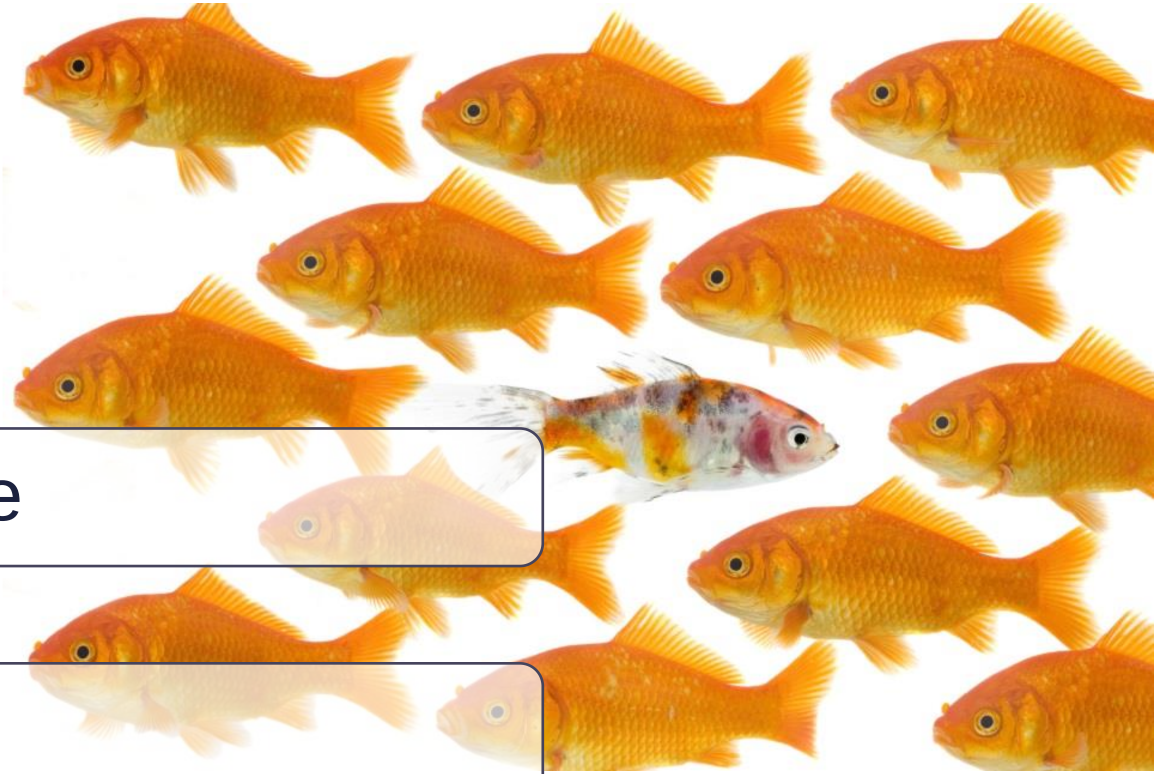
Same financial year, accounting standard, currency

Empirical tests for applicability

Description power:

What sample size is needed?

What is a high/low value?



Distribution of historical multiple value

Is the population homogeneous?

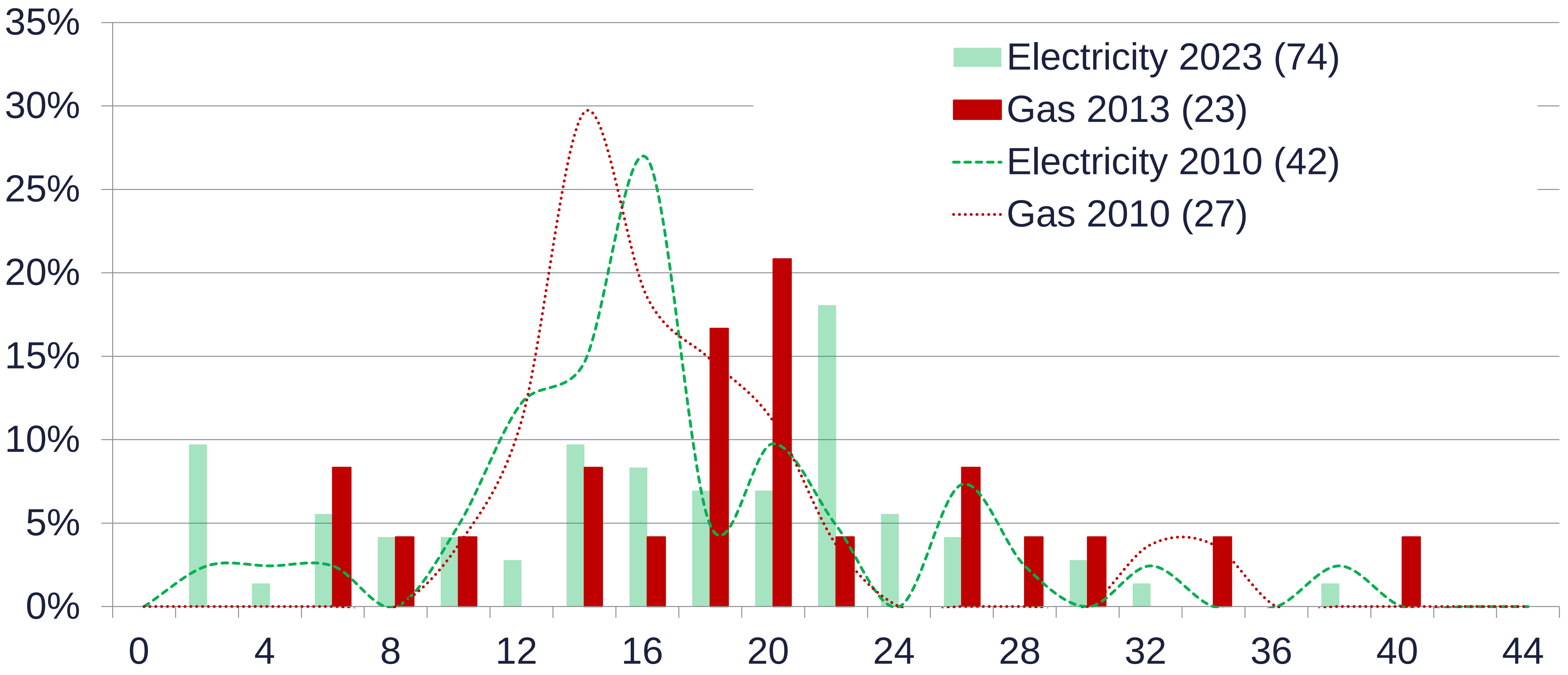
Outliers, averages

What needs to be deleted? What would the average show?

Calculation challenges

Negative profit or equity → distortions

P/E rates (USA, March)



Source: Yahoo! Finance

Content control

What fundamentals influence the value of the multiple?
What are the prerequisites of comparability?

$$P_0 = \frac{DIV_1}{(r_E - g)} = \frac{EPS_1 * div\%}{(r_E - g)}$$

Dividend pay-off ratio

$$\frac{P_0}{EPS_1} = \frac{P}{E} = \frac{div\%}{r_E - g}$$

Growth outlook

Risk of equity

$$\frac{EV}{EBIT} = \frac{E_0 + D_0 - Cash_0}{EBIT_1} = \frac{(1-t)*(1-K)}{WACC - g}$$

$$\frac{P}{S} = \frac{P_0}{Sales_1} = \frac{\frac{EPS}{Sales} * div\%}{r_E - g}$$

$$\frac{EV}{Sales} = \frac{E_0 + D_0 - Cash_0}{Sales_1} = \frac{\frac{EBIT}{Sales} * (1-t)*(1-K)}{WACC - g}$$

$$\frac{P}{BV} = \frac{P_0}{BV(E)_0} = \frac{ROE * div\%}{r_E - g}$$

$$\frac{EV}{BV} = \frac{EV_0}{BV(E+D)_0} = \frac{ROI * (1-t)*(1-K)}{WACC - g}$$

The process of multiple-based valuation

0. Pick a multiple.

1. Choose comparable firms. Similar risk, growth, profit generation (No industry or size requirement theoretically.)

2. Control for differences. (Similar but not the same.)

Subjective corrections (weighted average)

Modified multiples (corrected for differences)

3. Check reasonability.

Industry regressions

Market regressions

Assumptions of multiple types 1

Profit-based (P/E, EV/EBIT): the same link between the right CF and the given profit. (g, CF-generation, risk)

Revenue-based (EV/S): from a given amount of revenue any firms generates the same profit. **(g, CF-generation, risk, „margin”: CF/S)**

Assumptions of multiple types 2

Asset-based (P/BV, EV/IC): the assets generate the same profit everywhere.

(g, CF-generation, risk, ROI(c)/ROE)

Industry-specific (EV/employee): the given specific variable creates the same profit everywhere (g, CF-generation, risk, „industry”: same technology and efficiency)

P/E effects of differences

$$\frac{P}{E} = \frac{div\%}{r_E - g}$$

Interest rates ↑ → Cost of capital ↑, P/E ↓

Risk taking willingness ↑ → r_E↓, P/E ↑

General market outlook → average g ↑, P/E ↑

Profitability ↑ → div% ↑, P/E ↑

Do not forget that in the long run

ROE * (1-div%) = g*

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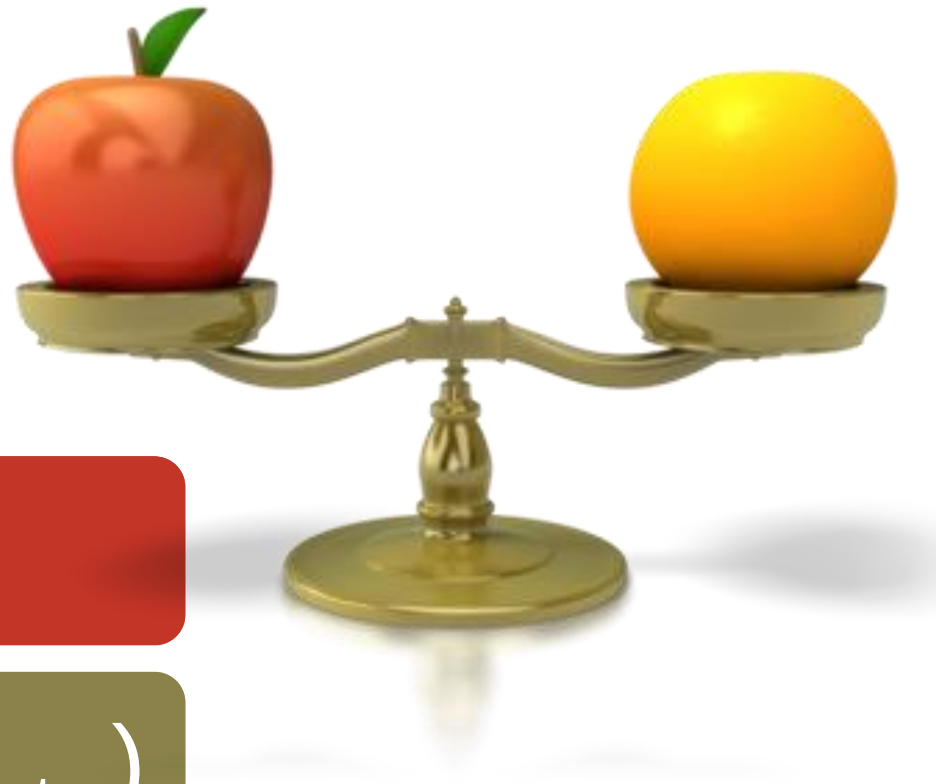
Interpreting P/E 1

High P/E:

High dividend pay-off

Low r-g

- r big & g huge
- r small & g tiny



P/E/g (PEG)

Relative P/E = $(P/E_{Firm})/(P/E_{Industry})$

Interpreting P/E 2

		PEG	
		Low	High
P/E	Low	r high, g low	r high, g very low
	High	r high, g high	r low, g low

Divided payoff ratio?

Only true for firms with the same dividend policy.

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Easy to use
Always positive
No accounting issues

Very far from the CF

P/S is inconsistent.

$$\frac{EV}{S} = \frac{\frac{EBIT}{Sales} * (1-t) * (1-K)}{WACC - g}$$

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Asset-based multiples 1

\+ Stable, easy-to-compare,
loss making firms create no issue

− Accounting policy differences,
off-balance sheet items, BV(E) < 0

P / BV(E)

P_0/BV_0 = ROE * $(1+g_n)$ * OFH / $(r-g_n)$ = ROE * P/E

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Asset-based multiples 2

EV / BV (D+E) (MV/BV)

$$= ROIC * (1-K) / (WACC – g)$$

$$ROIC = NOPLAT / BV(D+E)$$

Stability: $ROIC * K = g \rightarrow$

$$K = g / ROIC$$

$$EV/BV = (ROIC – g) / (WACC – g)$$

James Tobin
(1918-2002)

Asset-based multiples 3

Tobin's Q

= Market value of Assets / Replacement value of Assets

Original use: When shell we purchase a stock?

NPV (added value) ratio

Importance of off-balance sheet items

Hard to use for valuation (no E or EV in the formula)

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Industry specific multiples

Value creation linked to operation

No accounting issues

No further information needed

Same technology and efficiency?

Examples

Number of subscribers, readers, visitors, buyers, patients, quantity produced, owned resource confirmed, sales area, number of rooms

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Special sector multiples

EV/EBITDAR	Retail, airlines	Without rental fees and lease
EV/EBITDAX	Oil and gas, mining	Without exploration expenses
EV/Resource	Oil and gas, mining, lumber	
EV/Capacity	Oil and gas refinery, hotels, restaurants	
EV/Production (Service)	Oil and gas, harbours, airports	
EV/Subscriber	Telecom, media, internet services	
P/tBV	Banks	Tangible book value

A real life example 1

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BANKS

	Price (local currency)	P/E 2013	P/E 2014F	P/E 2015F	P/tBV 2014F	P/tBV 2015F	ROE 2014F	Div. yield 2014F
OTP	4,285	18.7	-11.5	11.1	1.0	1.0	-7.3	3.9
Erste Bank	23.4	158.3	-6.7	12.9	1.2	1.1	-14.2	0.0
FHB	797.0	n.a.	n.a.	95.9	3.1	3.1	-22.2	0.0
PKO BP	32.6	12.6	12.1	11.6	1.6	1.5	12.7	4.5
Pekao	185.55	17.5	16.7	15.5	2.1	2.1	12.4	5.4
Raiffeisen	13.5	4.7	-10.6	8.7	0.5	0.4	-3.9	0.0
Bank Handlowy	106	14.6	15.8	14.3	n.a.	n.a.	11.8	6.2
Komercni Banka	5,304	15.5	15.7	14.9	1.8	1.6	12.4	5.6
Unicredit	5.95	16.1	12.4	10.0	0.7	0.7	5.3	2.7
Intesa	2.98	36.9	16.1	13.0	n.a.	n.a.	6.5	3.8
Peer group`s average		34.5	11.2	20.8	1.5	1.4	1.3	3.2
Peer group`s average excl. FHB		34.5	11.2	12.4	1.3	1.2	4.0	3.6

Source: Bloomberg, Concorde estimates (OTP, Erste Bank, FHB, PKO BP, Pekao, Raiffeisen)

Source: Concorde, 2 March 2015

A real life example 2

UTILITY

	Price	P/E			P/BV	EV/EBITDA		Div yield
	(local curr.)	2013	2014F	2015F	2014F	2014F	2015F	2014F
CEZ	**621.1**	**9.2**	**11.2**	**16.9**	**1.2**	**7.2**	**8.6**	**5.2**
E.ON	14.3	n.a.	16.3	16.5	0.9	5.4	5.7	3.5
RWE	25.0	n.a.	11.5	11.8	2.2	4.5	4.5	3.9
Enel	4.1	13.4	12.9	12.8	1.1	7.0	7.1	3.2
Endesa	18.1	64.9	18.3	17.4	2.2	7.8	7.7	4.9
Iberdrola	6.1	16.5	16.2	15.2	1.1	8.9	8.6	4.5
EdP	3.5	15.9	13.7	14.0	1.5	9.5	9.5	5.4
EdF	24.7	12.4	11.9	12.2	1.3	5.0	5.0	5.1
PGE	20.7	10.6	11.8	12.3	0.9	5.2	5.1	4.3
Tauron	4.9	7.5	7.2	8.3	0.5	4.1	4.2	3.6
Enea	16.2	8.1	8.3	10.6	0.6	4.3	4.5	3.7
Peer group`s average		**17.6**	**12.7**	**13.5**	**1.2**	**6.3**	**6.4**	**4.3**

Source: Bloomberg, Concorde estimates (CEZ)

Source: Concorde, 2 March 2015

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A real life example 3

SOTP VALUATION

	EV/Share [HUF]	EV [HUF million]	Multiple	EBITDA [HUF million]	Margin Forecast
Hungarian Fix	257	267,504	4.7x	56,916	33%
Hungarian Mobile	549	572,508	5.5x	104,092	36%
SI/IT	127	132,030	9.0x	14,670	22%
Macedonia	69	71,915	3.8x	18,925	38%
EV	**1,001**	**1,043,957**	**5.8x**	**194,603**	**32%**
Net debt	-297	-309,976			
Minority	-47	-49,317			
Dividend	-25	1,043			
Spectrum CAPEX	-91	-95,000			
Fair value	**540**	**590,707**			
Upside w dividend	27%				

REGIONAL PEERS

	EV/EBITDA	EBITDA margin
Hrvatski Telekom	3.8x	39%
DIGI	4.0x	31%
Telekom Slovenije	3.8x	29%
Telekom Austria	5.5x	32%
Hellenic Telecom	4.6x	32%
Orange Polska	4.6x	25%
Netia	6.3x	27%
Cyfrowy Polsat	6.9x	37%
Megafon	4.1x	32%
Rostelecom	3.6x	31%
Turk Telekom	4.8x	34%
AVG	**4.7x**	**32%**

Source: Bloomberg

Source: Concorde, 2 March 2018

What if not all similarity requirements are fulfilled?

Recalculated multiples

Our firm differs from the average (g, r, CF, β). Is that maintainable in the long run?

Discounts & premiums

The peers are different in a given perspective (other markets, countries, liquidity).

Recalculated multiple

The average forward P/E of European food wholesalers was **15.2** at the beginning of 2019. (Div%=93.07%, g=7.30%*) The firm to value has the same risk, but div%=80%, g=8%.

$$div\% \,/\, P/E = r-g = 93.07\%/15.20 = 6.12\%$$

$$r-g_{mod} = 6.12\%-(8.00\%-7.30\%) = 5.42\%$$

$$P/E_{mod} = 80.00\%/5.42\% = 14.76$$

*Source: Damodaran Online, 2019

The operator of a 20-room hotel with unique regional traits is to value. Similar quality hotels are quoted at the stock exchange at 8 M HUF/room.

20*8 = 160 million HUF

Long-run competitive advantage: + 10%
Illiquidity of the investment: – 30%

160 * 1.1 * 0.7 = 123.2 million HUF

Typical mistakes when using multiples

- Description power of multiples not checked.
- Calculation of multiples is not standardised.
- Peers are competitors rather than similar firms.
- Necessary discounts not applied.

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Thank you for your attention. Any questions?

peter.juhasz@uni-corvinus.hu

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Asset-based valuation

Péter Juhász, PhD, CFA

What we still have to cover …

$$V = V_{OP} + V_{Other}$$

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Typical „other" assets

Surplus cash

„Cash not needed for the operation"

Securities

Stocks, bonds, options. Use the market price.

Investments

Strategic: synergies, consolidation

Unused assets

Assets not used by own operation

Over- and underfinanced pension plans

What makes an asset? – Classic Accounting

Owned
by the firm

Expected
incomes exceed
expected costs

Nearly no
uncertainty

SFAS (until 2009) See Munter, P., & Ratcliffe, T. A. (1980). On the Attributes of an Asset. *Abacus*, *16*(2), 116–123. https://doi.org/10.1111/j.1467-6281.1980.tb00091.x

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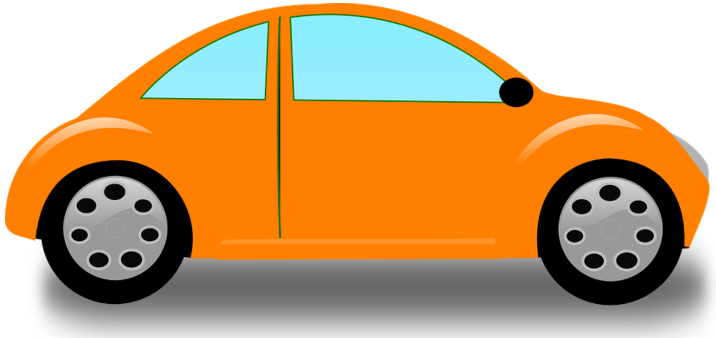
What makes an asset? – Modern Accounting

Controlled by the firm

Will exchange for cash in the future

Manageable uncertainty



IFRS: "An asset is a present economic resource controlled by the entity as a result of past events. An economic resource is a **right** that has the potential to produce economic benefits."
US GAAP: "An asset is a present **right** of an entity to an economic benefit."

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What makes an asset? - Finance

Assets generating cash flows

DCF

Assets not generating cash flow

Replacement value, comparable assets

Option-like assets

Real options

Quasi assets

Asset manageable but not ownable

Asset valuation methods

Type of value	Time orientation of data used		
	Past	**Present**	**Future**
Entry	Purchase cost	Substitution value (repurchase, recreation) Replacement value	–
Exit	–	Net realisable value, Liquidation value	DCF, Capitalisation, Real options

Based on Wilson (1986)

What is our warehouse worth?

We have bought it five years ago for 5 M HUF.

It was renovated last year for 2 M HUF.

Current book value is 5.6 M HUF.

A similar one can be bought for 5.9 M HUF in the neighbourhood.

Last week, 6 M HUF has been offered for it.

We could build a similar one over 18 month for 7.1 M HUF.

The present value of future rent income is 7.4 M HUF.

Valuation issues

Valuation issues 1 – Current Assets

- Non-operating cash (missing from the DCF value)
- Securities (revaluation)
- Receivables (payment risk)
- Other lending (will we get that back ever?)
- Inventory (FIFO, LIFO, guaranteed purchase price)

Value of cash and securities

Generally, these investments have an NPV of zero, but those could be worth less.

- Double taxation
- Investments below required rate (effective market guarantee zero NPV only at the time of the purchase)
- Investor anxiety about poor M&As (e.g. take-over protection)

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Valuation issues 2 – Invested assets

Intangibles: R&D, brand name, franchise, achieves, lists, transferable contracts (purchased vs created, going concern vs liquidation)

Financial assets: shares, bonds, futures, forwards, options, warrants, convertible bonds, convertible shares, swaps, swaptions

Stakes in other firms

Valuation could be challenging (no data):
- market price
- relative pricing
- simplified DCF (target firm cost of capital needed)

<u>May be valued below their market value …</u>

Information gaps (structure not transparent)

Market scepticism

Is the mother company able to manage the cross-owned portfolio optimally? The holding discount may even amount to 5-10%.)

Plant, Property & Equipment

2. Invested assets (cont.)
   c) Machinery
      Real obsolescence (vs D&A):
         1, economic (market situation),
         2, operational (wear & tear),
         3, functional (technology)
   d) Real estate (long useful lifetime, obsolescence, market need, inflation vs D&A)

Consider tax effect if asset sale is assumed.

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Valuation issues 4 - Liabilities

- Payables (for each BU)
- Lease (maintenance cost as expense)
- Non-consolidated affiliates
- Liabilities to employees
- Shareholder loans

Valuation issues 4 – Off Balance Sheet

Pension funds (asset or liability)

Postponed tax payments (equity or liability?)

Guarantee, warranty, suretyship

Environmental obligations

Additional challenges

- Litigations
- Compliance
- Special challenges
  - Preferred equity, convertible equity, various types of shares

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The aim of asset valuation is to estimate the amount of the invested capital.

Valuation methods used to estimate capital employed by business units

Net book value	87%
Replacement value	4%
Gross book value	3%
Market value	1%
Other	5%

Source: Arnold-Davies (2000, p. 159.)

Proportion of book value in % of the market value for US listed firms

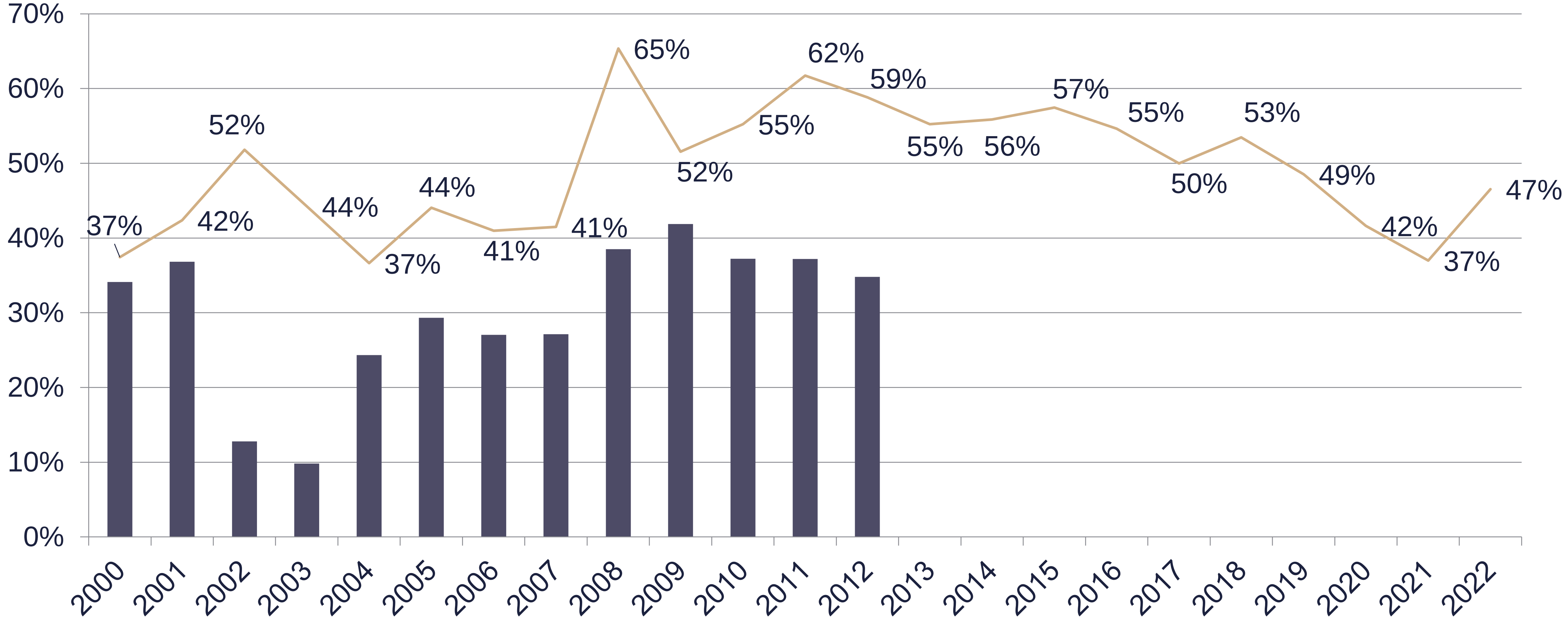
1978	95%
1988	**28%**
2002	**< 20%**

Source: Personnel Today, 2002

Hungary

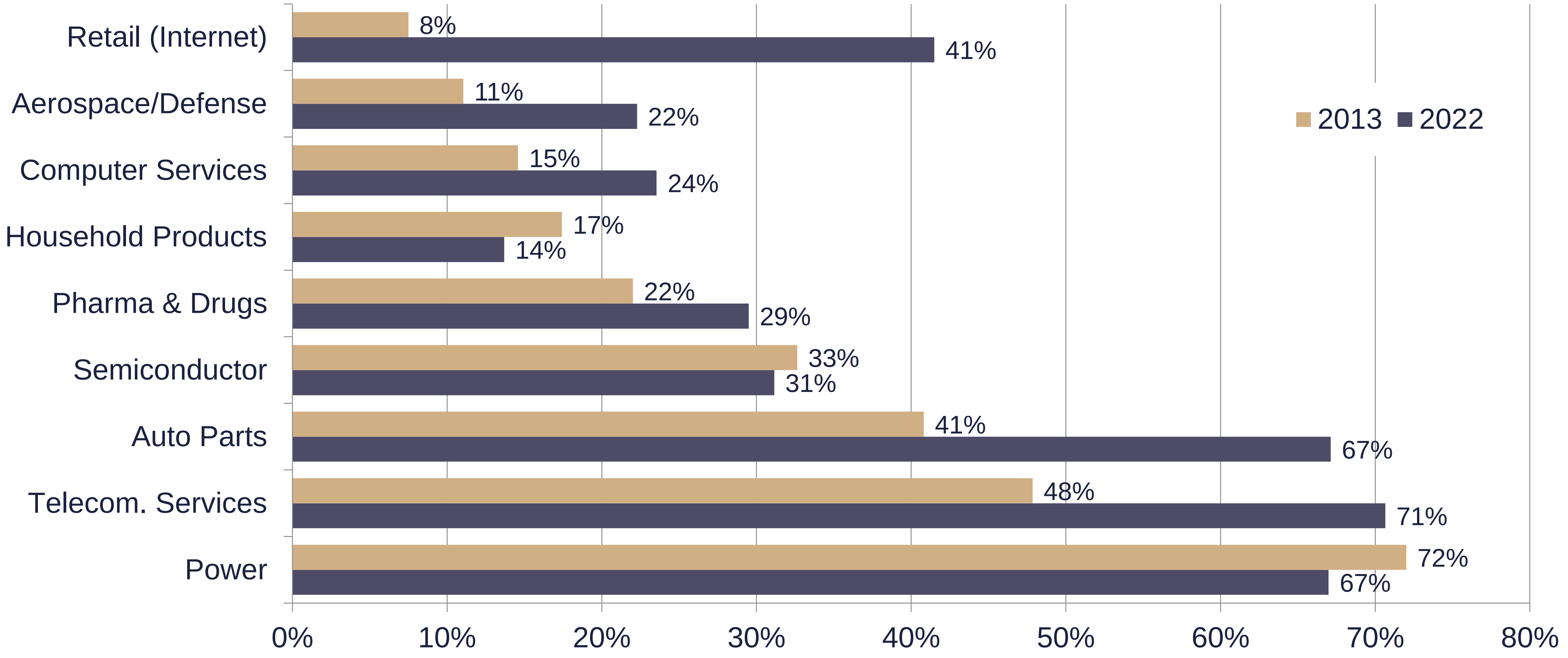
	2004	2009	2013	2019
Net book value	59.3%	53.3%	45.8%	60.7%
Replacement value	5.5%	3.2%	12.0%	3.8%
Gross book value	12.3%	17.8%	7.3%	15.6%
Market value	22.0%	23.6%	33.3%	19.0%
Other	1.0%	2.2%	1.6%	0.9%

The ratio of the book value and the market value of the invested capital in the US*



*December
Source: based on Damodaran (2023)

IC/EV in US*



*December

Source: based on Damodaran (2023)

Value perceptions

Accounting value

Past oriented

Based on documents

Rule of prudence

Aims to inform outside stakeholders

Business value

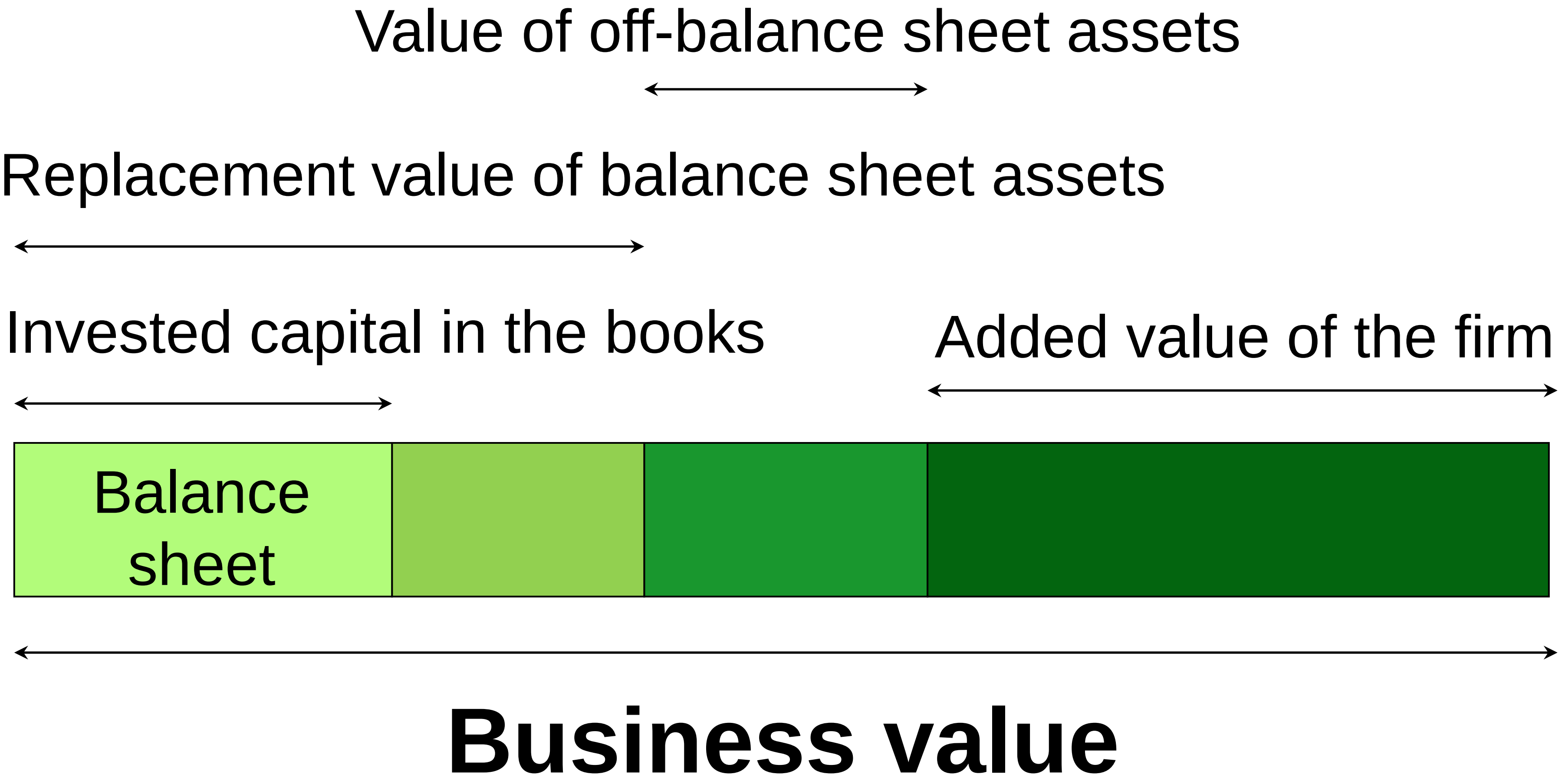
Present and future focused

Predictions, estimations

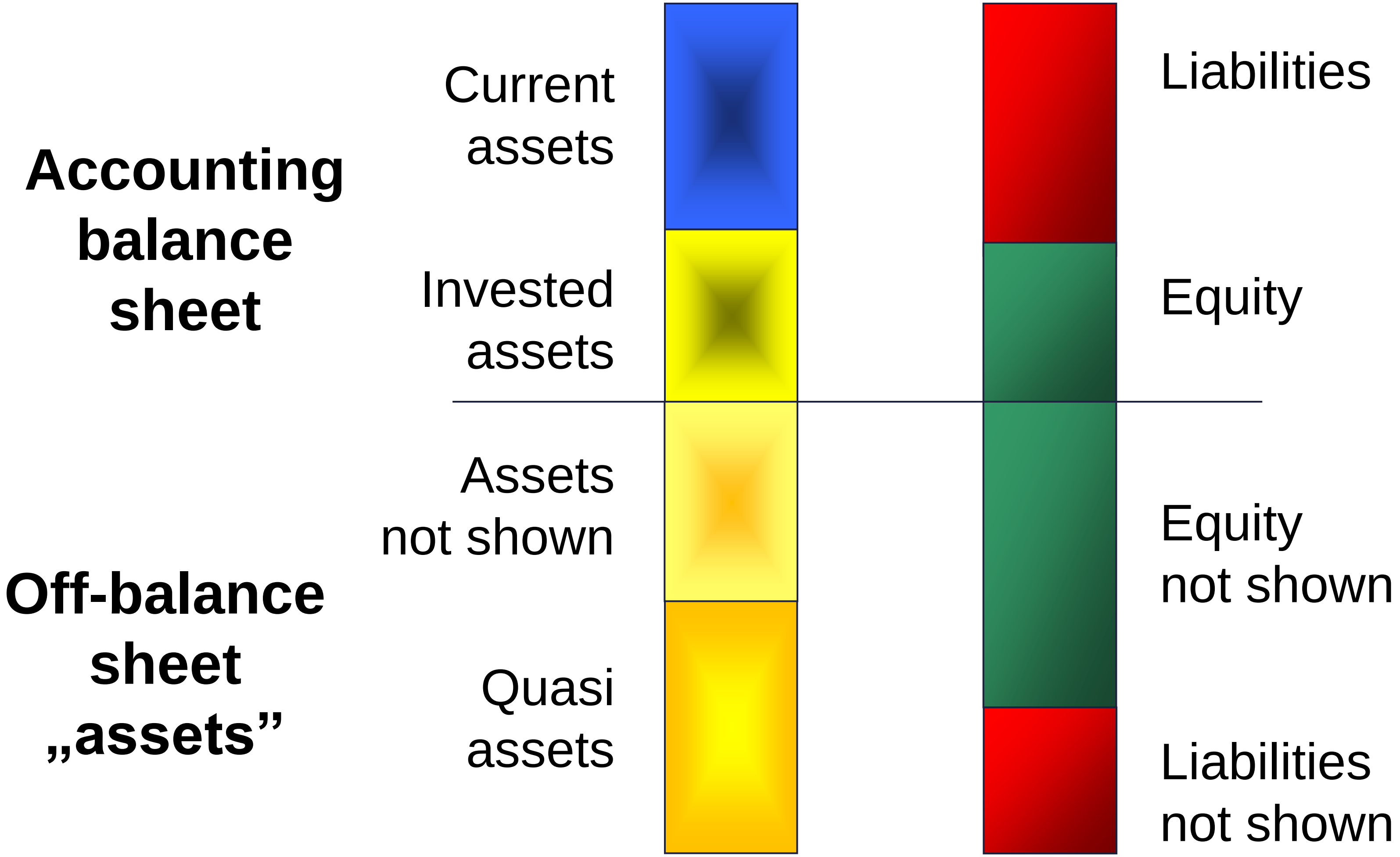
Expected value

Made for owners and managers

Types of off-balance sheet items



The extended balance sheet



Grouping off-balance sheet items based on type and connectedness

	Ownable	Manageable only
Organisation linked	Brands, transferable contracts	Supply contracts, documented knowledge
Employee linked	– (Playing rights)	Work connections, non-recorded knowledge

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Stakeholder	Advantage of good reputation
Investors	Lower price volatility, higher quotes
Employees, management	Lower recruitment cost, higher loyalty (less fluctuation), lower wages
Creditors	Lower rates, more attractive credit terms
Suppliers	Reduced prices, longer payment terms
Retail chain	Lower fees, preferential treatment, wider spatial coverage, more flexible connection
Customers	Premium price, more frequent purchases, higher volume
Competitors	Entry barrier, comparative advantage, openness to cooperation

What is the aim measuring?

Do we measure anything? What? How often?

Use of money-based measurement

Tracking changes (numbers)

Value comparison (measurement unit)

The value is in the eye of the beholder. (strategy, synergy)

Quasi assets can not be sold on their own.

ScoreCards 1

1985 Skandia (Sweden)

Five dimensions: finance, customers, employees, processes, renewal

73 traditional + 91 new (intellectual capital) ratios (164 ratios)

1992 Kaplan and Norton: Balanced Scorecard

finance, customers, operational processes, learning-development

1993 Dow Chemical

First in the USA

ScoreCards 2

Mid 1990s PWC

ValueReporting: market, strategy, value creating activities, financial performance

1999 Edvinsson and Malone

Skandia system corrected for redundancies

112 ratios remained even afterwards

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Thank you for your attention. Any questions?

peter.juhasz@uni-corvinus.hu

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Discounts and Premiums

Péter Juhász, PhD, CFA

Why do we need D&Ps?

To include effects into our valuation that the given method has not yet (fully) considered.

The use and extent of discounts and premiums vary across valuation methods even within the same valuation task.

Inclusion of key effects

Effects	DCF	Multiples	Asset-based
Business outlook	CF plan	Industry average	–
Market judgement	Discount rate	Market ratios	Market prices
Liquidity	**Premiums and discounts**		Different!
Ownership structure			None!
Off-balance sheet items, others			Should be already included

Each approach calls for a different correction.

DCF: Theoretical intrinsic value under perfect conditions (liquid markets, no information asymmetry)

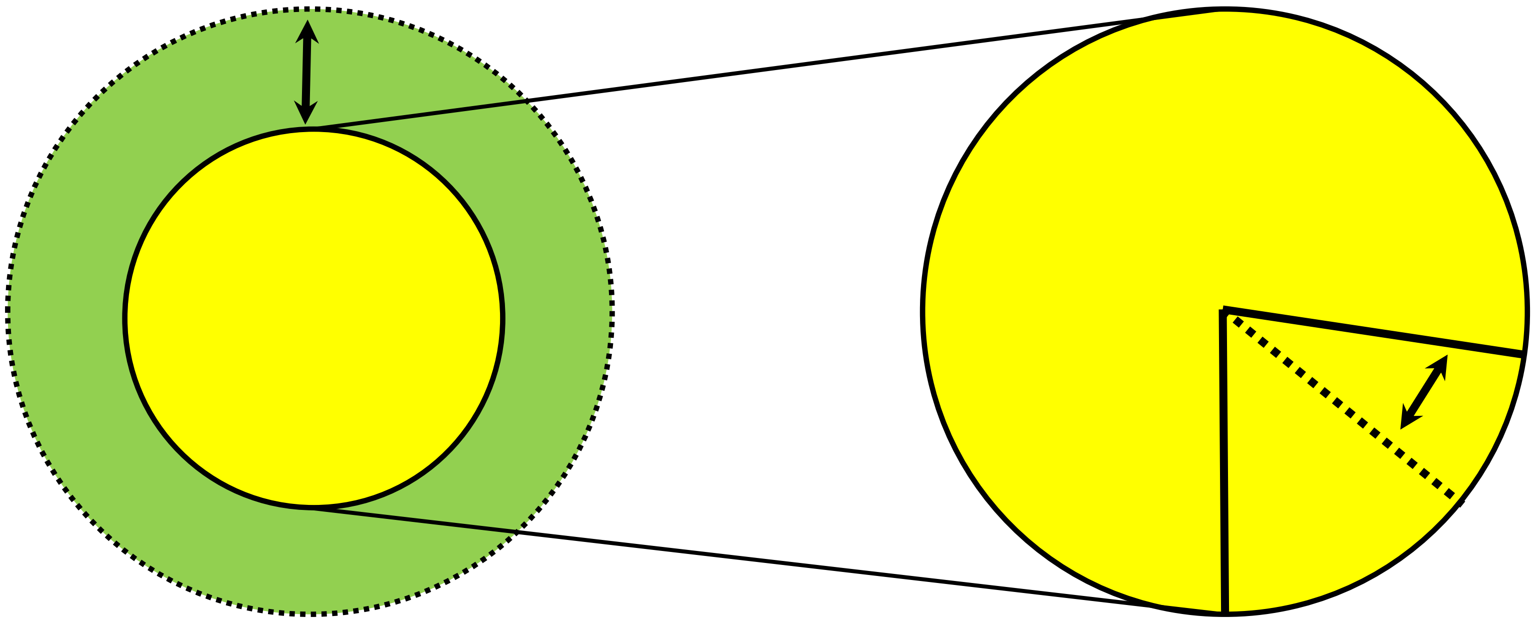
Market multiples: Average value of similar firms (typical, listed, liquid firm)

Comparable transactions: We are exactly like the average of the peers (same country, markets, size, liquidity)

Corrections should only hit the value of operation.

Corrections for equity value

Corrections for a given stake



D&Ps for total equity

Liquidity („illiquidity") discount (Several meanings: assets, size, financial power, chance of going public.)

Key person discount

Complex portfolio without synergy

Environmental, legal, and dependent liabilities

Long term liabilities (contracts, warranties, guaranties, warrants)

D&Ps for an equity stake

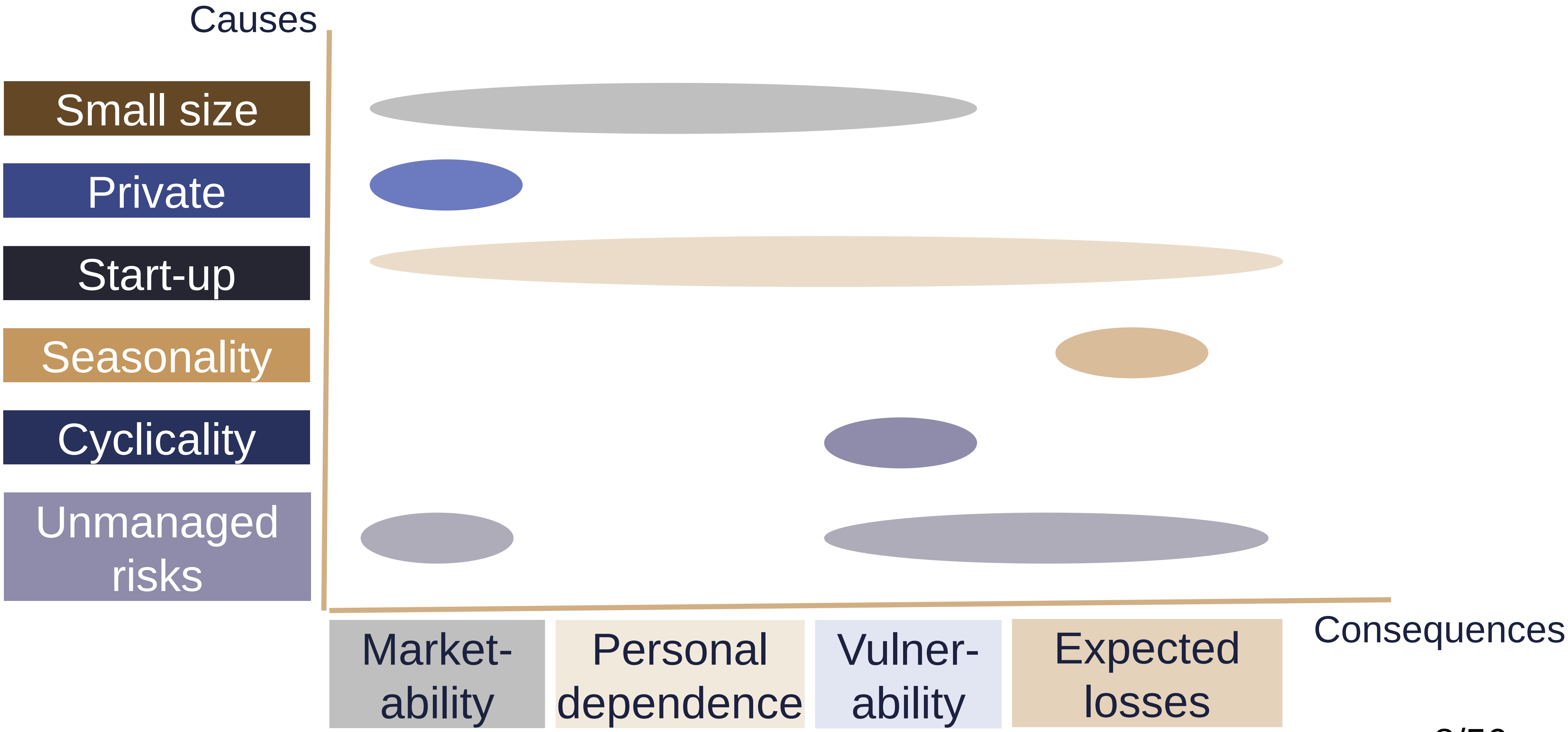
Liquidity discount (no buyer e.g. due to the ownership structure)

Control rights (minority, blocking minority, majority, strategic majority)

Additional voting right (preferred shares)

Big package discount (if sold current price would fall)

Do not mix the dimensions



The valuation process

1. Determining the value of operation (V_{op})
2. Valuing non-operative assets (NOA)
3. Calculating the total firm value ($V=V_{op}+NOA$)
4. Valuing the foreign capital (D)
5. Determining the market value of equity ($E=V-D$)
6. **Finding the price of a share or stake**

The value of a firm is 1200, its debt amounts to 200 and there are 2000 shares issues. What is the market price of a single share?

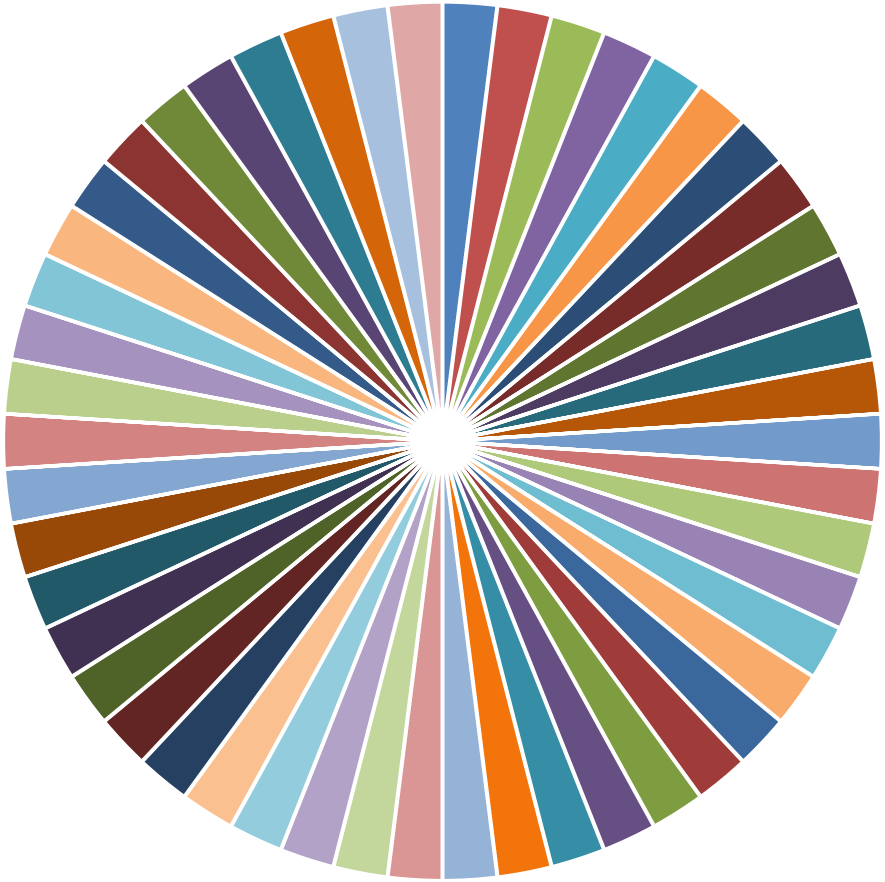
MV(D+E)-MV(D)= MV(E)

1200-200=1000

? P=1000/2000=0.5 ?

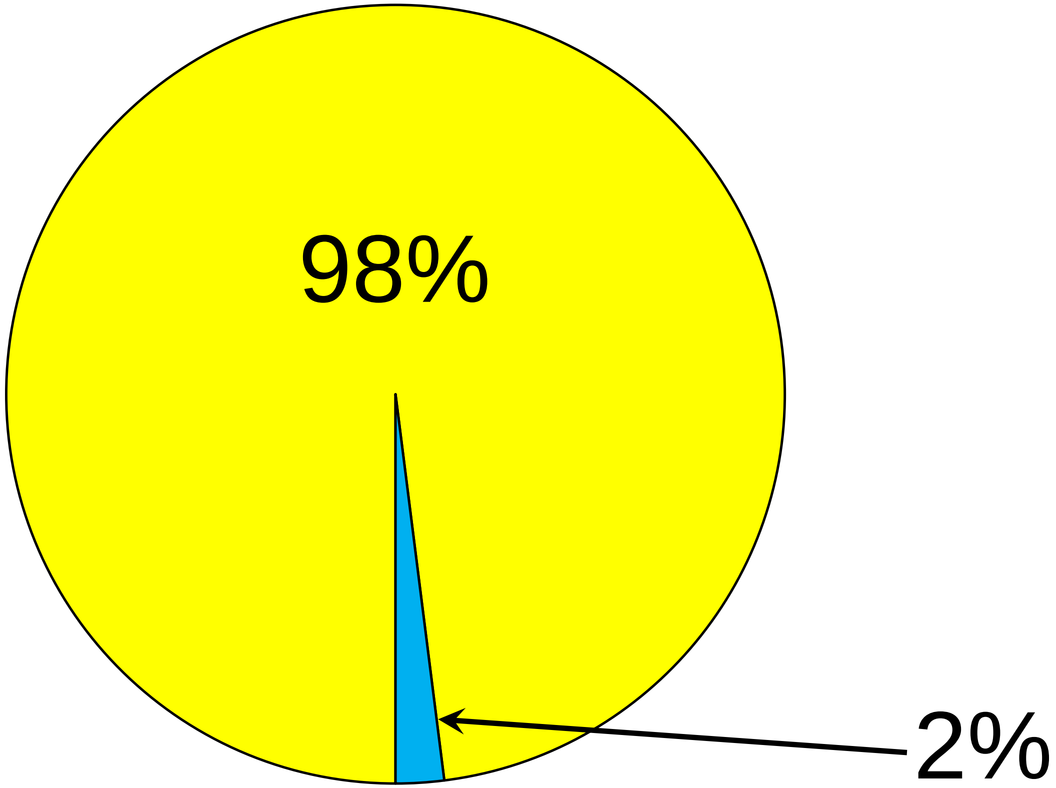
MV(E)*q <?> MV(E*q)

The value of single share 1



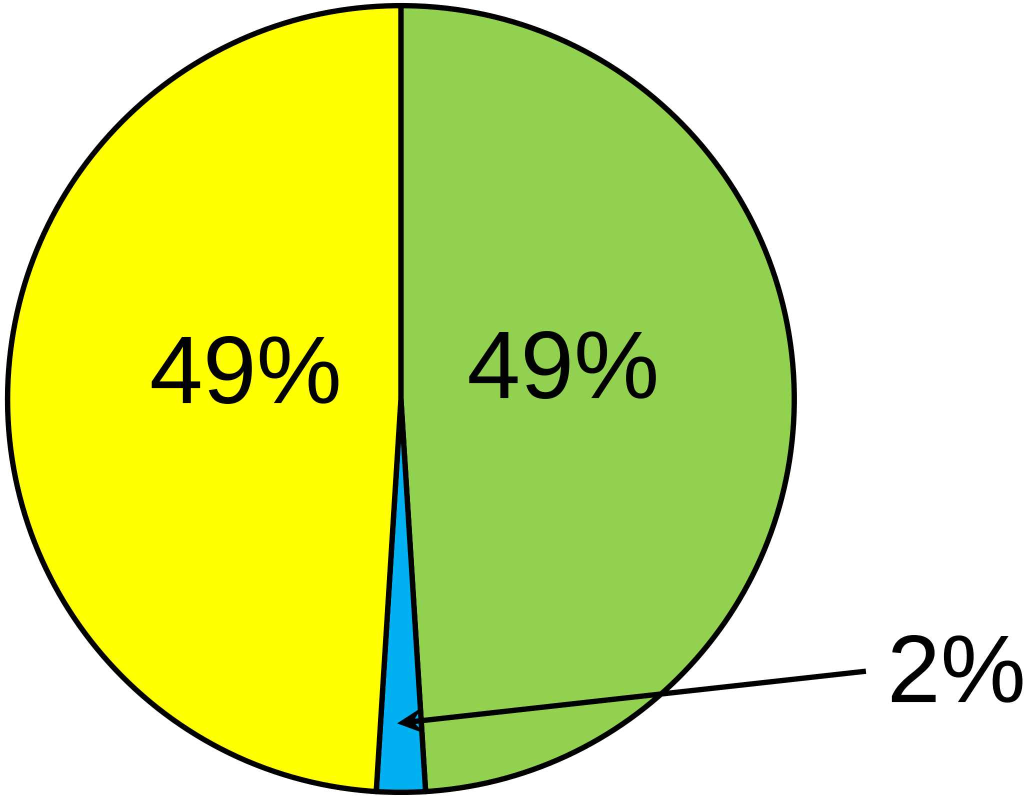
MV(E)*q = MV(E*q)

The value of single share 2



MV(E)*q >> MV(E*q)

The value of single share 3

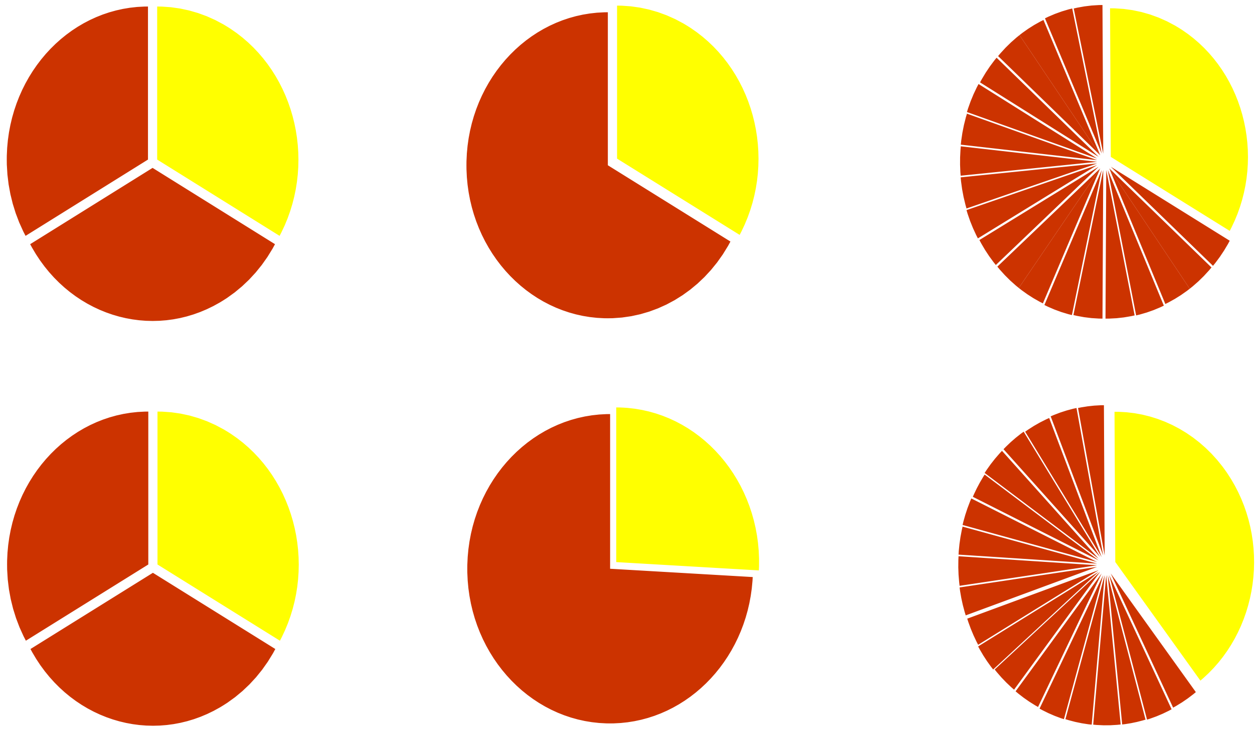


$MV(E)*q \ll MV(E*q)$

The value of a 33% stake

Ownership stake

Value stake



-30%
(or -22.1%)
discount

+30%
(or +18.2%)
premium

The sum is always 100%

60%-40% division of the ownership rights

Control premium (%)		Minority discount (%)
10.0	60*0.10/40=	15.0
15.0	60*0.15/40=	22.5
20.0	60*0.20/40=	30.0
25.0	60*0.25/40=	37.5

Ownership-structure effect

What kind of opportunity to influence the operation of the firm is offered by the given stake? (Why you pay for the control is that you can make the firm more valuable using it.)

The exact size of the control premium/discount can only be estimated on a case-by-case basis.

The value of the control

1. The likelihood of a change in the control of the firm

2. The value added by the change in the control:
   quality of the current management,
   difficulty of exercising the control

When to look for a control premium? 1

- Poor asset management
- Too conservative investment policy
- Extreme desire for investment
- Missed strategic opportunities
- M&As without synergies

When to look for a control premium? 2

- Too conservative financing structure
- Extreme indebtedness
- Poor choice of financing forms and sources
- No trust of spending the accumulated cash correctly

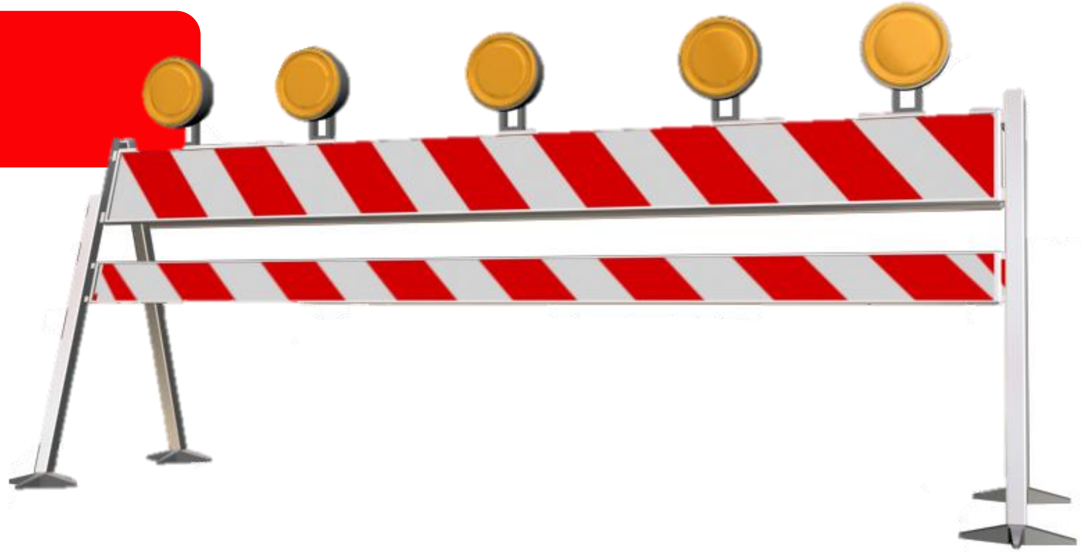
Likelihood of a change in control

Institutional barriers

- Takeover regulations
- Limitations of ownership
- Evading conflicts of interest

Firm-specific barriers

- Voting preferred shares
- Cross-ownership holding structure
- Significant management ownership
- Very diversified ownership
- Base rules limiting ownership



Value of control for listed firms

Market value = current value +

+ probability in change of control *

* (optimal value – current value)

We could do it better, but it is not for sure that we may.

Control premium for public firms

A premium may be paid if a subjective valuation dictates a higher than current market price. (Information on mistakes)

Quotes may move significantly if the estimated likelihood of a change in control would modify. (News on possible M&As)

Weaker investor protection leads to lower share prices.

Private firms: Control premium 1

For private firms, control may have a significant value as usually management would only change if ownership changes.

Minority discount depends on the ratio of the optimal and the current equity value.

Control premium 2

Example on the difference between the valuation of stakes of 49.9%:

Current equity value= 100
Optimal equity value= 150

optimal value 49.9% = 74.85
current value 49.9 % = 49.90

Minority discount = $1 - \frac{49.90}{74.85} = 33\%$

Control premium 3

Minority discounts are inversely linked to management abilities.

You do not always need a 50%+1 stake for the control.

The value of any stake depends heavily on whether the owner is capable and willing to be involved into the management of the firm .

Deviations across the world

More efficient investment protection: 2-5% (e.g. US, UK)

Less strict: 60-65% (e.g. Brazil or Czechia)

Median: 10-12%.

What is liquidity? 1

Everything can be sold at a low-enough price.

The level of liquidity is different across assets.

- Price effect (timing, information effect, continuity of prices)
- Alternative cost of waiting
- Transaction costs

High transaction fees:

real estate agents: 1-4%

action houses for paintings: 15-30%

Reasons for higher fees:
fewer agents,
non-standardised products,
more personal knowledge needed

Marketability discount

Dividend payment	Availability of information	Accessibility of potential buyers
Size of the stake	Put option	Limitations on sale

Measuring liquidity discount 1

Non-liquide investments have no well-organised markets – higher selling costs, less stable sales price

Measuring liquidity discount 2

How to quantify illiquidity discount?

- Liquidity discount in the stake or share price at the end
- Increasing required yield for liquidity risks
- Multiple-based valuation based on firms with similar liquidity

Use one method only at a time.

Setting the liquidity discount 1

Fixed discount (range) – strongly relying on former studies that have set 25-35% for shares with limited liquidity.

Many have shown that the samples were distorted and the realistic value is far smaller.

Setting the liquidity discount 2

Firm-specific discount – depends on the possible number of buyers and the difficulty of the sale. (Real estates are easier, collections are harder to sell.)

The liquidity of private firms may change from firm to firm and buyer to buyer.

Lower liquidity discount

More liquid assets	Stable, positive cash flow	Going public is near
Majority stake	Bigger firm (Costs fixed)	Operational and asset value are similar

Liquidity discount in the yields

Liquidity discount appears as a premium in required yield. **The premium desired changes over time.**

Liquidity discount may increase during times the liquidity is needed. (When market liquidity is poor asset liquidity becomes more important.)

In the long run everything is liquid. How long may you add a liquidity correction?

Liquidity discount in the yields

Returns of liquid and less liquid shares show a historical average difference of 3.0-3.5%.

Smaller firms on average had a 4.0% higher yield than S&P companies (15.7% vs 11.7%, 1984-2004).

Life cycle matters: younger companies yielded 19.9% while more mature firms offered 13.7% (vs 11.7% S&P500) (1984-2004)

Links among discounts

More liquid markets have lower lack-of-control discounts

It is easy to sell if the majority owner does not do a good job.

Discounts and premiums

R

Liquidity discount (assets, size, financial power, chance of going public)

Control rights (blocking minority)

Additional voting rights

Big package discount

Key person discount

Discount on portfolios without synergy

Environmental, legal, dependent liabilities discounts

Long term liabilities (contracts, warranties, guaranties, warrants)

Court decisions and market transaction, US (1980-1997)

Minority discount:

21-31%

Marketability discount:

(10) 20-60 (80) %

Big stake discount:

0-20%

Example

The HardWork Co. has three owners (40%-35%-25%). What is the value of the 35% stake, if the DCF value of the total equity amounts to 100 under the current strategy? (All parties agree on the valuation.)

If investors only focus on their own intention to buy…

Outside buyer

Own stake after the deal:		
	100 * 35% =	35.0
Private firm (total of equity)		
(liquidity, transparency)		-10%
Strategic minority		+ 5%
	35 * 0.90 * 1.05 =	33.1
Premium used	33.1 / 35.0 – 1 =	-5.4%

25% owner

Own stake until now:	100 * 25% =	25.0
Illiquidity of the stake:		-10%
	25 * 0.90 =	22.5
Own stake after the deal:		
	100 * 0.60 =	60.0
Control premium		+17%
	60 * 1.166 =	70.0
Value of 35% stake	70 – 22.5 =	47.5
Premium used	47.5 / 35 – 1 =	35.7%

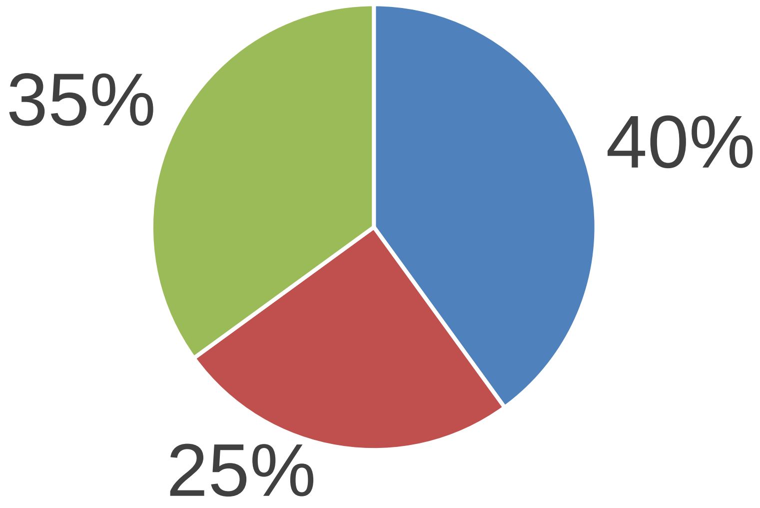
40% owner

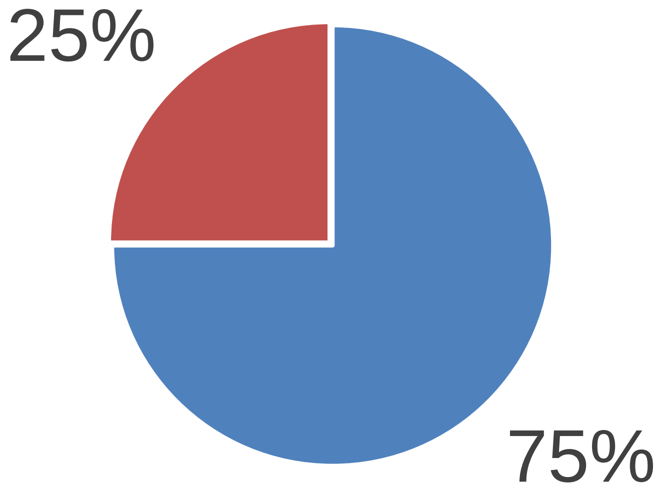
Own stake until now:	100 * 40% =	40.0
Own stake after the deal:		
	100 * 0.75 =	75.0
Strategic control premium		+13%
	60 * 1.1333 =	85.0
Value of 35% stake	85.0 – 40.0 =	45.0
Premium used	45 / 35 – 1 =	28.6%

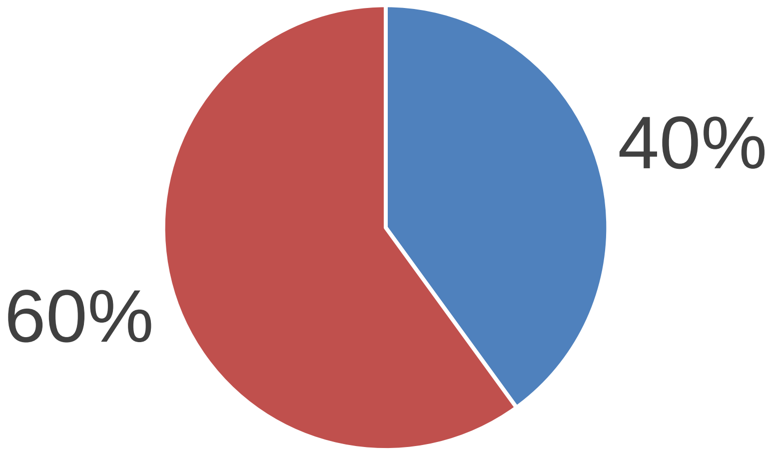
What if everyone is interested?

Owners do not contrast to the current status rather than to **the future status they will be in if not purchasing**.

Possible outcomes







For the 25% owner

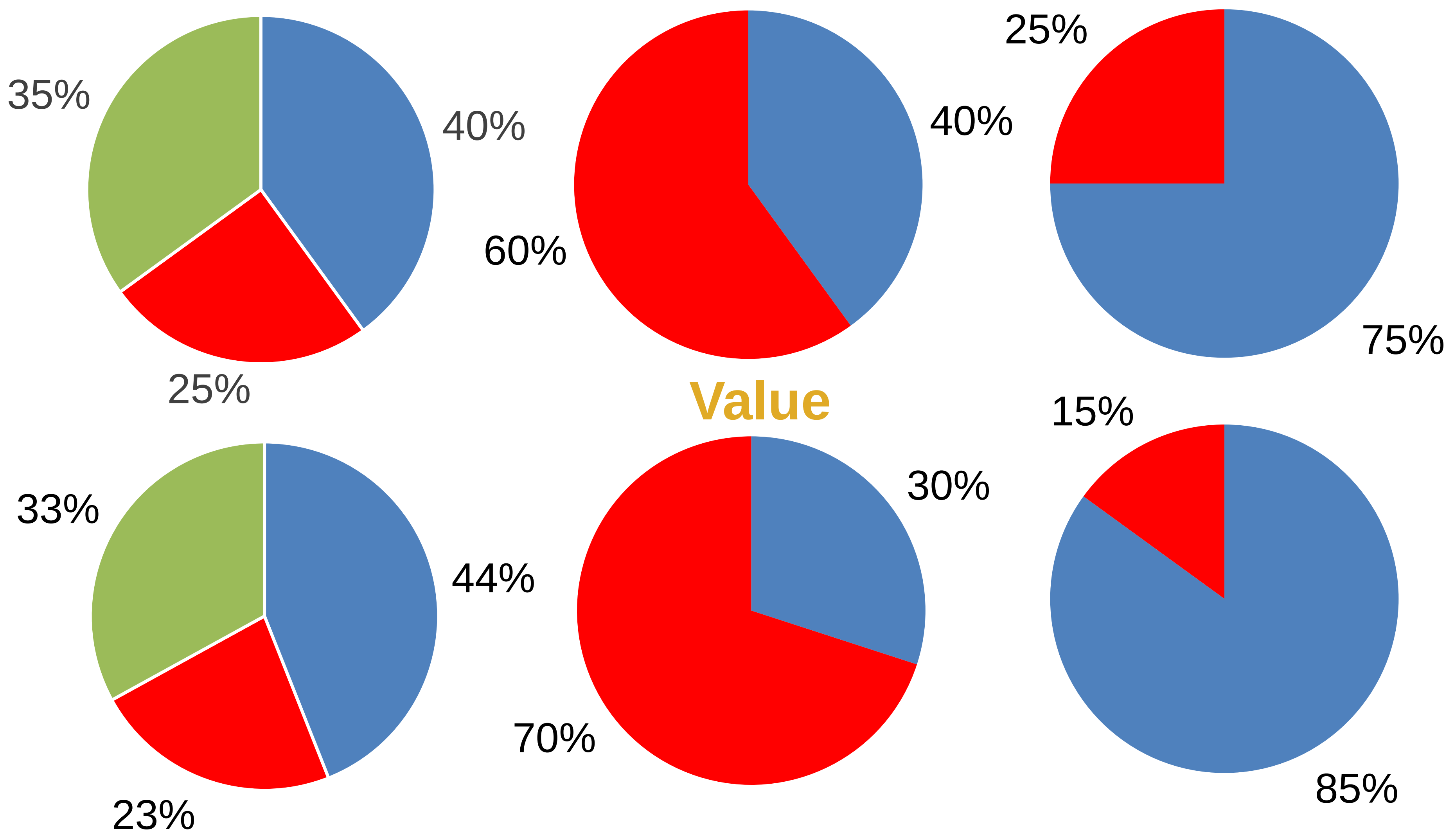
Own stake if 40% owner buys:		25.0
Liquidity and minority discount		-40%
	25 * 0.60 =	15.0
Own stake (25%+35%) after the deal:		
	100 * 0.60 =	60.0
Control premium		+17%
	60 * 1.166 =	70.0
Value of 35% stake	70.0 – 15.0 =	55.0
Premium used	55 / 35 – 1 =	57.1%

For the 40% owner

Own stake if 25% owner buys :		40.0
Liquidity and minority discount		-25%
	40 * 0.75 =	30.0
Own stake (40%+35%) after the deal :		
	100 * 0.75 =	75.0
Control premium		+13%
	60 * 1.20 =	85.0
Value of 35% stake	85.0 – 30.0 =	55.0
Premium used	55 / 35 – 1 =	57.1%

Dividing value across stakes

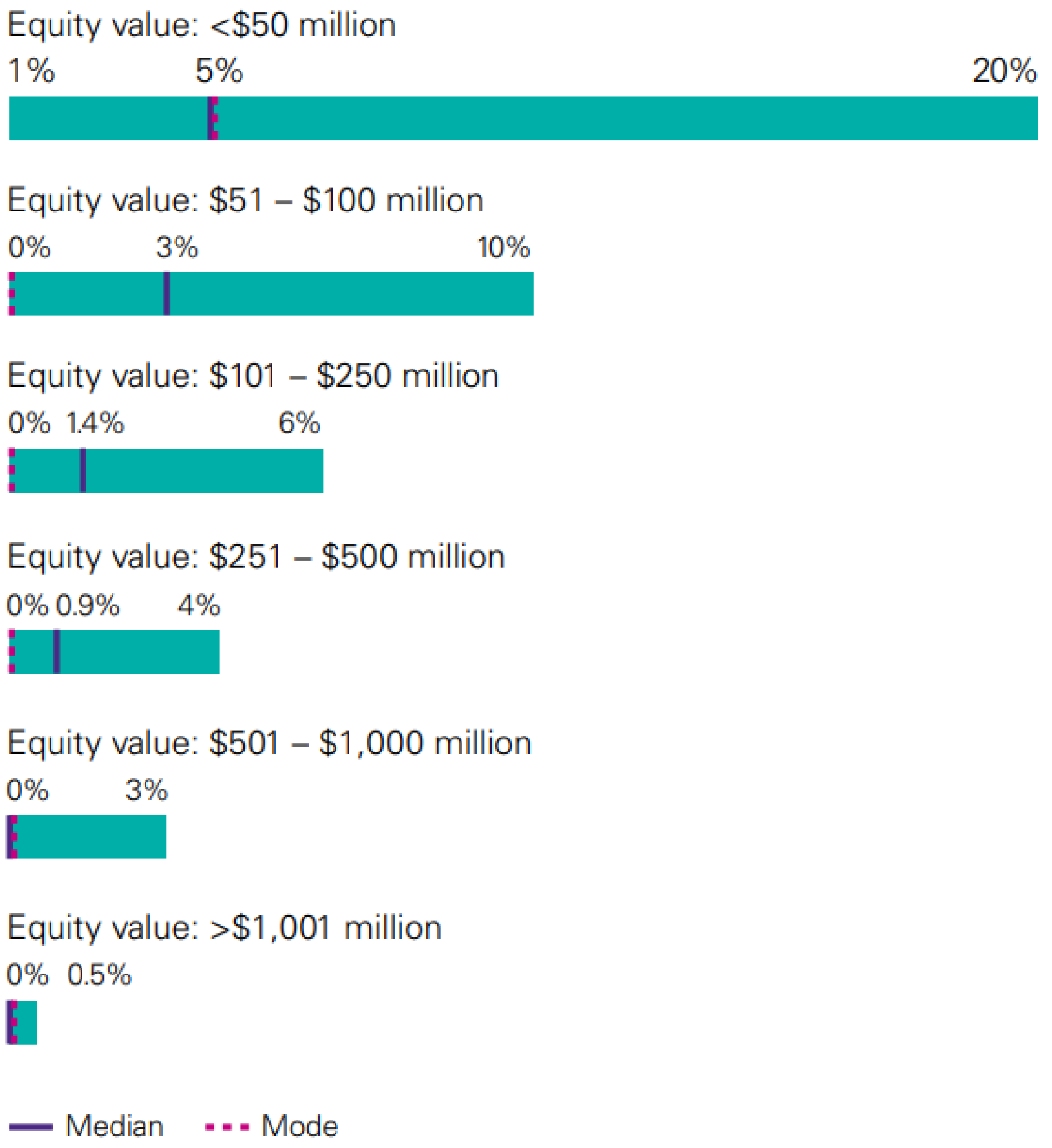
Ownership



BUDAPESTI CORVINUS EGYETEM

How is it done in the industry?

What premium is added to the discount rate to account for the size of the entity being valued?



Source: For all it's worth – KPMG Valuation Practices Survey 2017 Australia https://assets.kpmg/content/dam/kpmg/au/pdf/2017/valuation-practices-survey-2017.pdf

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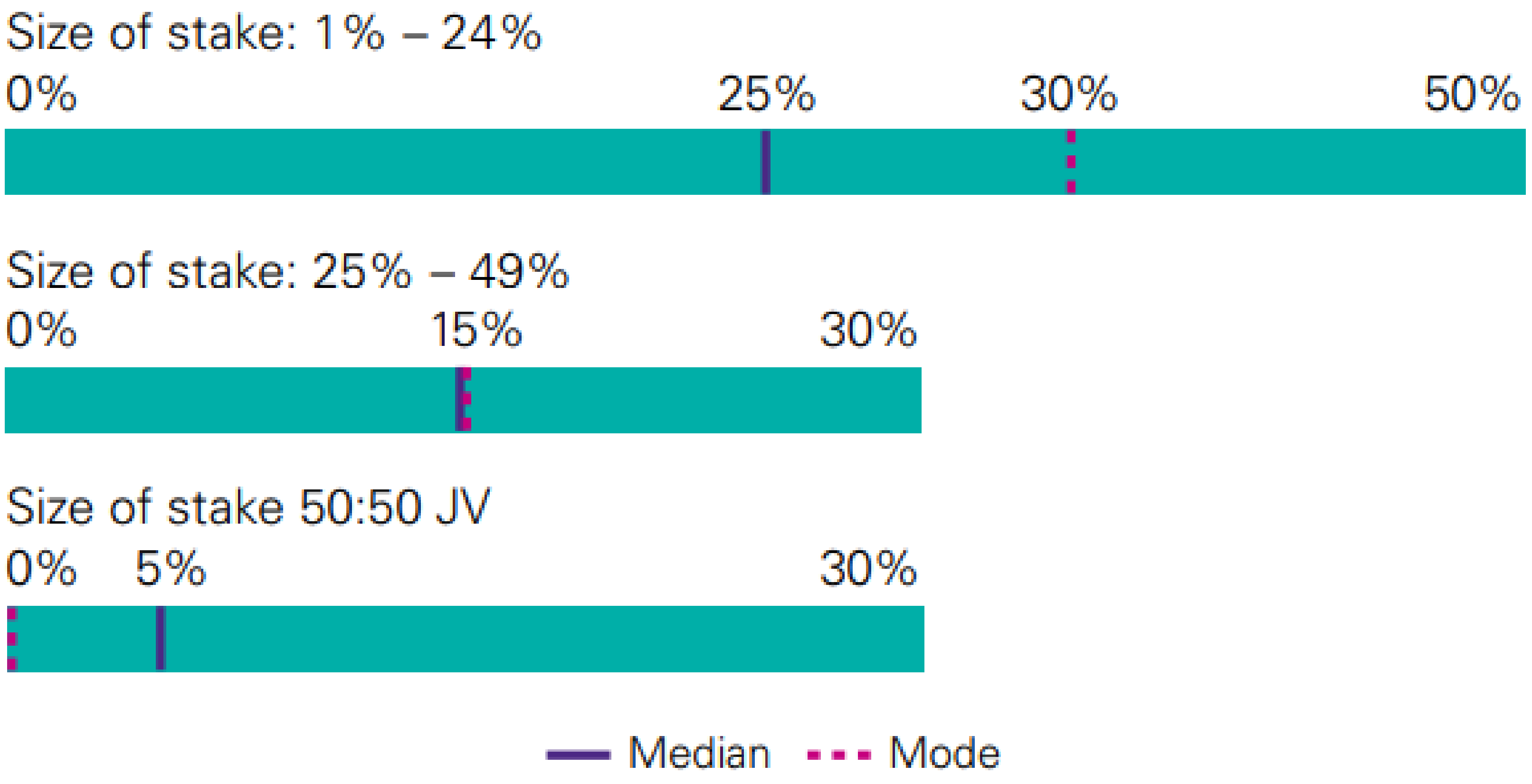
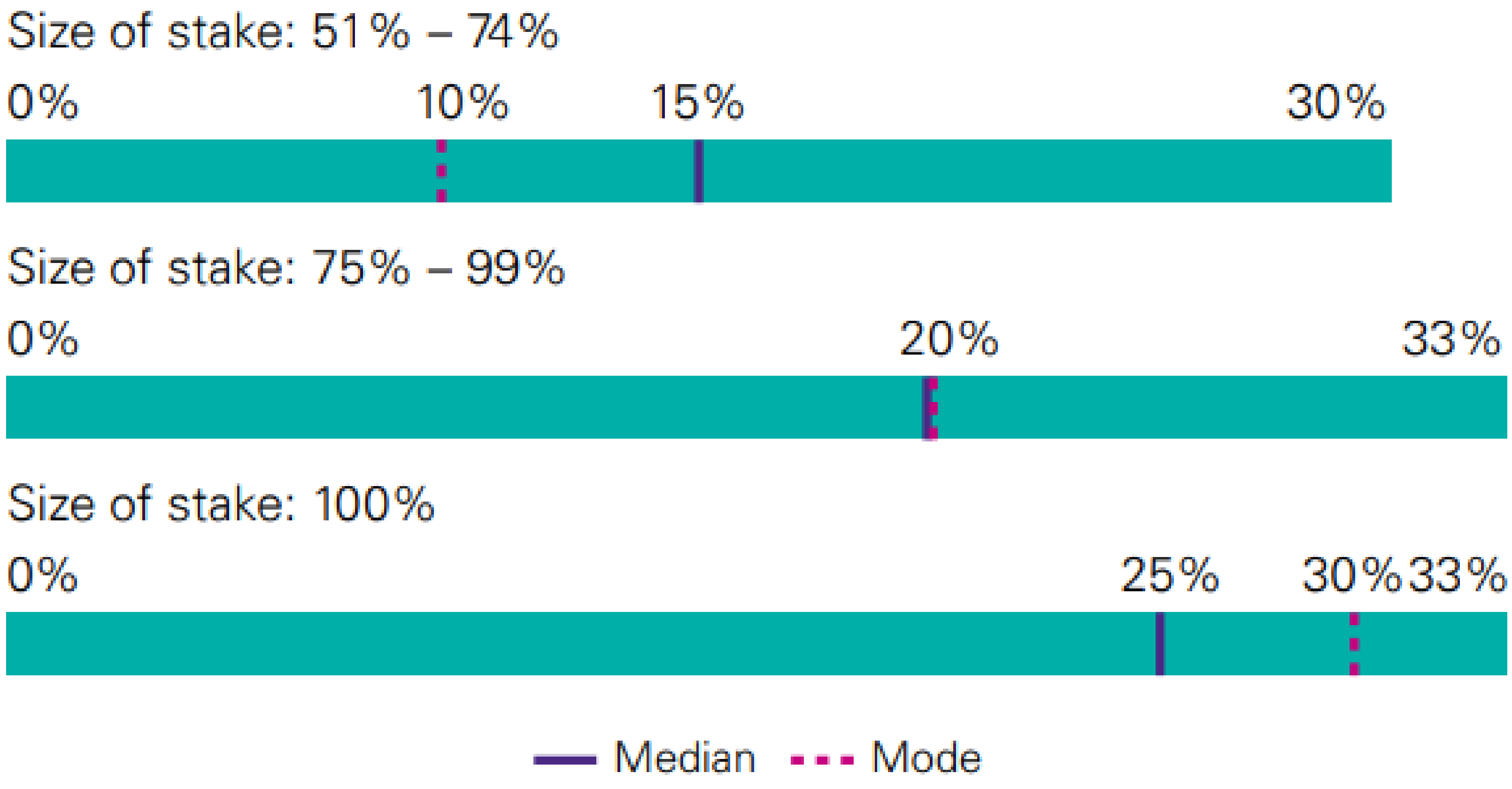
What discount do you apply in valuing a minority interest?



Source: For all it’s worth –
KPMG Valuation Practices Survey 2017 Australia

What premium do you apply to reflect the level of control offered by the size of the stake being valued?



Source: For all it's worth –
KPMG Valuation Practices Survey 2017 Australia

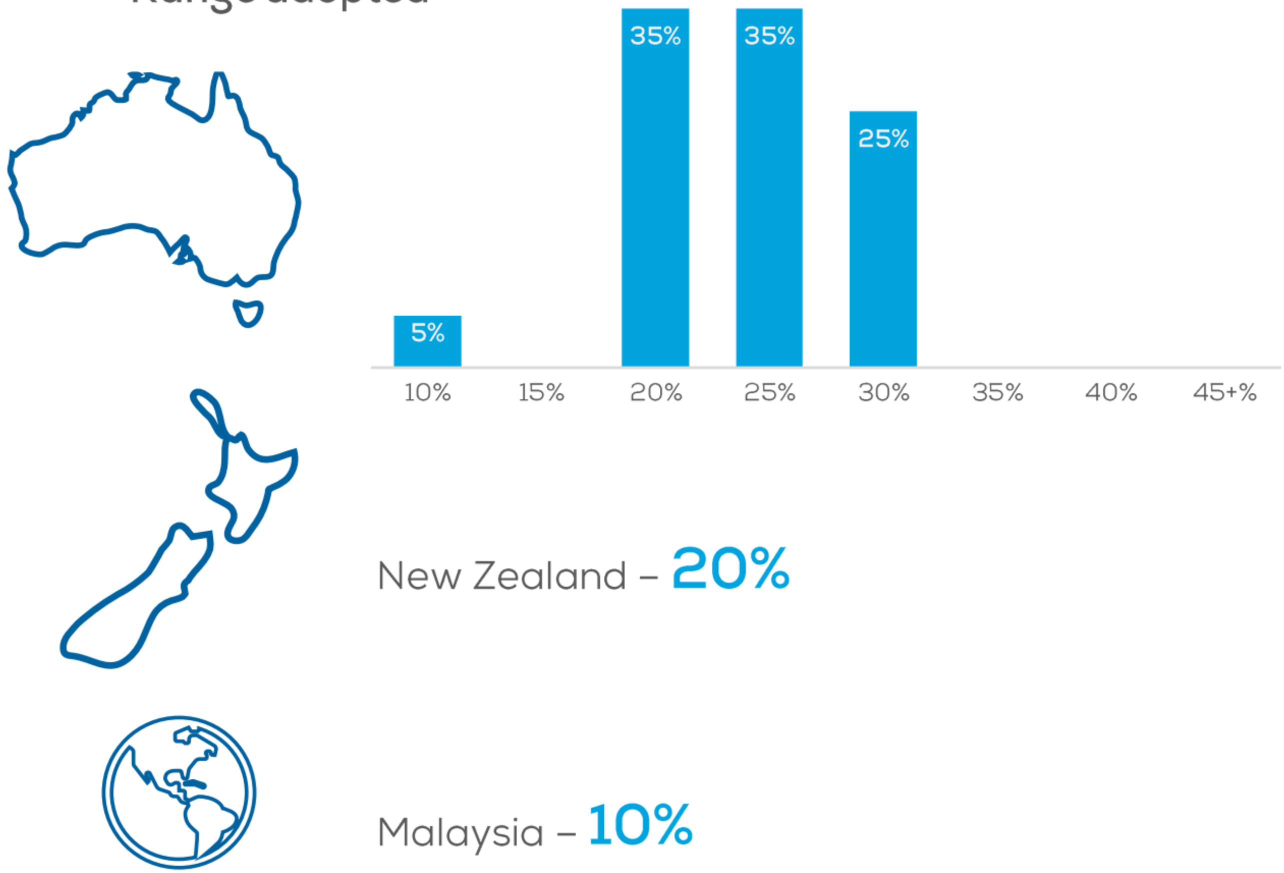
For those using a standard control premium, the most common range adopted is 20 - 25%

Use of standard control premium

57%

43%

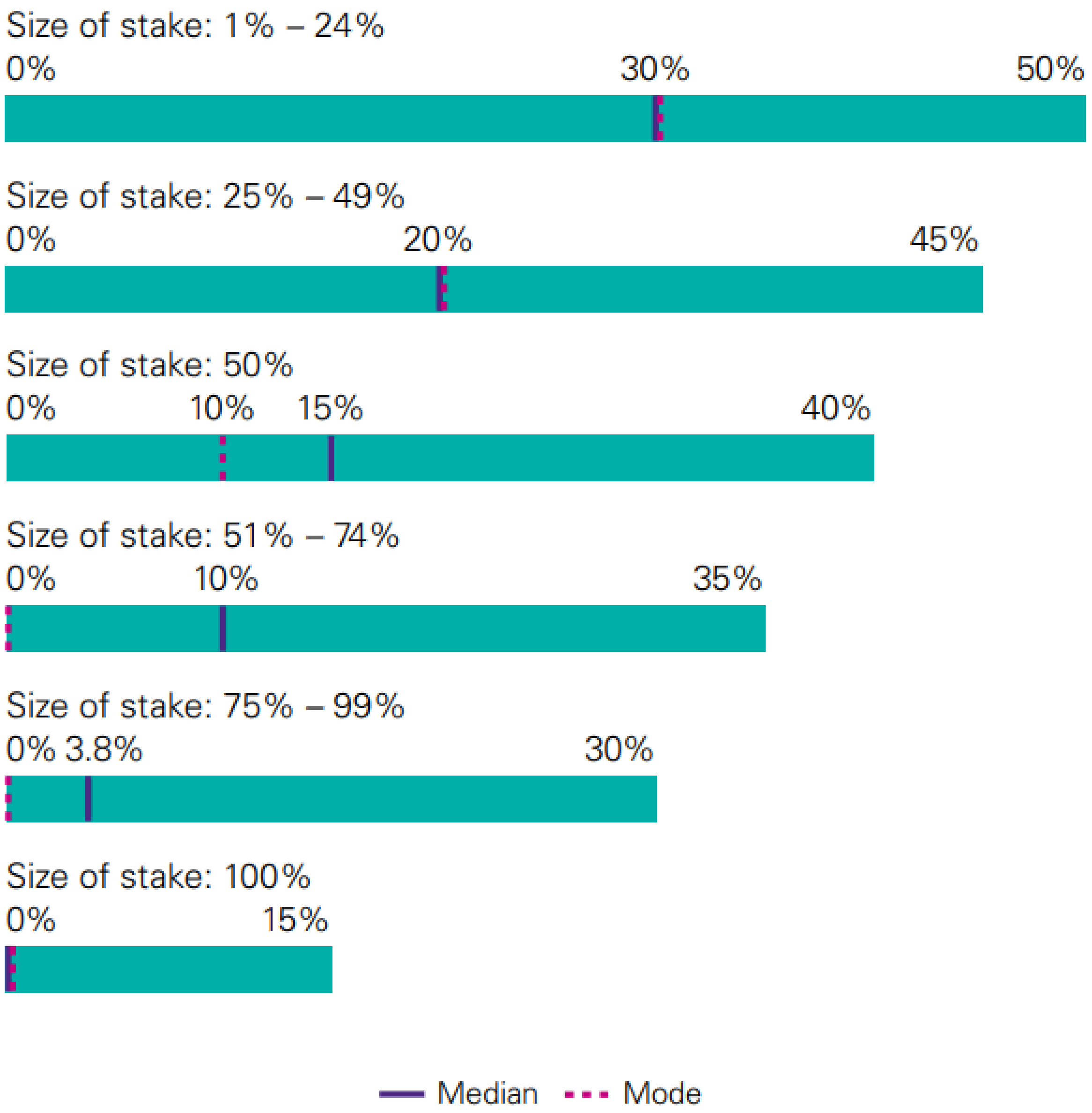
Range adopted



Notes: sample size n = 10.1: 51, 10.2: 22
Q10.1: Do you use a standard control premium?
Q10.2: What range do you currently adopt as the standard control premium in the {Country} market?

https://www.charteredaccountantsanz.com/-/media/23f66cbca26749b4a3564376525ccac2.ashx

When a marketability discount is required, what sized discount do you apply?



Source: For all it's worth – KPMG Valuation Practices Survey 2017 Australia

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Thank you for your attention!

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Mergers and Acquisitions

Péter Juhász, PhD, CFA

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Value creation areas

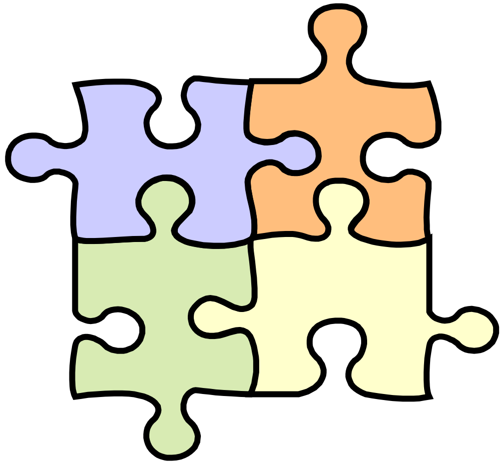
Current market value

Communication and information gap

Maximal value creation

True business value

Total potential value



Interior efficiency

Financial restructuring

Value with enhanced efficiency

Business value after interior enhancements, asset sales, investments and M&As.

Value after creative disruptions

Layoffs, asset sales

New growth opportunities

Source: McKinsey

Ways of creating value

Increase cash flows

Reduce risks

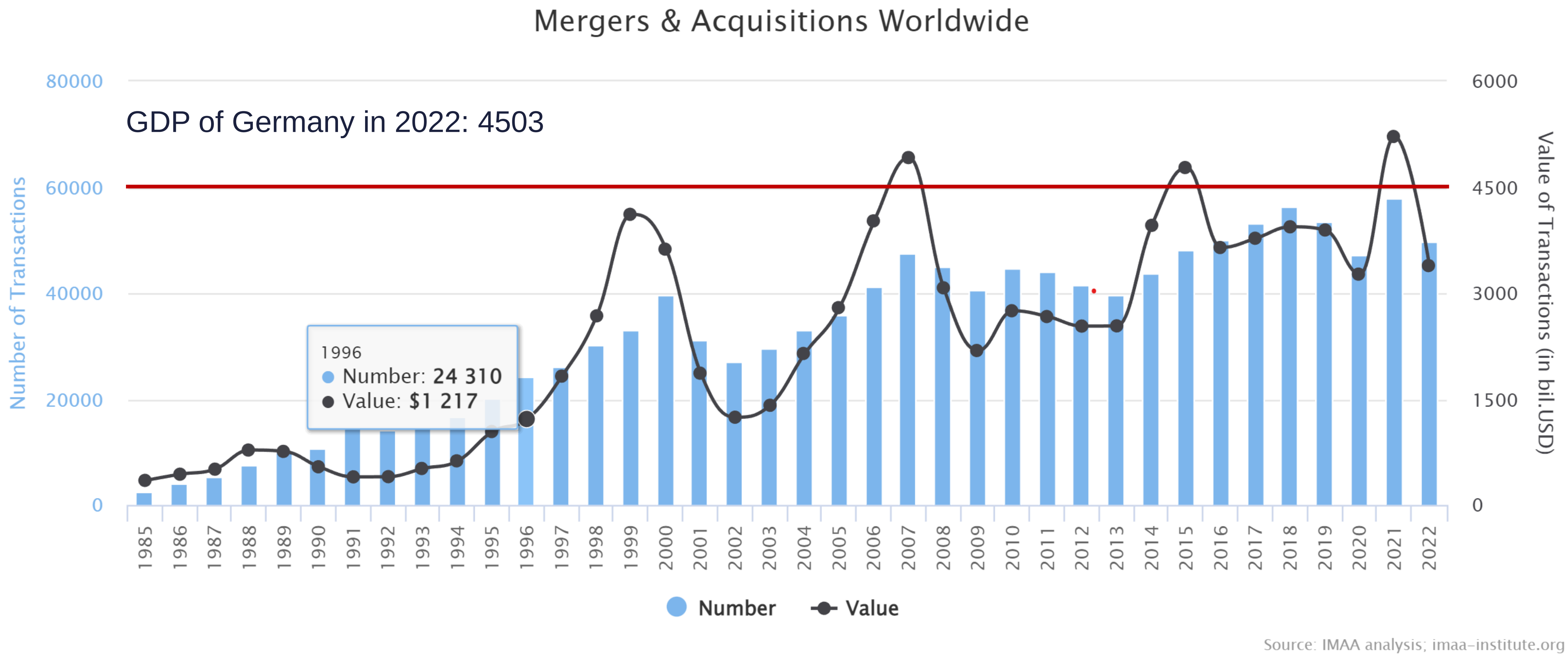
Cut back on capital need

$$MVA = \sum_{i=1}^{\infty} \frac{FCFF_i}{\prod_{j=1}^{i}(1 + WACC_j)} - IC_0$$

Tools:

- New, positive NPV projects
- Growth,
- Sell-off of poor projects

Importance of M&As



Source: https://imaa-institute.org/mergers-and-acquisitions-statistics/

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Sell-side motives

Management issues (inheritance)

Liquidity issues, lack of capital (bankruptcy, growth)

Stress among owners (divorce)

Strategic opportunities (better investments available)

exit

(venture capital)

Buy-side motives

- Extensive growth
- Efficiency of scale
- Market force and marketing considerations
- Vertical integration
- Diversification
- Taxation
- Access cash
- EPS increase
- Prestige
- Complementary resources
- Acquiring crown jewels
- Undervaluation
- Management considerations

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YOU'RE AN INCOMPETENT CEO, BUT THE DOGBERT INVESTMENT BANK CAN HELP YOU PRETEND TO UNLOCK SHARE-HOLDER VALUE.
Dilbert.com DilbertCartoonist@gmail.com
I'LL ARRANGE AN UNWISE MERGER SO YOU CAN CASH OUT WHILE I COLLECT AN OBSCENE COMMISSION.
1-30-10 ©2010 Scott Adams, Inc./Dist. by UFS, Inc.
IT'S LIKE A BRIBE, BUT INSTEAD OF GOING TO JAIL, A STRANGER WILL WRITE A BEST-SELLING BOOK WITH YOUR NAME ON IT.
CAN I READ IT?

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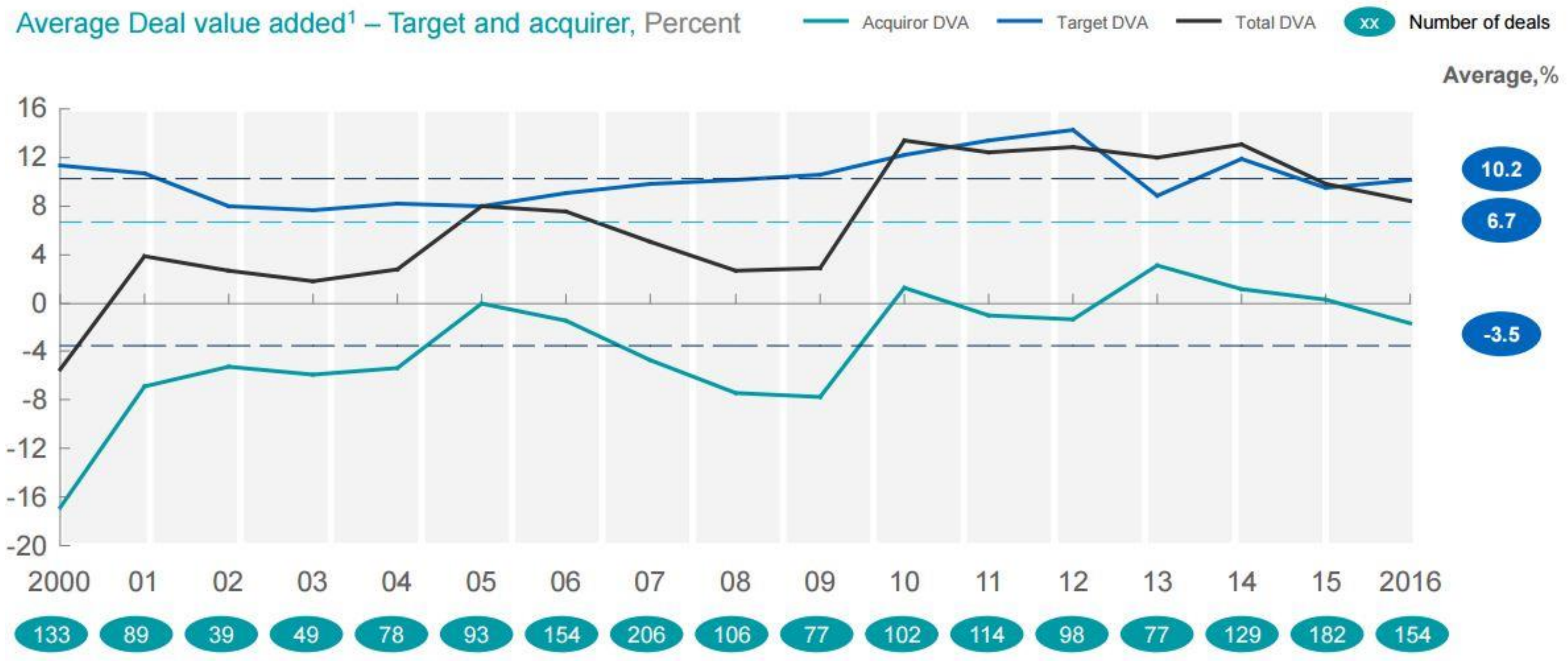
Do M&As create value?

For the sellers, absolutely. The average takeover premium is 30% in the US.

There is a huge deviation across buyer effects – usually they lose value.

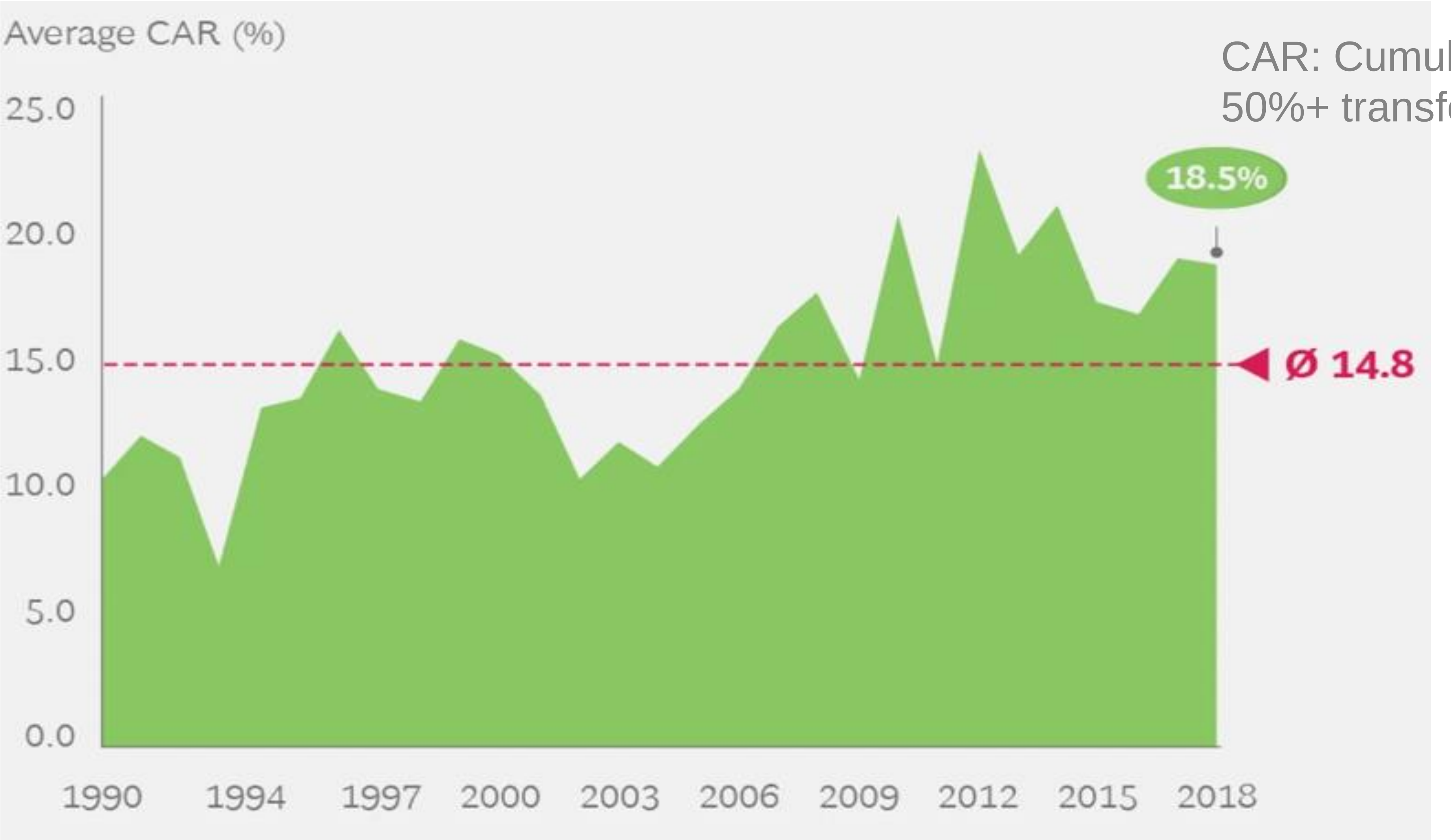
Creating value by takeovers



Source: McKinsey (2017): Overview of M&A, 2016

DVA: Deal Value Added, the change of joint market value of the target and the buyer, -2/+2 days around the announcement.

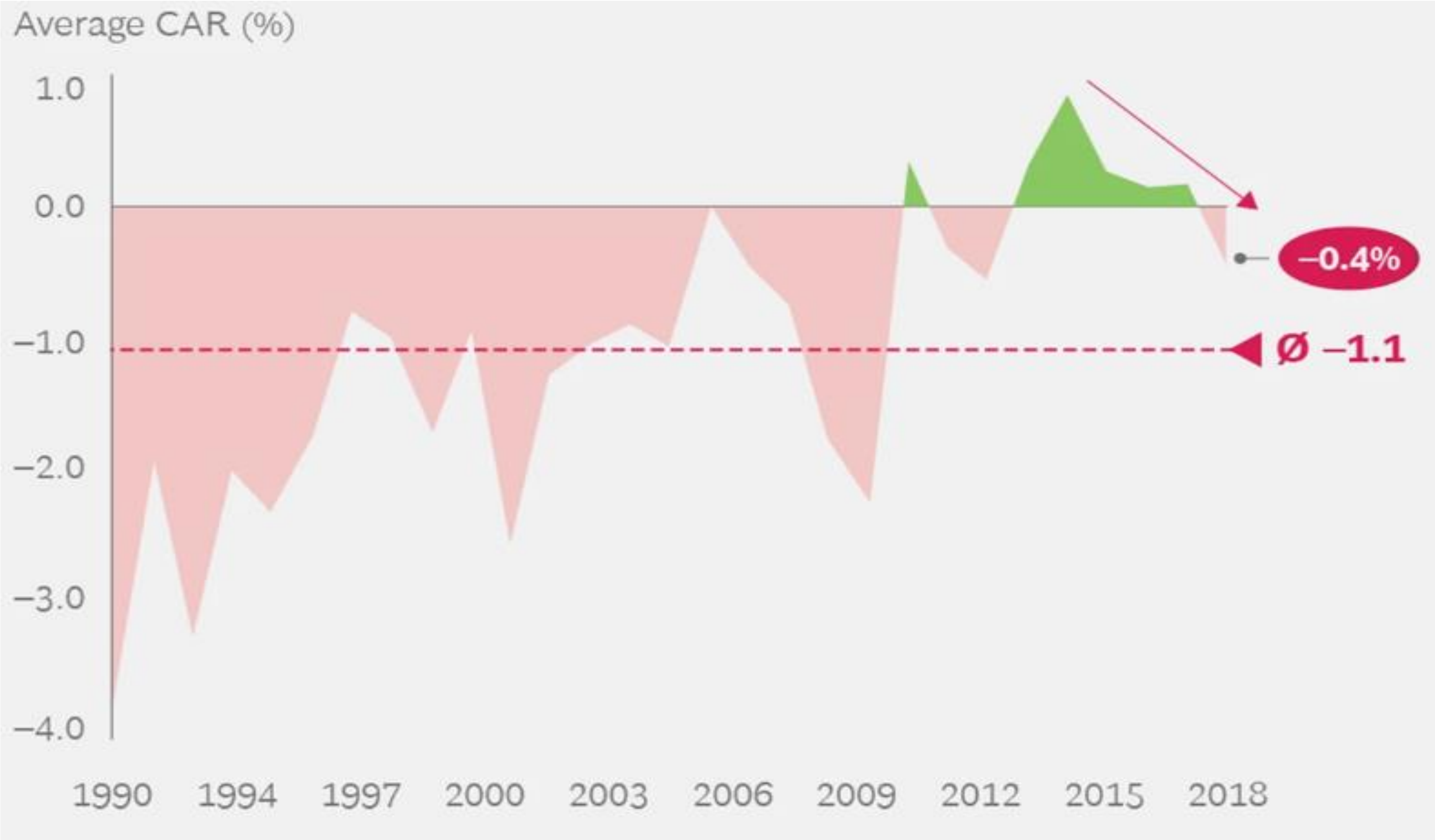
Creating value by takeovers (sellers)



CAR: Cumulative Abnormal Return,
50%+ transfers

Source: BCG, 2019 Mergers & Acquisitions (M&A) Report

Creating value by takeovers (buyers)



Source: BCG, 2019 Mergers & Acquisitions (M&A) Report

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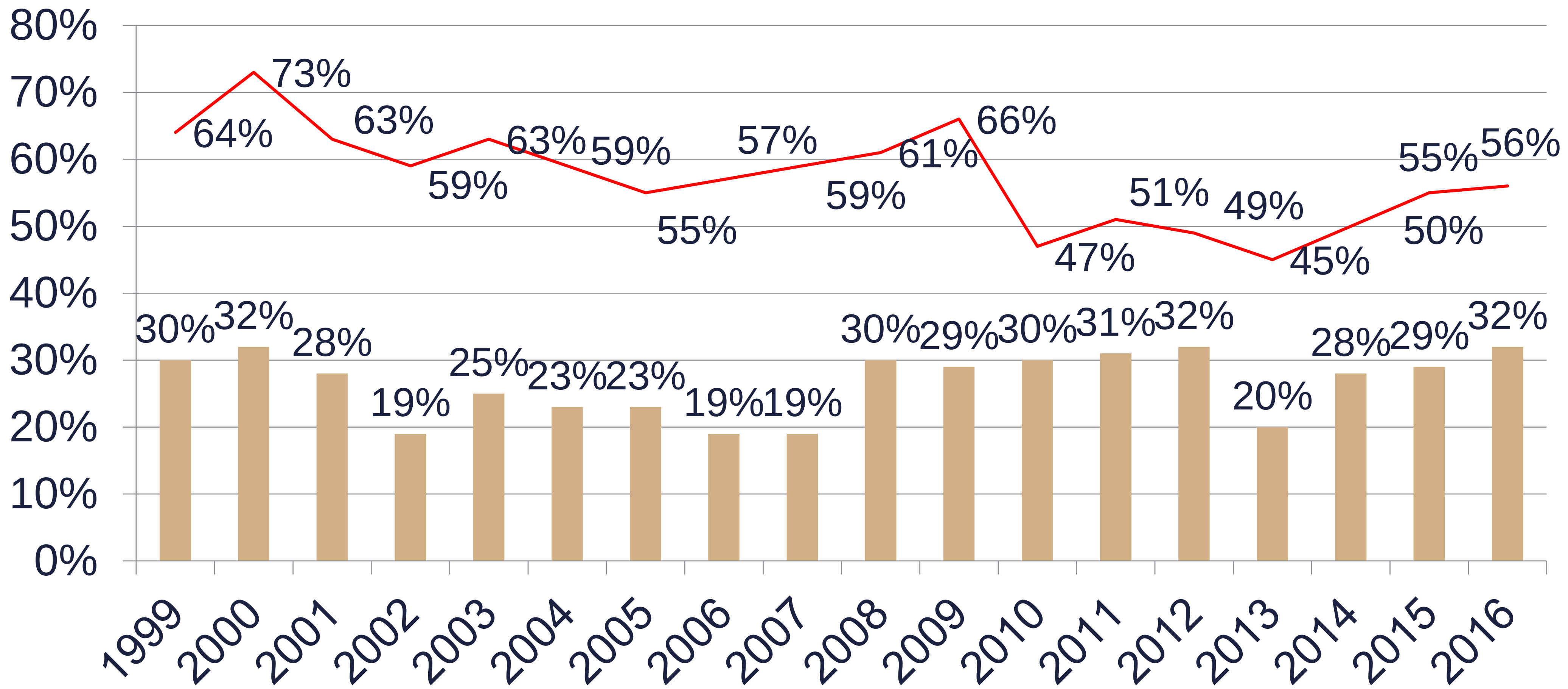
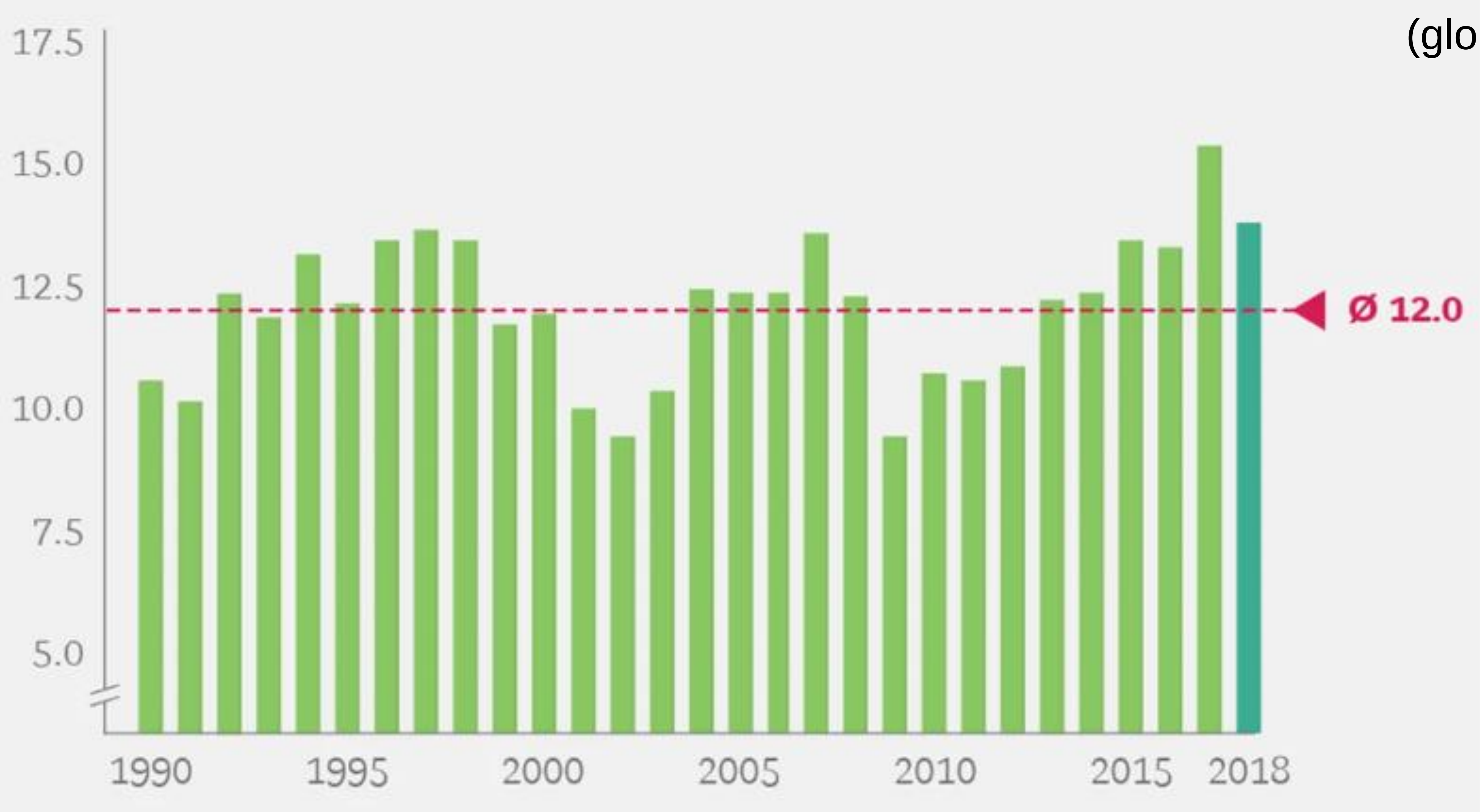
Premiums on the global M&A market



Source: McKinsey, 2017, Premium: -7/0 days, median
POP: Percent of Overpayers, rate of buyers seeing their value falling due to the announcement, -2/+2 days around the announcement.

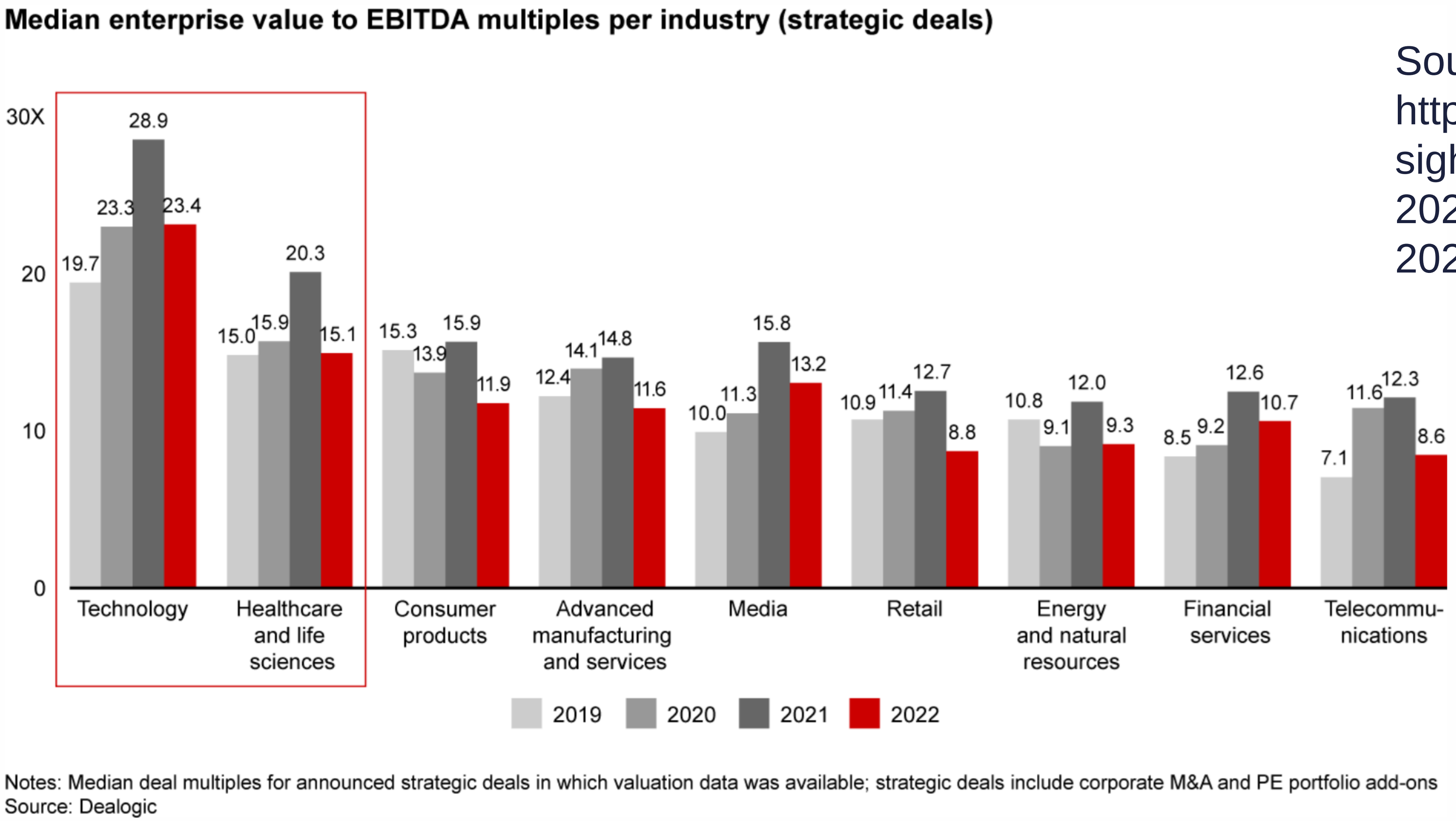
EV/EBITDA multiples

(global median)



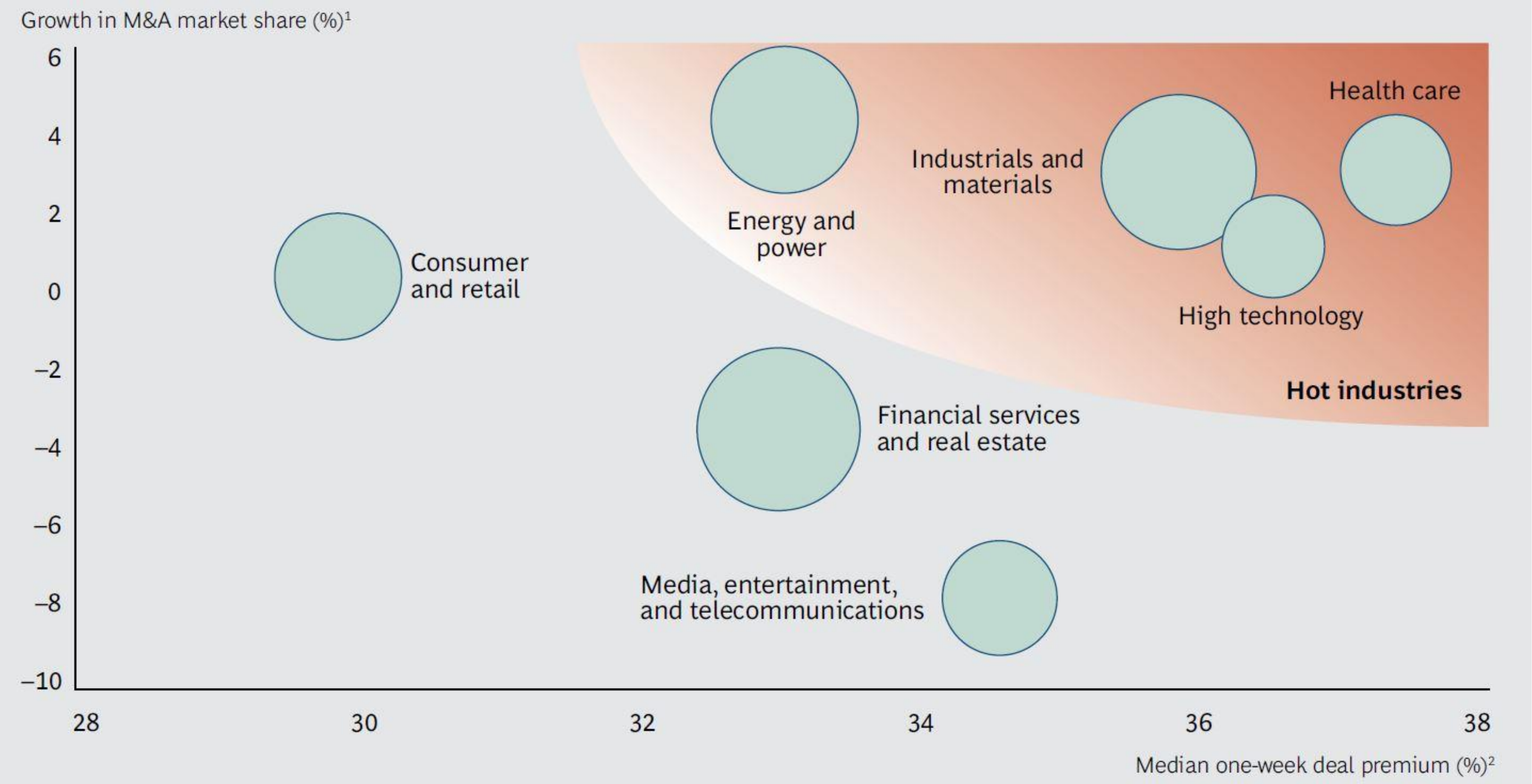
Source: BCG, 2019 Mergers & Acquisitions (M&A) Report

EV/EBITDA multiples (global median)



Source: https://www.bain.com/insights/looking-back-at-2022-m-and-a-report-2023/

M&A premium across industries



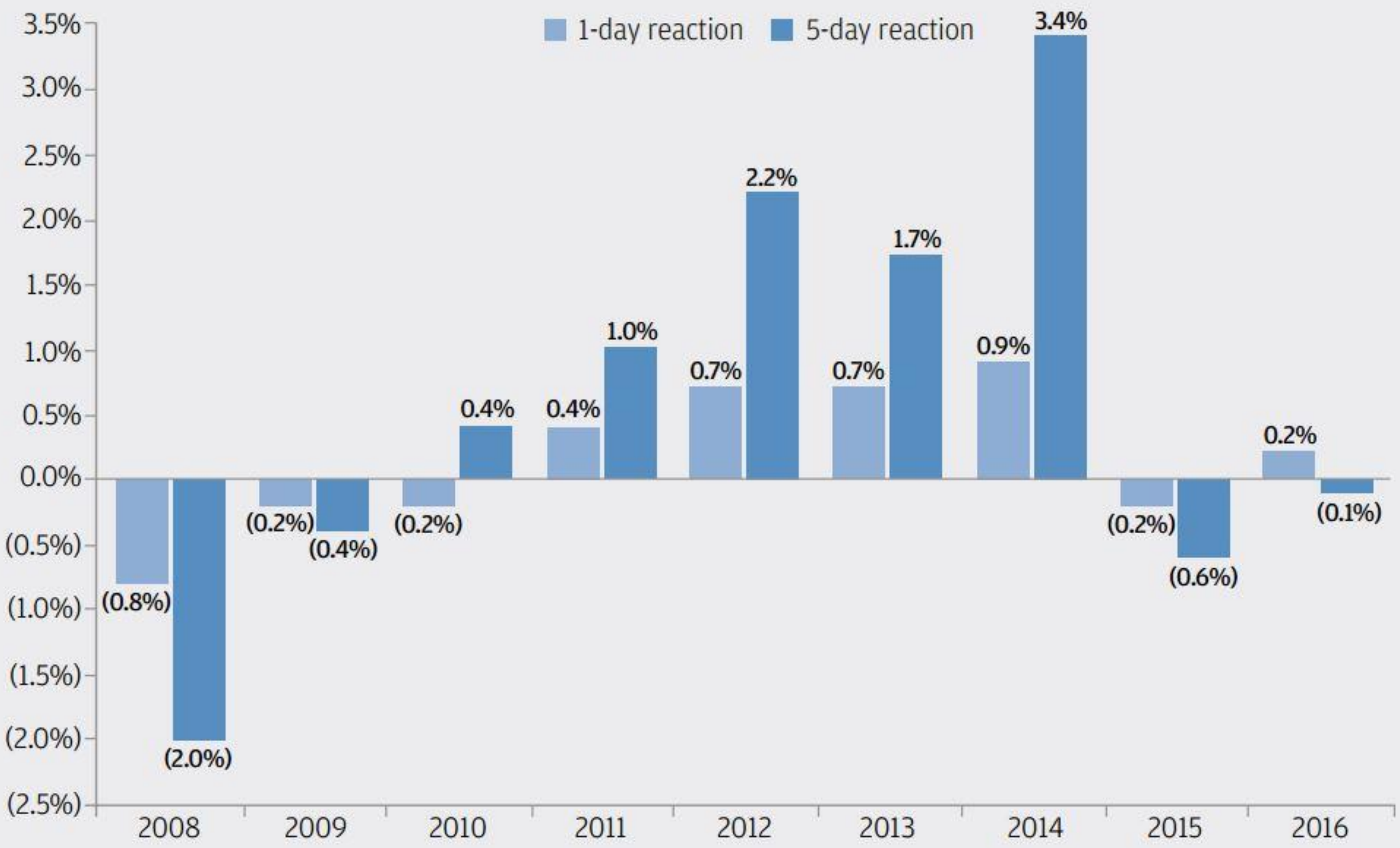
Source: BCG, 2014, 2011-2014 H1 transactions, median

Creating value by M&As

Survey	Positive	Neutral	Negative
Based on financial ratios			
Mercer, 1995	25%	30%	45%
KPMG, 1999	17%	30%	53%
A.T. Kearney, 1999	27%	19%	54%
KPMG, 2001	30%	39%	31%
McKinsey, 2001	12%	46%	42%
Based on management interviews			
MAPI, 1999	61%	27%	12%

Source: Pautler, 2003

Shares price reaction (buyers)



Sources: FactSet and company filings as of December 31, 2016

Note: Includes acquisitions by U.S. buyers with minimum deal value of $500mm, % owned < 20%, % acquired > 80% and target > 5% of the size of the acquirer

[a] Excess return over S&P 500 returns times acquirer's beta from unaffected date prior to announcement

Source: JP Morgan, 2017, median, US

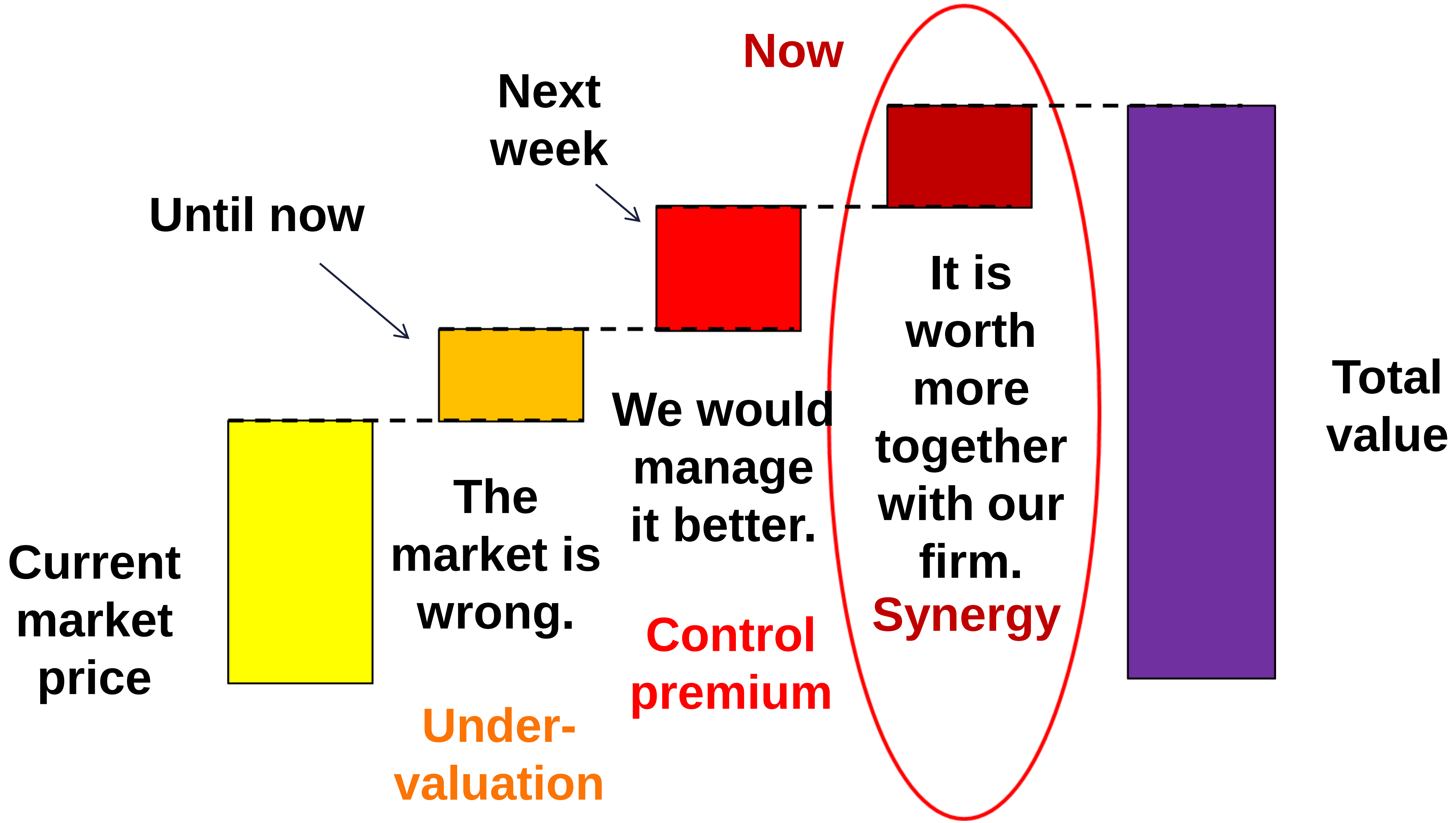
Why paying more than the current price?

Undervaluation „premium”: The fair firm value is 3 today, but the market price is only 2. We pay a maximum of 3; no need to own another firm.

Control premium: Should we control the firm, the value will climb from 2 to 3. No need for owning another firm.

Synergy: Two firms together add up for more: 2+2=5, thus we pay a maximum of 3 for the firm.

The structure of the total value



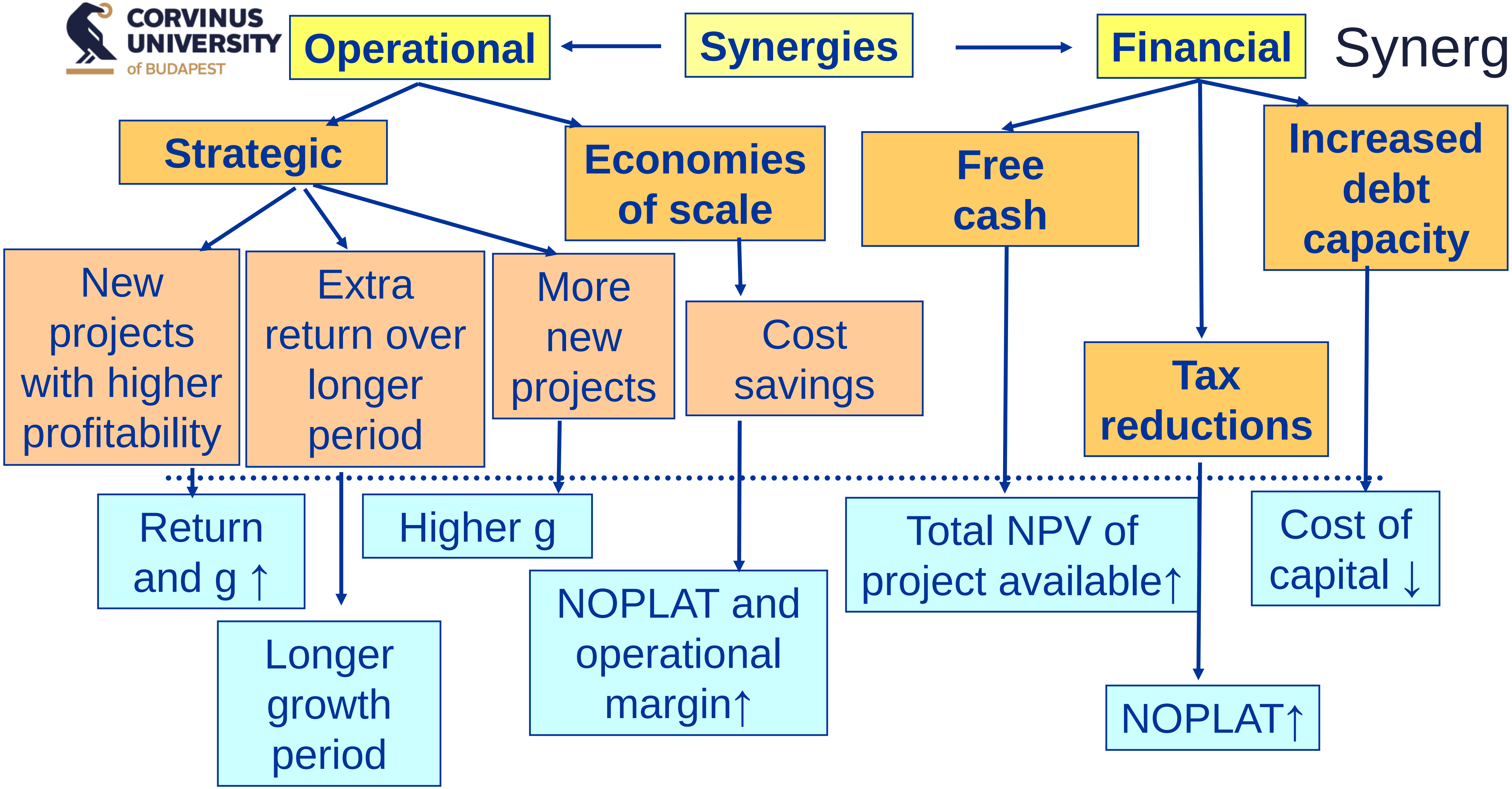
The effect due to which parts together are more valuable than separately.

Car – ignition key

Cell phone – SIM card

Computer – software

Synergies



Source: Damodaran online

Assessment of synergies

What form do those show?
How do those affect value drivers?
When do those start having an impact?

Modelling operational synergies (higher FCFF):

Cost efficiency – naturally limited
Sales growth synergies – only limited by common sense (Country and market growth? Maintainable?)

Doubtful synergies

EPS growth – if a firm with a P/E = 30 buy another with P/E = 20, EPS would grow (And what then? – this is not a value focused approach)

Faster growth – it is of no value alone. We need to review:

- WACC <> ROIC, RONIC?
- Premium <> Synergy?

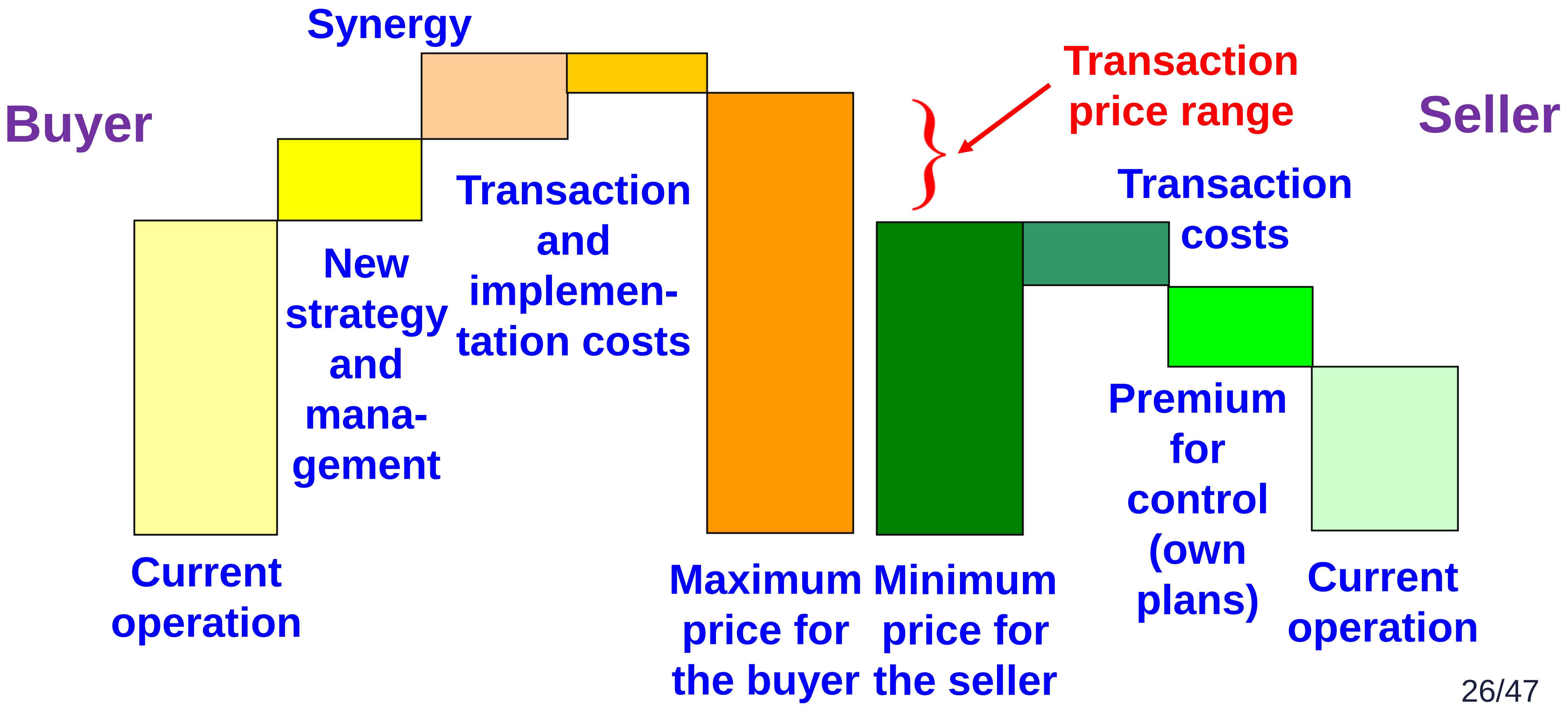
Valid motives for acquisitions 1

Motive	Explanation	Limit price	Comment
Undervalued stocks	$P < E_0$	P (no premium)	Undervaluation at the stock exchange or forced sale and liquidation.
Diversification	More stable CF, less expected risk	E_0 (no premium)	Could not be done on the stock exchange.
Operational synergies	g↑, return to scale, market share↑, complementary activities	P + Synergy	V(A+B)>V(A)+V(B) Added value from growth in sales or margins

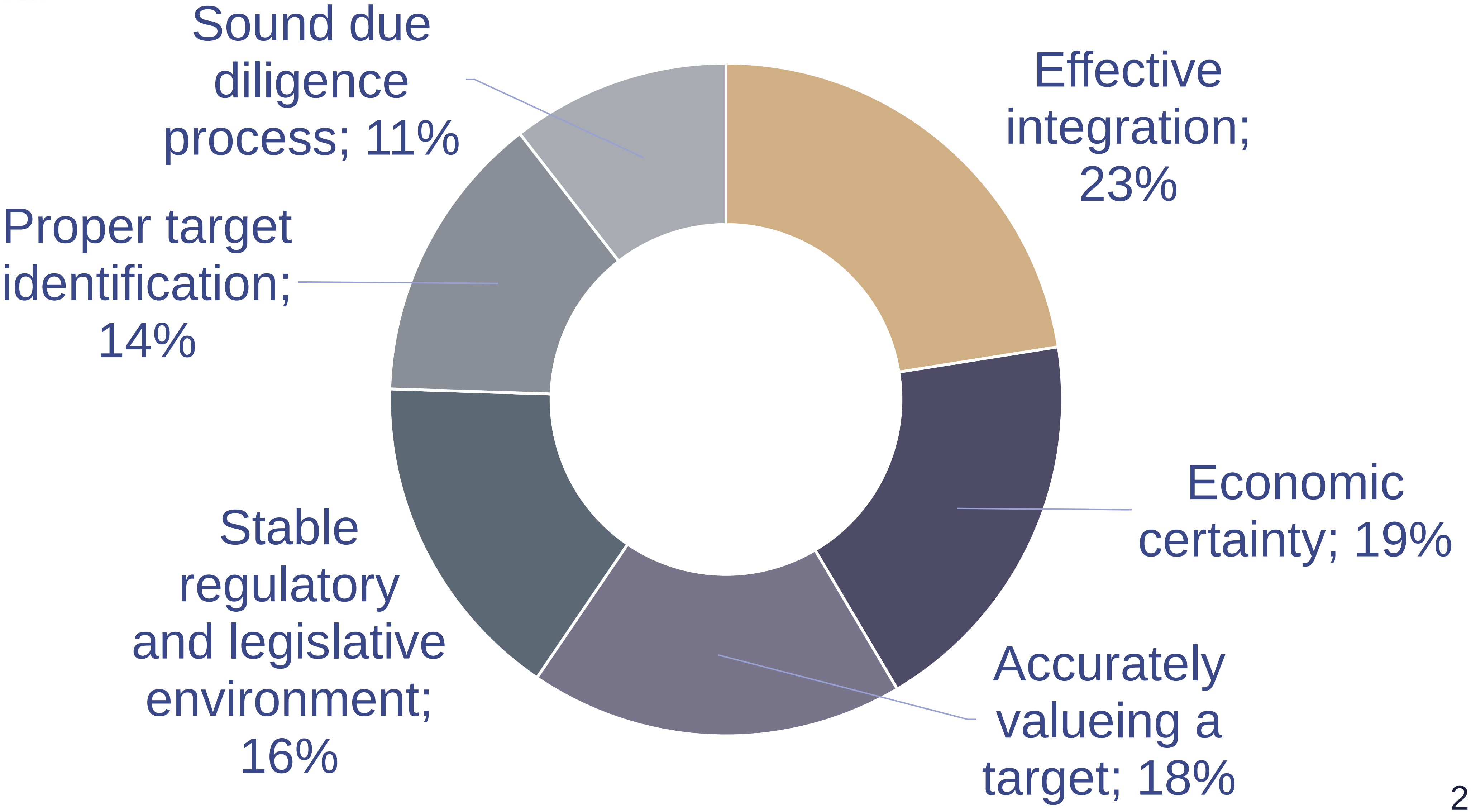
Valid motives for acquisitions 2

Motive	Explanation	Limit price	Comment
Financial synergies	Free cash, leverage opportunity, potential tax savings	P + Synergy	Higher CF, lower WACC or added TS.
Poor management	Resources not optimaly used	Value under optimal managment: P + control premium	Management issues reflected in price. (Mostly private firms.)

Setting the price



Success factors of M&As



Source: Deloitte, M&A trends 2019, 2018

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Why would mergers fail?

1. Overpayment (extreme optimism)

Overoptimism on market opportunities

Overestimating synergies

Incomplete due diligence, hidden or unhandled issues

Overbidding other offers, management ego („the winners' curse")

2. Poor integration management

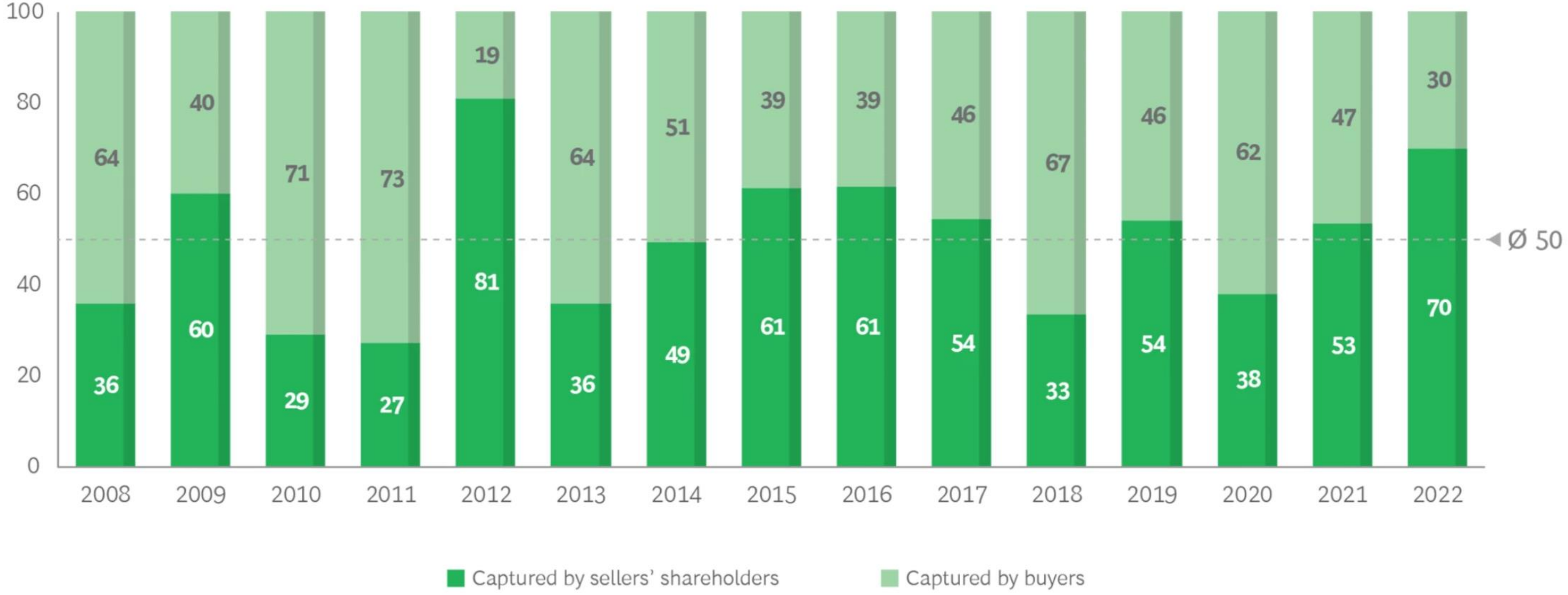
Lack of setting a new business model

Uncertainties and conflicts remain unsolved

Late reactions to market pressures

Premium vs Synergy

Median value of synergies captured by sellers' shareholders and buyers (%)[1]



Sources: Refinitiv; company press releases; BCG analysis.

Note: Analysis based on 1,385 public-to-public M&A transactions from 2008 through 2022 with a majority stake acquired by nonfinancial buyers.

[1]Value captured by sellers' shareholders is based on cumulative abnormal returns during the period from three days before to three days after announcement of the deal.

Source: BCG, 2023 Mergers & Acquisitions (M&A) Report

Dia címsor


MY STRATEGY IS TO WORK LIKE A PARA- NOID BEAVER DURING THE MERGER.
www.dilbert.com scottadams@aol.com
SURELY MY PRODUC- TIVITY WILL BE NOTICED AND REWARD- ED WHEN THE DOWN- SIZING BEGINS.
© 2003 Scott Adams, Inc. Dist. by UFS, Inc.
AND WHO IS ALICE?
I'M PRETTY SURE THAT'S THE BALD GUY WITH GLASSES.

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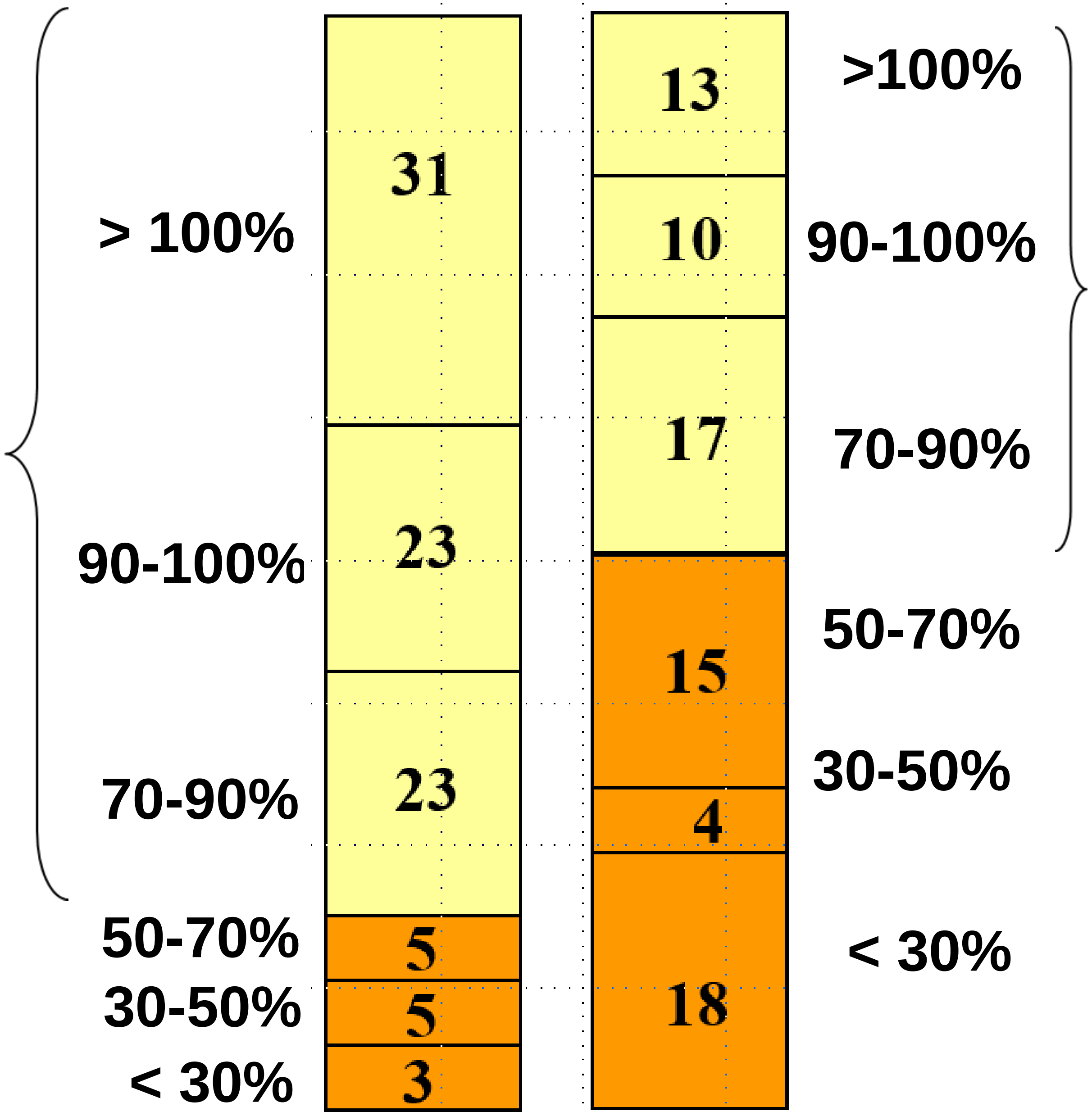
Plans vs Reality

In **85 %** of the transactions, at least 70% of cost synergies were achieved.

N=90

Cost synergy

Source: McKinsey

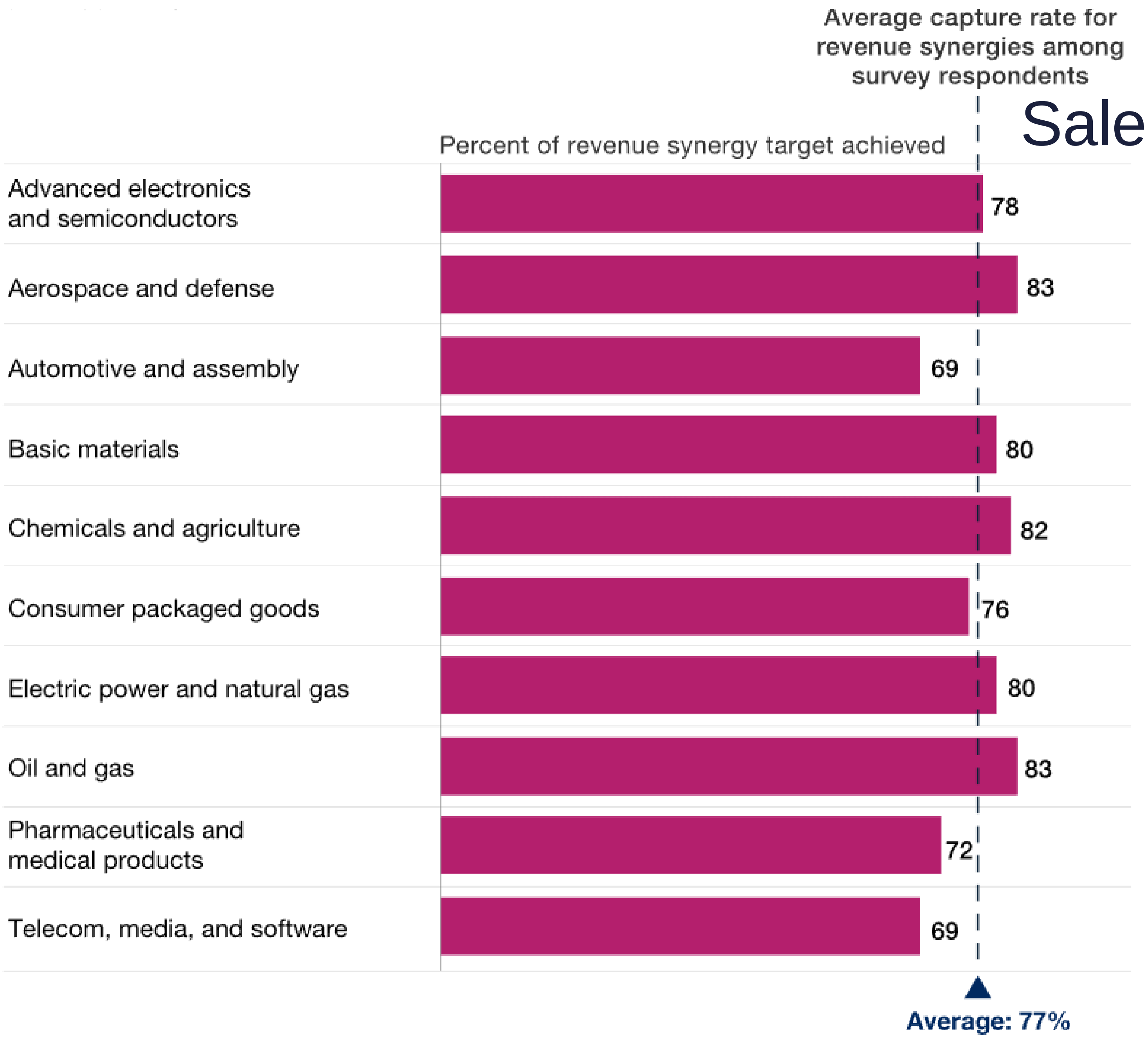


In **52%** of the transactions at least 70% of the sales synergies were realised.

N=77

Sales synergy

Sales synergies realised



Source: McKinsey, 2018

The risk of synergies 1

The likelihood of realising the different types of synergies is not the same.

The joint sales will not start growing immediately as

1) the transaction process jeopardises the customer connections.

2) competitors aim immediately at our best staff,

3) customers buying from both firms hit take some of their purchase to a competitor,

4) the remaining customers ask for a reduction as thanks to their loyalty

The risk of synergies 2

Costs of realisation

lay-off expenses, restructuring, repositioning, change of names, IT integration

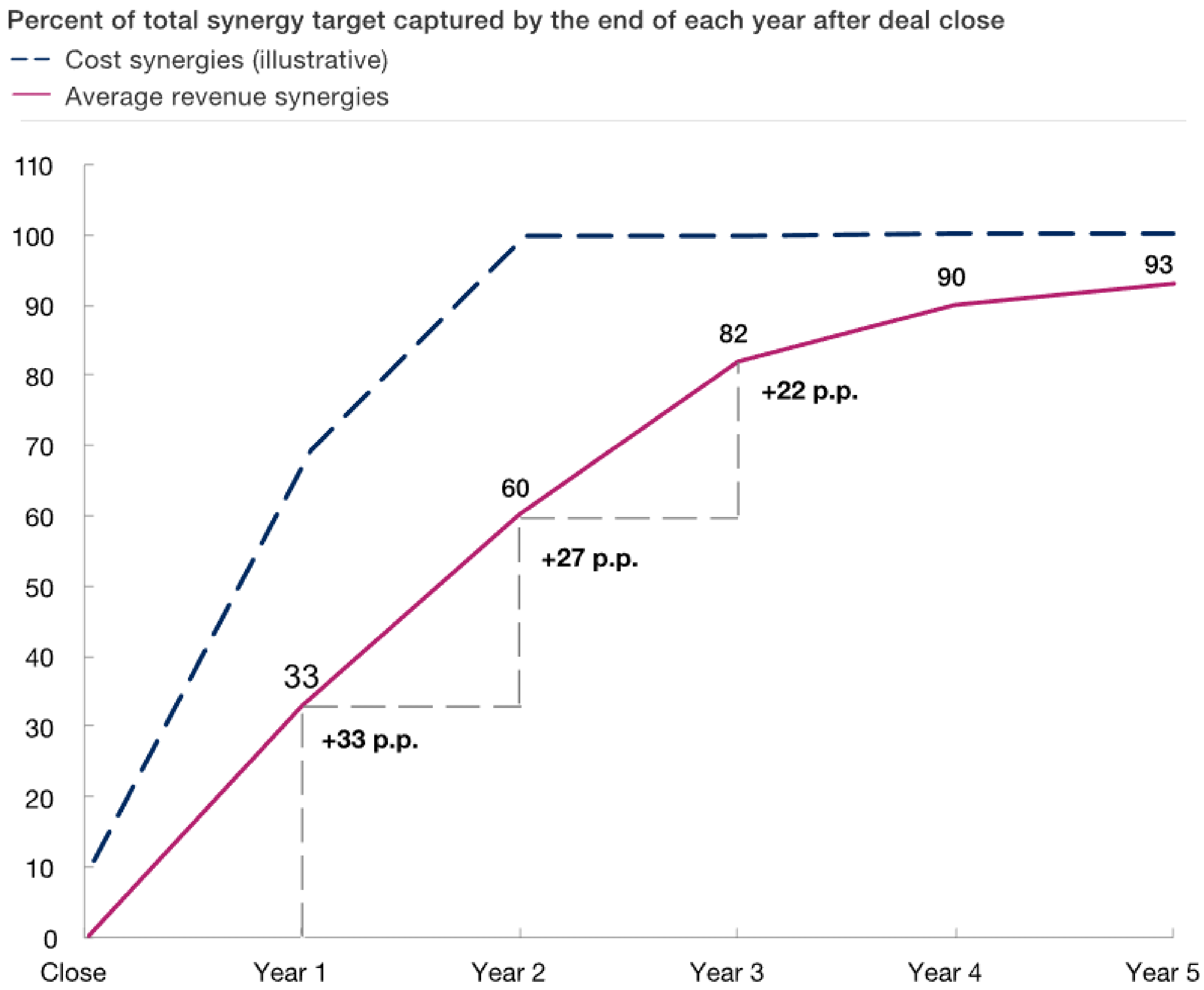
These can add-up to more than one year of synergy

Time need of realisation

Different customer management systems, trainings, IT integration

Synergies that could not be realised during the first complete business year after the transaction are usually completely lost.

The time need of realising the synergy



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How to pay for a firm?

Cash

- Liquidity issues
- All risk remains with the buyer
- All profit stays with the seller

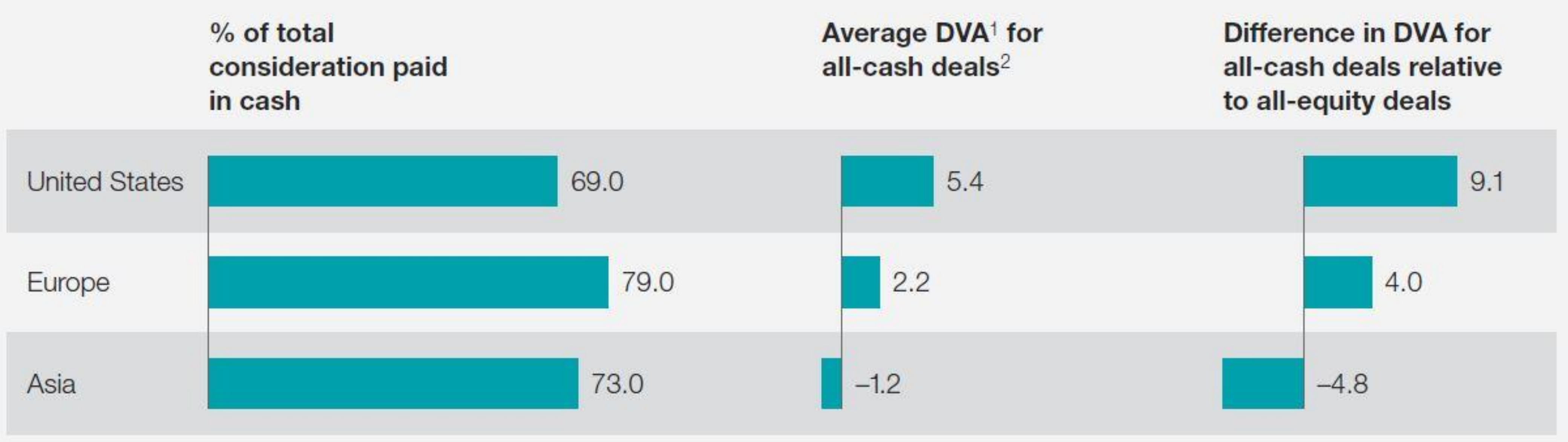
Shares

- The seller can not exit
- Shared risks
- No liquidity shock

Payment methods worldwide

Source: McKinsey, 2017

Market view of payment methods

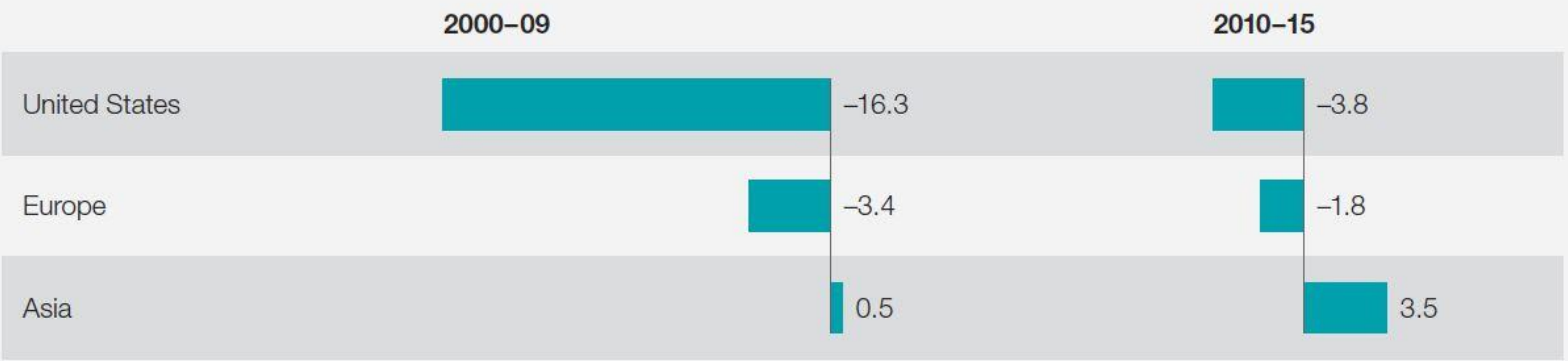


[1] Deal value added.
[2] For M&A involving publicly traded companies; defined as combined (acquirer and target) change in market capitalization, adjusted for market movements, from 2 days prior to 2 days after announcement, as % of transaction value.

Source: Datastream; Dealogic; McKinsey analysis

Market view of shares-only payment

Average DVA[1] index for all-stock deals,[2] %



[1] Deal value added.

[2] For M&A involving publicly traded companies; defined as combined (acquirer and target) change in market capitalization, adjusted for market movements, from 2 days prior to 2 days after announcement, as % of transaction value.

Source: Datastream; Dealogic; McKinsey analysis

Lessons from the history 1

1. The fusion of **similar-sized firms** is less likely to be successful.

2. Synergies from **immediate cost reductions** (e.g. downsizing) are easier to capture than those linked to future growth, new products, or joint R&D.

3. **Hostile takeovers** promise better management and results than friendly transactions.

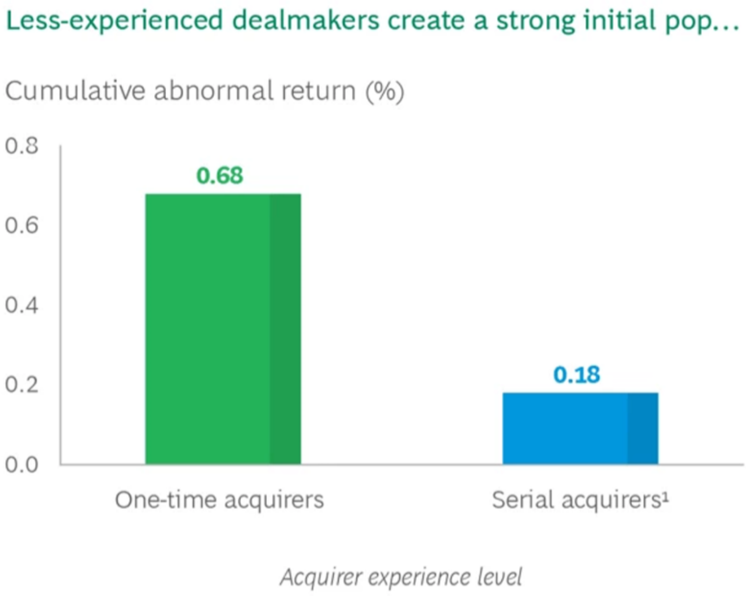
3. Acquisitions focused on **smaller private firms** are more likely to be successful. The fiercer the competition is, the more likely the loss in value.

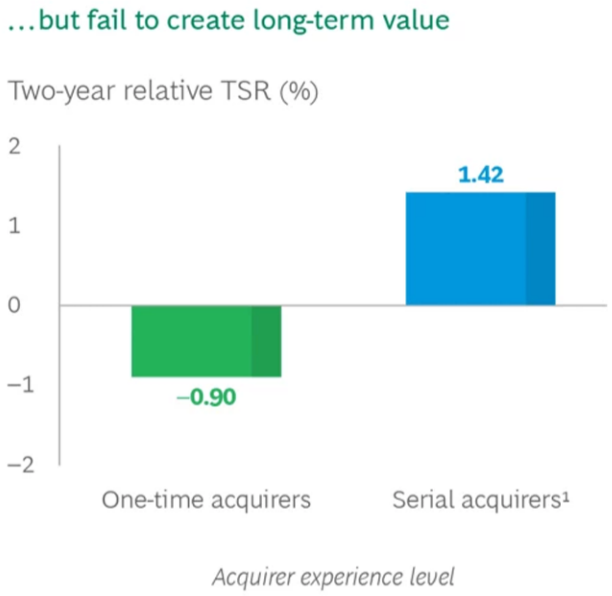
4. **Firms carefully considering** synergies and M&As are more likely to be successful.

Lessons from the history 3

5. **Companies with outstanding efficiency** are more likely to create value by acquiring firms.

6. **Firms that gained experience in M&A** are more likely to be successful in the long run.





Source: BCG, 2023 Mergers & Acquisitions (M&A) Report

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Thank you for your attention. Any questions?

peter.juhasz@uni-corvinus.hu

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Example

ABC Co. purchases BCD Co. They pass on 100 unit of synergy, one third of the price is paid cash.

How will the price of ABC shares change due to the transaction, if there are 2000 pieces issued?

Data

ABC current value: V=1000 E=800 D=200

BCD current value: V=300 E=200 D=100

Synergy=300

Joint value (before payment): V=1600 E=1300 D=300

BCD shares purchase price: 200+100=300

Cash: 300*1/3=100

Shares: 300*2/3=200

Together (after the payment): V=1500 E=1200 D=300

Results

ABC current quote: 800/2000=0.4

Synergy to current owners: 300-100=200

Required values of ABC shares after the transaction:

800+200=1000

Share price: 1000/2000=0.5

To be paid for BCD: 200

Number of shares to issue: 200/0.5=400

Summary

Number of shares after the transaction: 2000+400=2400

Price quote: 1200/2400=0.5

Attention! If during the transaction the amount of debt changes of extraordinary dividend is paid, you also need to consider those effects before calculating the share prices and number of shares to issue!

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Private Equity

Péter Juhász, PhD, CFA

Sources of financing

Internal	Debt	Equity
Profit reinvestment	Bank loan	Current investors
Unneeded assets	Bonds	New owners

What is private equity?

Specialised investor targeting private companies, willing to obtain control over the firms and exit within some time specified in advance.

How is private equity different?

Projects are considered on a portfolio basis not individually.

Successful projects compensate for capital and return lost in failure projects.

Investor provides knowledge next to capital.

The exit is pre-planned.

Benefits

Management knowhow

Strategic view

Efficiency increase

Synergies

M&A opportunities

Limit on agency problem

Value based compensation

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Where does it come from?

Private equity

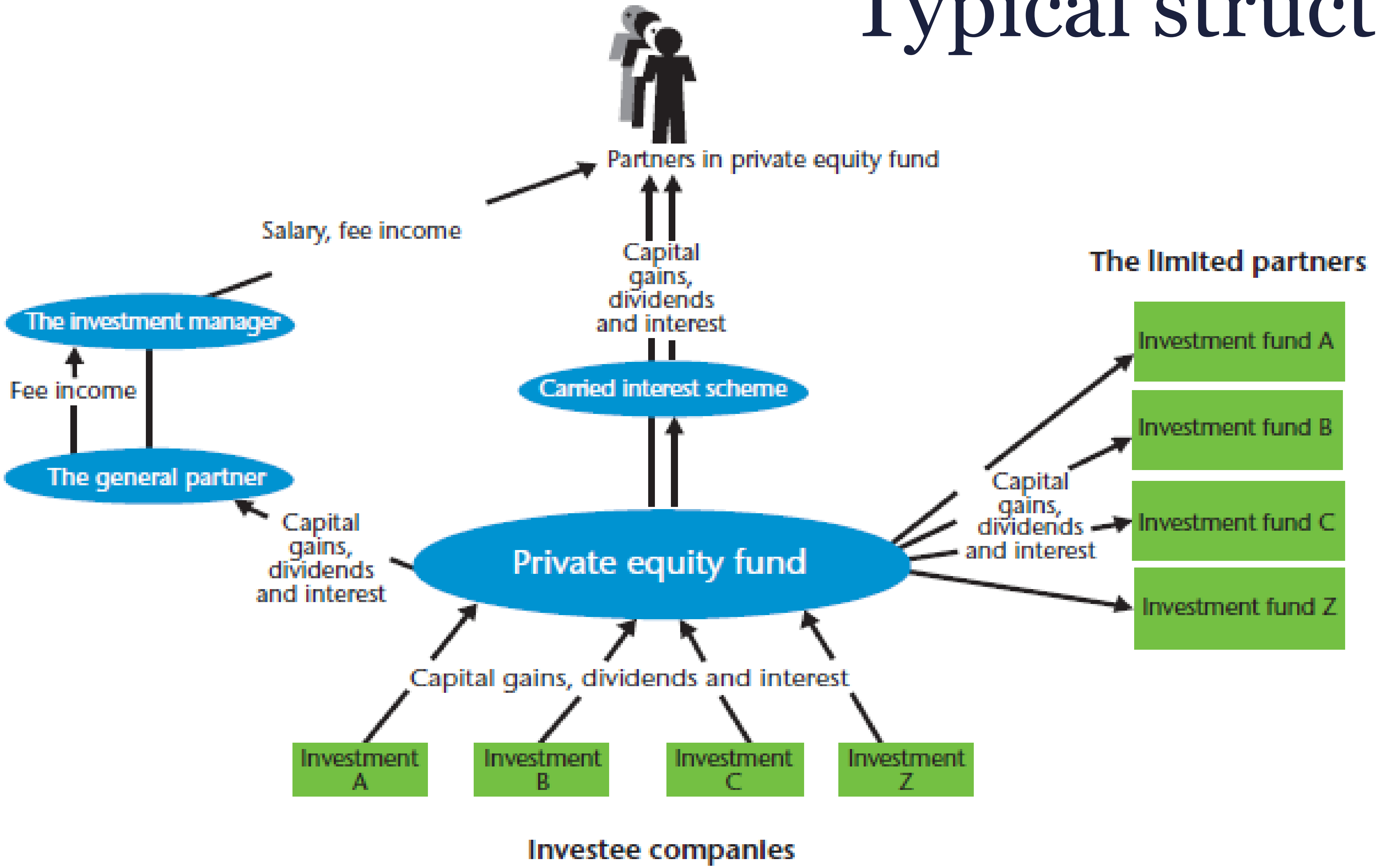
Strategic investors

Holding companies, selling and buying their sectorial subsidiaries

Financial investors specialised for sector

Financial investors

Typical structure



Source: Gilligan-Wright (2010): Private Equity demystified

Forms of private equity

Direct	Indirect
Investor owns the shares directly	Investor and fund manager are different entities

Typical investors

	Direct	Indirect
Individuals	✓	✓
Companies	✓	✓
Banks		✓
Insurance companies		✓
Investment funds		✓
Pension funds		✓
Governmental institutions		✓

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Investment policy

- Written statement
- Specifies target sectors, investment size, selection and decision processes
- Investment committee
- Investors’ council

Categorisation

Sector focus

Sector knowledge

High exposure to a specific sector

Duration of investment and cycle of sector to be harmonised

Sectors before expansion or consolidation

Region focus

Countries in similar economic cycle

Size focus

1/5/20/100/500 M euros

Life cycle focus

Phase of life cycle	Phase 1 Start-up	Phase 2 Expansion	Phase 3 Fast growth	Phase 4 Mature growth	Phase 5 Decline
External financial need	Remar-kable, invest-ment cont-rolled	High com-pared to the value of com-pany	Modest com-pared to the value of com-pany	Des-cending com-pared to the value of com-pany	Low, because projects are ending

Forms of PE

- →Fund financing
- →Early financing
- →Expansion financing
- →Merchant bank financing
- →Buy-out financing
- →Turn-around financing

Other categorisations

- Leverage
- Acquired stake

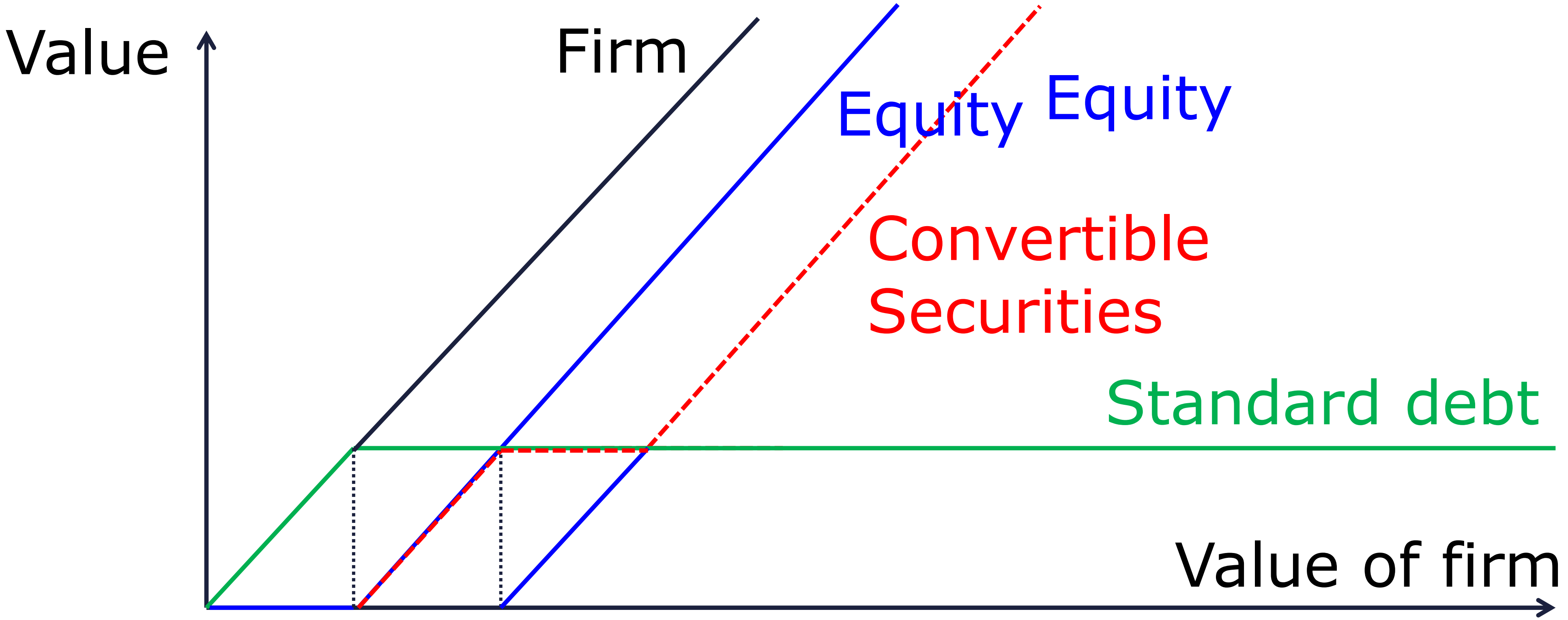
Steps of PE investments

Fund raising	Investment	Portfolio management	Exit
Limited partnership Committed capital 10-15 years	Selection Due diligence Exclusivity $1^{st}/2^{nd}$ round	Controlling Stepwise investment	5-7 years IPO Drag along, tag along

Convertible securities

Convertible loans, bonds + warrants

Convertible preferred shares



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Venture capital

Private Equity that invests in start-ups and early-stage deals.

Capital for Start-ups

FFF

Family, Friends, and Fools

Business Angels

Wealthy individuals who invest in start-ups and early-stage deals.

Equity crowdfunding

A group of individuals who invest in start-ups and early-stage deals.

Venture Capital

Private Equity that invest in start-ups and early-stage deals.

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Types of Venture capital

Early-stage

Seed, start-up, first-stage

Expansion

Later stage: Acquisition, Buy-out

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Dark side of private equity

- Growth pressure
- Efficiency pressure
- Exit pressure

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Ethical issues

Conflicts of interest	Buyout/exit not always in the interest of all owners
Vulnerability of employees	Restructuring may mean lay-offs
Capital markets	Public-to-private deals, Insider info
Tax avoidance	PE firms and management usually very good at tax optimising

Required return for VC

At a VC fund with an average investment horizon of 4 years 70% of similar projects fail. In case of default, 110% of initial investment can be collected at the end. Cost of financing is 20%.

Required return for projects surviving:

$$70\%*110\%+30\%*(1+r)^4=1.2^4$$

$$30\%*(1+r)^4=1.3036$$

r=44.38%

Drivers of investment success

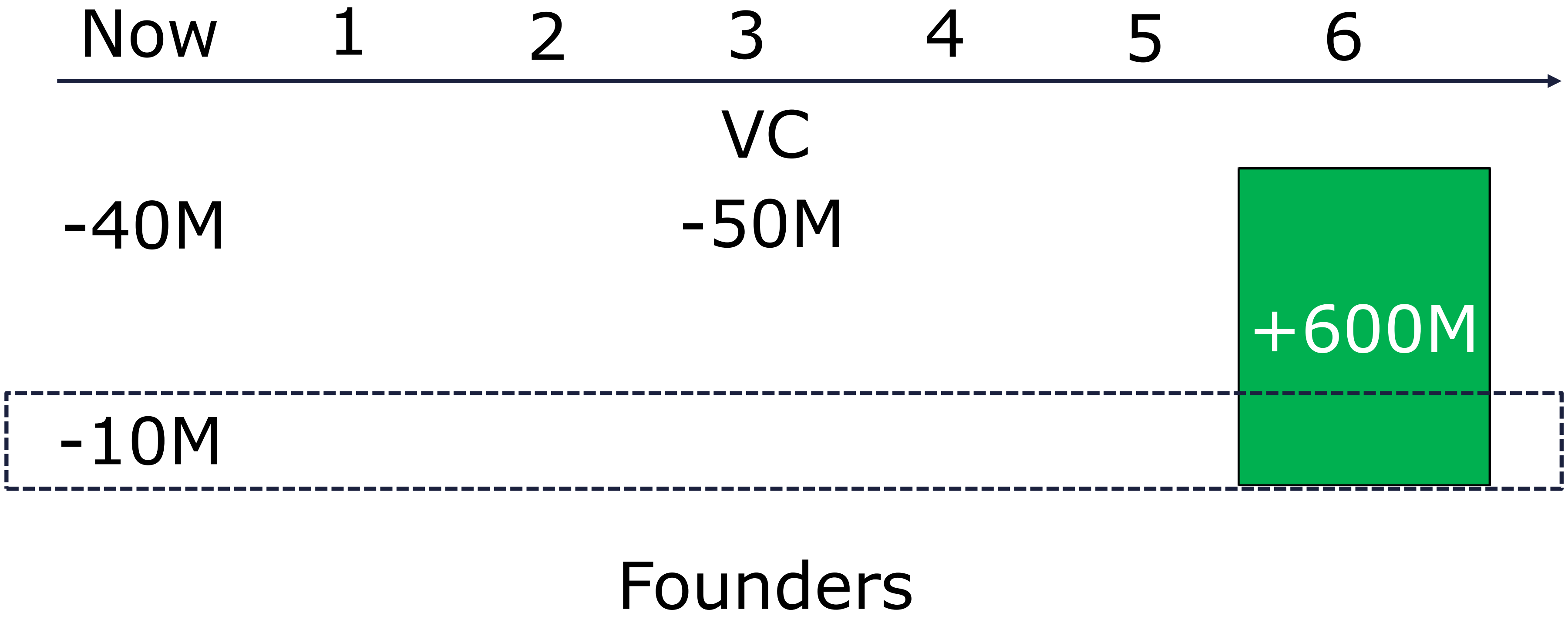
Better selection (specialisation)	Close control
Professional assistance	Covenants and collaterals

Fair price and stake 1.

A firm with a capital of 10 M needs an immediate investment of 40 M. In case of success after 3 years another 50 M will be needed and after 6 years the total of equity may be sold for 600 M.

What stake should VC receive for the equity increases, if required rate of return is 40%?

Investment plan



Fair price and stake 2.

Equity value (E) in 6 years: 600.00 M

In 3 years, after capital: $600.00/1.4^3=218.66$

In 3 years, before capital:

$$218.66-50.00=168.66$$

Stake for investment of 50 M in 3 years:

$$50.00/218.66=22.87\%$$

Calculations

Now 1 2 3 4 5 6

Post-Money	219M
VC	-50M
Pre-Money	169M

23%

+600M

VC	-40M
Founders	-10M

Fair price and stake 3.

In 3 years, before capital: 218.66-50.00=168.66

Today, after capital: $168.66/1.4^3=61.46$

Today, before capital: 61.46-40.00=21.46

Stake for current investment:

40.00/61.46=65.08%

Calculations

Now 1 2 3 4 5 6

Post-Money 219M

-50M

Pre-Money 169M

23%

77%

50%

+600M

73%

Post-Money 61M

VC -40M

Pre-Money 21M

Founders -10M

65%

* 77%

35%

27%

Fair price and stake 4.

	VC	Founders
Now	0.00%	100.00%
After investment	65.08%	34.92%
In 3 years, after investment	73.07%	26.93%

Stake for capital in year 3: 22.87%

VC total:

65.08%*(1-22.87%)+22.87%=73.07%

Stake of founders:

34.92%*(1-22.87%)=26.93%

Fair price and stake 5.

Return to founders:

$$600.00 * 26.93\% = 161.58$$

$$161.58 / 21.46 = (1+r)^6$$

$$\mathbf{r = 40.00\%}$$

Measuring value created

EVA:
IC*(ROIC-WACC)

Residual income:
PAT – St*r_E

ΔMVA:
$(V_1-IC_1)-(V_0-IC_0)$

SVA:
ΔE±CF

CFROI:
IRR on FCFE

TSR:
$$\frac{(P_1 - P_0) + Div_1}{P_0}$$

Balanced Scorecard

The secret of successful VBM

Top management support	Including operational decision	Ability development
Practice focus	Individual KPIs	Provide data access
Standardised format for reports and analysis	Bonuses linked to value creation	Obligatory value-based justification of resource needs

Source: McKinsey, 1994

Thank you for your attention!

Business valuation specialities of the Financial Service Industry

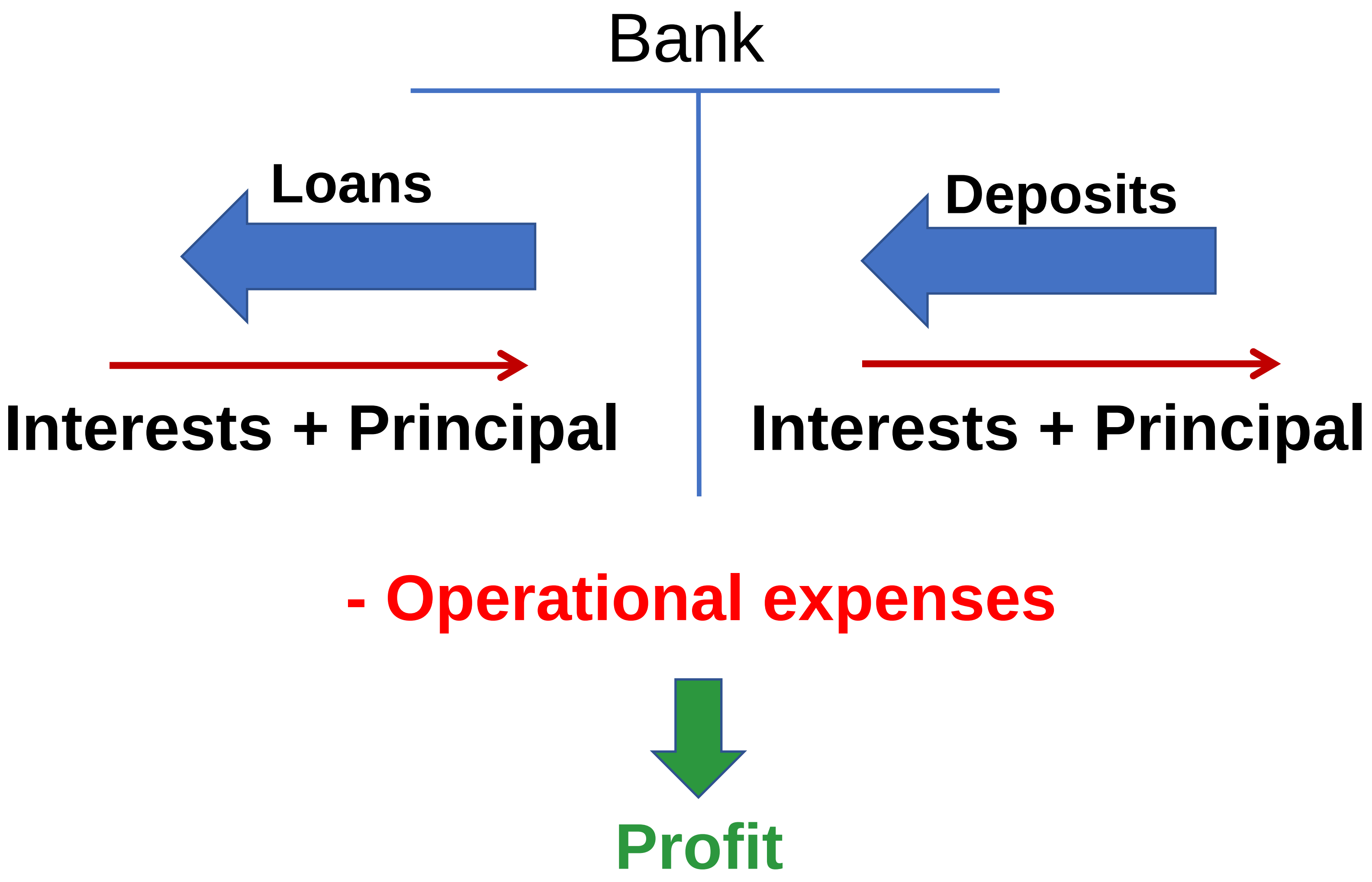
Péter Juhász, PhD, CFA

CORVINUS UNIVERSITY of BUDAPEST

Commercial & Investment banks

CORVINUS UNIVERSITY of BUDAPEST

Value creation at banks



Fund conversion

Quantity

Lifetime

Risk

Basic business logic

Loans	Deposits	Profit
100*8%=8	100*3%=3	8-3=5

Mistakes:

- Cost of operation?
- Liquidity?
- Loan defaults?

CORVINUS UNIVERSITY of BUDAPEST

Improved business logic

Assets (100)

Loans	Liquid assets	Cash
90*8%*95%=6.84	6*5%=0.30	4*0%=0.00

↗
Default loss

Deposits	Cost of operation
100*3%=3.00	4.00

Assume Equity = 10% * Assets

Profit: 6.84+0.30+0.00-3.00-4.00=0.14

Profit/Assets: 0.14 / 100 = 0.14%

Profit/Equity: 0.14 / 10 = 1.40%

Alternative scenario

Assets

Loans	Liquid assets	Cash
91*8.2%*98%=7.31	7*5.5%=0.39	2*0.5%=0.01

Deposits	Cost of operation
100*2.8%=2.80	3.60

Assume Equity = 10% * Assets

Profit:	7.31+0.39+0.01-2.80-3.60=1.31
Profit/Assets:	1.31 / 100 = 1.31%
Profit/Equity:	1.31 / 10 = 13.10%

Types of banks

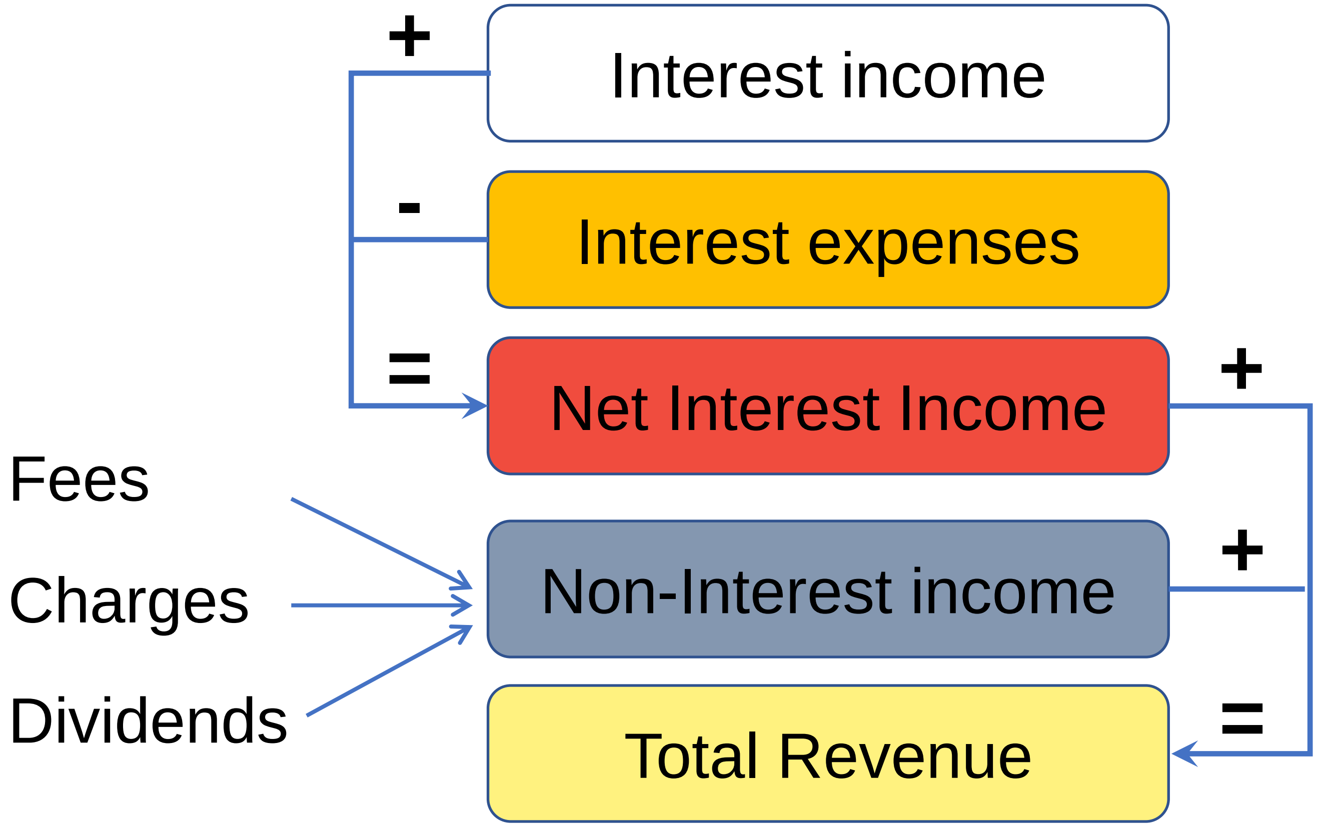
Commercial banks	Investment banks
Deposits Loans Accounts	Funds Trading Security issue

Universal banks

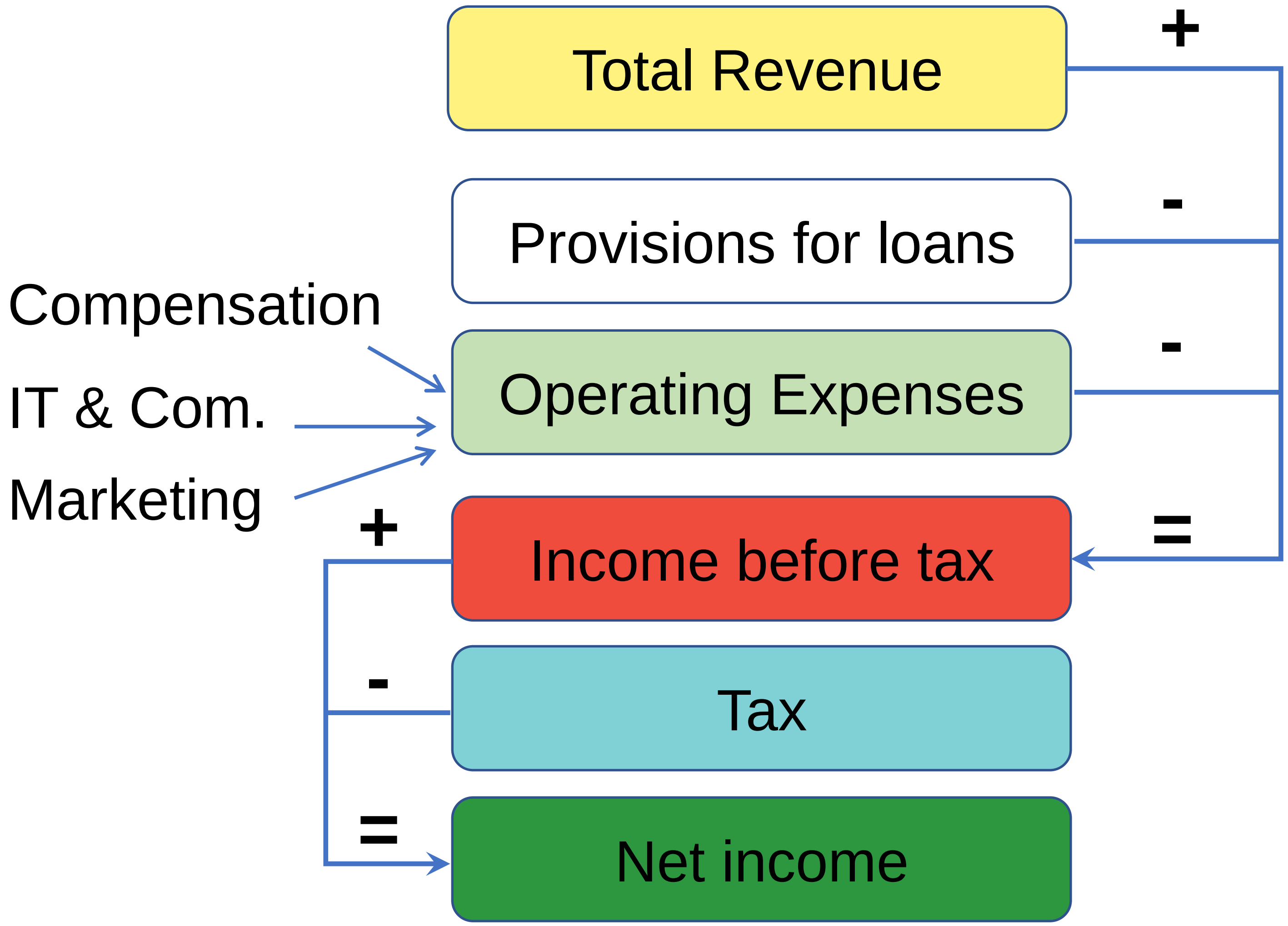
Balance sheet for banks

Cash	Money market deposits
Liquid assets	Time deposits
Loans	CDs, short term loans
Long term investments	Long term debt
Real estate & equipment	Equity

Income statement for banks 1



Income statement for banks 2



Challenges of Analysis

One-time effects

(extraordinary items)

Change in strategy / activity mix

Differences / changes in accounting policy

Off-balance sheet activities

Cosmetics (Window dressing)

CAMELS analysis

Capital Adequacy

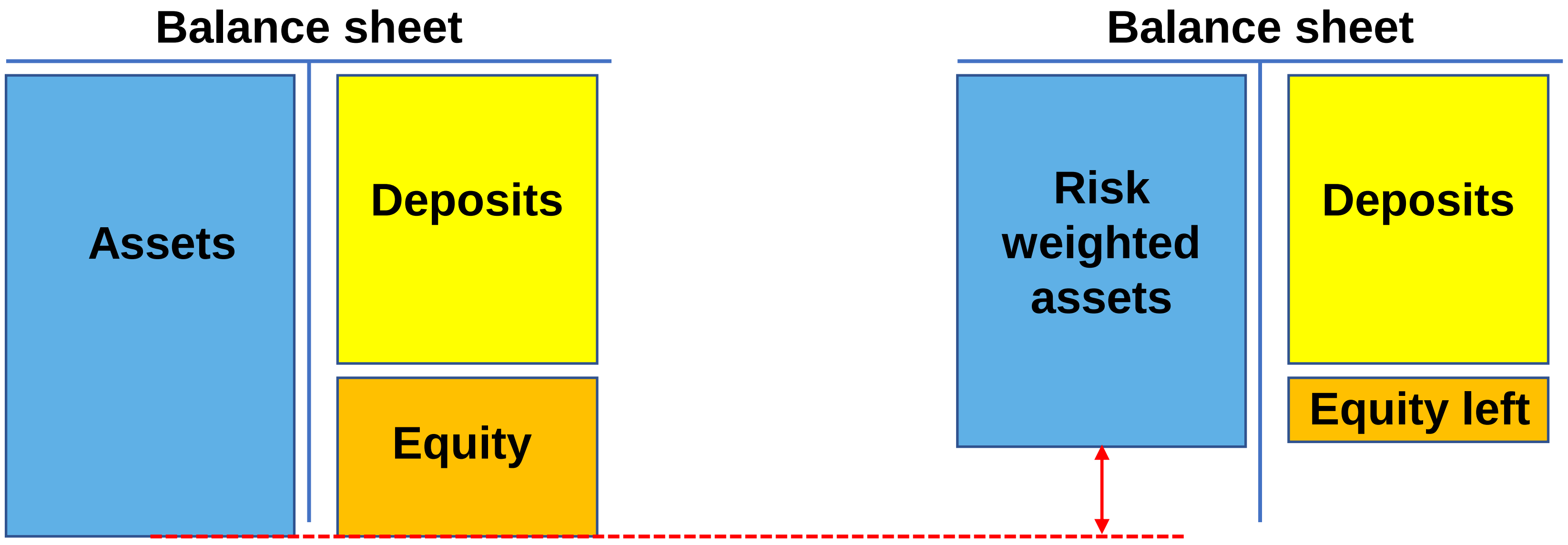
Asset Quality

Management

Earnings (Profitability)

Liquidity & Funding

Sensitivity to Market Risks



BIS Tier 1 Capital: at least 4%

BIS Tier 2 Capital: at least 8%

Capital Adequacy 2

Basel / U.S. baking regulation: „Well capitalised”:

Tier 1 leverage ratio > 5%

$$\frac{Tier\ 1\ capital}{Average\ total\ assets}$$

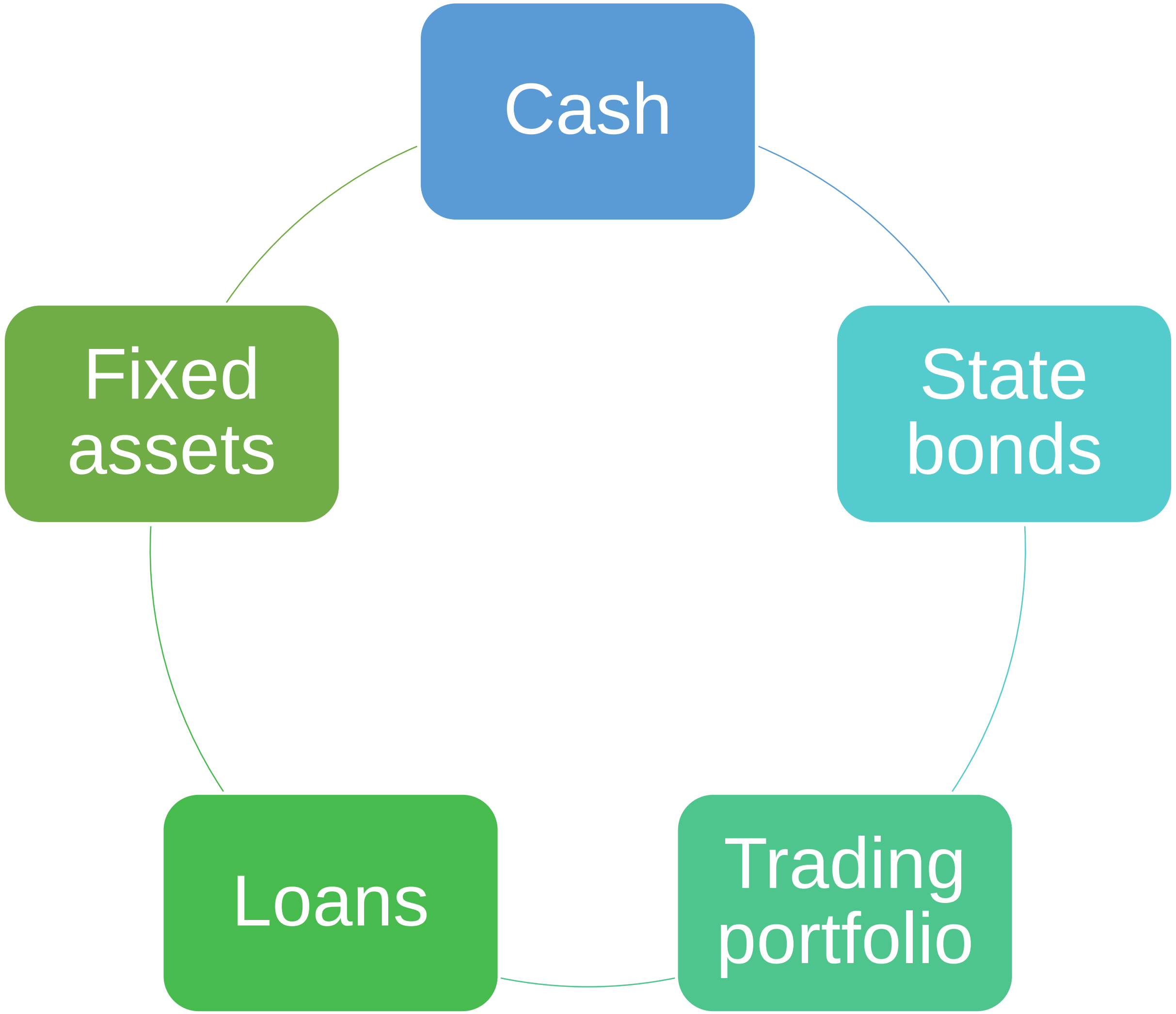
Tier 1 risk-based ratio > 6%

$$\frac{Tier\ 1\ capital}{Total\ risk\ adjusted\ assets}$$

Total risk-based ratio > 10%

$$\frac{Total\ capital}{Total\ risk\ adjusted\ assets}$$

Asset quality 1



Asset quality 2

- Loan structure:
  - Consumer vs Commercial
  - Term Structure
  - Industry Exposure
- Loan loss reserve
- Asset growth rate

How to measure asset quality?

Loan loss reserve / Total loans

Coverage ratio (> 1.5)

$$\frac{\text{Loan Loss Reserves}}{\text{Non}-\text{Performing or Non}-\text{current Loans and leases}}$$

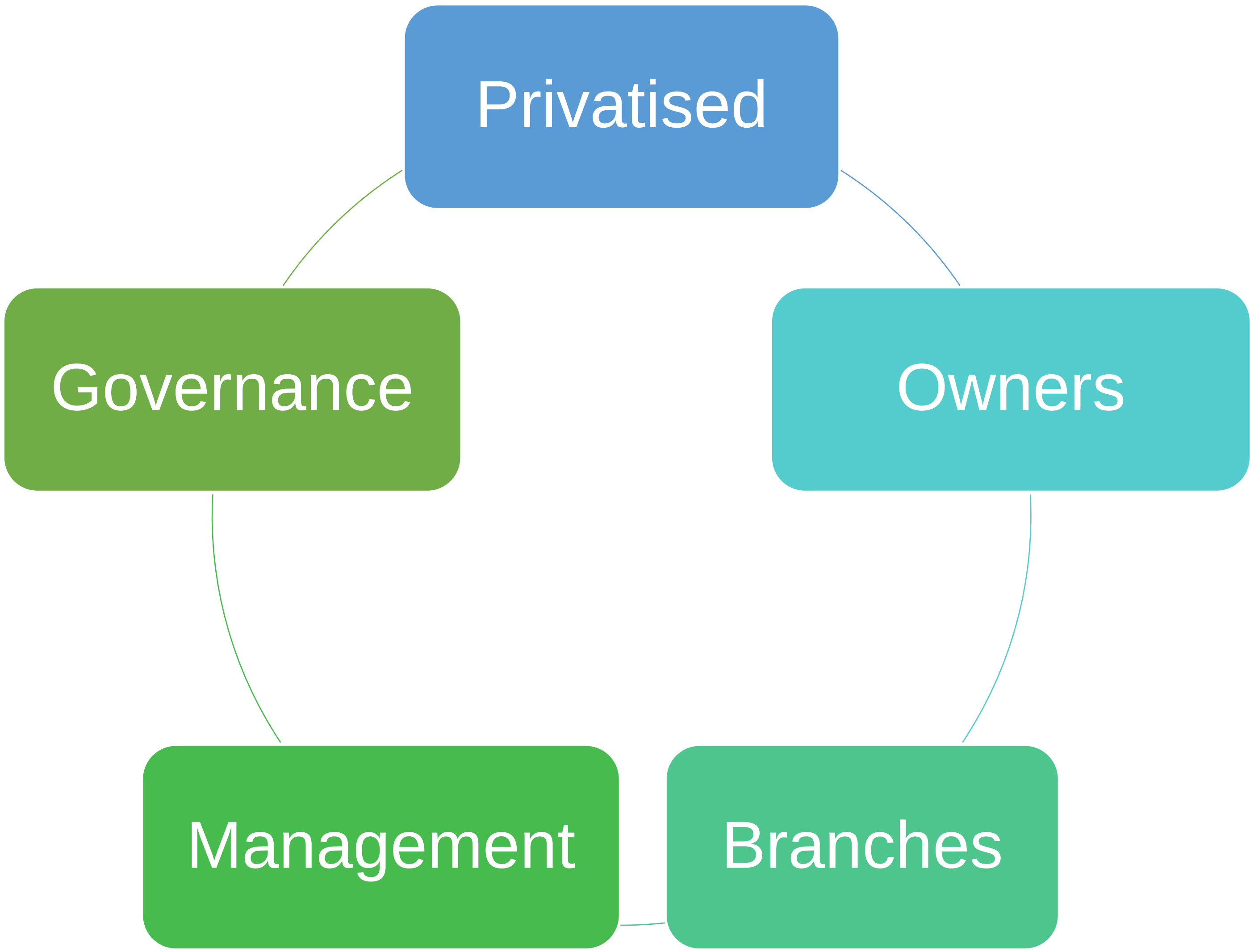
Overdue Loans to Total Loan Ratio (30-89 days)

90-Day Overdue Loans to Total Loans Ratio (90 days)

Exposure Structure

%	Total Exp.	Uncovered (Risk)	Uncov./ Total exp.
Agriculture	0.01	0.03	**52.14**
Oil, Gas, Mining	1.61	1.51	17.39
Manufacturing	3.82	6.99	**33.74**
Real Estate & Construction	**19.50**	**36.29**	**34.30**
Trade	7.47	**16.27**	**40.16**
Transp., Storage & Com.	0.41	0.01	0.50
Financial Institutions	**16.07**	11.91	13.67
Services	11.36	**13.58**	22.05
Government	**16.52**	12.38	13.82
Banking	**21.19**	0.06	0.05
Other	3.81	0.09	0.44
Total	**100.00**	**100.00**	**18.44**

Management



Earnings & Profitability

Strategic focus	Macro situation	Regional structure
Interest vs non-interest income	Growth trends	Extraordinary events

How to increase interest income?

Provide loan to more risky customers (quantity)

Provide more loan to riskier customers (structure)

How to measure profitability? 1

Net interest margin

$$\frac{Net\ Interest\ Income}{Average\ Interest\ Earning\ Assets}$$

Return on (Average) Assets (ROAA)

(US: 0.6-2.0%)

$$\frac{Net\ Operating\ Income}{Total\ (Average)\ Assets}$$

Return on (Average) Equity (ROAE)

(US: 15-17%)

$$\frac{Net\ Income\ After\ Tax}{Total\ (Average)\ Equity}$$

How to measure profitability? 2

Return on Earning Assets (ROEA)

$$\frac{Revenues\ Before\ Interest\ Expense}{Earning\ Assets}$$

Operating Profit Margin

$$\frac{Operating\ Profits}{Net\ operating\ revenues}$$

Overhead ratio

$$\frac{Non-Interest\ Expenses}{Total\ (Average)\ Assets}$$

How to measure profitability? 3

Average Collection Days of Interests

$$\frac{Accured\ Interest\ Receivables}{Interest\ income} * 365$$

Efficiency Ratio

(below 40% very efficient, above 75% very expensive)

$$\frac{Non-interest\ Expenses}{Total\ Income\ (Net\ int.+Non-int.)}$$

Du-Pont analysis for banks

$$ROE = \frac{Net\ Income}{Equity} = \underbrace{\frac{\text{Total Assets}}{\text{Equity}} * \frac{\text{Total Revenue}}{\text{Total Assets}}}_{ROA} \overbrace{\frac{\text{Income Before Tax}}{\text{Total Revenue}} \frac{\text{Net Income}}{\text{Income Before Tax}}}^{\text{Net profit margin}}$$

Net profit margin

Equity multiplier | Asset Utilisation | Gross Profit Margin | Tax burden

ROA

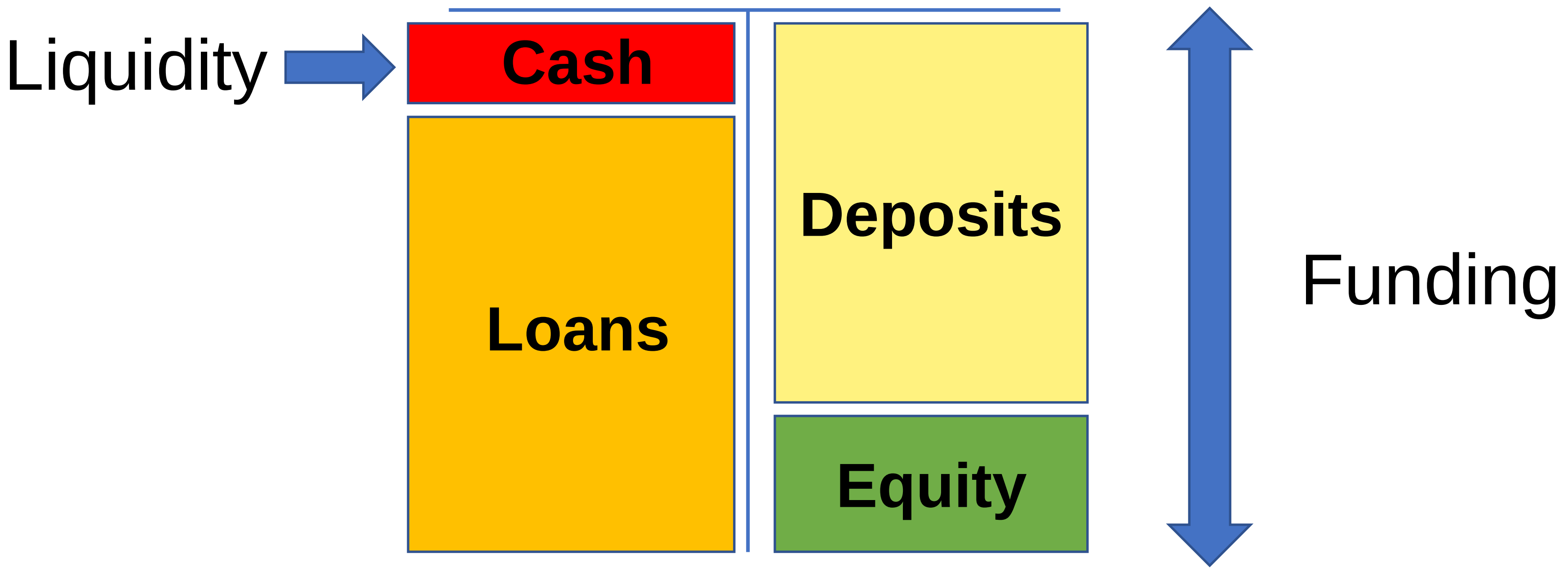
$$\frac{Interest\ Inc.}{Assets} + \frac{Non - Int.\,Inc.}{Assets} + \frac{Gain\ on\ Sec.}{Assets}$$

Which bank is better?

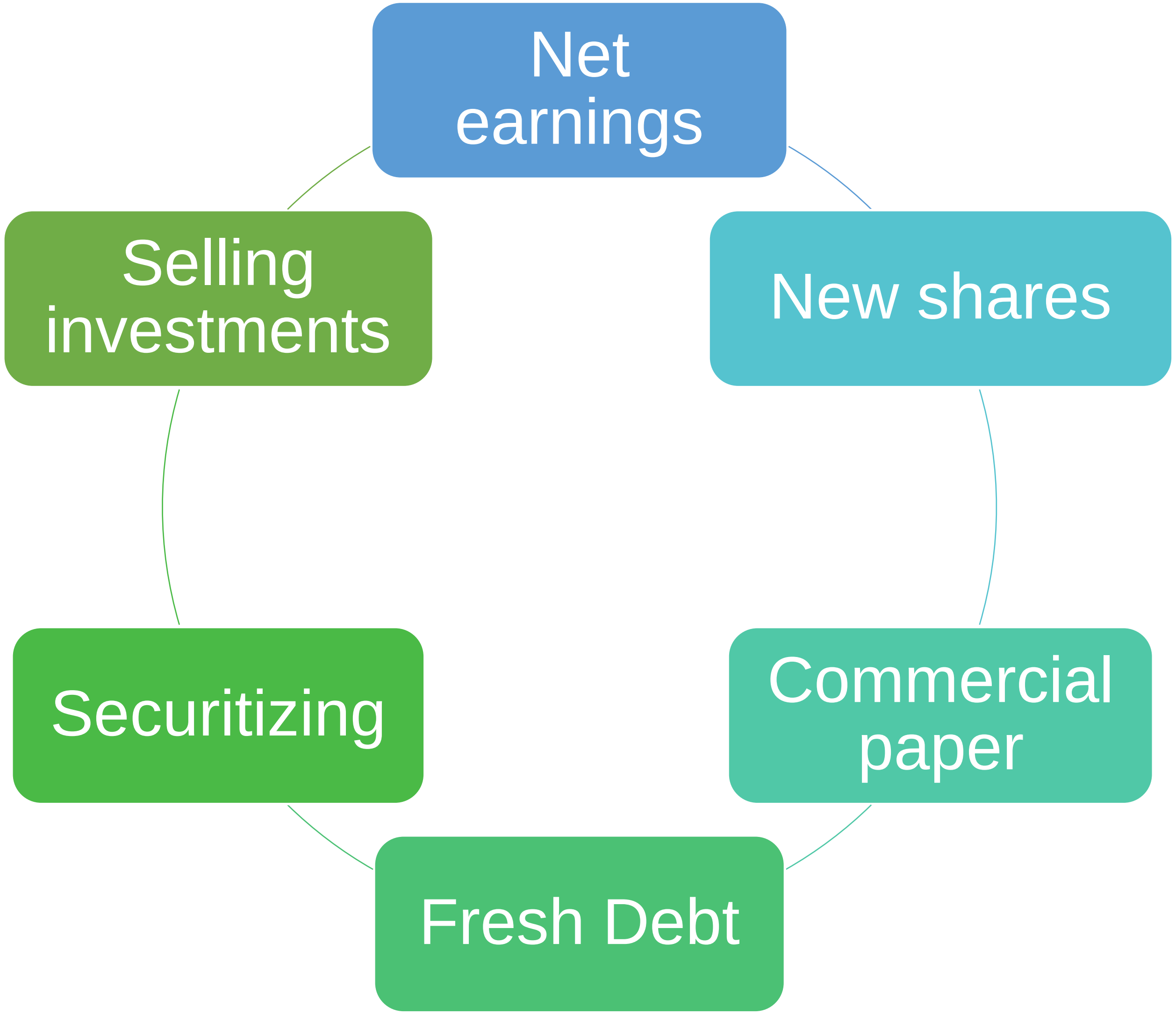
	A	B	C
ROAE	13.75%	12.38%	14.40%
Equity multiplier	5.00	5.50	6.00
Asset utilisation	5.00%	4.50%	4.00%
Net profit margin	55.00%	50.00%	60.00%

More profitable ≠ Better

Liquidity & Funding



Sources of funding

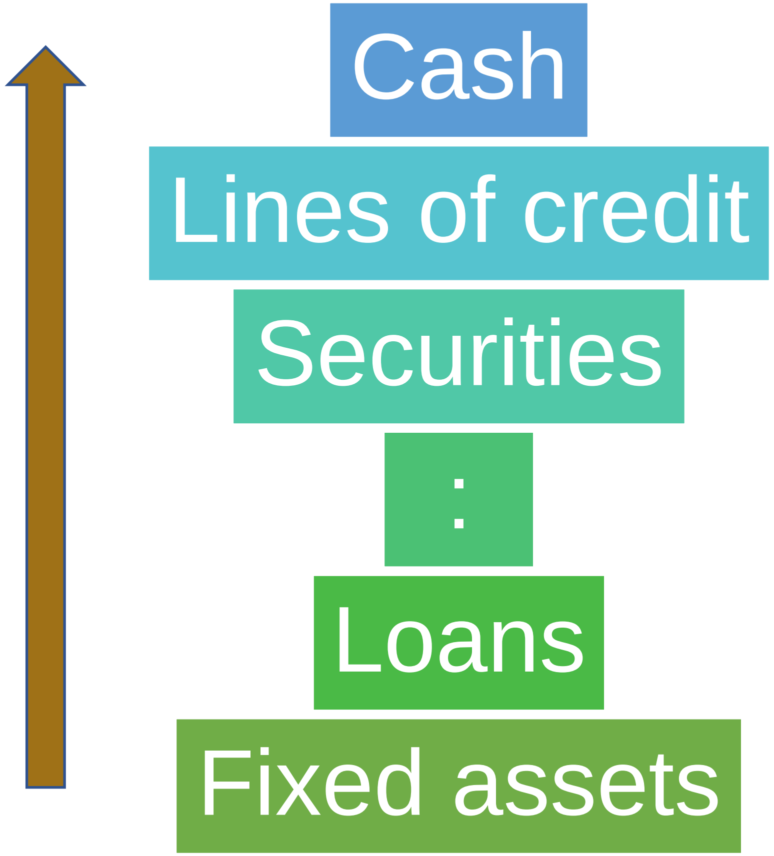


To check for funding…

- External / Internal (maturing loans) sources
- Dependency on large deposits (customers)
- No source should exceed 10%.

Liquidity…

…is the bank’s ability
to convert an asset
into cash.
… is a question of time.



Liquidity gap analysis

Predicting direct cash flow:

+ Assets maturing (principal payments)
- Assets increasing (loan demand, investment)
- Liabilities maturing (deposit withdrawals)
+ Liabilities increasing (new deposits, fresh capital)

Sensitivity to Market (Risk)

Interest rate	Operation	Credit	Funding
Liquidity	Market	Country	Strategic
	Compliance	Reputation	

BUDAPESTI CORVINUS EGYETEM

Property and Casualty Insurance

How do P&C Insurance firm function? 1.

Premiums

1% * 1000 = 10 + 4 =14

Proceeds on claims

Probability * Amount

1% * 1000 = 10

Profit

4

How do P&C Insurance firm function? 2.

Premiums

1% * 1000 = 10 + 4 =14

Proceeds on claims

Probability * Amount

1% * 1000 = 10

- Operational expenses

3

Profit

1

How do P&C Insurance firm function? 3.

Premiums

1% * 1000 = 10 + 3 = 13

Return on reserves ← Interests + Dividend + Price gain = 2

Proceeds on claims

1% * 1000 = 10

- Operational expenses

3

Profit

2

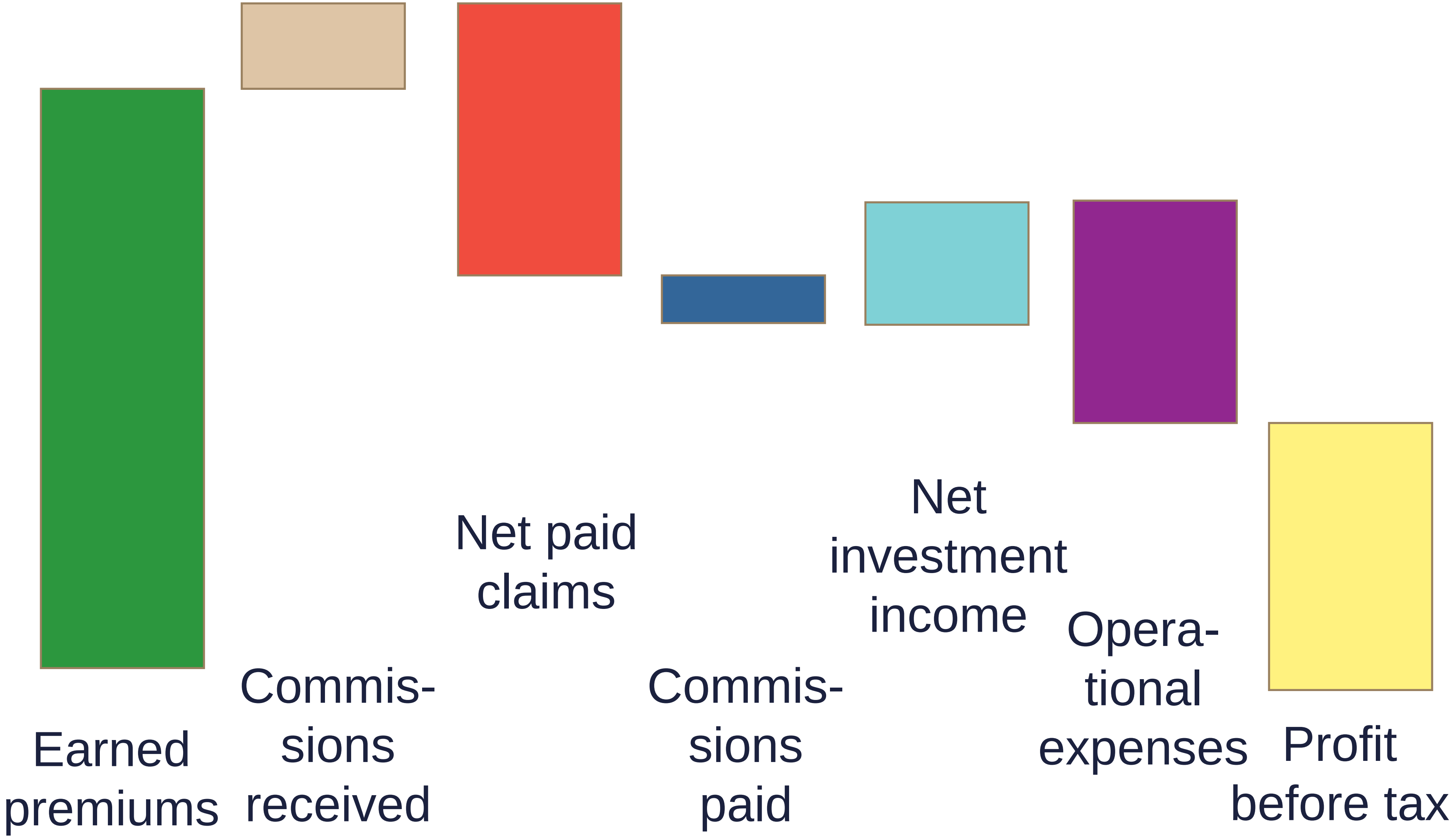
BUDAPESTI CORVINUS EGYETEM

Balance sheet of insurance companies

- Cash and cash like items
- Receivables
- Reinsurance assets
- Investments
- Plant, Property & Equipment

- Equity
- Insurance liabilities
- Provisions

Base case



Earned premiums
Commis-sions received
Net paid claims
Commis-sions paid
Net investment income
Opera-tional expenses
Profit before tax

Caramels – Insurance companies

Capital Adequacy
Asset Quality
Reinsurance
Actuarial
Management
Earnings (Profitability)
Liquidity & Funding
Sensitivity to Market Risks

Capital Adequacy

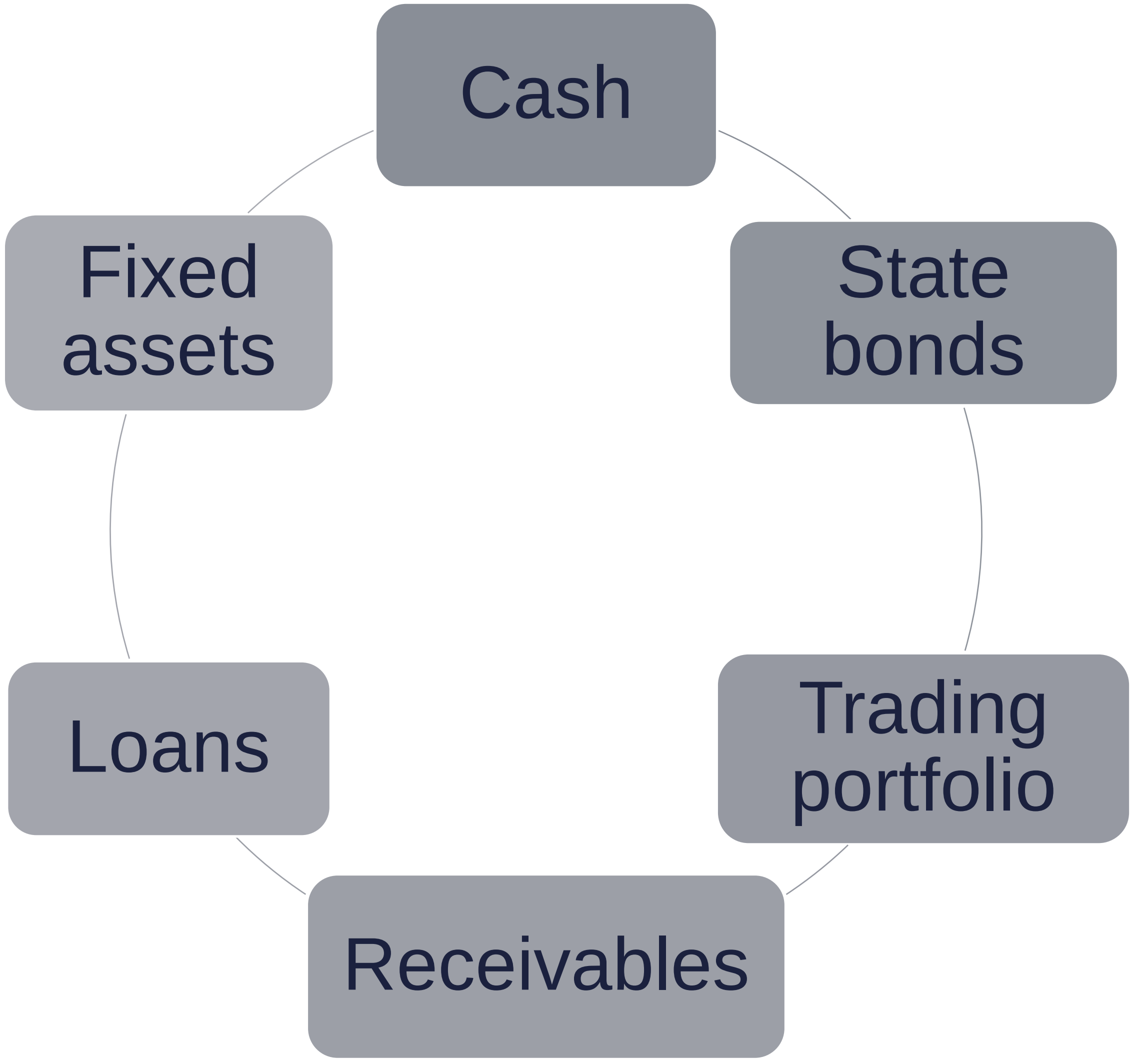
Minimum is subject to local regulation.

Extra reserve reflects more conservative strategy.

$$\frac{Capital}{Total\ Assets}$$

$$\frac{Capital}{Total\ (or\ Net)\ claims}$$

Asset quality



Reinsurance

Significant for non-life insurance business.

Reinsurance’s share of premiums and claims may be very different in each period.

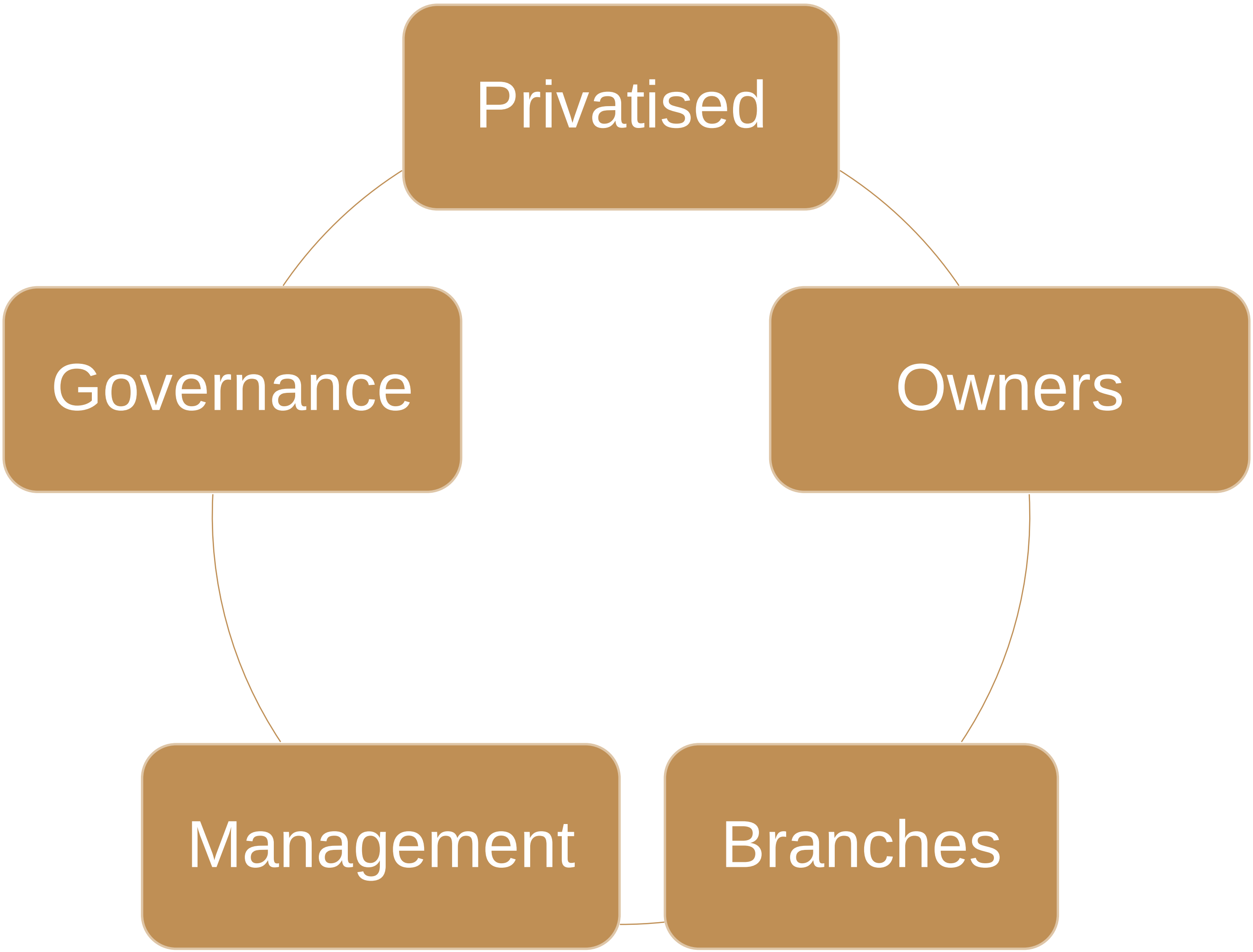
$$retention\ ratio = \frac{Net\ Premiums}{Gross\ premiums}$$

What insurance risk remains with the insurance firm?

$$\frac{Net\ technical\ reserves}{Average\ Net\ Claims\ over\ the\ last\ 3\ years}$$

$$\frac{Net\ technical\ reserves}{Average\ Net\ Premiums\ over\ the\ last\ 3\ years}$$

Management



Earnings & Profitability

Strategic focus	Macro situation	Regional structure
Interest vs non-interest income	Growth trends	Extraordinary events

How to measure profitability?

$$Combined\,ratio = \frac{Gross\ Claims}{Gross\ premiums} + \frac{Operating\ expenses}{Gross\ premiums}$$

$$Technical\,ratio = \frac{Expected\ loss}{Gross\ premiums} + \frac{Commissions}{Gross\ premiums}$$

Valuation issues

The strategic view of ROE

$$ROE = \boxed{(1 - T)} * \boxed{\frac{PAT}{Sales}} * \boxed{\frac{Sales}{Equity}}$$

Minimising tax

Maximising profit

Optimising capital need

DCF valuation methods

Method	To discount	Discount factor	Assessment
DDM	Dividend	Required equity return (r_E)	For easy to predict dividends (announced policy) on a mature market.
Direct equity DCF valuation	FCFE	r_E	Detailed hectic financing plan, bottom-up CF.
Economic profit	RI	r_E	Shows when and how much value is created.

Leading back multiples to the fundamentals

$$P_0 = \frac{DIV_1}{r_e - g}$$

$$P_0 / E_1 = \frac{(1-b)}{r_e - g}$$

$$P_0 / BV_0 = \frac{ROE_1 * (1-b)}{r_e - g}$$

$$P_0 / S_1 = \frac{\text{Net margin} * (1-b)}{r_e - g}$$

where

$$\text{Net margin} = \frac{\text{Profit after tax}}{\text{Sales}}$$

Sector-specific multiples

P/DIV

P/tBV

P/Deposits

Multiples' Summary Ranking (1990-2018)

		EU Investment Banks	EU Large Commercial Banks	EU Small Commercial Banks	US Investment Banks	US Large Commercial Banks	US Small Commercial Banks
Best choice	1	P/E (FY2)	P/E (FY2)	P/E (FY1)	P/E (FY2)	P/E (FY2)	P/E (FY2)
	2	P/E (FY1)	P/E (FY1)	P/E (FY2)	P/E (FY1)	P/E (FY1)	P/E (FY1)
	3	P/E (LTM Basic no Extra)	P/E (LTM Basic no Extra)	P/E (LTM Basic no Extra)	P/Common Dividends	P/E (LTM Diluted no Extra)	P/BV
	4	P/BV	P/Common Dividends	P/Common Dividends	P/E (LTM Diluted no Extra)	P/BV	P/TBV
	5	P/TBV	P/Deposits	P/Revenues	P/Deposits	P/TBV	P/E (LTM Diluted no Extra)
	6	P/Common Dividends	P/Revenues	P/Deposits	P/Revenues	P/Deposits	P/Revenues
	7	P/Deposits	P/BV	P/BV	P/BV	P/Revenues	P/Deposits
Worst choice	8	P/Revenues	P/TBV	P/TBV	P/TBV	P/Common Dividends	P/Common Dividends

https://www.fondazioneoiv.it/wp-content/uploads/2018/06/bank-valuation-using-multiples-in-us-and-europe.pdf

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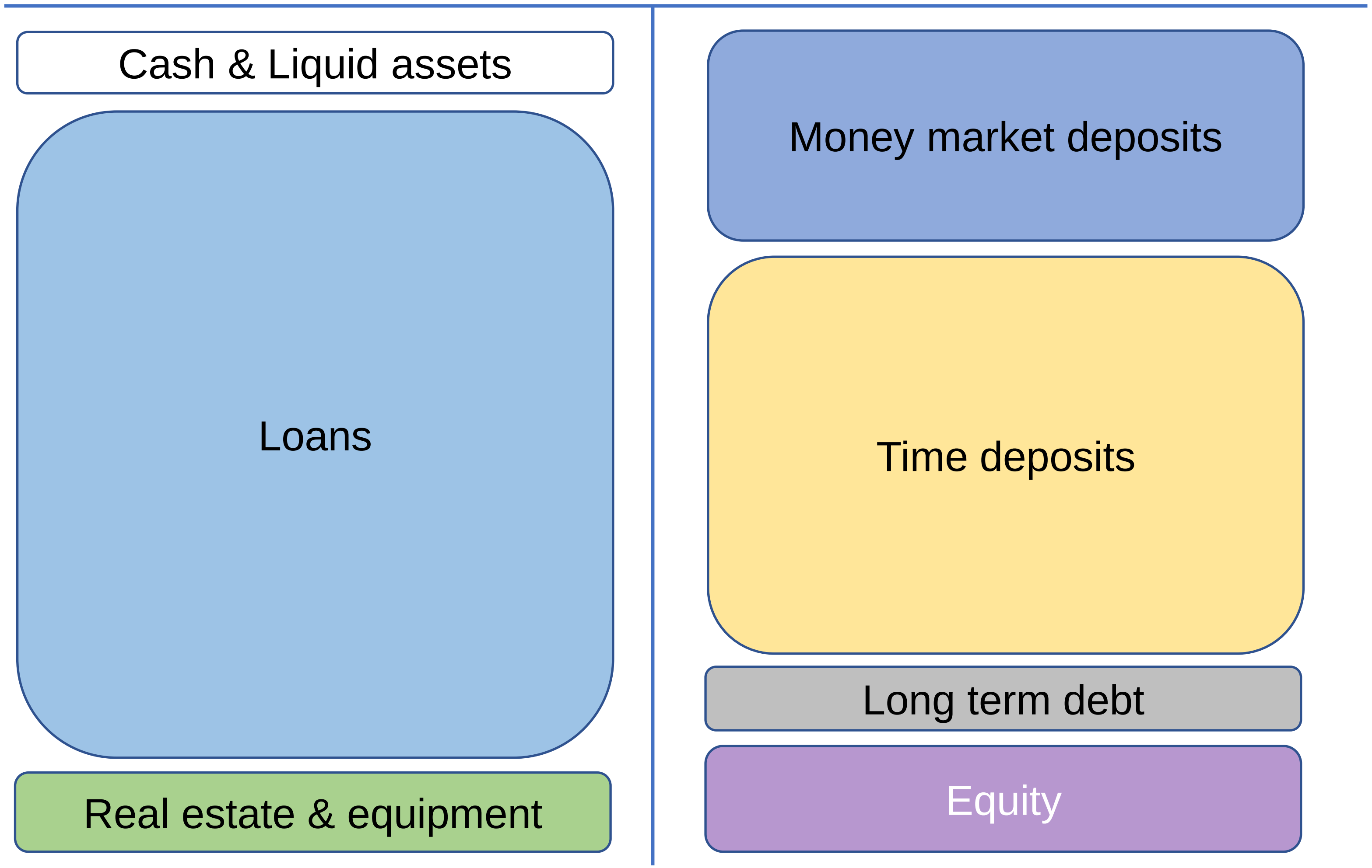
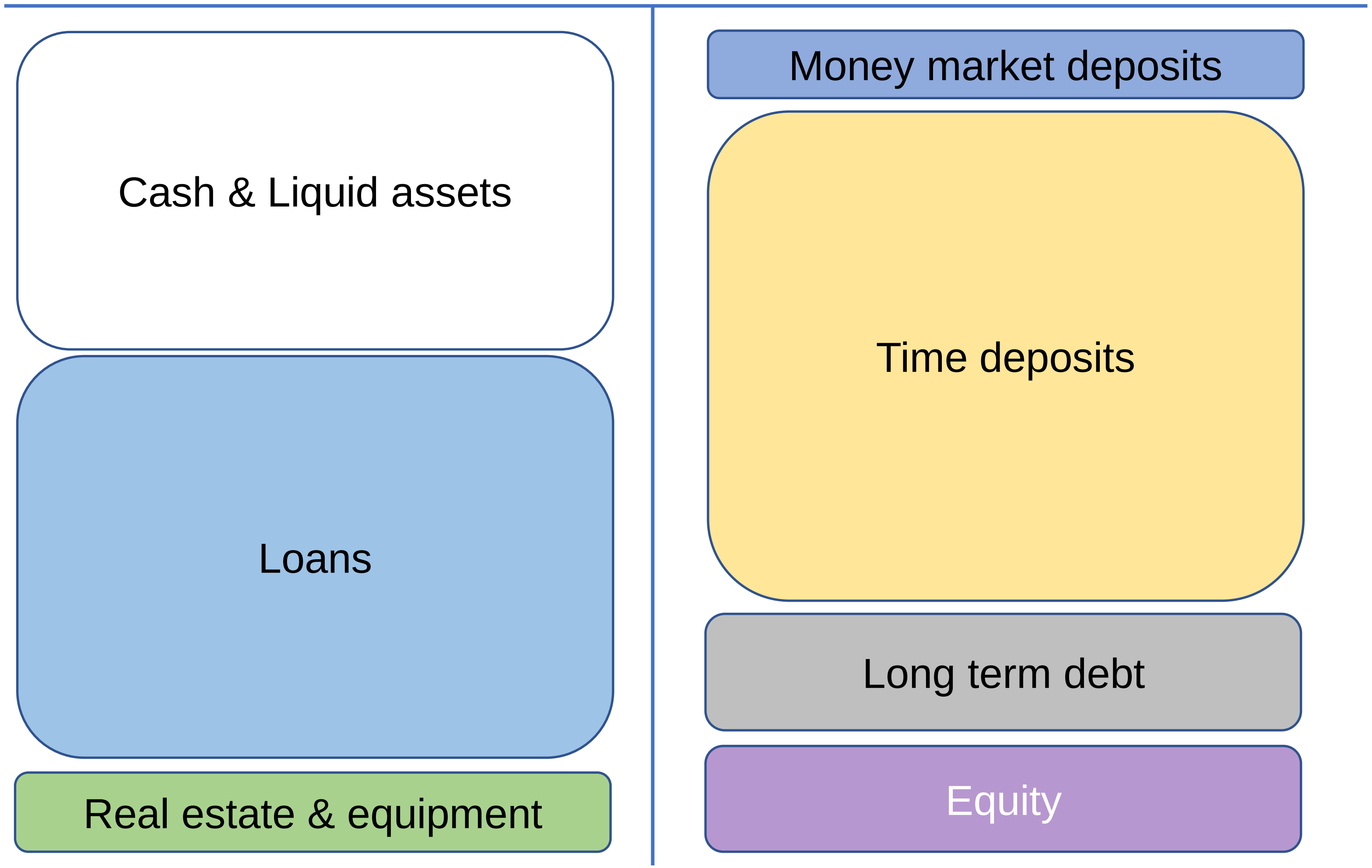
Thank you for your attention. Any questions?

Reserve slides for Banks

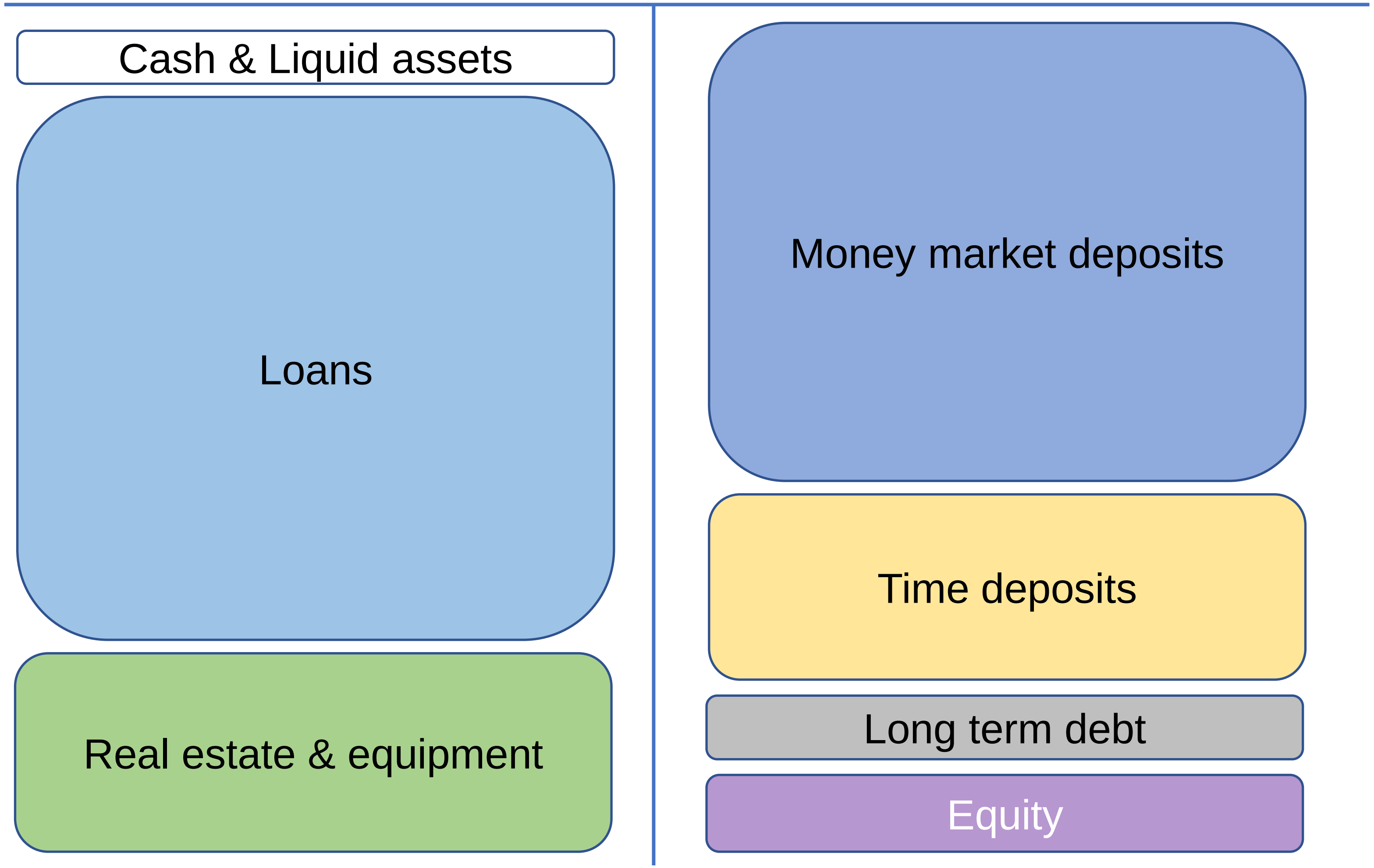
Base case


Cash & Liquid assets
Loans
Real estate & equipment
Money market deposits
Time deposits
Long term debt
Equity

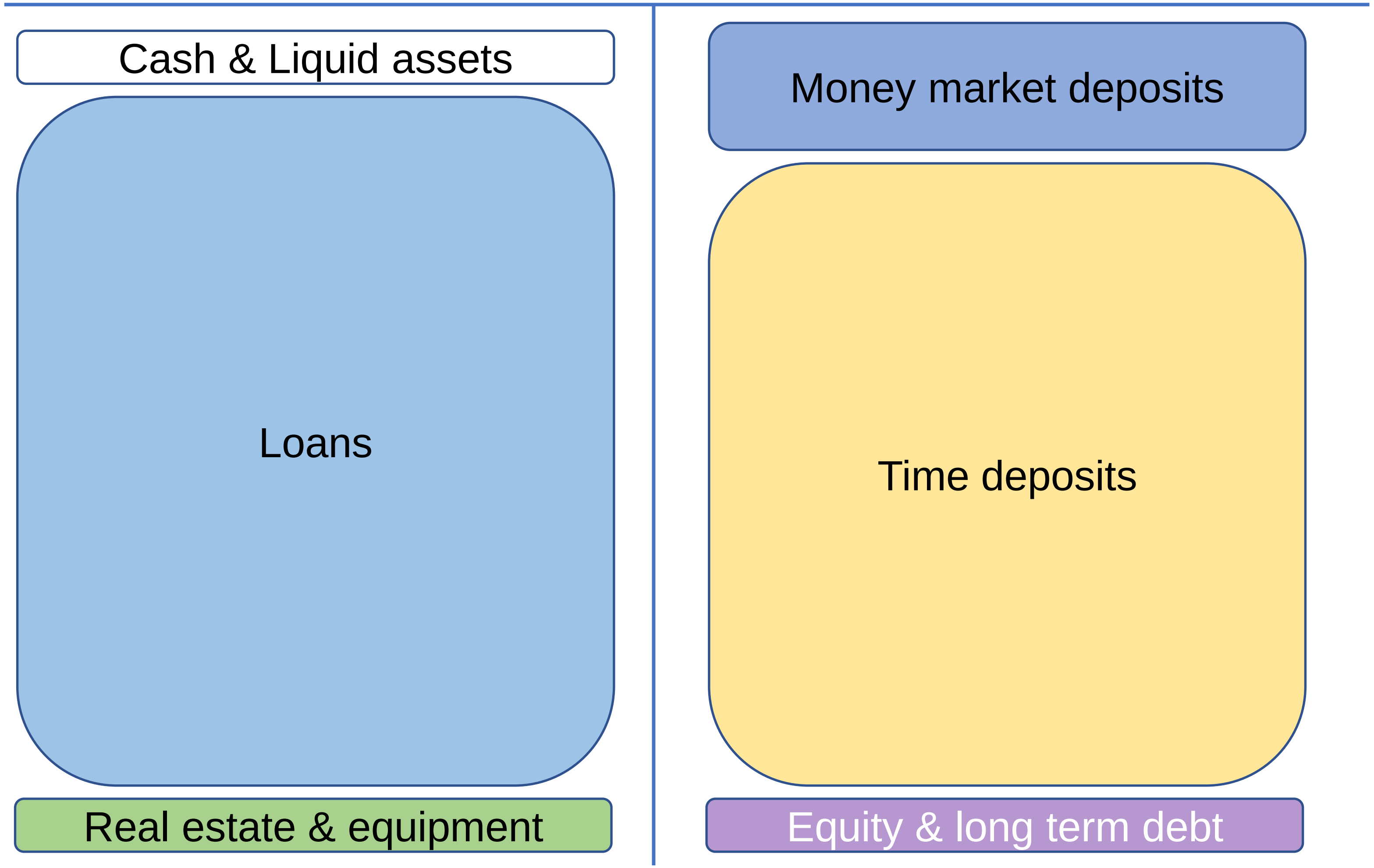
Low profitability



Low liquidity

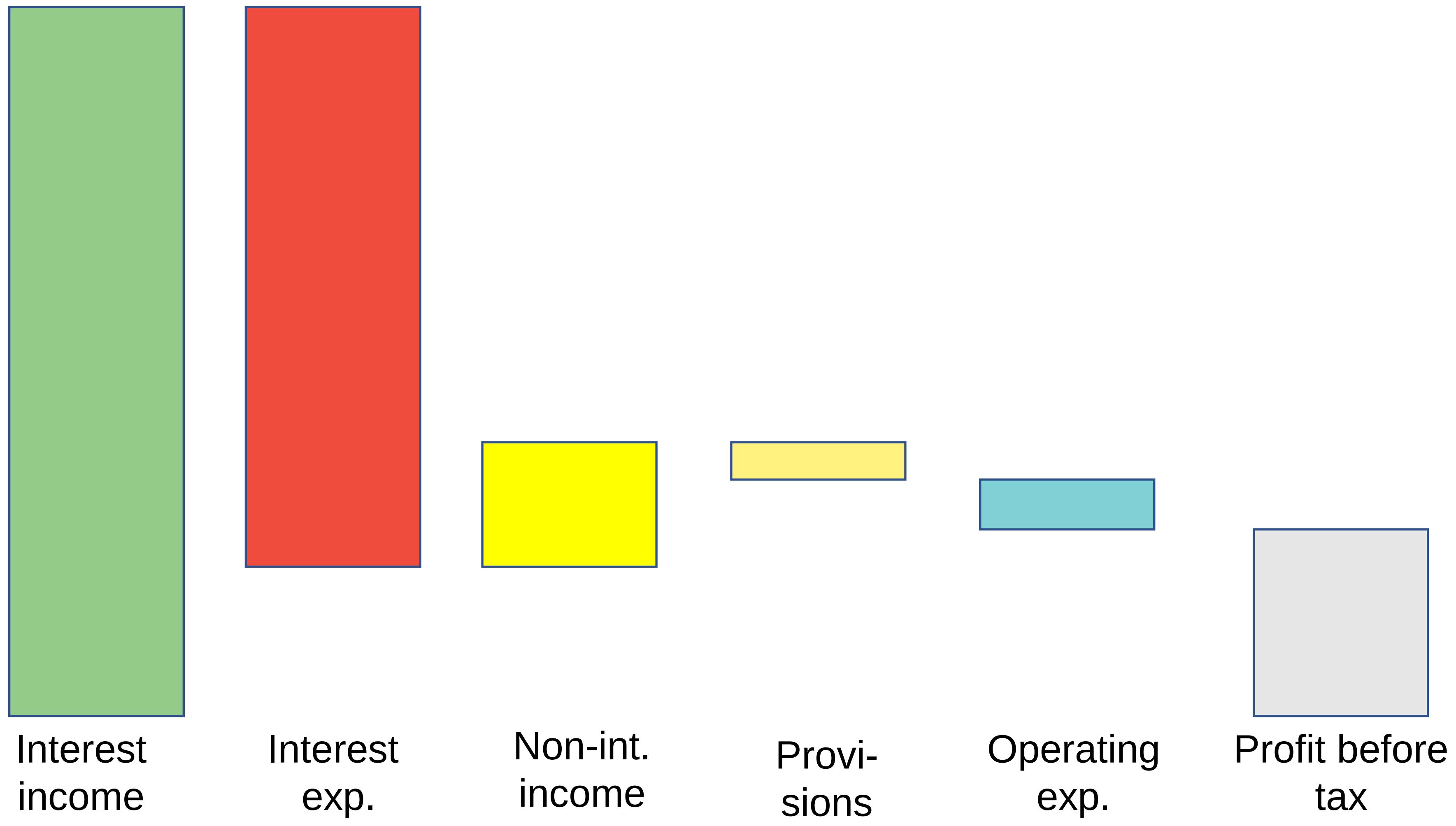


Low capital

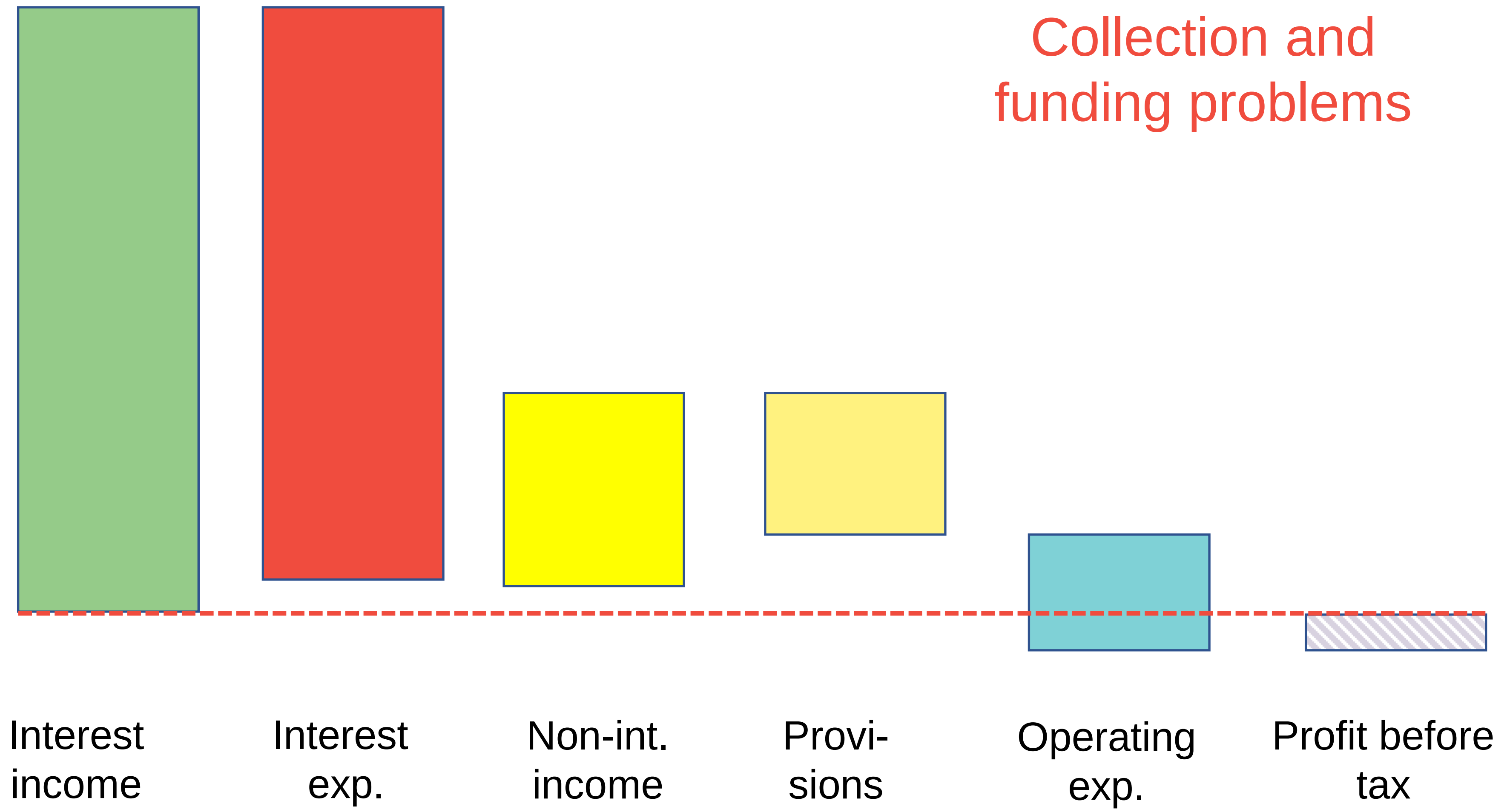


Cash & Liquid assets
Loans
Real estate & equipment
Money market deposits
Time deposits
Equity & long term debt

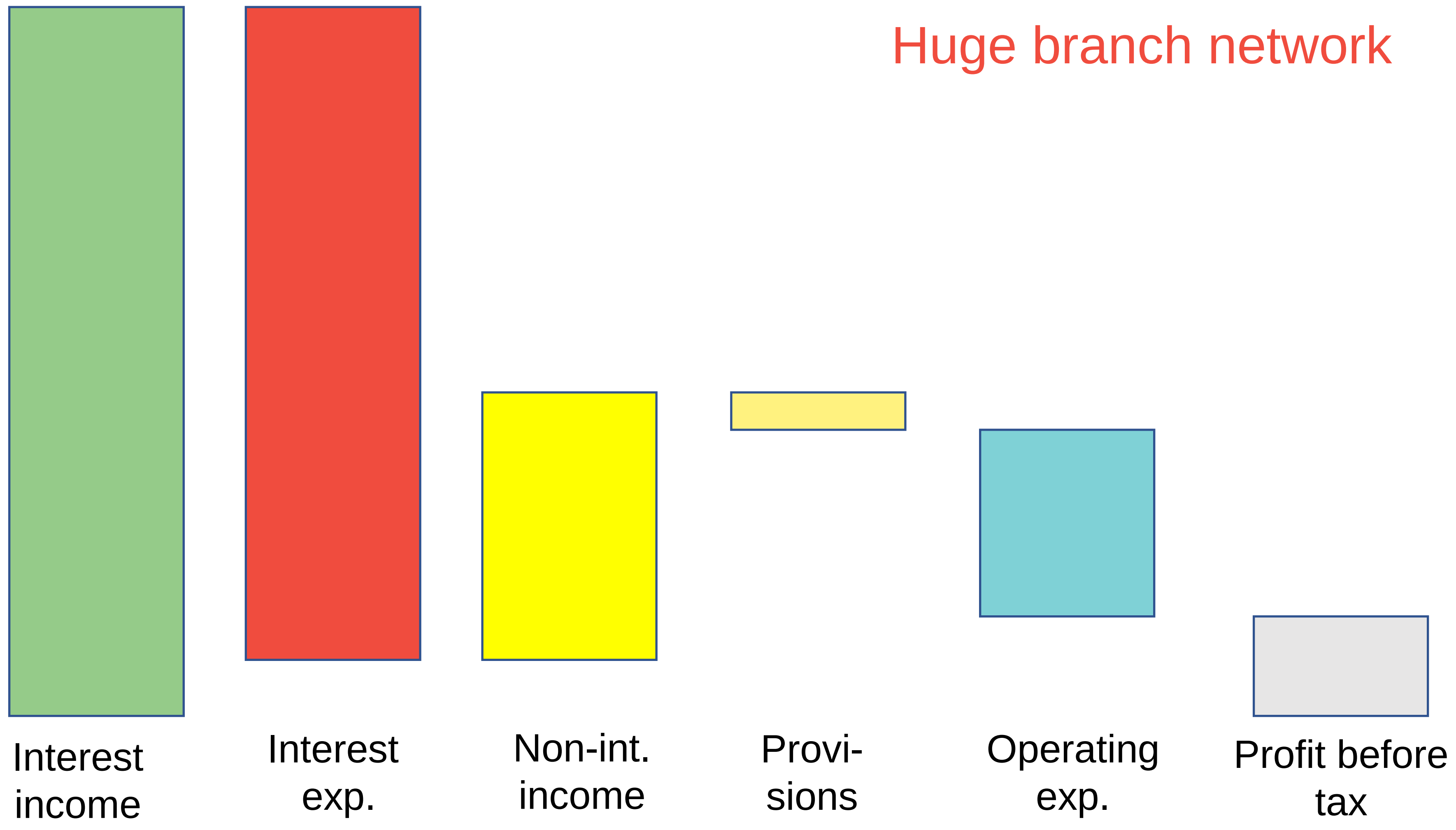
Base case


Interest income
Interest exp.
Non-int. income
Provi-sions
Operating exp.
Profit before tax

What is the issue? 1

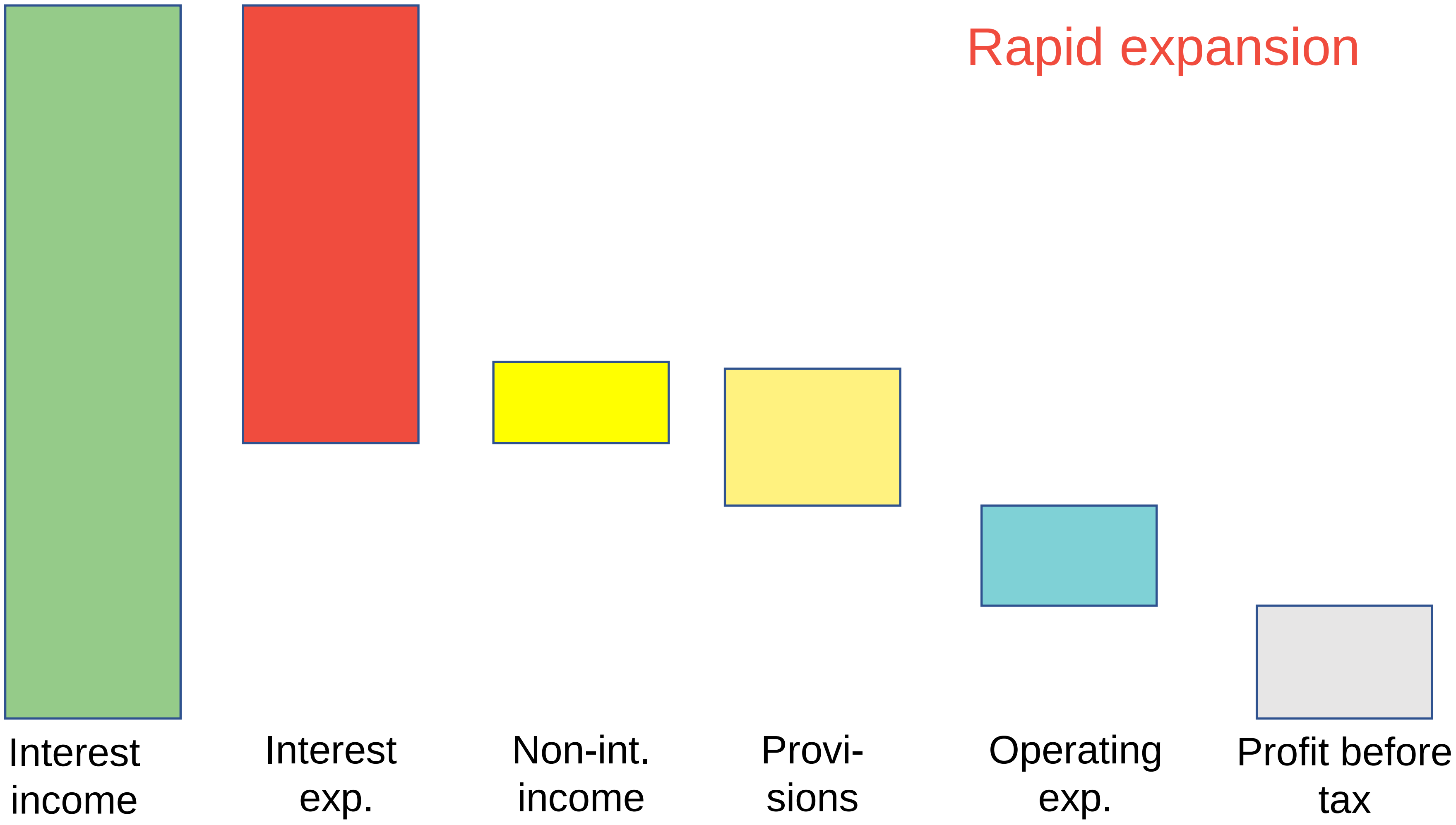


What is the issue? 2



What is the issue? 3

Rapid expansion



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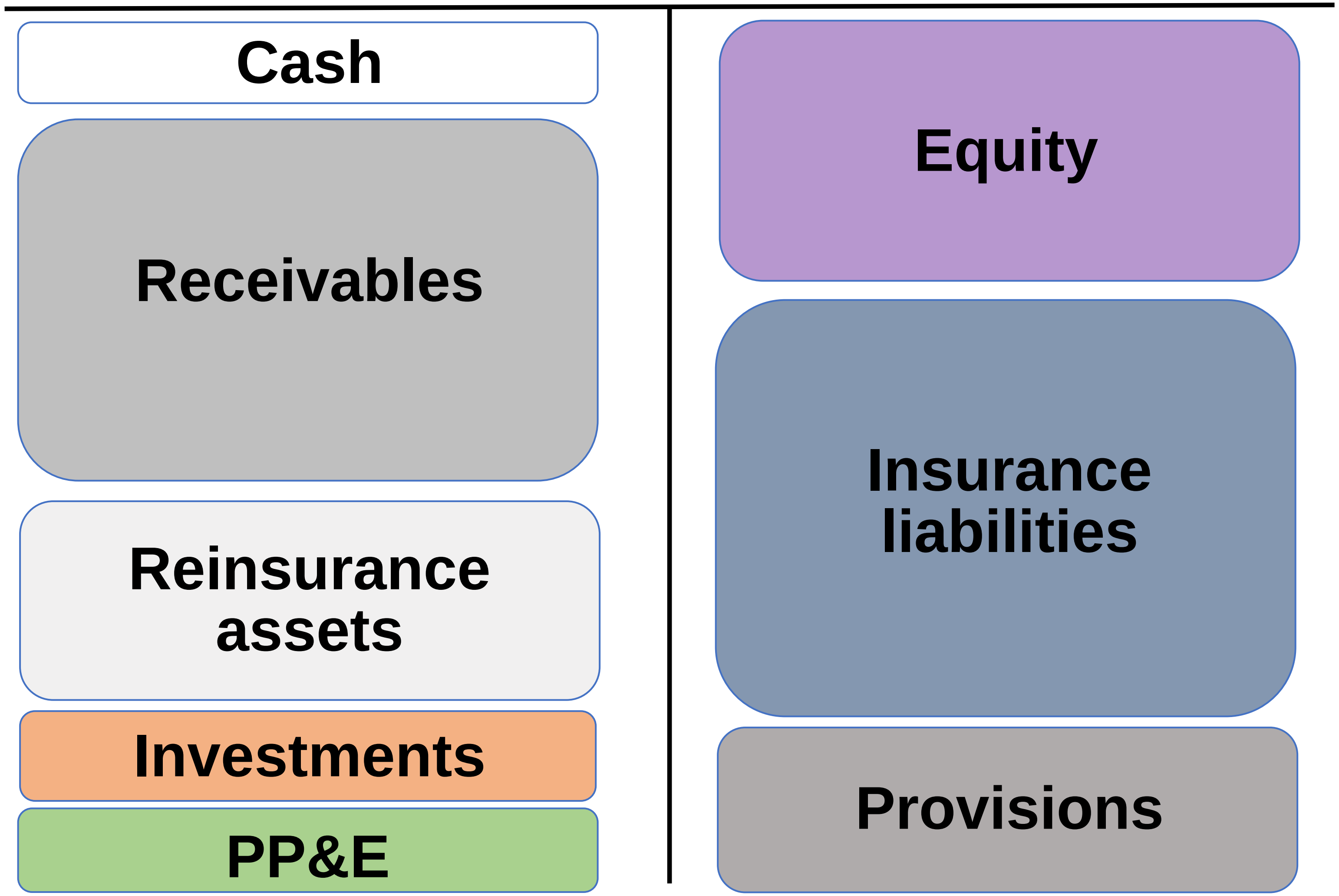
Reserve slides for Insurance Companies

CORVINUS UNIVERSITY of BUDAPEST

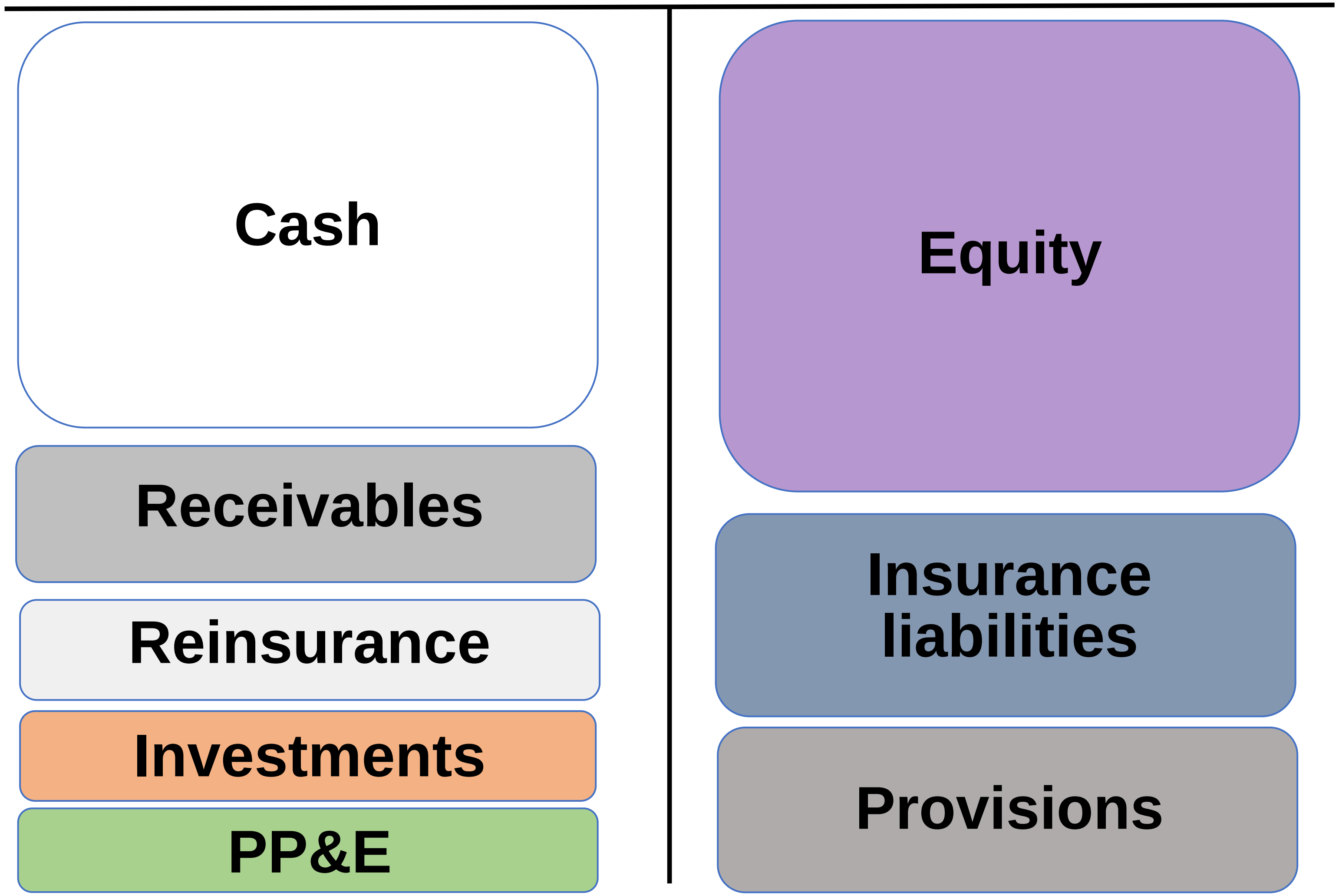
Base case

Cash	Equity
Receivables	Insurance liabilities
Reinsurance assets	Provisions
Investments	
PP&E	

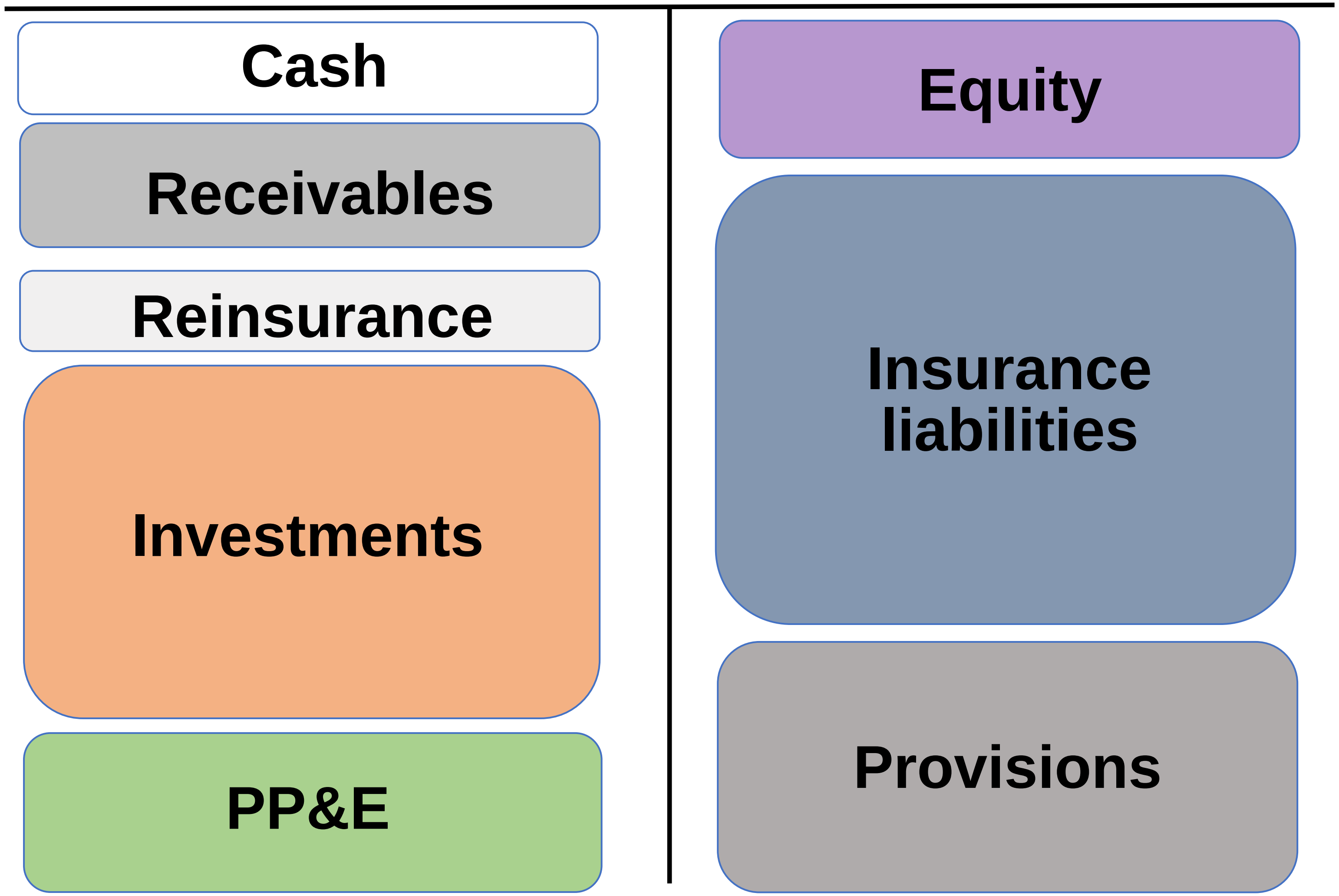
Liquidity problems



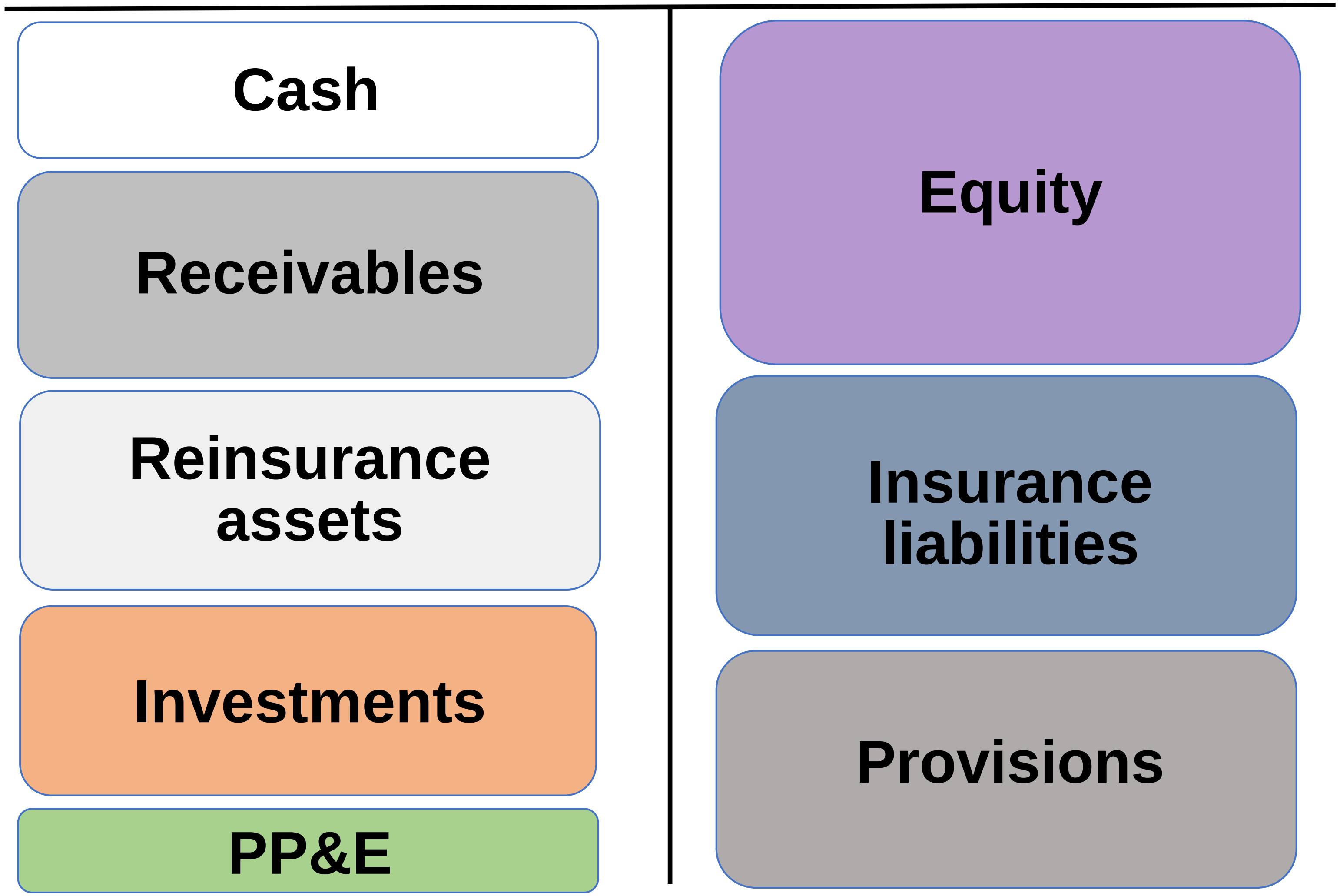
No operation?



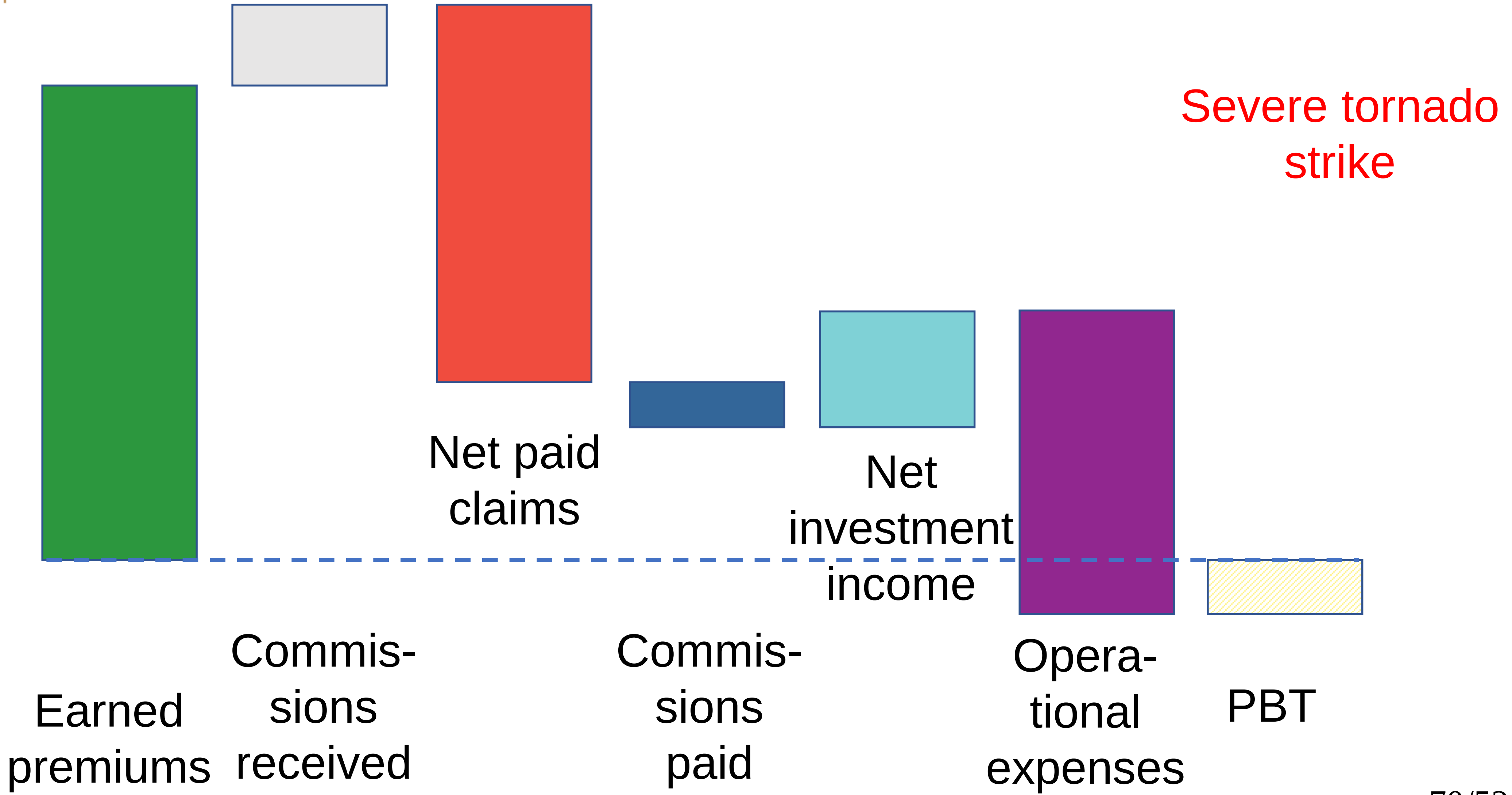
Risky operation



After good years

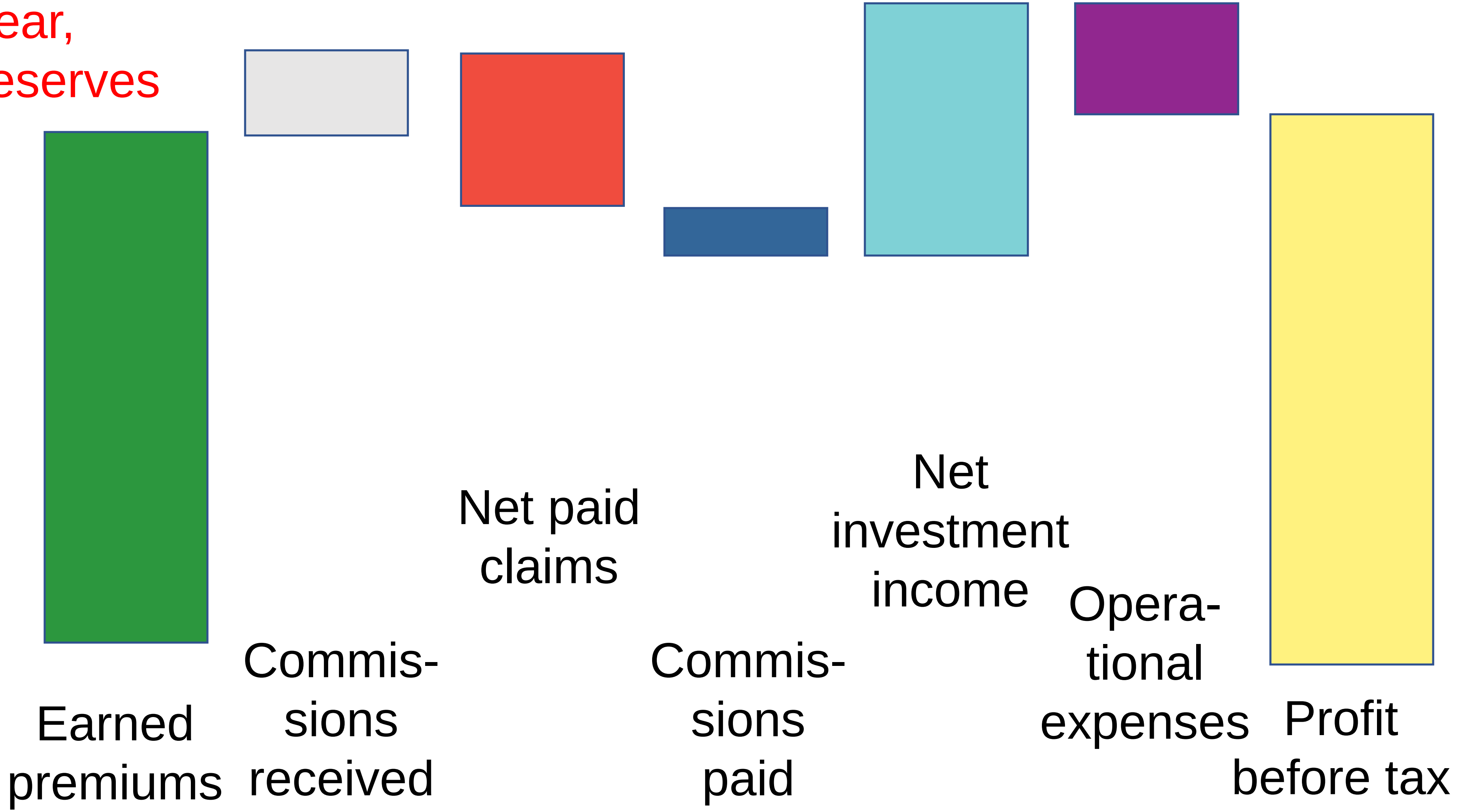


What happened? 2



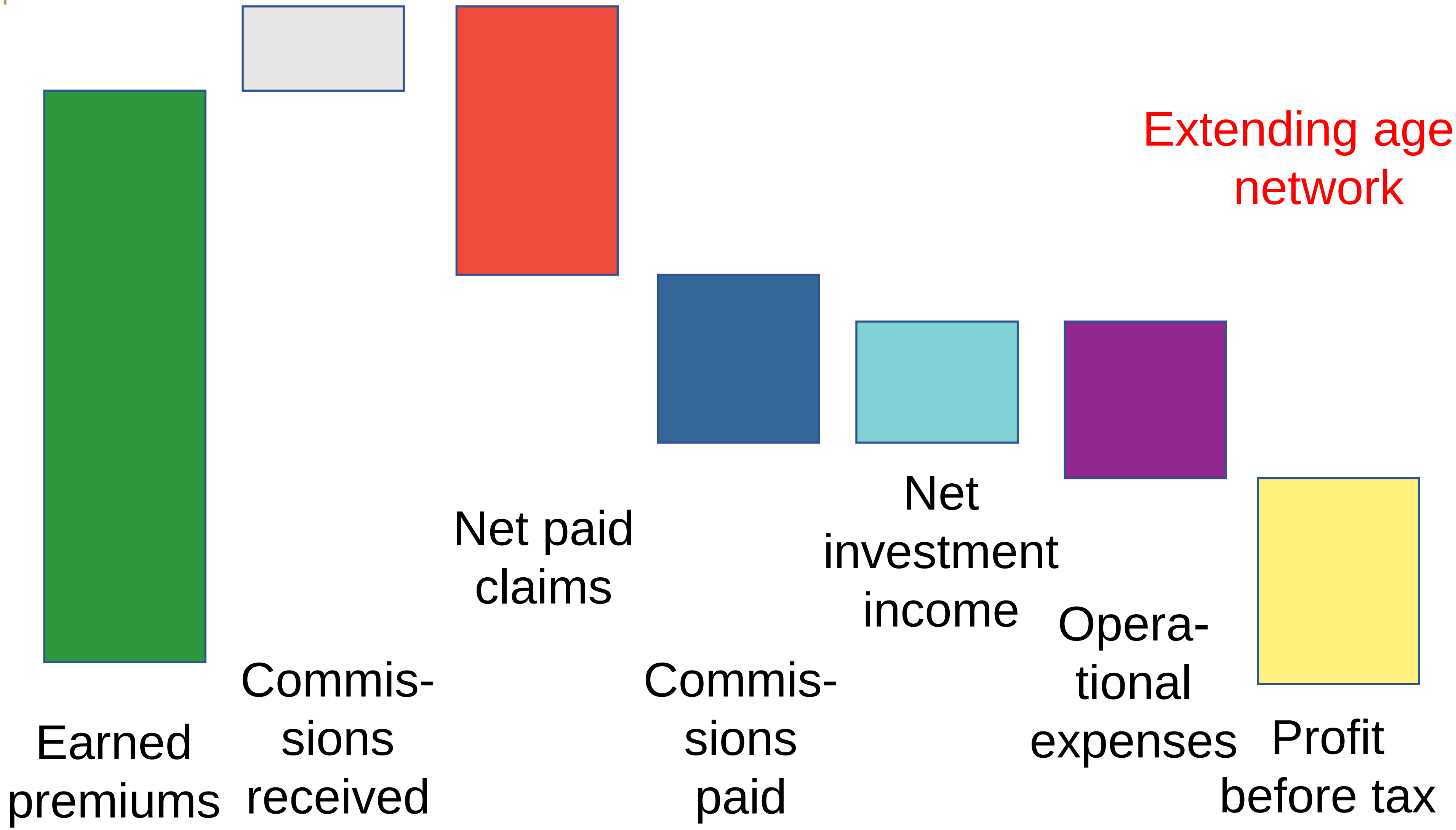
What happened? 3

Quiet year,
collecting reserves



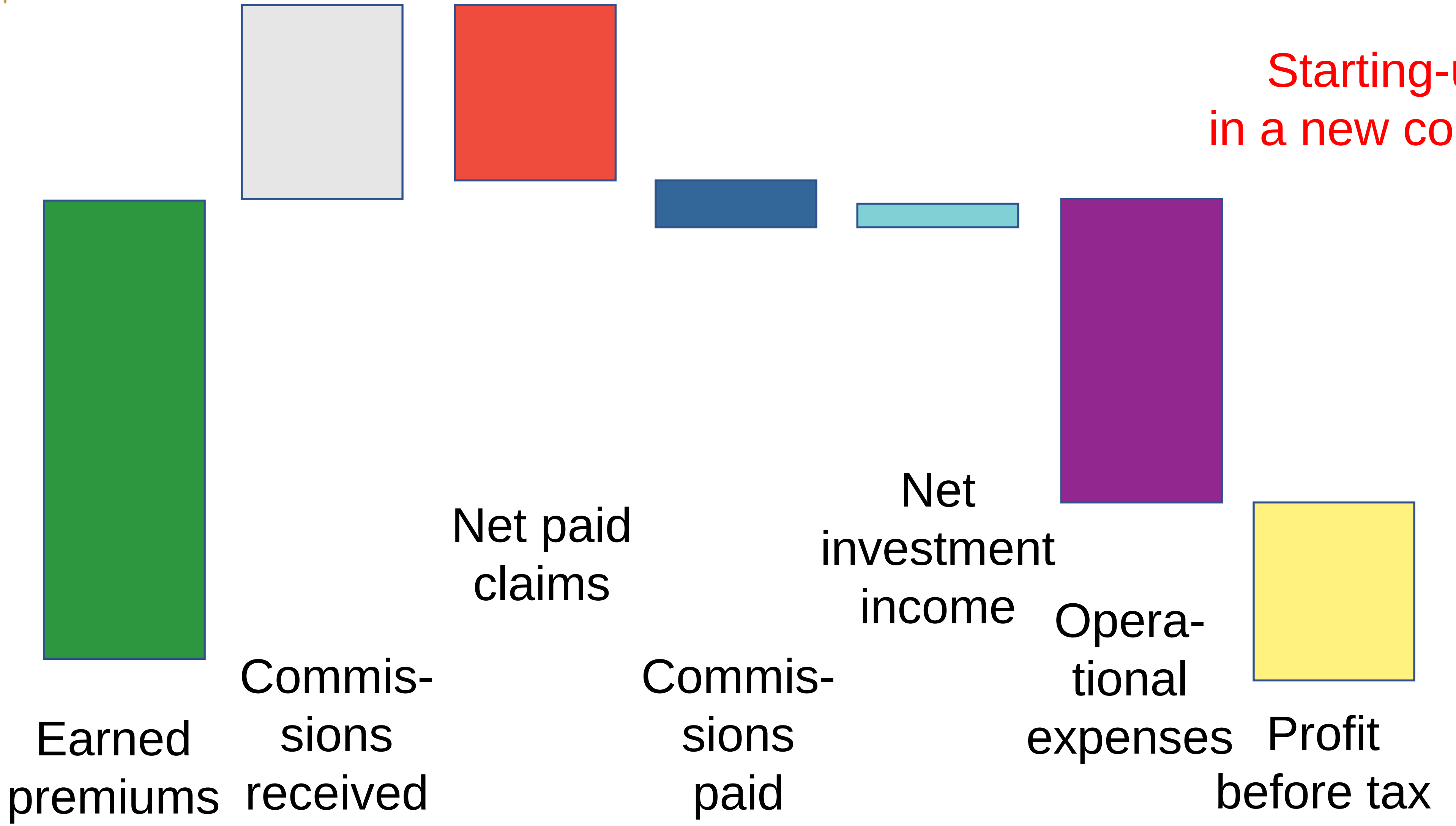
What happened? 4

Extending agent network

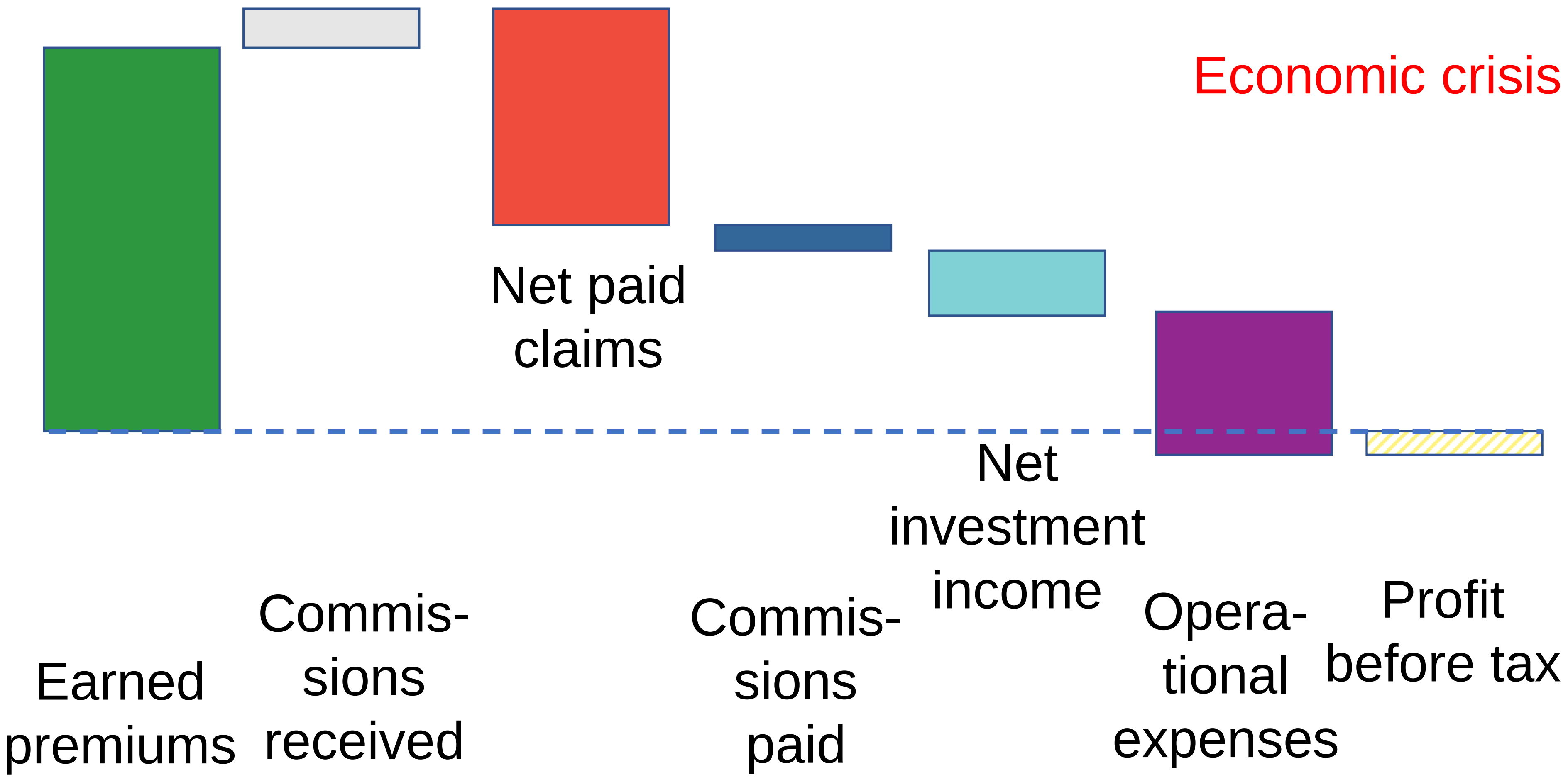


What happened? 5

Starting-up
in a new country



What happened? 6



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Business valuation: Closing revision

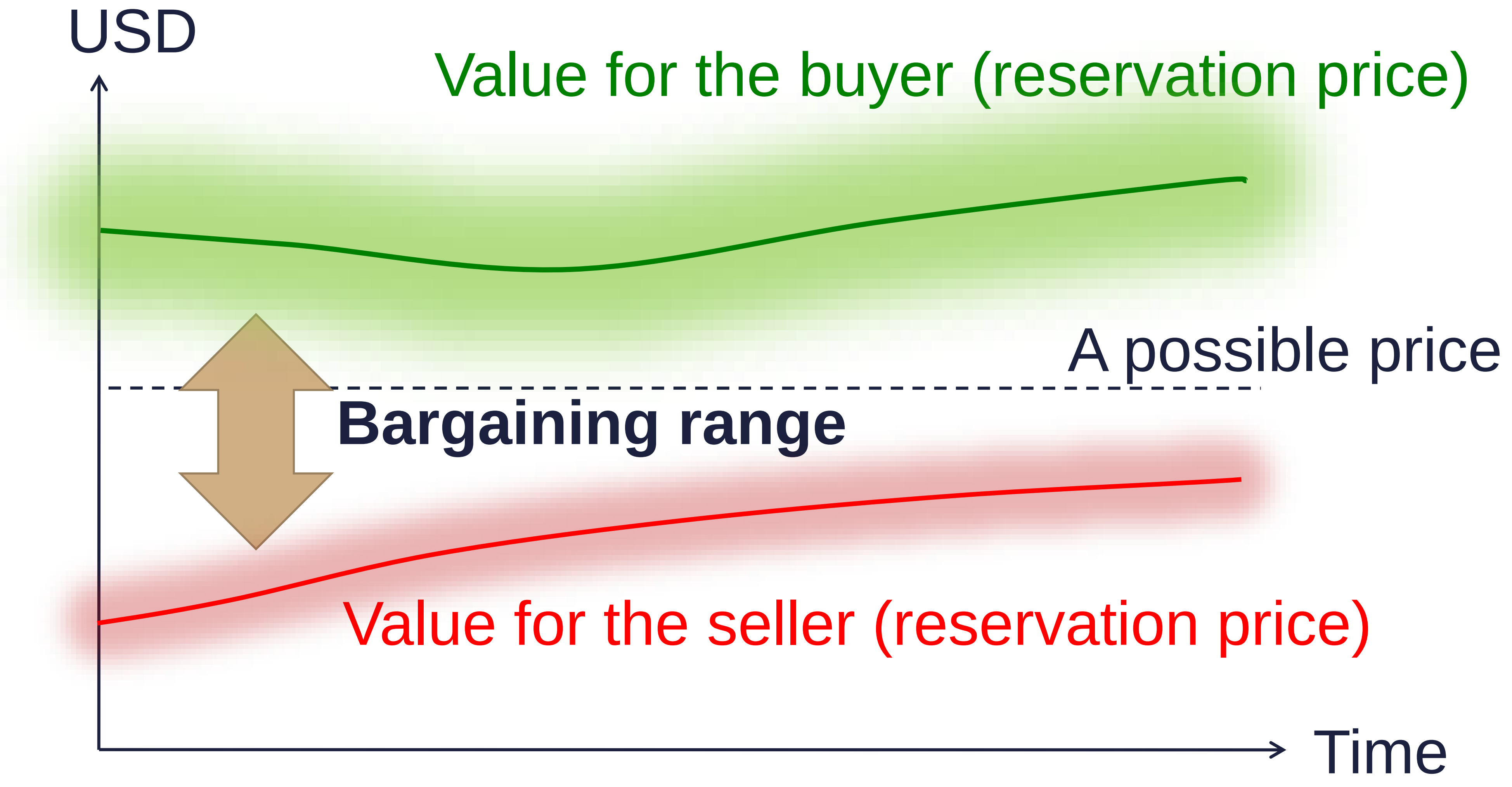
Péter Juhász, PhD, CFA

The value is different for everyone

What?	When?
Whom for?	Why?

Some value is not to be measured by money

All we may give is a range



We have at least three opportunities

Operating

Selling on the market

Liquidating

It is not only the NPV (MVA) that counts.

- Capital need
- Alternative investments
- Advantages not measurable by money

There are no universal rules.

Industry

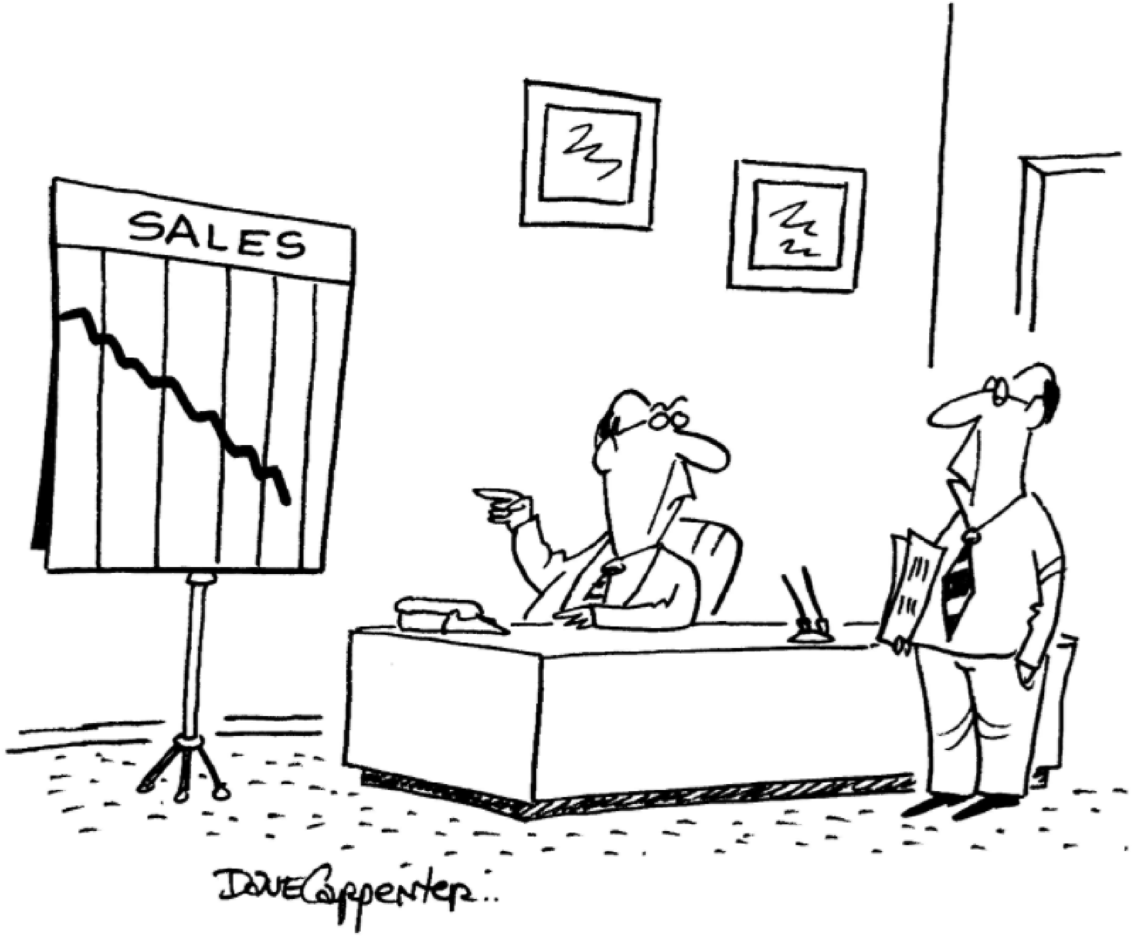
Market

Owner

Regulations

Management

Value creation lies in the culture.



" The sales team did all they could, so I'd have to say the blame for that must fall on the consumer."

Source: stevensonfinancialmarketing.wordpress.com/tag/customer-service/

The process of prediction

1. Correcting historical statements
2. Analysing statements
3. Financial planning
4. Reality check

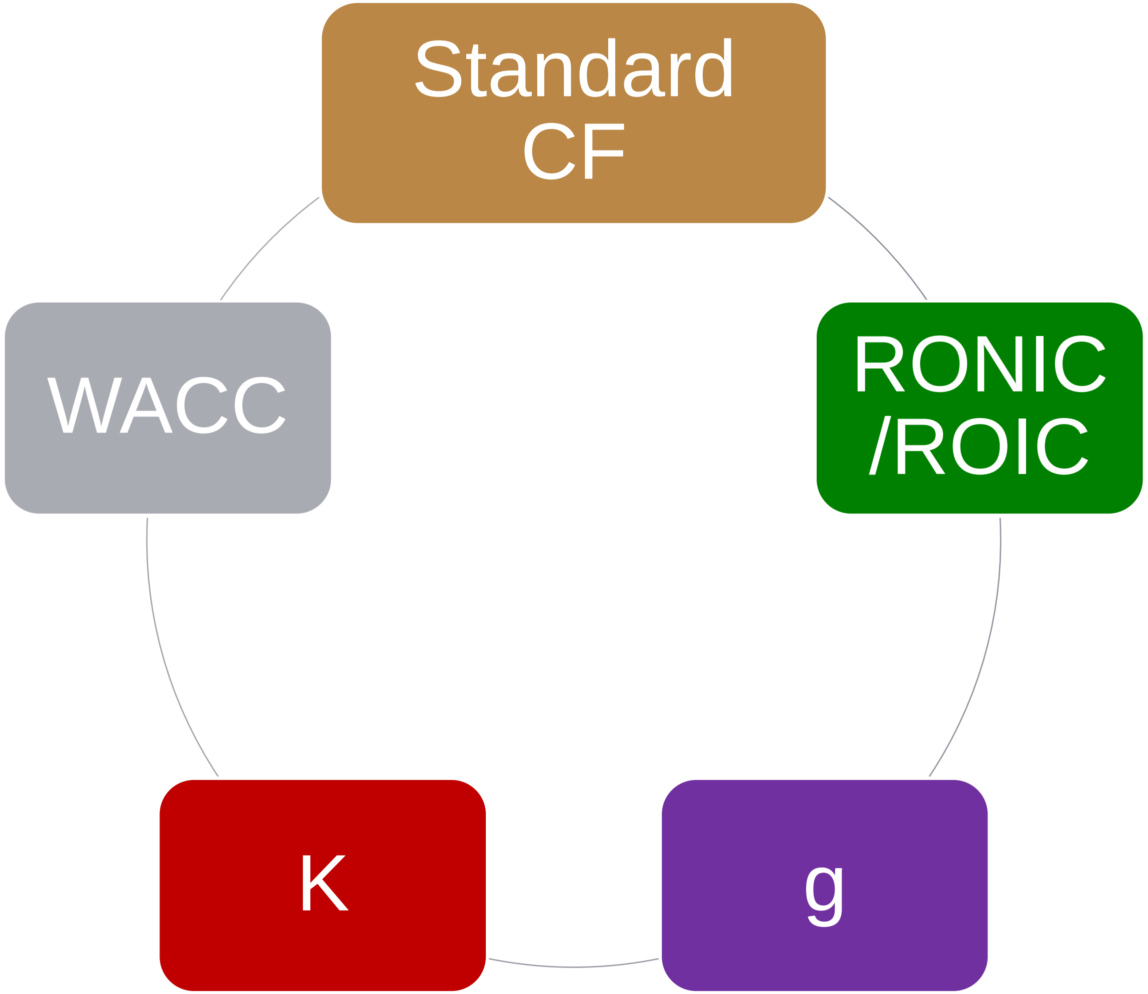
Excel calculates everything, but ...

Use only ratios for which you know

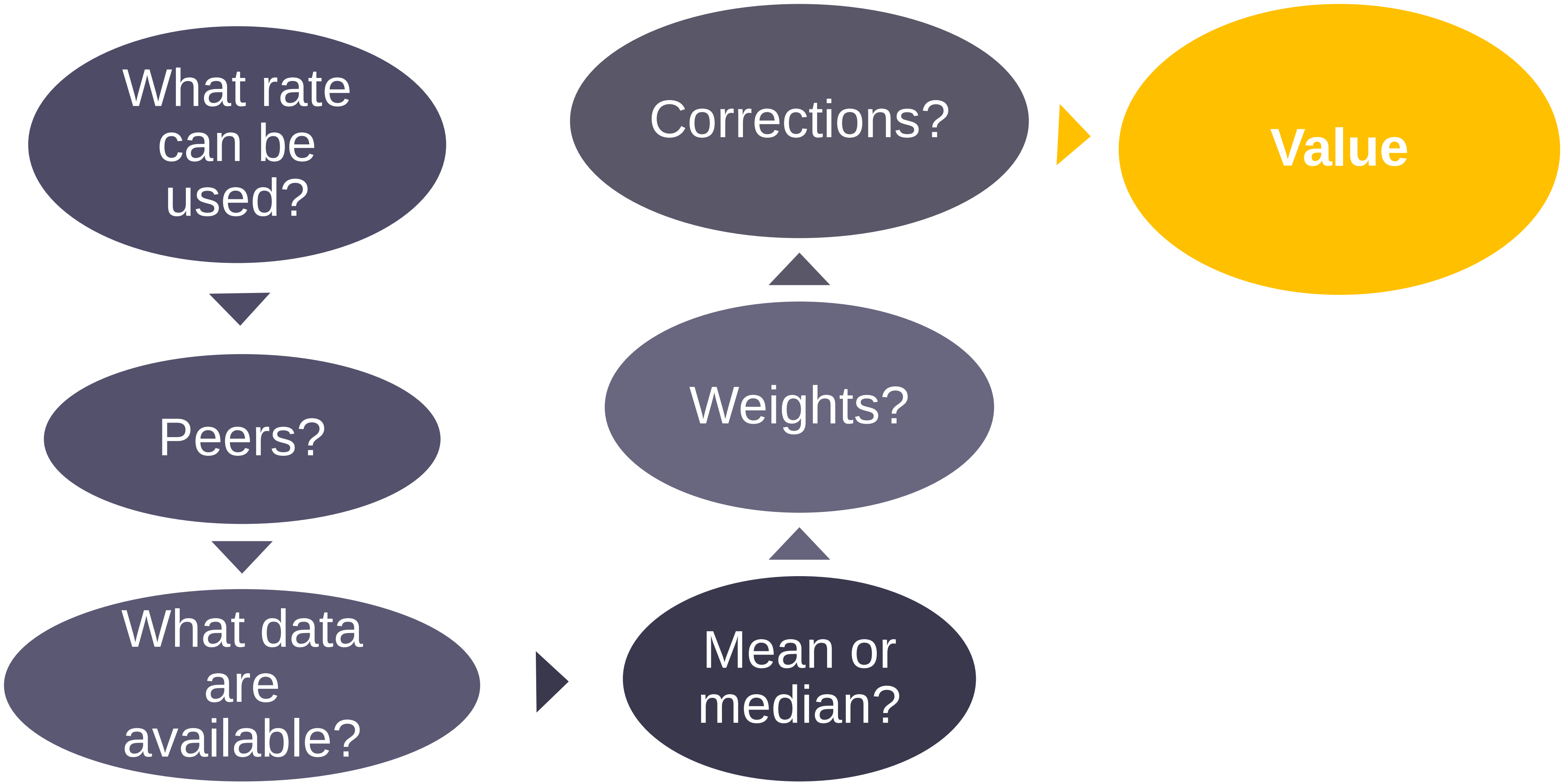
the formula, the meaning, and the limits.

(It does not matter whether it has a name or not.)

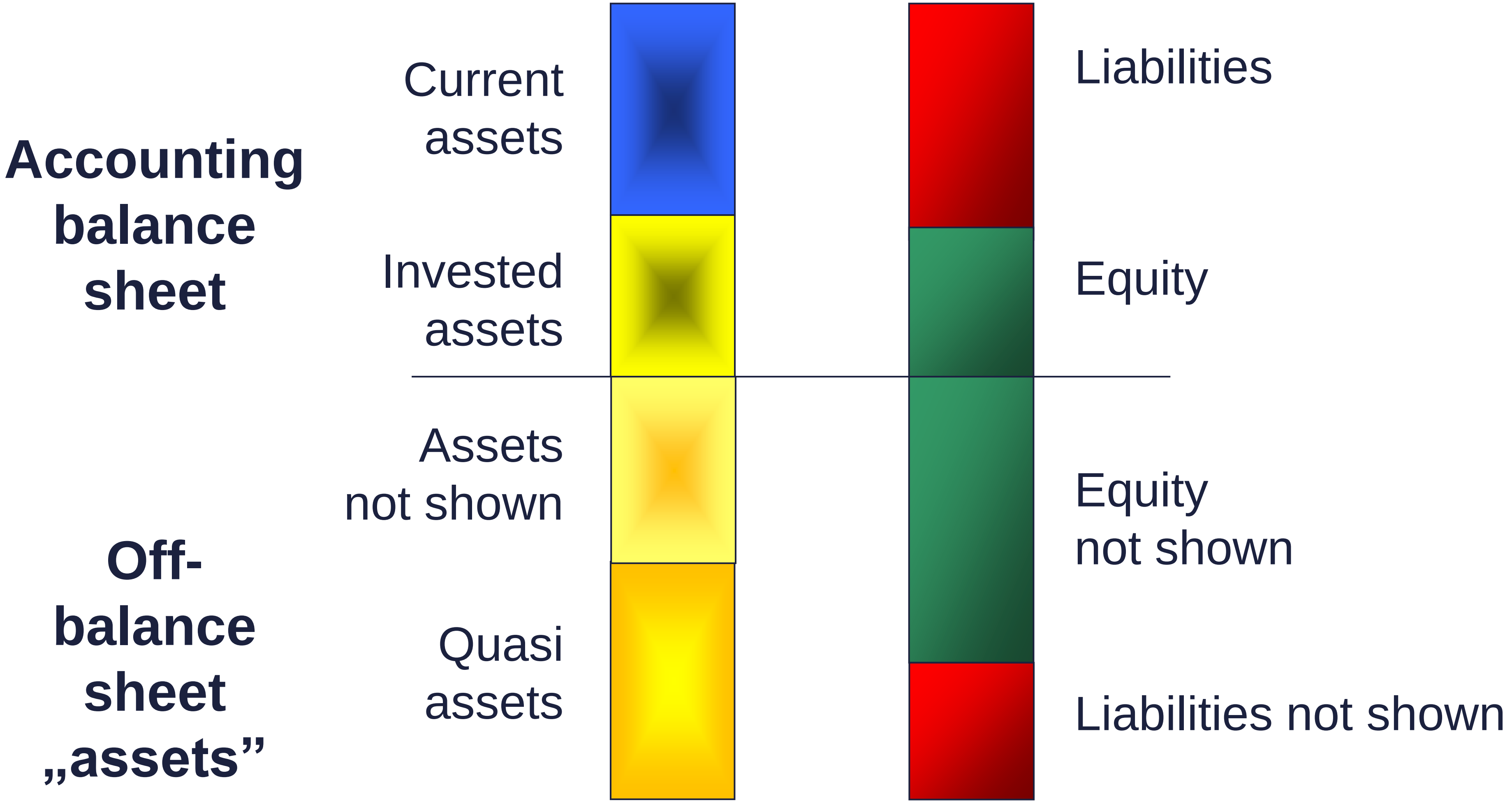
The terminal value



Multiple-based valuation



Asset-based valuation



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Business valuation approaches

	DCF	Multiples	Asset-based	Real options
Aim	Intrinsic value	Market price	Liquidation value	Decision alternatives
Key inputs	CF, Cost of capital, Planing possible	Peers with values	Asset prices, complete list of valuables	Distributions, expected values
Advan-tages	Performance clearly stated, long term focus	Quick, market judgement	Document-based	More realistic view of future
Dangers	Discounting dreams	Implicit assumptions	Quasi assets	Risk ads value

Bonus content 1

Synergy (Only quantifiable at company level, risk differs across types)

Discounts & premiums (different for each method, either on equity of for a stake)

Value of stakes The value of a single share is not simple the proportional part of equity: $MV(E/N) * N \neq MV(E)$

Private Equity (portfolio view, required rate depends on the investor not the target, pre/post money value)

Valuation in the financial service industry (different logic for planning, huge differences across sectors, equity valuation only)

Sample task 1

Calculate the business cash flow based on the information provided.

Key points:

- Corrections (liquidity debt, management fees, non-operational assets, Balance sheet and P/L effects)
- Operational cash, tax shield
- Structure

Sample task 2

Predict the financial statements of the firm based on the value drivers provided. How realistic do you think the plan is?

Key points:

- Use and understanding of the value drivers
- Links among statements (IC, dividend, retained earnings, interest)
- ROIC-RONIC-WACC; g-K-RONIC; r_D, r_A, r_E, D/V, t, WACC; g-market-GDP-financing

Sample task 3

Based on the data from the market and our competitors estimate the cost of capital for our firm.

Key points:

- r_A, WACC formulas
- CAPM, build-up models
- Turn competitor β_A into own WACC

Sample task 4

Prepare a DCF valuation based on the data provided.

Key points:

- What is the right discount rate?
- Start with the method not requiring iteration.
- Terminal value formulas

Sample task 5

Prepare (correct) the multiple-based valuation of this firm.

Key points:

- Consistency, Relevance, Uniformity.
- Is that EV (V) or P (E) that the ratio estimates?
- Peers' selection criteria are different for the four ratio groups.

Sample task 6

What is the asset-based value of the firm?

Key points:

- You should not only consider assets and liabilities in the balance sheet.
- We use market values.
- If the operational value (DCF, multiples) is below the asset value, we have to liquidate the business.

Sample task 7

Firm A wishes to buy Firm B. What is the maximum price (realistic price range)? What stake of teh joint firm do you have to offer if you can only pay X million in cash?

Key points:

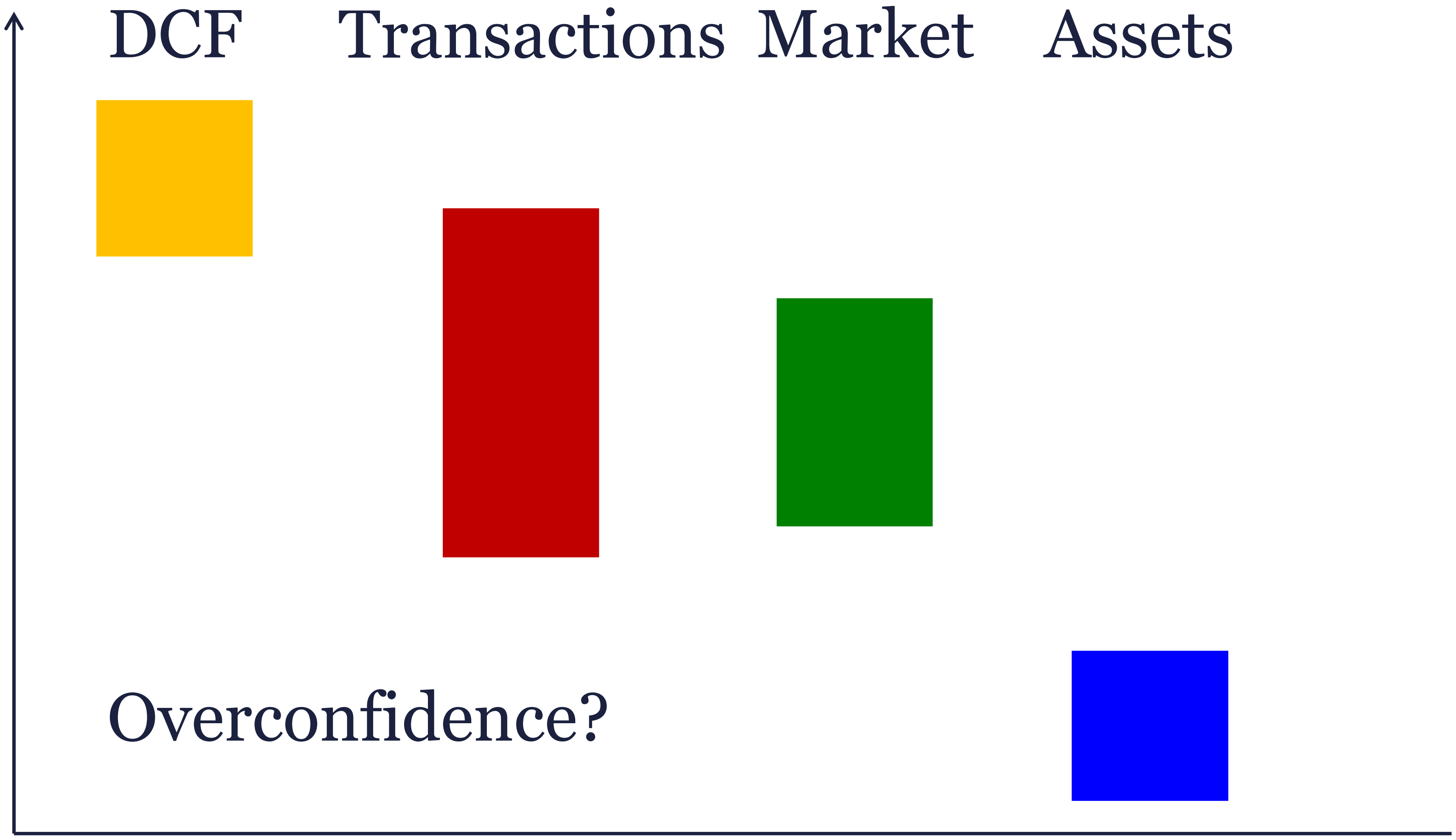
- **Synergy=V_{A+B}-V_A-V_B.**
- **Max price=E_B+Synergy=V_{A+B}-V_A-D_B**
- Buyer deducts, sellers adds transaction costs.
- If you have synergy and D/V is fix, there is a leverage effect.

Sample task 8

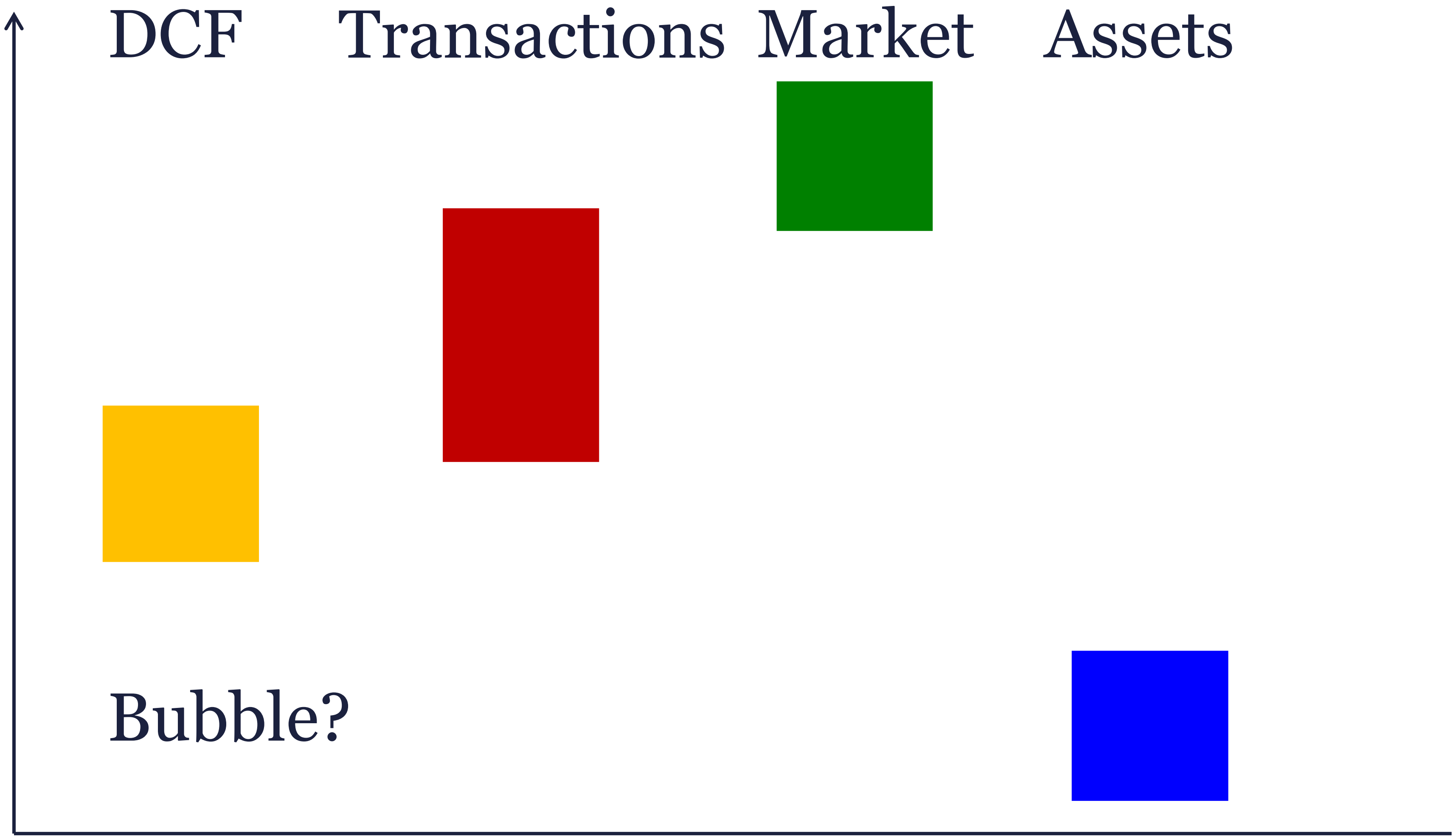
At the end of a valuation process we received the following result (E). What conclusions could you draw? What could be the fair price range?

DCF: 1000
Comparable transactions: 1200
Market multiples: 900
Asset-based: 700

What is the fair value 1?



What is the fair value 2?



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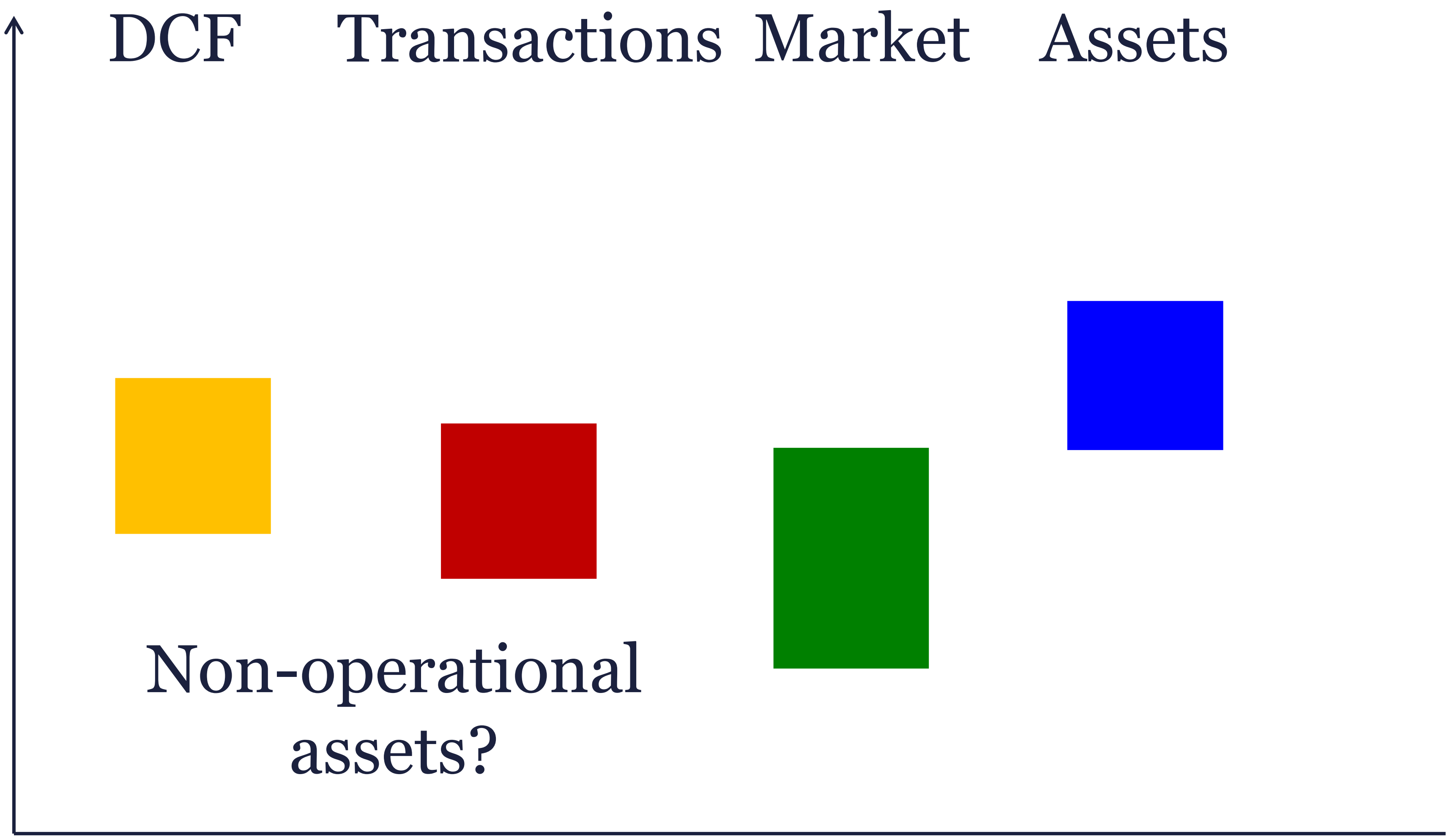
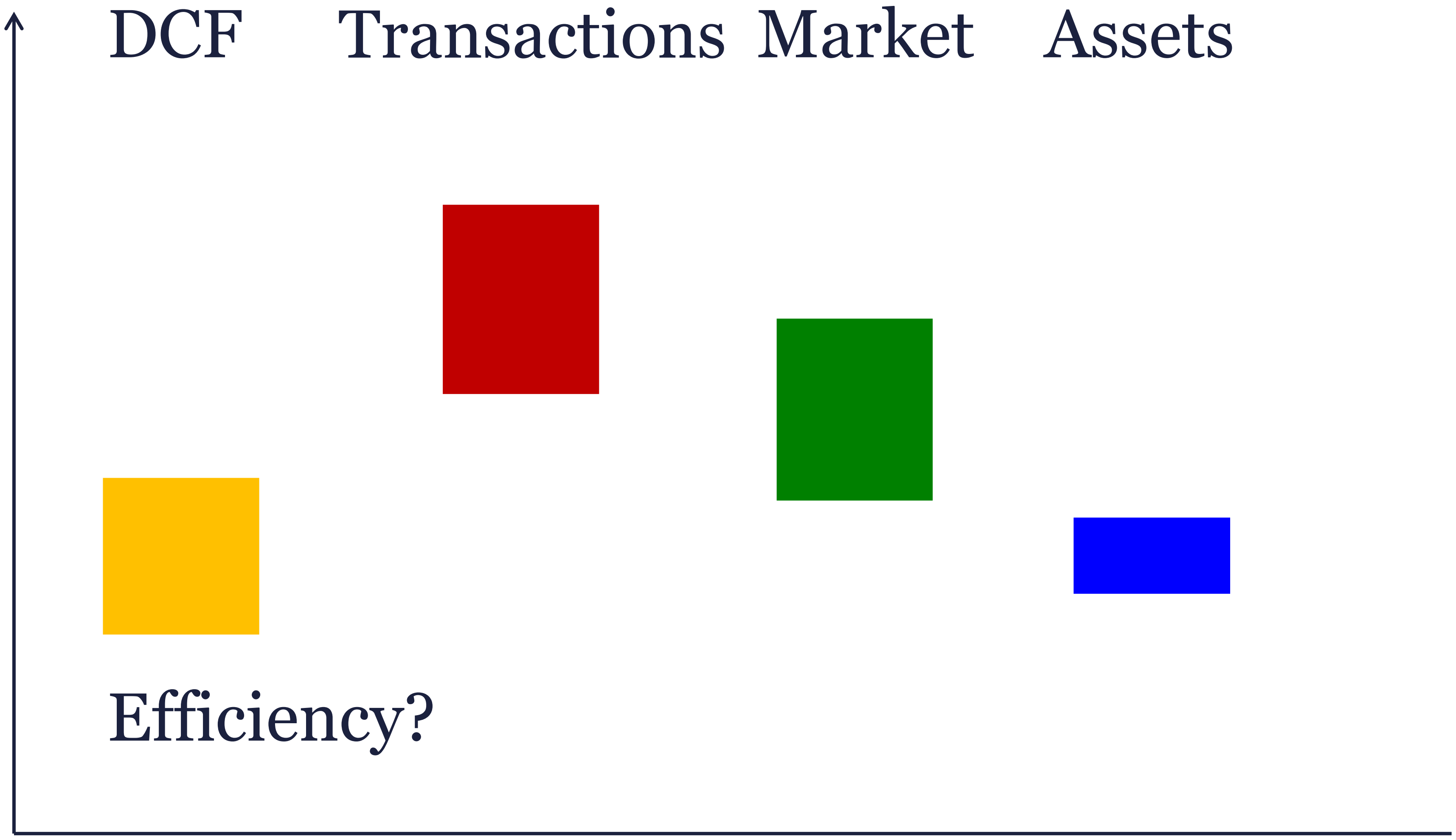
What is the fair value 3?



What is the fair value 4?



Takeaway

We have to sail.

Stay ethical. The customer comes first.

Declaration of a new doctor

Now, as a new doctor, I solemnly promise that **I will to the best of my ability serve humanity** — caring for the sick, promoting good health, and alleviating pain and suffering.

I recognise that the practice of medicine is a privilege with which comes considerable responsibility and **I will not abuse my position**. **I will practise medicine with integrity, humility, honesty, and compassion** — working with my fellow doctors and other colleagues to meet the needs of my patients.

I shall never intentionally do or administer anything to the overall harm of my patients. I will not permit considerations of gender, race, religion, political affiliation, sexual orientation, nationality, or social standing to influence my duty of care.

I will oppose policies in breach of human rights and will not participate in them**. I will strive to change laws that are contrary to my profession's ethics** and will work towards a fairer distribution of health resources.

I will assist my patients to make informed decisions that coincide with their own values and beliefs and will uphold patient confidentiality.

I will recognise the limits of my knowledge and seek to maintain and increase my understanding and skills throughout my professional life. I will acknowledge and try to remedy my own mistakes and honestly assess and respond to those of others.

I will seek to promote the advancement of medical knowledge through teaching and research. I make this declaration solemnly, freely, and upon my honour.

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1121898/

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Thank you for your attention. Any questions?